

Probability and Statistics for IC

Lecture notes

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Preface

These are notes for the course “Probability and Statistics for IC” given at EPFL during the Fall 2026. This is the second time they are used, yet they are likely to still contain typos or small blunders. In case you find some, I would be glad that you communicate them to me at: sebastien.ott@epfl.ch.

The chapters/sections/subsections/Theorems/Lemmas/proofs/... which names is preceded by the symbol “ * ” **are not part of the material which will be asked at the exam**, and will not necessarily be covered in class. They are here for those who want to go slightly deeper in the topic than what the course will cover.

Many of the examples will seem “too easy” for some people. It is intended that way: the idea behind taking “easy” examples is **not** to find interesting applications, but rather to **understand the formalism** in simple and intuitive examples.

Finally, as this course is *not* intended for mathematicians, rigour will often be put aside, and not all manipulations will duly be justified. Moreover, we will not be discussing measure theory/measurability issues, so some definitions/statements will not be entirely rigorous. In the same line of idea, many proofs are either only sketched, presented in a restricted case, or even omitted altogether, and some notions will be present only as supplementary material. The reader is encouraged to not bother with subtle questions such as “is this object well defined?”, or “does this space exist?” the answer will often be “no, but it is possible to make so that everything works out as if it was”.

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Chapter 0

Notations, glossary

0.1 Some frequently used notations

0.1.1 Summation symbol

We will frequently use the symbol $\sum$ for “summation”. Recall that one has that for numbers $a_1, a_2, \dots \in \mathbb{R}$, we can write

$$\sum_{k=1}^n a_k = a_1 + a_2 + \dots + a_{n-1} + a_n,$$

and that we can define the “infinite sum” version of this by

$$\sum_{k=1}^{\infty} a_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k,$$

provided the limit exists. We will use a small extension of that notation: if we have a finite set I , and numbers indexed by I , $a_i \in \mathbb{R}, i \in I$, we will use

$$\sum_{i \in I} a_i$$

for the “sum of all elements in the set $\{a_i : i \in I\}$ ”. Note that it is a well defined object as the order in which we sum the elements is not important: we sum only finitely many numbers.

Example 0.1.1. Let $I = \{\circ, \square, \triangle\}$. Let $a_{\circ} = 1, a_{\triangle} = 3.14, a_{\square} = 12.7$. We then have

$$\begin{aligned} \sum_{i \in I} a_i &= a_{\triangle} + a_{\circ} + a_{\square} = a_{\circ} + a_{\triangle} + a_{\square} = a_{\square} + a_{\triangle} + a_{\circ} \\ &= \dots \text{ any other permutation of } \square, \triangle, \circ \dots = 16.84. \end{aligned}$$

Defining a similar sum for infinite I 's is not so straightforward. We will say that the family of numbers $a_i, i \in I$ is *absolutely summable* if

$$\sup_{J \subset I: |J| < \infty} \sum_{j \in J} |a_j| < \infty.$$

In that case, one can define:

1. In the case where $a_i \geq 0$ for all i 's,

$$\sum_{i \in I} a_i = \sup_{J \subset I: |J| < \infty} \sum_{j \in J} a_j, \quad (1)$$

2. and in the general case $a_i \in \mathbb{R}$ for all i 's,

$$\sum_{i \in I} a_i = \sup_{J \subset I: |J| < \infty} \sum_{j \in J} a_j + \inf_{J \subset I: |J| < \infty} \sum_{j \in J} a_j.$$

In words: the supremum is “positive part of the mass” (note that it is positive), and the infimum is “the negative part of the mass” (note that it is negative). The condition of absolute summability guarantees that both are well defined.

Example 0.1.2. Let $I = \mathbb{Z}$. For $n \in \mathbb{Z}$, let $a_n = (-1)^n$. We then have

$$\sup_{J \subset \mathbb{Z}: |J| < \infty} \sum_{j \in J} |a_j| = \sup_{J \subset \mathbb{Z}: |J| < \infty} \sum_{j \in J} 1 = \sup_{J \subset \mathbb{Z}: |J| < \infty} |J| = +\infty,$$

so $\sum_{n \in \mathbb{Z}} a_n$ is *not well defined*.

We will mostly deal with finite sums, but it is worthwhile to remember how to work with infinite sums (and when it is possible to do so).

0.1.2 Integration symbol

For integrals, we will often put the dx right next to the integral sign:

$$\int_a^b dx f(x) := \int_a^b f(x) dx.$$

The advantage of this notation is that when doing integrations over multiple variables, it is easier to keep track of the domain over which each of the variables is integrated.

0.2 Symbols

$\emptyset$	Empty set
Ω	Set of realisations
$\mathcal{P}(A)$	Set of subsets of A
P	A probability measure
$P(A, B)$	The probability of A and B : $P(A, B) = P(A \cap B)$
$P(A B)$	The probability of A given B : $P(A B) = P(A \cap B)/P(B)$

0.3 Abbreviations

(s)LLN	(strong) Law of Large Numbers
CLT	Central Limit Theorem
i.i.d.	Independent Identically Distributed
CDF	Cumulative Distribution Function
MGF	Moment Generating Function
i.e.	<i>id est</i> , Latin for “that is”

0.4 English-French translations

Adequacy	Adéquation
Bias	Biais
Conditional	Conditionnel
Confidence interval	Intervalle de confiance
Correlation	Corrélation
Covariance	Covariance
Cumulative distribution function	Fonction de répartition
Density	Densité
Distribution	Distribution
Estimator	Estimateur
Expectation	Espérance
Function	Fonction
Independence	Indépendance
Law	Loi
Likelihood	Vraisemblance
Mass	Masse
Mean	Moyenne
Mean squared error	Erreur quadratique moyenne
Moment	Moment
Moment Generating Function	Fonction génératrice des moments
Random	Aléatoire
Random variable	Variable aléatoire
Sample	Échantillon
Statistic	Statistique
Variance	Variance

Chapter 1

Introduction and basics

1.1 Random phenomena, modelization

1.1.1 A first example

Let us start with an example. I roll a fair 6-faces dice. What will be the result? In theory, if one has a *perfect and complete* knowledge of all the physical system (the exact hand movement, force, skin moisture, air density, wind, composition of the table,...) one could determine the obtained value *before* the dice stops and gives its result. The problem being that obtaining all the required information is almost impossible! The result is therefore *practically un-predictable*. From our experience of the world, we thus say that “there is a probability $\frac{1}{6}$ to make a 6, a probability $\frac{1}{6}$ to make a 5, etc.”. What do we mean by this? One possible way to interpret this is to say that if we throw the dice 10'000 times, the dice will make a 6 about 1660 times ($\frac{1660}{10'000} \approx \frac{1}{6}$). And the same for 5, and for 4, etc.. More formally, our prediction for the result is based on repeated experiments, and on the *frequency* with which we see each results.

Probability theory will be the *language* we will use to speak about randomness: it will contain the description of the objects that will encode concepts such as “rolling a dice”, “looking at the result”, “the probability of making a 6”. It will give us ways to describe various phenomena by probabilistic models.

Statistics will be about *real world applications* of probability theory. Probability will permit us to define models, statistics will help us *chose a good model* for a given phenomena.

As a first contact with probability and statistics, let us formalize the example of rolling a dice. First, we describe our model (i.e.: we do probability theory).

- We first look at what are the possible realisations of our dice roll: if is one of the 6 faces of the dice. Define the *set of realisations* to be

$$\Omega_{\text{Dice}} = \{F_1, F_2, \dots, F_6\}$$

where F_i means “face with i dots”.

- We then define the probability of each realisations. If it was the first time we ever saw a dice, we would have no idea what is the probability/frequency of making a 3 or a 4, so we simply put:

$$p_i = P(\text{Dice} = F_i), \quad i = 1, \dots, 6.$$

We should have $p_i \in [0, 1]$, as an event cannot occur with negative frequency, and cannot occur more than every time. Moreover, as we get exactly one of the faces each time we throw the dice, we should have that $\sum_{i=1}^6 p_i = 1$.

- Finally, we define the observation we want to make on our experiment. In other words *what quantity do we want to measure*. We can for example take the measurement “number of dots on the face”, see Figure 1.1:

$$X : \Omega \rightarrow \mathbb{R}, \quad X(F_i) = i, \quad i = 1, \dots, 6.$$

Note that the last step might seem stupid or an un-necessary complication. Yet it is a step of fundamental importance: an experiment usually gives us a complicated object (even “the picture on the face of a dice” is not a simple object). Looking at numeric quantities gives us a much simpler object to handle (we can do more operations on a number than on “the face of a dice”). This would be the setup for a probabilistic model of the experiment “rolling a dice”. But we still have to fix something to be able to do calculations/predictions: the values of the p_i ’s!!!

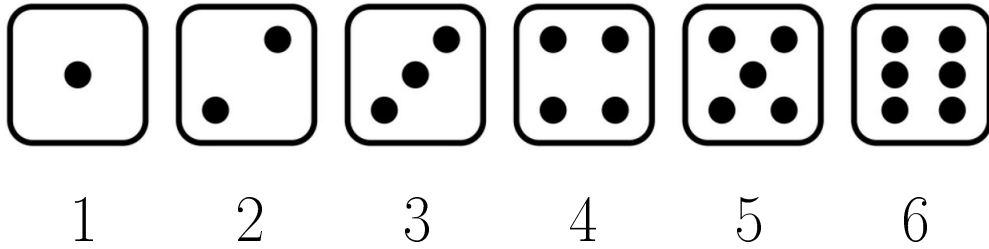


Figure 1.1: The realisations “faces of a dice” and the corresponding values of the measure “number of points on a face”

This is where statistics enter the game. We intuitively want to say that $p_i = \frac{1}{6}$ because we saw many dice roll, and we “know” that each face shows with equal frequency. Let us formalize this a bit. To measure the frequencies of each numbers, we start by performing many times the experiment: we roll the dice 10’000 times and write the results:

$F_2, F_6, F_3, F_3, F_1, F_1, F_6, F_6, F_6, F_4, F_4, F_6, F_1, F_5, F_6, F_5, F_4, F_2, F_2, F_4, F_2, F_2, F_2,$
 $F_4, F_4, F_4, F_4, F_6, F_6, F_6, F_5, F_4, F_1, F_6, F_6, F_6, F_2, F_5, F_6, F_5, F_5, F_1, F_3, F_6, F_3, F_4,$
 $F_2, F_1, F_2, F_6, \dots$

Then, we look at the number of times we did $F_1, F_2, \dots, F_6$:

$$\#F_1 = 1646, \#F_2 = 1680, \#F_3 = 1681, \#F_4 = 1666, \#F_5 = 1683, \#F_6 = 1644.$$

We can visualize them in a graphic, see Figure 1.2. We can now use our postulate that “the probability should represent the frequency of occurrence” and set

$$p_1 \approx \frac{1646}{10000}, \quad p_2 \approx \frac{1680}{10000}, \quad p_3 \approx \frac{1681}{10000}, \quad p_4 \approx \frac{1646}{10000}, \quad p_5 \approx \frac{1666}{10000}, \quad p_6 \approx \frac{1644}{10000}.$$

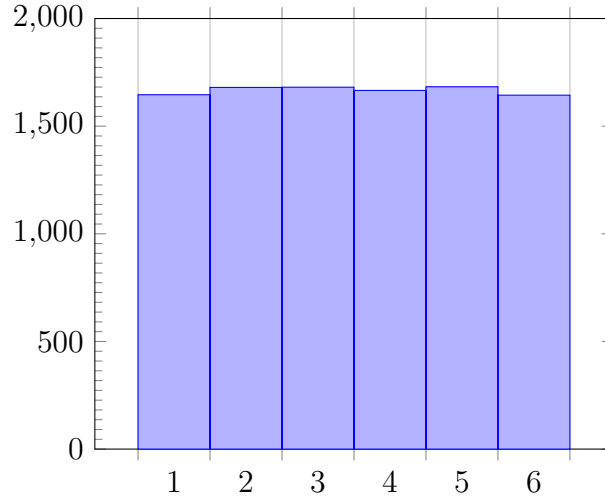


Figure 1.2: Number of occurrence of each face in 10'000 rolls of a 6-face fair dice.

We can then refine these values by trying 100'000, 1'000'000, 10'000'000,... rolls, and find that they all converge to $\frac{1}{6}$. This procedure of “finding the parameters” of the model from experiments is what is called *parametric inference*. It is a formalization of our intuition that the faces each appear with equal probability. In fact, our intuition on random phenomena is usually driven by “statistical inference at a non-conscious level”.

We can now do some calculation with our “tuned” model. For example, we cannot ask “what will be the result of the next throw”, but we could ask “what is the *average number of points* of the next throw”. This number is

$$\sum_{i=1}^6 X(F_i)P(\text{Dice} = F_i) = \sum_{i=1}^6 ip_i = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5.$$

In words: the average of a measurement is the sum of the values of the measurement weighted by the frequency at which we see the said value. We can note that it corresponds closely to the actual average of the values we observe in our 10'000 rolls: if we add up all the numbers we got, we obtain

$$1646 \cdot 1 + 1680 \cdot 2 + 1681 \cdot 3 + 1646 \cdot 4 + 1666 \cdot 5 + 1644 \cdot 6 = 34827$$

Dividing by the number of rolls, we obtain that the average roll had value

$$\frac{34827}{10000} \approx 3.48.$$

The formalism of probability theory will be the content of Chapter 2, whilst the problem of statistical inference will be discussed in Chapter 4, relying on the tools and concepts presented in Chapters 2 and 3.

1.1.2 Examples of random phenomena

The motivation behind putting some abstract formalism on something as simple as “rolling a dice”, is that we can use the same formalism to study far more complex

things. We will see here a few examples of phenomena that are well described using the language of probability. We will not touch most of these more advanced topics in this course, as they would each require a full semester to be studied. Probabilistic models are for example used to model

- **Weather evolution:** movements of large volumes of gas is highly un-predictable. So, it is often of interest to forget about trying to solve flow equations *exactly* whilst having insufficiently precise data to do so accurately, and to swap for *random evolution* with an *average movement* given by the flow equations. These models are less sensitive to data measurement errors.
- **Opinion spread:** many probabilistic models have been designed to understand the way opinion travels in a population, see for example the *voter model* https://en.wikipedia.org/wiki/Voter_model.
- **Heights/masses of individuals:** the distribution over a population of quantities such as height, mass, eyesight sharpness, is distributed in a *universal* fashion: all follow the same kind of bell-shaped curves, see Figure 1.3.
- **Insurance claims:** modelling the risk of many claims arriving in a short period of time was one of the problems leading to the development of *large deviations* in probability.
- **Stock market:** the evolution of the price of a given share when no world-changing event is occurring is very well modelled using Brownian motion (a very well studied model of particle moving on $\mathbb{R}$). See Figure 1.4.
- **Noised signal:** de-noising a signal borrows many tools from the theory of *Gaussian processes*, an important object of probability theory.

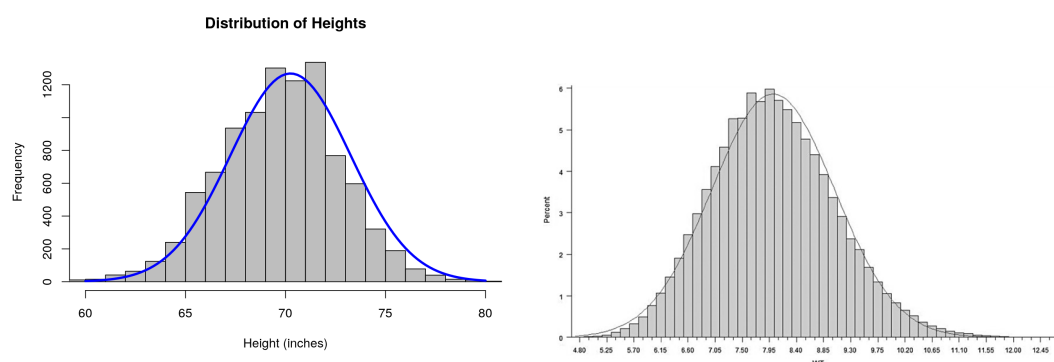


Figure 1.3: Left: Height repartition in the US (source: CDC survey). Right: weight repartition of 6 to 8 month infants in the world (source: WHO).

1.1.3 Some famous questions, problem, and observations

Monty Hall problem

The name comes from the American television game show *Let's Make a Deal* and its original host, Monty Hall. The problem was originally posed (and solved) in a letter



Figure 1.4: Top: sample of a Brownian motion. Bottom: IBM stock price evolution.

by Steve Selvin¹ to the *American Statistician* in 1975. The problem can be stated as follows:

Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?

(A reader of *Parade magazine*)

This question became famous in 1990 following the question of a reader of *Parade magazine*. The answer given to the reader was that one should systematically switch door, which is the correct answer. Yet, it triggered thousands of letters from reader (many of whom with a PhD degree...) saying that the answer was incorrect. Even amongst mathematician, there were people strongly believing that the answer was incorrect. We will formalize the question (and verify the answer) using the concept of *conditional probability*.

¹American statistician.

Martingale problem

The martingale problem was to find a strategy which would allow to systematically win at fair games (e.g.: each player bet the same amount in a coin toss game). The first recorded such strategy was to double the bet until one wins, leaving with a sure gain. It is possible to show that for *any* fair game (zero average gain/loss), *any* strategy has always a zero-gain average. We will not prove this as it requires theory that will not be part of the course, but we will quickly be able to show that the above strategy (doubling when losing) still has a zero-average gain.

Law of Large numbers

The law of large number is the result that underlies the way we found $p_1 = \dots = p_6 = \frac{1}{6}$ in the dice roll example of Section 1.1.1. In spirit it says: if one performs many times the same random experiment, the frequency with which a given event A is realized is converging to the probability of A as the number of repetitions augments.

Central Limit Theorem

The Central Limit Theorem is the formalization of the observation that for many measurements (height of individuals in a population, stock market prices, products with production errors in a factory) the *fluctuations around the average behaviour are the same*. In this sense, it is a refinement of the Law of Large Numbers: the LLN says that we are typically close to the average when looking at large samples, the CLT tells us how close for a fixed (large) sample size.

1.2 Visualizing probabilities in simple cases

1.2.1 Frequency histogram

The first tool one can use is the one we used in Section 1.1.1: we put all the measurements we made on successive realisations of an experiment in a histogram, see for example Figure 1.5. We then normalize everything by the number of observations, and we then obtain the *frequencies* of the different measurements. The mathematical concept related to this is called a *probability mass function*, see Definition 2.1.2. In many situations, the measurements can be any real number; in that case, we can discretize $\mathbb{R}$ and plot the number of measurements in intervals of size 0.1 or 0.01. Refining enough the discretization, we will often have the phenomenon that the histogram will converge to a continuous curve, the mathematical concept related to this curve is called a *probability density function*, see Definition 2.1.3.

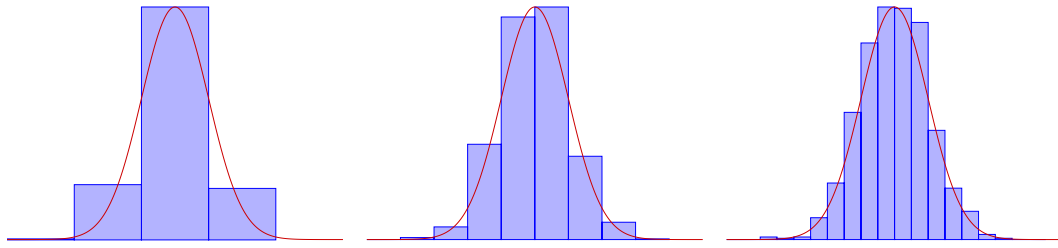


Figure 1.5: Each bar represents the proportion of measurement falling in an interval of size a amongst 1000 measurements of the same experiment. The red curve is the curve the data seem to approach. From left to right, a is equal to 2, 1, 0.5.

1.2.2 Probability/contingency table

When looking at, for example, realisations of tossing *both* a fair coin and a dice, we can represent the frequencies of each realisations in a table, see Figure 1.6. This allows to find the frequency of things such as “the coin is head” by looking at sums of columns/lines. This procedure is called *taking a marginal* and will be studied in the section devoted to random vectors, see for example Definition 2.2.9.

Dice \ Coin	Head	Tail	Marginal Dice
1	1/12	1/12	$1/12 + 1/12 = 1/6$
2	1/12	1/12	1/6
3	1/12	1/12	1/6
4	1/12	1/12	1/6
5	1/12	1/12	1/6
6	1/12	1/12	1/6
Marginal Coin	$6 \cdot 1/12 = 1/2$	1/2	

Figure 1.6: Table corresponding to the frequencies of seeing realisations of tossing a fair coin and a dice.

1.2.3 Probability tree

In many experiments, we perform (random) actions one after the other, and the outcome of an action depends on what happened before. For example, when playing Texas hold'em poker, the winning probabilities at each round strongly depends on the cards revealed so far (flop, turn and river). We can then represent the different possibilities at each stages in a *probability tree*. See a simple example in Figure 1.7.

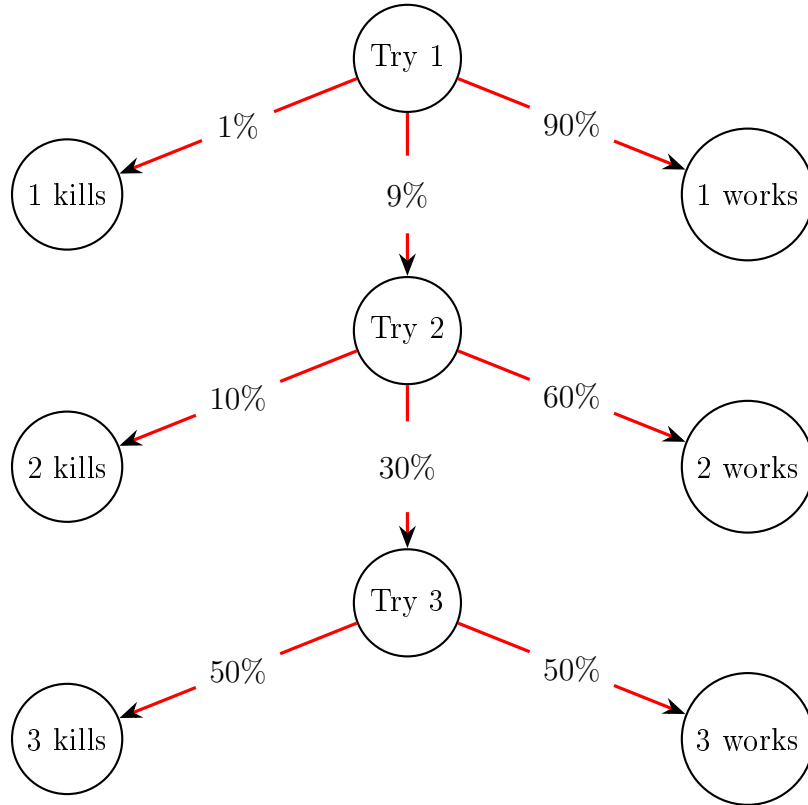


Figure 1.7: This example represent the following situation. A patient has a very severe illness. There are 3 treatments, each having some risks of killing the patient. The doctor attempt each treatment from the less risky to the most risky. Treatment 1 heals in 90% of cases, does nothing in 9% of cases, and kills the patient in 1% of cases. Treatment 2 heals in 60% of cases, does nothing in 30% and kills in 10%, and treatment 3 heals 50% of patients and kills the others. We can then compute the probability that the patient will be healed by the second treatment by multiplying the probabilities along the arrows: treatment 1 has to do nothing and treatment 2 has to work, which gives $\frac{9}{100} \cdot \frac{60}{100} = \frac{27}{500} = 5.4\%$.

1.2.4 Some introductory exercises

Exercise 1.1. We observed the eyes and hair colour of a sample of students in the University of Delaware^a. The results are summarized in the next table.

Hair \ Eyes	Brown	Blue	Hazel	Green
Black	68	20	15	5
Brown	119	84	54	29
Ginger	26	17	14	14
Blond	7	94	10	16

1. How many students participated in the study?
2. Create a table representing the frequency of seeing each hair-eyes colour combination.
3. What is the proportion of these students having black hair? and the proportion having blue eyes?
4. Consider black and brown to be dark colours and ginger, blond, blue, hazel, and green to be light colours. Amongst the students with dark hair, are there more with light or dark eyes?

Solution is given in [A.1](#).

^aData are taken from “Snee, R. D. (1974). Graphical display of two-way contingency tables. *The American Statistician*, 28, 9–12.”

Exercise 1.2. You need to buy toothpaste. There are 3 stores reasonably close to your home, numbered 1,2,3 in order of distances to your home (closest is 1). Store 1 has toothpaste in stock 90% of the time, Store 2 has toothpaste in stock 50% of the time, and Store 3 has toothpaste in stock 80% of the time. You start from the closest store, and you go back home as soon as you found your toothpaste. Represent using a tree the probabilities that you will end up in each store, or without any toothpaste. What is the probability of not finding toothpaste at all?

Solution is given in [A.1](#).

Exercise 1.3. A study on 1000 women between 25 and 29 years old asked what was their marital status. The proportions of answers for “never married”, “married”, “widow”, and “divorced” are recorded in the next table.

Marital status	never married	married	widow	divorced
Proportion	0.353	0.574	0.002	0.071

1. According to these proportions, are there women in this study with a different status than the 4 mentioned above?
2. What is the proportion of women in the study that are either divorced or never were married?
3. What is the proportion of women in the study that are not currently married?

Solution is given in [A.1](#).

1.3 Reminder I: a bit of set theory

We will rely on some mathematics notions. Most of what we will need is contained in the first semester of Analysis. The first important concept is *sets*.

Definition 1.3.1. A *set* A is a container containing *elements*. A set is entirely characterized by the elements it contains. We denote $x \in A$ for “the element x is in the set A ”, and $x \notin A$ for “the element x is not in the set A ”. Two sets are equal if they contain the same elements: $A = B$ if

for every $x \in A$, one has that $x \in B$, and for every $x \in B$, one has that $x \in A$.

Sets can be defined either by enumerating their elements, for example:

$$A = \{1, 2, 3\}, \quad A = \{\square, \triangle, \circ\},$$

or by specifying the condition an object has to fulfil to be an element of the set, for example:

$$A = \{x : x \text{ is an animal}\}, \quad A = \{n : n \text{ is an integer number}\}.$$

We will denote $A \subset B$ for “the set A is included in the set B ”, or more formally

$$A \subset B \text{ if for every } x \in A, \text{ one has that } x \in B.$$

I.e.: every element of A is also an element of B (in particular, $A \subset A$).

Definition 1.3.2. One can perform operations on sets: for two sets A, B , define the following operations.

- **Union:** $A \cup B = \{x : x \in A \text{ or } x \in B\}$.
- **Intersection:** $A \cap B = \{x : x \in A \text{ and } x \in B\}$.
- **Difference:** $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$.
- **Product:** $A \times B = \{(x, y) : x \in A \text{ and } y \in B\}$.

Some classical examples are recalled in the next table.

$\mathbb{N} = \{0, 1, 2, \dots\}$	The set of natural numbers
$\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$	The set of integers
$\mathbb{Q}$	The set of rational numbers
$\mathbb{R}$	The set of real numbers
$\mathbb{C}$	The set of complex numbers
$\mathbb{N}^*$	$\mathbb{N} \setminus \{0\}$, the same notation holds for $\mathbb{Z}, \mathbb{Q}, \dots$
$C^0(\mathbb{R}, \mathbb{R})$	The set of continuous functions from $\mathbb{R}$ to $\mathbb{R}$
$C^n(\mathbb{R}, \mathbb{R})$	The set of functions from $\mathbb{R}$ to $\mathbb{R}$ that are n times differentiable, and with $f^{(n)}$ continuous

Note that we will use the convention to include 0 in the natural numbers. We will also use two special type of set. The first one is the *empty set*:

$$\emptyset = \{\},$$

which is the set containing no element. The second one is obtained from another set: the *set of subsets of A*.

Definition 1.3.3. Let A be a set. Define

$$\mathcal{P}(A) = \{B : B \subset A\},$$

the set which elements are the sets included in A .

Lastly, we will need to talk about the size of sets.

Definition 1.3.4. Let A be a set.

- A is *finite* if contains a finite number of elements. In this case we write $|A|$ the number of elements in that set. When A is not finite, we denote $|A| = +\infty$.
- A is *countable* if there is a bijection^a $f : A \rightarrow \mathbb{N}$.
- A is *uncountable* if it is neither finite nor countable.

^aRecall that a bijection between A and B is a function $f : A \rightarrow B$ such that for every element $y \in B$, there is **exactly** one element $x \in A$ with $f(x) = y$.

Example 1.3.1. • Examples of countable sets include

$$\mathbb{N}, \quad \mathbb{Z}, \quad \mathbb{Z}^2, \quad \mathbb{Q}, \quad \mathbb{Z}^{45} \dots$$

- Examples of uncountable sets include

$$\mathbb{R}, \quad \mathbb{R}^2, \quad \mathbb{C}, \quad \mathbb{R}^{4569}, \quad \mathcal{P}(\mathbb{N}) \dots$$

1.4 Reminder II: functions

Definition 1.4.1 (Functions). Let A, B be two sets. A *function* $f : A \rightarrow B$ is a rule assigning *each* element $x \in A$ to an element $f(x) \in B$. A is called the *domain* of f , and B is called the *co-domain* of f .

Definition 1.4.2 (Indicator function). Let $A \subset B$ be two sets. The *indicator function* of A , denoted $\mathbb{1}_A : B \rightarrow \mathbb{R}$, is the function giving 1 on A and 0 elsewhere:

$$\mathbb{1}_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{else.} \end{cases}$$

Definition 1.4.3 (Image, pre-image). Let A, B be two sets and $f : A \rightarrow B$ be a function. The *image* of f is the set of elements in B that can be obtained as the image by f of an element in A :

$$\text{Image}(f) = \{y \in B : \exists x \in A, f(x) = y\}.$$

For $C \subset B$, the *pre-image* of C by f is the set

$$f^{-1}(C) = \{x \in A : f(x) \in C\}.$$

Definition 1.4.4. Let A, B be two sets and $f : A \rightarrow B$ be a function. We say that

- f is *injective* if $f(x) \neq f(y)$ whenever $x \neq y$ (different elements of A have distinct images by f).
- f is *surjective* if for any $y \in B$, there is $x \in A$ such that $f(x) = y$ (i.e.: $\text{Image}(f) = B$).
- f is *bijective* if it is both injective and surjective.

Definition 1.4.5. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two functions. We say that f is *smaller than* g , written $f \leq g$, if $f(x) \leq g(x)$ for every $x \in \mathbb{R}$. Use the same notation for functions $f, g : D \rightarrow \mathbb{R}$ with $D \subset \mathbb{R}$.

1.5 Uniform sampling and some combinatorics

1.5.1 Uniform sampling

Uniform sampling is probably the most intuitive “random phenomena” one can imagine: take n identical balls. Paint them so that they are all distinguishable (for example: write $1, 2, \dots, n$ on them, or paint them with n different colours, or put n different sky pictures on them). Then, put them all in a huge bag, shake it, and pick a ball in the bag without looking. As all the balls are identical, picking any fixed one has

probability $1/n$ (any two balls have the same probability and there are n of them). This is uniform sampling.

More formally, we can say that the *possible realisations* of the experience are the n possible balls, and that the *probability of each realisation* is $1/n$.

We can then look at more complicated probabilities. Suppose that we wrote the numbers $1, \dots, n$ on the balls. Then, what is the probability to draw either the ball 1 or the ball 2? Well, only one of them can occur, and they cannot occur at the same time, so we have

$$P(\text{Ball}_1 \text{ or } \text{Ball}_2) = P(\text{Ball}_1) + P(\text{Ball}_2) = \frac{2}{n}.$$

More generally, for $A \subset \{1, \dots, n\}$, what is the probability that we obtained a ball which number is in A ? The same reasoning as above gives $|A|/n$. A set A as above is called an *event* for the experiment: it is a subset of the set of possible realisations. We can for example formalize “getting a ball with an even number” by looking at the event

$$A_{\text{even}} = \{k \in \{1, \dots, n\} : k \text{ is even}\}.$$

The collection of all the probabilities of all the events we can look at is called a *probability measure*: it is a function from the set of events to $[0, 1]$ which attributes to every event its *probability*:

$$P(A) = \sum_{i \in A} P(\text{Ball}_i) = \frac{|A|}{n}, \quad \forall A \subset \{1, \dots, n\}.$$

We can deduce a few properties of this function.

- First of all, one has $P(\{1, \dots, n\}) = 1$ which translates into “the probability to draw a ball is 1”.
- Second, one has $P(\emptyset) = 0/n = 0$, which translates into “the probability to draw no ball at all is 0”.
- Finally, for two events $A, B \subset \{1, \dots, n\}$ with $A \cap B = \emptyset$, we have

$$P(A \cup B) = \frac{|A \cup B|}{n} = \frac{|A|}{n} + \frac{|B|}{n} = P(A) + P(B).$$

This translates into “if two events cannot occur simultaneously, the probability that at least one of them occurs is the sum of the probability of the two events”. From the probability measure point of view, this property is called *additivity of the measure*.

We will see that these three natural properties are in fact the defining properties of a probability measure in a much more general setup!

Remark 1.5.1. When computing probabilities in uniform sampling, we usually refer to the elements of the event A of which we want to compute the probability as *favourable cases* whilst the total number of elements in the set in which we do uniform sampling is usually referred to as *total number of cases*. The formula for the probability of an event is then

$$\frac{\text{number of favourable cases}}{\text{total number of cases}}.$$

Remark 1.5.2. A general principle in prediction theory is to consider that when no further information is available, the outcome of an experiment is uniformly distributed amongst the possible outcomes. One can make sense of the fact (and prove) that the uniform distribution is “the one with the least predictive power”, so we are indeed doing a “worst case scenario” when assuming things to be uniformly distributed.

1.5.2 Probabilities and some counting problems

We now introduce some classical quantities corresponding to counting problems.

Definition 1.5.1 (Powers, lists with repetition). For a set A finite, and $n \in \mathbb{N}$, the number of lists of n entries having an element of A in each entry is

$$|A^n| = |A|^n.$$

Indeed, there are $|A|$ possible choices for each entry.

Definition 1.5.2 (Factorial, list without repetitions). For $n \in \mathbb{N}$, define $0! = 1$ and for $n > 0$,

$$n! = n \cdot (n-1) \cdot (n-2) \cdots \cdots 3 \cdot 2 \cdot 1.$$

$n!$ counts the number of ways to arrange n distinguishable objects in a list of length n : there are n choices for the first element of the list (the n objects), $n-1$ choices for the second (the n objects minus the one the we put in the first place), and so on. Equivalently, $n!$ counts the number of ways to order n objects.

With the same reasoning as in the above definition, we can express the number of lists with k unrepeated entries one can form with n objects: if $k > n$ that number is 0 as we do not have enough objects to create the list. If $k \leq n$, one has n choices for the first entry, $(n-1)$ for the second, and so on up to entry number k which has $(n-k+1)$ leftover possible objects. The number of lists with k unrepeated entries one can form with n objects is thus

$$\begin{aligned} \frac{n!}{(n-k)!} &= \frac{n \cdot (n-1) \cdot (n-2) \cdots \cdots 3 \cdot 2 \cdot 1}{(n-k) \cdot (n-k-1) \cdots \cdots 2 \cdot 1} \\ &= n \cdot (n-1) \cdot (n-2) \cdots \cdots (n-k+2) \cdot (n-k+1). \end{aligned}$$

Definition 1.5.3 (Binomial coefficient). For $n, k \in \mathbb{N}$, with $k \leq n$ define

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$\binom{n}{k}$ counts the number of ways to pick k objects in a set of n distinguishable objects. One can see this as follows: there are $\frac{n!}{(n-k)!}$ lists of length k formed with the n objects, so there are $\frac{n!}{(n-k)!}$ ways to pick k objects from n objects *in an ordered fashion*. As we just want to pick k objects without any particular order, and there are $k!$ ways to order k objects, we counted every choice of k object $k!$ times. We thus need to divide $\frac{n!}{(n-k)!}$ by $k!$ to find the number of ways to pick k objects in a set of n distinguishable objects.

Exercise 1.4. In straight poker, a player receives 5 cards from a deck of 52 cards: 4 colours, 13 values (2,3,4,5,6,7,8,9,10,J,Q,K,A) for each colour. We suppose that all possible hands have the same probability. The A can also play the role of a 1 in sequences of successive values. Compute the probabilities of the following hands.

1. Royal straight flush (from 10 to A in the same colour).
2. Straight flush (5 cards of the same colour with successive values, which do not form a royal straight flush).
3. Four of a Kind (4 cards with the same value).
4. Full house (3 with the same value + 2 cards with the same value).
5. Flush (5 cards of the same colour which do not form a straight flush nor a royal straight flush).
6. Straight (5 cards with successive values which do not form a straight flush nor a royal straight flush).
7. Three of a kind (3 cards with the same value, but not a Four of a Kind and not a Full house).
8. Two pair (2 pairs of cards with the same value which do not form a Four of a Kind, and 1 card with a different value than the two pairs).
9. One pair (2 cards with the same value and 3 cards with distinct values, different from the pair).
10. High card (5 cards of distinct values which do not have all the same colour, and which do not form a straight).

Solution is given in [A.1](#)

Exercise 1.5. Consider a room with 23 people. What is the probability that two people in the room have the same birthday?

To simplify, we assume that every year has 365 days, and that the births are evenly distributed amongst these days.

Solution is given in [A.1](#)

Exercise 1.6. During a wedding, the 60 guests are seated uniformly at random in the 60 available seats. There are 10 round tables with 6 seats each. The following questions are all independent from one another.

1. What is the probability that the 5 members of family A. are seated at the same table?
2. The car of family B. broke down. The 7 members of this family did not make it to the wedding. Given that the 53 other guests are all present, what is the probability that at most one person is missing at each table?
3. What is the probability that Mr. and Mrs. C. are seated one next to the other at the same table?
4. Given that 3 men and 3 women are seated at the third table, what is the probability that there are no two men seated next to one another?

Solution is given in [A.1](#)

Chapter 2

Probabilities: starter guide

2.1 Probability spaces and measures

2.1.1 Probability spaces

We saw in Chapter 1 that the natural situation is to have a certain experiment we want to observe, that the outcome of the experiment is “random” (or non-predictable), and that there are things that we can *observe* on our experiment. Any “thing that we can observe” can be formalised by a subset of Ω , the set of realisations.

We can therefore introduce

- Ω the set of possible outcomes of the experiments, the *set of realisations* of the experiment;
- $\mathcal{F} \subset \mathcal{P}(\Omega)$ the set of *events* (corresponding to what we can observe).

A probability measure is then a consistent way to assign to each event $A \in \mathcal{F}$ its probability $P(A) \in [0, 1]$. We will see two ways to do so, corresponding to two classes of realisation sets. There is a general way to define formally what is meant by “set of events” and “consistent way to assign probabilities to events”, but it relies on measure theory, and goes beyond the scope of these notes. Those interested can get a sneak peek of the general theory in Section 2.1.4.

We will use the following “definition”. Its main issue is that we can’t take $\mathcal{P}(\Omega)$ as the set of events in general, but we will ignore that problem and work as if any subset of the realisation set was a “bona fide” event. Those interested by this issue can have a look at Section 2.1.4.

Definition 2.1.1 (Probability space). A probability space is a set of realisations, denoted Ω , together with a probability measure on Ω . A *probability measure* on Ω is a function $P : \mathcal{P}(\Omega) \rightarrow [0, 1]$ such that

1. $P(\emptyset) = 0$, $P(\Omega) = 1$.
2. If $A_i \in \mathcal{P}(\Omega)$, $i \in \mathbb{N}$ is a sequence of events with $A_i \cap A_j = \emptyset$ for $i \neq j$, then

$$P\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i \in \mathbb{N}} P(A_i).$$

The first property assesses that the probability that something happens is 1 and that nothing happens is 0, and the second says that the probability of events that cannot occur simultaneously is the sum of the probabilities of the events. From the two properties, we can deduce many more that are summarized in the next Theorem.

Theorem 2.1.1. *Let P be a probability measure on some realisation set Ω . Then,*

1. $P(\emptyset) = 0$, $P(\Omega) = 1$;
2. for any event A , $P(\Omega \setminus A) = 1 - P(A)$;
3. if two events A, B are such that $A \subset B$, $P(B) = P(A) + P(B \setminus A)$. In particular, $P(A) \leq P(B)$;
4. for two events A, B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B);$$

5. finite σ -additivity: if $n \geq 2$, and $A_1, \dots, A_n$ are events such that $A_i \cap A_j = \emptyset$ for $i \neq j$, then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i);$$

6. countable σ -additivity: if $A_1, A_2, \dots$ are events such that $A_i \cap A_j = \emptyset$ for $i \neq j$, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i);$$

7. finite σ -sub-additivity: if $n \geq 2$, and $A_1, \dots, A_n$ are events, then

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i);$$

8. countable σ -sub-additivity: if $A_1, A_2, \dots$ are events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i);$$

9. monotone convergence, increasing sequences: if $A_1, A_2, \dots$ are events such that $A_i \subset A_{i+1}$ for all i 's, then

$$\lim_{n \rightarrow \infty} P(A_n) = P\left(\bigcup_{n=1}^{\infty} A_n\right);$$

10. monotone convergence, decreasing sequences: if $A_1, A_2, \dots$ are events such that $A_i \supset A_{i+1}$ for all i 's, then

$$\lim_{n \rightarrow \infty} P(A_n) = P\left(\bigcap_{n=1}^{\infty} A_n\right).$$

It is not necessary to learn this list by heart, but it is worthwhile to convince oneself that the properties are natural and to get some intuition as to why they are true when solving the exercises. Some of them will be done in Exercise 2.3 in a restricted framework. The fourth point is a particular case of the *inclusion-exclusion principle*, see Section 2.1.5. The fifth and sixth points are a generalization of the *additivity property* that we encountered in Chapter 1: it is called *sigma-additivity* or *countable-additivity*.

We will encounter two main types of probability spaces in these notes:

- **Discrete probability spaces:** in that case, Ω is a finite or a countable set, and the set of events really is $\mathcal{F} = \mathcal{P}(\Omega)$. I.e.: we can check the realisation of any subset of the realisations space.
- **Continuous probability spaces:** in that case, $\Omega = \mathbb{R}^d$ with $d \geq 1$ integer. The set of events is called *the set of Borel sets*, and is denoted $\mathcal{F} = \mathcal{B}(\mathbb{R}^d)$. We will not go into a formal definition of this set, and we will do as if we could take $\mathcal{F} = \mathcal{P}(\mathbb{R}^d)$.

Exercise 2.1. Which statement(s) is (are) true?

1. A probability measure on a realisation space Ω is a function from Ω to $[0, 1]$.
2. The probability of the union of two disjoint events is the sum of the probabilities of the individual events.
3. It is possible to find an event A such that the probability of both the occurrence of A and of its complement is strictly greater than $1/2$.

Solution is given in A.2.

2.1.2 Discrete probability measures

The simplest case is the one of probabilities on finite or countable sets.

Definition 2.1.2. Let Ω be a finite or countable set. A *probability mass function* (or *discrete probability density function*) on Ω is a function $p : \Omega \rightarrow [0, 1]$ such that

$$\sum_{\omega \in \Omega} p(\omega) = 1.$$

The *probability measure* associated to a probability mass function p is the function $P_p : \mathcal{P}(\Omega) \rightarrow [0, 1]$ given by

$$P_p(A) = \sum_{\omega \in A} p(\omega).$$

In words: in finite or countable cases, one can specify the individual probabilities of “elementary realisations”: $p(\omega)$ is the probability that the realisation ω occurs. The probability of a given event is then the probability that one of the realisations constituting the event occurs.

Example 2.1.1. • The probability space corresponding to throwing a fair 6-faces dice with face numbered $1, 2, \dots, 6$ is $\Omega = \{1, \dots, 6\}$. Finally, the mass function is given by

$$p(i) = \frac{1}{6}, \quad i = 1, \dots, 6.$$

- The probability space corresponding to throwing a fair coin is $\Omega = \{\text{Head}, \text{Tail}\}$. Finally, the mass function is given by

$$p(\text{Head}) = p(\text{Tail}) = \frac{1}{2}.$$

Exercise 2.2. In each of the following cases, give a realisation set Ω and a probability mass function $p : \Omega \rightarrow [0, 1]$ modelling the situation.

1. Simultaneous tossing of a 1CHF coin, a 2CHF coin and a 5CHF coin.
2. We do the following procedure: first toss a biased coin that does head twice as often as it does tail. If the coin shows “head”, roll a fair 4 faces dice. If the coin shows “tail”, roll a fair 6 faces dice.
3. Toss two fair dice and look at the total number of points obtained.

Solution is given in [A.2](#).

Exercise 2.3. Let Ω be finite and $p : \Omega \rightarrow [0, 1]$ be a probability mass function.

1. Show that $P_p(\Omega) = 1$, $P_p(\emptyset) = 0$.
2. Show that if $A, B \subset \Omega$ are such that $A \cap B = \emptyset$, $P_p(A \cup B) = P_p(A) + P_p(B)$.
3. Let $n \geq 2$. Show that if $A_1, \dots, A_n \subset \Omega$ are such that $A_i \cap A_j = \emptyset$ for all $i \neq j$, then $P_p(A_1 \cup A_2 \cup \dots \cup A_n) = P_p(A_1) + P_p(A_2) + \dots + P_p(A_n)$. *Hint: remember what is a proof by induction.*
4. Show that for any $A \subset \Omega$, $P_p(A) = 1 - P_p(\Omega \setminus A)$.
5. Show that for any $A, B \subset \Omega$, $P_p(A \cup B) = P_p(A) + P_p(B) - P_p(A \cap B)$.

Solution is given in [A.2](#).

Exercise 2.4. Which of the following p are probability mass functions^a?

1. $\Omega = \{0, 1, \dots, L\}$, $p(n) = \frac{1}{L}$ for $n \in \Omega$.
2. $\Omega = \mathbb{N}^*$, $p(n) = \frac{6}{\pi^2 n^2}$ for $n \in \Omega$.
3. $\Omega = \mathbb{Z}^*$, $p(n) = \frac{1}{n^3}$ for $n \in \Omega$.
4. $\Omega = \mathbb{N}$, $p(n) = 2^{-n-1}$ for $n \in \Omega$.

Solution is given in [A.2](#).

^aRecall that $\sum_{n=1}^{\infty} n^{-2} = \frac{\pi^2}{6}$.

Exercise 2.5. We have a bag containing 10 blue balls and 15 red ones. We pick four of them out of the bag. All the balls are assumed to have the same shape and weight (uniform sampling).

1. What is the probability that we picked two blue and two red balls?
2. What is the probability that we picked **at most** one red ball?
3. What is the probability that we picked **at least** one blue ball?

Solution is given in [A.2](#).

2.1.3 Continuous probability measures

First note that in the situation where there are strictly more than countably many possible realisations (for example: measuring the height of an individual gives a real number), any attempts to proceed as we did in the previous section will result in a total fiasco: even making sense of a “sum over all reals” is not obvious. The problem is solved as follows: one cannot make sense of the probability that a drop of water falls at *precisely* one point x , but it is relatively easy to make sense of the probability that the drop falls *is a small disk* around x . This is the essence of the next definition. One can think of the infinitesimal quantity $f(x)dx$ as being the “probability that x occurs”.

Definition 2.1.3. Let $d \geq 1$. A *probability density function* on $\mathbb{R}^d$ is a Riemann integrable function $f : \mathbb{R}^d \rightarrow [0, +\infty)$ such that

$$\int_{-\infty}^{+\infty} dx_1 \cdots \int_{-\infty}^{+\infty} dx_d f(x_1, \dots, x_d) = 1.$$

The *probability measure* associated to a probability density function f is the $[0, 1]$ -valued function P_f given by

$$P_f(A) = \int_{-\infty}^{+\infty} dx_1 \cdots \int_{-\infty}^{+\infty} dx_d f(x_1, \dots, x_d) \mathbb{1}_A(x_1, \dots, x_d).$$

The above definition is slightly non-rigorous: the problem being the structure of the domain of the probability measure P_f . We will not worry too much about this problem as it will not be of fundamental importance for what we will study. Those interested by a formal solution to this issue can have a look at Section [2.1.4](#).

Remark 2.1.1. It is a good time to make a remark about what are the important objects depending on what one is trying to achieve. If we are concerned with “being able to define things properly and in the greatest generality”, looking at mass or density functions is **not** a good way to go. The best way is to **directly** define the probability measure P by specifying its value on (in theory) **every** event. This is what is explained in Section [2.1.4](#).

But from a “realistic” point of view, this is a terrible way to go: in any real world application, it is much harder to compute/evaluate/approximate **all** the numbers $P(A)$ than “just” a mass function p or a density function f . In all these notes, we will only use probability measures that are induced by a mass/density function.

Example 2.1.2. • The probability space corresponding at the experiment “sampling a uniform point on $[-1, 1]$ ” is given by

$$\Omega = \mathbb{R}, \quad f(x) = \frac{1}{2} \mathbb{1}_{[-1,1]}(x).$$

- The probability space corresponding at the experiment “looking at the point where the first rain drop of the year falls on Lausanne” is given by

$$\Omega = \mathbb{R}^2, \quad f(x) = \frac{1}{\text{Area}(\text{Lausanne})} \mathbb{1}_{\text{Lausanne}}(x),$$

where $\text{Lausanne} \subset \mathbb{R}^2$ is given by Figure 2.1. Here one could argue that if we know where the wind comes from, the distribution of the first rain drop is not uniform over the area of Lausanne. We make here the assumption that we have no information whatsoever about the situation.

- The probability space corresponding at the situation where we have 5 computers in an office, and we look at the times before the first hardware incident, is $\Omega = \mathbb{R}^5$, and f is some function which one has to chose or estimate. We could for example reasonably assume that
 - the time a computer survives is ≥ 0 ;
 - the fact that a given computer still works at time t does not depend on whether the others are still working or not;
 - at each “instant” there is a “positive probability” that a given computer breaks.

This would lead to

$$f(t_1, \dots, t_5) = \prod_{i=1}^5 \lambda e^{-\lambda t_i} \mathbb{1}_{t_i \geq 0}.$$

The $\mathbb{1}_{t_i \geq 0}$ is for the first point, the product structure reflects the second (see Exercise 2.7), $e^{-\lambda t_i}$, $\lambda > 0$, reflects that if at each instant a computer can fail, the probability that it survives for a long time decays exponentially, and the factor λ is a normalization constant so that $\int_{\mathbb{R}^5} dx f(x) = 1$.

Exercise 2.6. In which cases is the function f a probability density function?

1. $\Omega = \mathbb{R}, f(x) = \mathbb{1}_{[0,3]}(x).$
2. $\Omega = \mathbb{R}, f(x) = \frac{x}{4} \mathbb{1}_{[-1,3]}.$
3. $\Omega = \mathbb{R}, f(x) = \frac{1}{\pi} \frac{1}{1+x^2}.$
4. $\Omega = \mathbb{R}^2, f(x, y) = e^{-\sqrt{x^2+y^2}}.$
5. $\Omega = \mathbb{R}^2, f(x, y) = \mathbb{1}_{x \geq 0} \mathbb{1}_{y \geq 0} e^{-x-y}.$

Solution is given in A.2.



Figure 2.1: Lausanne district (in orange).

Exercise 2.7. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x, y) = \frac{1}{2\pi} \frac{1}{1+x^2} e^{-|y|}.$$

1. Show that f is a probability density function on $\Omega = \mathbb{R}^2$.
2. Show that for any events $A, B \subset \mathbb{R}$, we have

$$P_f(A \times B) = P_f(A \times \mathbb{R})P_f(\mathbb{R} \times B).$$

In particular, “the first and second coordinates are independent”.

Solution is given in [A.2](#).

2.1.4 *General probability spaces and measures

In the general setup, the first object of interest is *the set of events*. It is a formalization of the set of “Yes-No” questions we can ask about our experiment and of which we can check the answer by doing some measurement.

Definition 2.1.4 (Sigma-algebra). Let Ω be a set. A *sigma-algebra* on Ω is a set $\mathcal{F} \subset \mathcal{P}(\Omega)$ which satisfies

1. $\mathcal{F}$ contains the empty set ($\emptyset \in \mathcal{F}$).
2. $\mathcal{F}$ is stable by taking the complement ($A \in \mathcal{F} \implies \Omega \setminus A \in \mathcal{F}$).
3. $\mathcal{F}$ is stable by countable unions (if for all $i \in \mathbb{N}$, $A_i \in \mathcal{F}$, then $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{F}$).

The algebraic properties asked about sigma-algebras reflect respectively:

1. One can check if something occurs.
2. If one can check if the answer to a “Yes-No” question is “Yes”, then one can also check if the answer is “No”.
3. If one can answer each questions in a sequence of “Yes-No” questions, one can answer to the question “is the answer to at least one of the questions “Yes”?”.

From the algebraic properties of sigma-algebra, we can deduce the following additional properties.

Theorem 2.1.2 (Algebraic properties of sigma-algebras). *Let Ω be a set and $\mathcal{F}$ a sigma-algebra on Ω . Then all of the following hold.*

1. $\Omega \in \mathcal{F}$.
2. $\mathcal{F}$ is stable by finite intersections: if $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$.
3. If $A, B \in \mathcal{F}$, then $A \setminus B \in \mathcal{F}$.
4. $\mathcal{F}$ is stable by countable intersections: if $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcap_{i \geq 1} A_i \in \mathcal{F}$.
5. $\mathcal{F}$ is stable by increasing limits: if $A_i \in \mathcal{F}, i \geq 1$ is such that $A_i \subset A_{i+1}$, then, $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.
6. $\mathcal{F}$ is stable by decreasing limits: if $A_i \in \mathcal{F}, i \geq 1$ is such that $A_i \supset A_{i+1}$, then, $\bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$.

Definition 2.1.5 (Probability measure). Let Ω be a set and $\mathcal{F}$ a sigma-algebra on Ω . A *probability measure* (or *probability distribution* or *probability* for short) on $(\Omega, \mathcal{F})$ is a function $P : \mathcal{F} \rightarrow [0, 1]$ such that

1. $P(\Omega) = 1$.
2. If $A_i \in \mathcal{F}, i \in \mathbb{N}$ is a sequence of events with $A_i \cap A_j = \emptyset$ for $i \neq j$, then

$$P\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i \in \mathbb{N}} P(A_i).$$

Definition 2.1.6 (Probability space). A *probability space* is a triplet $(\Omega, \mathcal{F}, P)$ where Ω is a set (the set of realisations), $\mathcal{F}$ is a sigma-algebra on Ω (the set of events), and P is a probability measure on $(\Omega, \mathcal{F})$.

Example 2.1.3. Discrete probability spaces correspond to Ω finite or countable with $\mathcal{F} = \mathcal{P}(\Omega)$, and continuous probability spaces correspond to $\Omega = \mathbb{R}^d$ and $\mathcal{F} = \mathcal{B}(\mathbb{R}^d)$ the *Borel sigma-algebra*.

2.1.5 *Inclusion-Exclusion

The final generic result about the algebraic properties of probability measures that we will see is the *inclusion-exclusion principle*. It is a generalisation of the following fact that we encounter when counting objects: to count the number of objects with

property A or property B , we can count the number of objects with property A add the number of objects with property B , and correct our over-counting by removing from this the number of objects with both property A and property B (which were counted twice). In combinatorics, the inclusion-exclusion principle deals with the case where there is more than one property. In probability, we replace the number of objects having such-and-such property with the probability that such-and-such events occurs.

Lemma 2.1.3 (Inclusion-Exclusion Principle). *Let P be a probability measure on some realisation set Ω . Let $n \geq 1$ and $A_1, \dots, A_n$ be events. Then,*

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P\left(\bigcap_{j=1}^k A_{i_j}\right).$$

Moreover, for $1 \leq l \leq n/2$ integer^a,

$$P\left(\bigcup_{i=1}^n A_i\right) \begin{cases} \leq \sum_{k=1}^{2l-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P\left(\bigcap_{j=1}^k A_{i_j}\right) \\ \geq \sum_{k=1}^{2l} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P\left(\bigcap_{j=1}^k A_{i_j}\right) \end{cases}.$$

^aThere successive majorations/minorations are called *Bonferroni's inequalities*.

Remark 2.1.2. A few remarks are in order.

1. The particular case $n = 2$ is the fourth point in Theorem 2.1.1. This is the case we will encounter most of the time.
2. The Bonferroni's inequalities allow to obtain rigorous approximations on the complicated probability $P(\bigcup_{i=1}^n A_i)$ by computing the (simpler) probabilities of fewer intersections.

A very famous application of the inclusion-exclusion principle is the computation of the asymptotic probability for a uniform random permutation to have no fixed point. It is the content of Exercise 2.8.

***Exercise 2.8.** Consider the set $\{1, \dots, n\}$. Let $\mathcal{S}_n$ be the set of permutation of $1, \dots, n$ (i.e.: the set of bijections from $\{1, \dots, n\}$ to $\{1, \dots, n\}$). Call $\phi \in \mathcal{S}_n$ a *derangement* if ϕ has no fixed point^a. Denote

$$D_{n,k} = \{\phi \in \mathcal{S}_n : \phi \text{ has exactly } k \text{ fixed points}\},$$

so that the set of derangements is $D_{n,0}$. Let P_n be the uniform probability measure on $\mathcal{S}_n$.

1. Enumerate all elements of $\mathcal{S}_3$. What are the values of $P_3(D_{3,0})$, $P_3(D_{3,1})$, $P_3(D_{3,2})$, $P_3(D_{3,3})$?
2. What is $|\mathcal{S}_n|$?

For $1 \leq k \leq n$ and $1 \leq i_1 < i_2 < \dots < i_k \leq n$, denote

$$A_{n;i_1, \dots, i_k} = \{\phi \in \mathcal{S}_n : \phi(i_j) = i_j, j = 1, \dots, k\},$$

the set of permutations which fix $i_1, \dots, i_k$.

3. Show that for any fixed $i_1, \dots, i_k$ as above,

$$P_n(A_{n;i_1, \dots, i_k}) = \frac{(n-k)!}{n!}.$$

4. Show that $\mathcal{S}_n \setminus D_{n,0} = \cup_{i=1}^n A_{n;i}$.
5. Show that

$$P_n(D_{n,0}) = 1 + \sum_{k=1}^n (-1)^k \frac{1}{k!},$$

and deduce that $\lim_{n \rightarrow \infty} P_n(D_{n,0}) = e^{-1}$.

Solution is given in [A.2](#).

^aA fixed point of ϕ is a number k such that $\phi(k) = k$.

2.2 Random variables and expectation

2.2.1 Random variables

Probability spaces are the objects that model situations/experiments involving uncertainty/randomness. Random variables correspond to the measurements that we can make in these situations.

Random variables are therefore functions going from the set of realisations to the real numbers: for example, if the “experiment” is looking at all people born in 2000, one could make the measurement of the height of the first individual born that year. In general, there are some restrictions on what can a random variable be, of technical nature and of -mostly- no interest in applications. The key problem lies in what is the structure of the set of events, but, as before, we shall not worry too much about this. Those interested by these issues can have a look at [Section 2.2.2](#).

Definition 2.2.1 (Random Variable). A (real) random variable is a function from the realisation space Ω to $\mathbb{R}$. The *probability that a random variable falls in a set A* is

$$P(X \in A) := P(X^{-1}(A)).$$

In words: it is the probability that the realisation of the experiment is such that the measurement X takes a value in A .

Remark 2.2.1. We will frequently use notations similar to the following:

$$\begin{aligned} P(X = x) &\equiv P(X \in \{x\}), \\ P(X \leq x) &\equiv P(X \in (-\infty, x]), \\ P(X > x) &\equiv P(X \in (x, +\infty)). \end{aligned}$$

Remark 2.2.2. Note that we have the intuitive fact that the variable X takes values in its image with probability one:

$$P(X \in \text{Image}(X)) = P(X^{-1}(\text{Image}(X))) = P(\Omega) = 1.$$

An recurrent point about the study of random variables is that *the whole probability space is not necessarily needed* to determine the behaviour of a given random variable. Before formalizing this, let us look at an example.

Example 2.2.1. Consider three different probability spaces:

- $\Omega_1 = \{0, 1\}$, $p_1(0) = p_1(1) = \frac{1}{2}$;
- $\Omega_2 = \mathbb{R}$, $f_2(x) = \frac{1}{2}\mathbb{1}_{[0,2]}$;
- $\Omega_3 = \{a, b, c, d, e\}$, $p_3(a) = p_3(b) = p_3(c) = \frac{1}{4}$, $p_3(d) = p_3(e) = \frac{1}{8}$.

Then look at the following random variables:

$$\begin{aligned} X_1 : \Omega_1 &\rightarrow \mathbb{R}, & X_1(0) &= 0, & X_1(1) &= 1, \\ X_2 : \Omega_2 &\rightarrow \mathbb{R}, & X_2(x) &= \mathbb{1}_{(-\infty, 1]}(x), \\ X_3 : \Omega_3 &\rightarrow \mathbb{R}, & X_3(\omega) &= \begin{cases} 1 & \text{if } \omega \in \{a, b\}, \\ 0 & \text{else.} \end{cases} \end{aligned}$$

In all three cases, we have

$$\text{Image}(X_1) = \text{Image}(X_2) = \text{Image}(X_3) = \{0, 1\}.$$

Moreover, by direct calculations,

$$\begin{aligned} P_{p_1}(X_1 = 1) &= P_{f_2}(X_2 = 1) = P_{p_3}(X_3 = 1) = \frac{1}{2}; \\ P_{p_1}(X_1 = 0) &= P_{f_2}(X_2 = 0) = P_{p_3}(X_3 = 0) = \frac{1}{2}. \end{aligned}$$

In all three cases, the considered random variable take value 0 or 1, each with probability $\frac{1}{2}$! We want to say that each of these *different* random variables, each defined on a *different space*, are somehow “the same”. This will be formalized with the concept of *the law of a random variable*.

Definition 2.2.2 (Discrete random variable). Let Ω be a set of realisations, and let P be a probability measure on Ω . Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. X is *discrete* if there is $\mathcal{D}_X \subset \mathbb{R}$ countable or finite such that $P(X \in \mathcal{D}_X) = 1$. The *law of X* is then the probability measure on $\mathbb{R}$ given by

$$P_X(A) = \sum_{x \in A \cap \mathcal{D}_X} P(X = x).$$

In words, a discrete random variable is a random variable that can take only finitely or countably many values with non-zero probability.

Remark 2.2.3. Note that the set $\mathcal{D}_X$ is not unique: we can add to it any fixed $x \in \mathbb{R}$ such that $P(X = x) = 0$. The property $P(X \in \mathcal{D}_X) = 1$ is called “ X is supported on $\mathcal{D}_X$ ”. In many cases, we will simply be able to take $\mathcal{D}_X = \text{Image}(X)$: as $P(X \in \text{Image}(X)) = 1$, X is discrete as soon as $\text{Image}(X)$ is finite or countable!

Example 2.2.2. In Example 2.2.1, we could take $\mathcal{D}_{X_1} = \mathcal{D}_{X_2} = \mathcal{D}_{X_3} = \{0, 1\}$, or $= \mathbb{Z}$, or $= \{-100, 0, 1, \pi, 3\pi, 64\} \dots$

Exercise 2.9. Suppose that $X : \Omega \rightarrow \mathbb{R}$ is a random variable such that $\text{Image}(X) = \{2, 5, 7\}$. Let $p_2 = P(X = 2)$, $p_5 = P(X = 5)$, $p_7 = P(X = 7)$.

1. Show that X is a discrete random variable (i.e.: what set $\mathcal{D}_X$ can we take?).
2. What is the value of $p_2 + p_5 + p_7$?
3. Check that the function $P_X : \mathcal{P}(\mathbb{R}) \rightarrow [0, 1]$ defined in Definition 2.2.2 is a probability measure on $\mathbb{R}$.

As functions of p_2, p_5, p_7 , what are the values of

4. $P(X \in [0, 6])$?
5. $P(X \leq 3)$?

Solution is given in A.2.

Exercise 2.10. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let P be the uniform probability measure on Ω . Let $X, Y : \Omega \rightarrow \mathbb{R}$ be given by $X(\omega) = 0$ if ω is even number and $X(\omega) = \omega$ else, and $Y(\omega) = 0$ if ω is an odd number and $Y(\omega) = \omega - 1$ else. Describe the laws of X and Y . What do you notice? Define $Z : \Omega \rightarrow \mathbb{R}$ by $Z = X - Y$. What is the law of Z ? What can you deduce from this about the equality between X and Y ?

Solution is given in A.2.

The second very important family of random variables are *continuous random variables*.

Definition 2.2.3. Let Ω be a set of realisation, and let P be a probability measure on Ω . Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. X is *continuous random variable* if there is a density function $f_X : \mathbb{R} \rightarrow [0, +\infty)$ such that

$$P(X \in A) = \int_{\mathbb{R}} \mathbf{1}_A(x) f_X(x) dx.$$

The *law of X* is then the probability measure on $\mathbb{R}$ given by P_{f_X} .

Exercise 2.11. Let $\Omega = \mathbb{R}$, $f(x) = \frac{1}{4} \mathbf{1}_{[0,4]}(x)$. Let $X : \mathbb{R} \rightarrow \mathbb{R}$ be given by $X(x) = x^2$.

1. What is the value of $P_f(X \leq 4)$? and of $P_f(X \leq 9)$?
2. Find a density function f_X such that $P_f(X \in A) = \int_{\mathbb{R}} \mathbf{1}_A(x) f_X(x) dx$.

Solution is given in A.2.

There is a similar notion of law for general random variables.

Definition 2.2.4. Let Ω be a set of realisation, and let P be a probability measure on Ω . Let $X : \Omega \rightarrow \mathbb{R}$. The *law of X* is the probability measure P_X on $\mathbb{R}$ given by

$$P_X(A) = P(X \in A).$$

Those interested by more details (in particular, what kind of restrictions should we put on X to have well defined objects) can have a look at Section 2.2.2. One can check that the particular cases of continuous and discrete random variables are particular cases of the general definition.

We denote

$$X \stackrel{\text{Law}}{=} Y$$

for “ X and Y have the same law”. There is a tool which allows to ensure that random variables have the same law: the *cumulative distribution function*.

Definition 2.2.5. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. The *cumulative distribution function of X* is defined by

$$F_X(t) = P(X \leq t).$$

Two random variables have the same law if and only if they have the same cumulative distribution functions. Note that if X is a continuous random variable with density f_X , one has that F_X is a primitive of f_X :

$$F'_X(t) = f_X(t) \quad \forall t \in \mathbb{R}.$$

From the properties of probability measures, we have that CDF are right-continuous, non-decreasing functions with $\lim_{t \rightarrow -\infty} F(t) = 0$ and $\lim_{t \rightarrow +\infty} F(t) = 1$.

The statement at the end of the definition is a non-trivial statement. We will show it in the particular case of $|\text{Image}(X)| < \infty$ in Exercise 2.14. Morally, the cumulative distribution function encodes all the information about a random variable: for each $t \in \mathbb{R}$, it gives the probability to fall on the left or right of t . We can then proceed by inclusion-exclusion to find the probability to fall in any reasonable set¹.

¹This statement is not true as such: there might be problems due to the difference between open-closed and closed-open intervals, but this is a good intuition to have in mind.

Exercise 2.12. For each of the following random variable, say whether it is discrete or continuous.

1. The number of rainy days in Lausanne during March.
2. The volume of rain in Lausanne during March.
3. The number of problems that you will correctly solve during the exam.
4. The number of points that you will get at the exam.
5. The time you will need to complete the exam.

Solution is given in [A.2](#).

Exercise 2.13. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Compute F_X in each of the following cases.

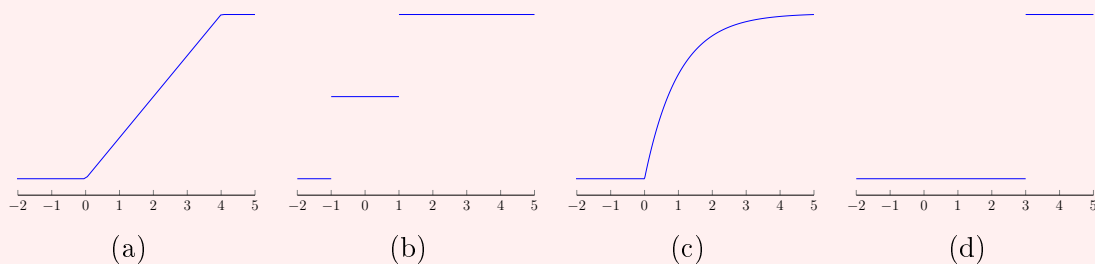
1. X is such that $P(X = -1) = P(X = 1) = \frac{1}{2}$.
2. X is such that $P(X = 3) = 1$.
3. X is a continuous random variable with density

$$f_X(x) = \frac{1}{4} \mathbf{1}_{[0,4]}(x).$$

4. X is a continuous random variable with density

$$f_X(x) = \mathbf{1}_{[0,+\infty)}(x)e^{-x}.$$

Associate each of the above with one of the following graphs.



Solution is given in [A.2](#).

***Exercise 2.14.** Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Suppose that $|\text{Image}(X)| < \infty$, and that for any $x \in \text{Image}(X)$, $P(X = x) > 0$.

1. Show that there is $\epsilon > 0$ such that for any $x \in \text{Image}(X)$,

$$P(X = x) = P(X \in (x - \epsilon, x + \epsilon)).$$

2. What are the discontinuity points of F_X , the cumulative distribution function of X ?
3. For every $x \in \text{Image}(X)$, express $P(X = x)$ in terms of F_X .
4. Let $Y : \Omega \rightarrow \mathbb{R}$ be a random variable. Suppose that $F_Y(t) = F_X(t)$ for every $t \in \mathbb{R}$. Show that for every event $A \subset \mathbb{R}$,

$$P(X \in A) = P(Y \in A).$$

Solution is given in [A.2](#).

Exercise 2.15. In each of the following situations, give a probability space associated to the situation (the probability measure does not need to be an explicit one), and random variables giving the quantities of interest.

1. We test a new model of graphic cards. We have 1000 cards of this model to perform our test on, and we want to know how likely it is for a card to reach 100 degrees during the test.
2. We pick 100 humans uniformly at random in the population and give them an IQ test. We wonder how what is the best and worst IQ amongst them.

Solution is given in [A.2](#).

2.2.2 *Measurable functions

Definition 2.2.6 (Measurable function). Let Ω_1, Ω_2 be two sets of realisations, and let $\mathcal{F}_1, \mathcal{F}_2$ be sigma-algebra on Ω_1, Ω_2 . A function $f : \Omega_1 \rightarrow \Omega_2$ is called $(\mathcal{F}_1, \mathcal{F}_2)$ -measurable if for any $A \in \mathcal{F}_2$, $f^{-1}(A) \in \mathcal{F}_1$.

A (real) random variable on a probability space $(\Omega, \mathcal{F}, P)$ is then a $(\mathcal{F}, \mathcal{B}(\mathbb{R}))$ -measurable function.

Definition 2.2.7 (Law of random variable). Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. The *law of X* is the probability measure^a P_X on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ given by

$$P_X(A) = P(X^{-1}(A)) \equiv P(X \in A).$$

^aThe fact that this is a probability measure is non-trivial, and it is very important that X is a measurable function to have this property.

2.2.3 Expectation

As everything we defined so far, expectation will come in two versions: discrete and continuous. As usual, it admits a general version which we will not discuss.

Definition 2.2.8 (Expectation). **Discrete case:** If Ω is a finite or countable set, $p : \Omega \rightarrow [0, 1]$ is a probability mass function, and $X : \Omega \rightarrow \mathbb{R}$ is a random variable such that

$$\sum_{\omega \in \Omega} |X(\omega)|p(\omega) < \infty,$$

the *expectation of X under P_p* is

$$E_p(X) := \sum_{\omega \in \Omega} X(\omega)p(\omega).$$

In this case, we say that X is P_p -integrable.

Continuous case: If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is a probability density function, and $X : \mathbb{R}^d \rightarrow \mathbb{R}$ is a random variable such that

$$\int_{\mathbb{R}^d} dx |X(x)|f(x) < \infty,$$

the *expectation of X under P_f* is

$$E_f(X) := \int_{\mathbb{R}^d} dx X(x)f(x).$$

In this case, we say that X is P_f -integrable.

General case: If Ω is a set of realisation, and P is a probability measure on Ω , we will denote $E_P(X)$ the *expectation of X under P* .

It is instructive to look at what these definitions mean in the case $\text{Image}(X)$ is finite or countable. In that case, one has

$$\begin{aligned} E_p(X) &= \sum_{\omega \in \Omega} p(\omega)X(\omega) = \sum_{\omega \in \Omega} \sum_{x \in \text{Image}(X)} p(\omega)X(\omega)\mathbb{1}_{X(\omega)=x} \\ &= \sum_{x \in \text{Image}(X)} x \sum_{\omega \in \Omega} p(\omega)\mathbb{1}_{X(\omega)=x} = \sum_{x \in \text{Image}(X)} xP_p(X = x), \end{aligned} \quad (2.1)$$

and

$$\begin{aligned} E_f(X) &= \int_{\mathbb{R}^d} dy f(y)X(y) = \int_{\mathbb{R}^d} dy \sum_{x \in \text{Image}(X)} f(y)X(y)\mathbb{1}_{X(y)=x} \\ &= \sum_{x \in \text{Image}(X)} x \int_{\mathbb{R}^d} dy f(y)\mathbb{1}_{X(y)=x} = \sum_{x \in \text{Image}(X)} xP_f(X = x). \end{aligned} \quad (2.2)$$

We recover the intuitive interpretation of what “expected value” should be: it is a weighted mean, with weight representing the frequency with which the variable X takes a given value.

Remark 2.2.4. Note that the expected value of a random variable depends only on its law!

Example 2.2.3. Let us look at a few examples.

1. Look at a fair coin toss: $\Omega = \{\text{Head}, \text{Tail}\}$, $p(\text{Head}) = p(\text{Tail}) = \frac{1}{2}$. Take the random variable $X(\text{Head}) = 1$, $X(\text{Tail}) = -1$. Then,

$$E_p(X) = p(\text{Head})X(\text{Head}) + p(\text{Tail})X(\text{Tail}) = \frac{1}{2} - \frac{1}{2} = 0$$

2. Look again at a fair coin toss. Take the random variable $X(\text{Head}) = 10$, $X(\text{Tail}) = -2$. Then,

$$E_p(X) = p(\text{Head})X(\text{Head}) + p(\text{Tail})X(\text{Tail}) = \frac{1}{2}10 + \frac{1}{2}(-2) = 5 - 1 = 4.$$

3. Look at a fair 6-face dice roll: $\Omega = \{F1, F2, \dots, F6\}$, $p(\omega) = \frac{1}{6}$ for every $\omega \in \Omega$. Take the random variable $X(F1) = X(F3) = X(F5) = -1$, $X(F2) = X(F4) = X(F6) = 2$, then

$$\begin{aligned} E_p(X) &= p(F1)X(F1) + p(F2)X(F2) + p(F3)X(F3) \\ &\quad + p(F4)X(F4) + p(F5)X(F5) + p(F6)X(F6) \\ &= -\frac{1}{6} + \frac{1}{6}2 - \frac{1}{6} + \frac{1}{6}2 - \frac{1}{6} + \frac{1}{6}2 = \frac{1}{2}. \end{aligned}$$

Note that we could also have used (2.1), and compute

$$\begin{aligned} P_p(X = -1) &= P_p(\{F1, F3, F5\}) = p(F1) + p(F3) + p(F5) = \frac{1}{2}, \\ P_p(X = 2) &= P_p(\{F2, F4, F6\}) = p(F2) + p(F4) + p(F6) = \frac{1}{2}. \end{aligned}$$

Which gives $E_p(X) = 2P_p(X = 2) - P_p(X = -1) = 1 - \frac{1}{2} = \frac{1}{2}$.

4. Finally, look at a $\Omega = \mathbb{R}$, $f = \frac{1}{4}\mathbb{1}_{[0,4]}$. Take the random variable $X : \mathbb{R} \rightarrow \mathbb{R}$, $X(x) = x^2$. Then,

$$E_f(X) = \int_{-\infty}^{+\infty} x^2 \frac{1}{4}\mathbb{1}_{[0,4]}(x)dx = \frac{1}{4} \int_0^4 x^2 dx = \frac{1}{4} \left(\frac{4^3}{3} - \frac{0^3}{3} \right) = \frac{16}{3}.$$

The properties of expectation (which are direct generalisations of the properties of sums and integrals) are summarized in the next Theorem.

Theorem 2.2.1. Let Ω be a realisation set, and P a probability measure on Ω . Let $X, Y : \Omega \rightarrow \mathbb{R}$ be two random variables. Then^a,

1. *linearity*: for any $a, b \in \mathbb{R}$, $E_P(aX + bY) = aE_P(X) + bE_P(Y)$;
2. *ordering*: if $P(X \geq Y) = 1$, $E_P(X) \geq E_P(Y)$. In particular,
 - if $P(X \geq 0) = 1$, then $E_P(X) \geq 0$;
 - if $P(a \leq X \leq b) = 1$, then $a \leq E_P(X) \leq b$;
 - $|E_P(X)| \leq E_P(|X|)$.

^aTo be more precise, we should additionally require that $E_P(|X|), E_P(|Y|) < \infty$.

In words: expectations behave exactly as (improper) integrals, or (infinite) sums. We will not worry too much about the usual mathematical problems encountered when exchanging order of limits/sums/integrals, in all the cases considered in this course, the exchange will be allowed and justifiable.

Exercise 2.16. Let Ω be a **finite** realisation set, $p : \Omega \rightarrow [0, 1]$ a probability mass function, and P_p the associated probability measure on Ω . Let $X, Y : \Omega \rightarrow \mathbb{R}$ be two random variables.

1. Show that for $a, b \in \mathbb{R}$, $E_p(aX + bY) = aE_p(X) + bE_p(Y)$.
2. Show that if $X \leq Y$, then $E_p(X) \leq E_p(Y)$.

Solution is given in [A.2](#).

Exercise 2.17. Let X the the random variable which gives the life duration of a television (in years). Assume that X is a continuous random variable with density given by

$$f_X(x) = \mathbf{1}_{[0,+\infty)}(x) \frac{1}{10} e^{-x/10}$$

1. What is the probability that the television lasts at least 10 years?
2. What is the average time you will be able to keep your television without repairing it?

Solution is given in [A.2](#).

Exercise 2.18. A casino wants to offer the following game. The player pays a sum of money, x CHF, to start playing. The player starts with a $G_0 = 2$ CHF initial gain. Then, at every step, the player can chose to either take the current gain, or to play a fair coin toss to double his gain or lose everything. The game stops when the player choses to exit or loses. Imagine the casino has infinite money, so that the game can carry on for as long as needed. What is the amount of money x that the player should pay in order for the casino to not be at lose on average?

Solution is given in [A.2](#).

2.2.4 Transfer Theorem

The Transfer Theorem (also called *Law of the unconscious statistician*) formalizes an intuitive fact: for example, if we have a random variable X that takes values x_1, x_2, x_3 with probabilities p_1, p_2, p_3 , and we have a function $g : \mathbb{R} \rightarrow \mathbb{R}$, then, intuitively, the expectation of the random variable $g(X)$ should be

$$E(g(X)) = p_1 g(x_1) + p_2 g(x_2) + p_3 g(x_3).$$

But there could be a subtlety hiding has, for example, $g(x_1) = g(x_2) \neq g(x_3)$ is possible and we do not get (a priori) the same expression as the definition: we would have $\text{Image}(g(X)) = \{g(x_1), g(x_3)\}$, and

$$E(g(X)) = g(x_1)P(g(X) = g(x_1)) + g(x_3)P(g(X) = g(x_3)).$$

The Transfer Theorem says that the definition of expectation is so that the intuition is correct and that no weird monsters are hiding in the closet.

Theorem 2.2.2 (Transfer Theorem). *Let X be a random variable. Then,*

1. *if X is a discrete random variable, for any $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(X)$ is a discrete random variable, and*

$$E(g(X)) = \sum_{x \in \text{Image}(X)} g(x)P(X = x)$$

as soon as the sum converges absolutely;

2. *if X is a continuous random variable, for any $g : \mathbb{R} \rightarrow \mathbb{R}$*

$$E(g(X)) = \int_{-\infty}^{+\infty} f_X(x)g(x)dx$$

as soon as the integral converges absolutely.

**Proof.* We present the proof in the case Ω finite. Extension to other cases uses the same idea but requires some care when exchanging order of summation/integration.

First note that in this case $|\text{Image}(X)| < \infty$, and $\text{Image}(g \circ X) < \infty$. We then have that by definition of the expectation

$$\begin{aligned} E_p(g(X)) &= \sum_{\omega \in \Omega} p(\omega)g(X(\omega)) = \sum_{\omega \in \Omega} \sum_{x \in \text{Image}(X)} \mathbb{1}_{X(\omega)=x} p(\omega)g(X(\omega)) \\ &= \sum_{\omega \in \Omega} \sum_{x \in \text{Image}(X)} \mathbb{1}_{X(\omega)=x} p(\omega)g(x) = \sum_{x \in \text{Image}(X)} g(x) \sum_{\omega \in \Omega} \mathbb{1}_{X(\omega)=x} p(\omega) \\ &= \sum_{x \in \text{Image}(X)} g(x)P_p(X = x), \end{aligned}$$

where we used that $\sum_{x \in \text{Image}(X)} \mathbb{1}_{X(\omega)=x} = 1$ for any $\omega \in \Omega$ as X is a function, we used that the order in which we perform a finite sum does not matter to swap the summation order, and we used that by definition

$$P_p(X = x) = \sum_{\omega \in \Omega} p(\omega) \mathbb{1}_{X(\omega)=x}.$$

□

2.2.5 Random vectors

Definition 2.2.9. Let Ω be a realisation set, and P a probability measure on Ω . Let $d \geq 1$. A *random vector of dimension d* is a function $X : \Omega \rightarrow \mathbb{R}^d$. We will denote

$$X(\omega) = (X_1(\omega), \dots, X_d(\omega)).$$

The functions $X_i : \Omega \rightarrow \mathbb{R}$ are random variables. They are called the *marginals* of X . The *cumulative distribution function* (CDF) of a random vector $X : \Omega \rightarrow \mathbb{R}^d$ is given by $F_X : \mathbb{R}^d \rightarrow [0, 1]$,

$$F_X(t_1, \dots, t_d) = P(X_1 \leq t_1, \dots, X_d \leq t_d).$$

Remark 2.2.5. Random vectors are simply a finite sequence of random variables arranged in an array.

In the same spirit, we can define complex random variables.

Definition 2.2.10. Let Ω be a realisation set, and P a probability measure on Ω . A *complex random variable* is a function $X : \Omega \rightarrow \mathbb{C}$. The real and imaginary parts of X are then random variables.

In the same way we defined discrete and continuous random variables, we can define discrete and continuous random vectors.

Definition 2.2.11. Let Ω be a realisation set, and P a probability measure on Ω . Let $X : \Omega \rightarrow \mathbb{R}^d$ be a random vector. We say that

- X is a *discrete random vector* if there is a finite or countable set $\mathcal{D}_X \subset \mathbb{R}^d$ with $P(X \in \mathcal{D}_X) = 1$;
- X is a *continuous random vector* if there is a density function $f_X : \mathbb{R}^d \rightarrow [0, +\infty)$ such that

$$P(X \in A) = \int_{\mathbb{R}^d} \mathbb{1}_A(x) f_X(x) dx$$

In the case of continuous random vector $X : \Omega \rightarrow \mathbb{R}^d$, we have that the marginals of X are themselves continuous random variables, and we can obtain a density for a fixed marginal by integrating the other coordinates: if $X : \Omega \rightarrow \mathbb{R}^d$ has density f_X , the density of the marginal X_i , f_{X_i} , is given by

$$f_{X_i}(x_i) = \int_{\mathbb{R}} dx_1 \cdots \int_{\mathbb{R}} dx_{i-1} \int_{\mathbb{R}} dx_{i+1} \cdots \int_{\mathbb{R}^d} dx_d f_X(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_d).$$

Example 2.2.4. Consider (X, Y) a uniform random vector in the unit disc:

$$f_{(X,Y)}(x, y) = \frac{1}{\pi} \mathbb{1}_{[0,1]}(x^2 + y^2).$$

The first marginal, X , of this random vector is then a continuous random variable with density given by $f_X(x) = 0$ for $|x| > 1$, and, for $|x| \leq 1$,

$$f_X(x) = \int_{-\infty}^{\infty} f_{(X,Y)}(x, y) dy = \frac{1}{\pi} \int_{-\infty}^{\infty} \mathbb{1}_{[0,1-x^2]}(y^2) dy = \frac{1}{\pi} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy = \frac{2}{\pi} \sqrt{1-x^2}.$$

So, we obtained

$$f_X(x) = \frac{2}{\pi} \mathbb{1}_{[-1,1]}(x) \sqrt{1-x^2}.$$

Exercise 2.19. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x, y) = \frac{1}{2\pi} e^{-(x-1.7)^2/0.08} e^{-(y-100(x-1))^2/50}.$$

1. Check that f is a probability density function.^a

Let $X = (X_1, X_2)$ be a continuous random vector with density f . One can think of X as modelling the height (in meters) and weight (in kilograms) of an individual in a given population (which “fact” about height and weight is not respected by this modelling?).

2. What is the density of the random variable X_1 ?
3. What is the average height in that population?
4. What is the average weight in that population?
5. What is the probability that an individual taken uniformly at random in the population is taller than 1.8m. whilst weighting less than 70kg.? (You can answer with a definite integral, your computer can estimate it later).

Solution is given in [A.2](#).

^aYou can use $\int_{-\infty}^{+\infty} e^{-x^2/2} dx = \sqrt{2\pi}$, it will be computed later in the notes.

Exercise 2.20. Consider a random vector $X = (X_1, X_2) : \Omega \rightarrow \mathbb{R}^2$ with law $P(X = (0, 1)) = P(X = (1, 1)) = P(X = (0, 0)) = \frac{1}{3}$.

1. What is the value of $P(X_1 \leq X_2)$?
2. What is the law of X_1 ? and of X_2 ?
3. What is the expected value of $Y = X_1 X_2$?

Solution is given in [A.2](#).

2.2.6 Change of variable formula

In the case of continuous probability spaces/random vectors, we can obtain expressions for the density of the law of a random vector $g(X)$ as a function of the density of the random vector X and of the function g .

Start by a few reminders. Let $d \geq 1$, $U \subset \mathbb{R}^d$ an open set, and $\varphi : U \rightarrow \mathbb{R}^d$, $\varphi(y) = (\varphi_1(y), \dots, \varphi_d(y))$, $y = (y_1, \dots, y_d)$. Then,

- we say that φ is *continuously differentiable* on U if the partial derivatives $\frac{\partial \varphi_i}{\partial y_j}$ exists and are continuous on U ;

- we define $D_\varphi(y)$ the *Jacobian matrix* of φ at y :

$$D_\varphi(y) = \begin{pmatrix} \frac{\partial \varphi_1}{\partial y_1}(y) & \cdots & \frac{\partial \varphi_1}{\partial y_d}(y) \\ \vdots & \cdots & \vdots \\ \frac{\partial \varphi_d}{\partial y_1}(y) & \cdots & \frac{\partial \varphi_d}{\partial y_d}(y) \end{pmatrix};$$

- we denote $\det$ the determinant.

With these in hand, we can state the *change of variable formula*.

Theorem 2.2.3. *Let $d \geq 1$, $U \subset \mathbb{R}^d$ an open set, $V \subset \mathbb{R}^d$, and $\varphi : U \rightarrow V$ a continuously differentiable bijection with $\det D_\varphi(y) \neq 0$ for all $y \in U$. Then, for any function^a $f : V \rightarrow \mathbb{R}$, we have*

$$\int_U dy_1 \dots dy_d f(\varphi(y)) |\det D_\varphi(y)| = \int_V dx_1 \dots dx_d f(x).$$

^aOne should add some restriction on the functions, but we will not bother with this.

From this Theorem, we can deduce something about continuous random vectors.

Corollary 2.2.4. *Let Ω be a realisation set, P be a probability measure on Ω , and $X : \Omega \rightarrow \mathbb{R}^d$ a random vector with a density f_X . Let $U, V, \varphi : U \rightarrow V$, be as in Theorem 2.2.3. Suppose that $P(X \in U) = 1$. We then have that $Y = \varphi \circ X : \Omega \rightarrow \mathbb{R}^d$ ($Y = \varphi(X)$) is a continuous random vector with density*

$$f_Y(y) = f_X(\varphi^{-1}(y)) |\det D_{\varphi^{-1}}(y)| = \frac{1}{|\det D_\varphi(\varphi^{-1}(y))|} f_X(\varphi^{-1}(y)).$$

In the case $d = 1$, this formula simplifies to

$$f_Y(y) = f_X(\varphi^{-1}(y)) |(\varphi^{-1})'(y)| = \frac{1}{|\varphi'(\varphi^{-1}(y))|} f_X(\varphi^{-1}(y)).$$

Example 2.2.5. We can take U a uniform random variable on $[0, 1]$: U is a continuous random variable with density $f_U(x) = \mathbb{1}_{[0,1]}(x)$. Then, for $a \in \mathbb{R}$ and $r > 0$, define $X = a + rU$. Using Corollary 2.2.4 with $\varphi(x) = a + rx$, $\varphi^{-1}(x) = \frac{x-a}{r}$, we get that X is a continuous random variable with density

$$f_X(x) = f_U(\varphi^{-1}(x)) \frac{1}{\varphi'(\varphi^{-1}(x))} = \mathbb{1}_{[0,1]}(\frac{x-a}{r}) \frac{1}{r} = \frac{1}{r} \mathbb{1}_{[a, a+r]}(x).$$

So, X is a uniform random variable on $[a, a+r]$.

Remark 2.2.6 (Polar variables). It is often very useful to remember the following change of coordinates: $\varphi : (-\pi, \pi] \times [0, +\infty) \rightarrow \mathbb{R}^2$, given by

$$\varphi(\theta, r) = (\varphi_1(\theta, r), \varphi_2(\theta, r)) = (r \cos(\theta), r \sin(\theta)),$$

(θ, r) are called the *polar coordinates* in $\mathbb{R}^2$. We then have

$$\varphi^{-1}(x, y) = (\text{atan2}(y, x), \sqrt{x^2 + y^2})$$

where

$$\text{atan2}(y, x) = \begin{cases} \arctan(y/x) & \text{if } x > 0, \\ \arctan(y/x) + \pi & \text{if } x < 0, y \geq 0, \\ \arctan(y/x) - \pi & \text{if } x < 0, y < 0, \\ \frac{\pi}{2} & \text{if } x = 0, y > 0, \\ -\frac{\pi}{2} & \text{if } x = 0, y < 0, \\ \text{undefined} & \text{if } x = y = 0. \end{cases}$$

In this case, the change of coordinate is given by

$$|\det D_\varphi(\theta, r)| = \begin{vmatrix} \frac{\partial \varphi_1}{\partial r} & \frac{\partial \varphi_1}{\partial \theta} \\ \frac{\partial \varphi_2}{\partial r} & \frac{\partial \varphi_2}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{vmatrix} = r \cos^2(\theta) + r \sin^2(\theta) = r,$$

which leads to the formula

$$\int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dy f(x, y) = \int_{-\pi}^{\pi} d\theta \int_0^{\infty} dr r f(r \cos(\theta), r \sin(\theta)),$$

which can be written for short $dx dy = r d\theta dr$

Exercise 2.21. Let $X = (X_1, X_2)$ be a random vector with density

$$f_X(x, y) = \frac{15}{3!} \mathbf{1}_{[0,1]}(x) \mathbf{1}_{[0,1]}(y) (x + y)^4.$$

1. What is the density of X_1 ?
2. Give a density for the random vector

$$Y = (Y_1, Y_2) := (X_1 + X_2, X_1 - X_2).$$

3. Give a density for the random vector

$$Z = (Z_1, Z_2) := ((X_1 + X_2)^2, (X_1 - X_2)^3).$$

Solution is given in [A.2](#).

Exercise 2.22. Let X be a continuous random variable with density f_X . In each of the following cases, compute the density f_Y of $Y = g(X)$.

1. $f_X(x) = \frac{1}{\pi(1+x^2)}$ and $g(x) = x^3$.
2. $f_X(x) = \mathbb{1}_{[0,+\infty)}(x)e^{-x}$ and $g(x) = 25x^2$.
3. $f_X(x) = \frac{2}{\pi}\mathbb{1}_{[0,\pi/2]}(x)$ and $g(x) = \sin(x)$.

Solution is given in [A.2](#).

2.2.7 Moments, and Moment Generating Function

We saw that one way to characterize the law of a random variable was its cumulative distribution function. In certain circumstances, there is another very useful object to do this: the *moment generating function*.

Definition 2.2.12. Let X be a random variable. Let $p > 0$. We say that X *admits a moment of order p* if

$$E(|X|^p) < \infty.$$

When X admits a moment of order p , we define

- the *p th moment* of X : $E(X^p)$;
- the *p th absolute moment* of X : $E(|X|^p)$.

When p is an integer, we generally simply speak of “moment” while $p \notin \mathbb{N}$ is usually referred to as “fractional moment”. The case $p \rightarrow \infty$ gives rise to the notion of *essentially bounded random variable*. X is *essentially bounded by $C \in (0, +\infty)$* if $P(|X| \leq C) = 1$.

We will only consider integer moments in this course. A good point to notice is that $E(|X|^p) < \infty$ already gives some information about the probability that X takes on large values. To see this on an example, take X to be a continuous random variable with density function

$$f_X(x) = c_\alpha \frac{1}{x^\alpha} \mathbb{1}_{[1,+\infty)}(x),$$

where c_α is the constant making f_X a probability density function. Then, if we try to compute the absolute p th moment of X , we have

$$E(|X|^p) = \int_{-\infty}^{\infty} |x|^p f_X(x) dx = c_\alpha \int_1^{\infty} x^{p-\alpha} dx.$$

As seen in Analysis I, this integral is finite if and only if $p - \alpha < -1$ which is equivalent to $\alpha > p + 1$. So having large moments implies that f_X decays fast enough at infinity.

Definition 2.2.13. Let X be a random variable. We say that X *admits exponential moments of order $\delta > 0$* if

$$E(e^{\delta|X|}) < \infty.$$

When X admits exponential moments, we define the *moment generating function* of X by

$$M_X(t) = E(e^{tX}), \quad t \in (-\delta, \delta).$$

Theorem 2.2.5. Let X, Y be two random variables. Suppose that there is $\delta > 0$ such that

$$E(e^{\delta|X|}) < \infty, \quad E(e^{\delta|Y|}) < \infty.$$

Then, we have the following properties.

- X admits moments of any integer order.
- M_X is analytic^a in a neighbourhood of 0, and for any $n \in \mathbb{N}$,

$$M_X^{(n)}(0) = E(X^n).$$

- M_X, M_Y characterise X, Y :

$$M_X(t) = M_Y(t) \text{ for all } t \in (-\delta, \delta) \implies X \stackrel{\text{Law}}{=} Y.$$

^aRecall that a function h is (real)-analytic in a neighbourhood of 0 if there is $r > 0$ such that $h(z) = \sum_{k \geq 0} a_k z^k$ for some coefficients $a_k \in \mathbb{R}$ and the series is absolutely convergent for $|z| < r$. Moreover, the coefficients are given by $a_k = \frac{h^{(k)}(0)}{k!}$.

The proof of the first point is in the next exercise. Try to see why the second part of the second point is true by replacing the exponential by its Taylor series. The proof of the third point goes beyond the scope of these notes and is therefore completely omitted. Theorem 2.2.2 is particularly useful when computing moments as is shown in the next Exercise.

Exercise 2.23. Compute the first three moments of the random variable X in each of the following cases.

1. X is a continuous random variable with density $f_X(x) = \frac{9}{10} \mathbb{1}_{[1,10]}(x) \frac{1}{x^2}$.
2. X is a discrete random variable with $P(X = -1) = P(X = +2) = P(X = 0) = \frac{1}{3}$.
3. X is a discrete random variable with $P(X = -1) = P(X = +1) = \frac{1}{2}$.
4. X is a continuous random variable with $f_X(x) = \frac{1}{2} e^{-|x|}$.

Solution is given in A.2.

Exercise 2.24. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Show that if there is $\delta > 0$ such that $E(e^{\delta|X|}) < \infty$, then, for any $p \in \mathbb{N}$,

$$E(|X|^p) < \infty.$$

Can you find explicit bounds on $E(|X|^p)$ depending on p and $C = E(e^{\delta|X|})$?

Solution is given in A.2.

2.2.8 *Characteristic Function

The moment generating function is a very powerful tool, sadly it is not defined for many random variables of interest due to the fact that $E(e^{\lambda X})$ might be infinite for any

$\lambda \neq 0$. There is another tool which palliates this problem: the characteristic function.

Definition 2.2.14. Let X be a random variable. We define the *characteristic function* of X by: for $\xi \in \mathbb{R}$,

$$\varphi_X(\xi) = E(e^{i\xi X}).$$

Formally, this is the moment generating function evaluated at a purely imaginary number. It is not plagued with the convergence issues of the moment generating function: as $|e^{it}| = 1$ for all $t \in \mathbb{R}$, $E(|e^{i\xi X}|) = 1$. As it has the same “exponential form” as the moment generating function, it shares many of its properties.

Theorem 2.2.6. Let X, Y be two random variables. We have the following properties.

- for any $n \in \mathbb{N}$, if $E(|X|^n) < \infty$, then

$$\varphi_X^{(n)}(0) = i^n E(X^n).$$

- φ_X, φ_Y characterise X, Y :

$$\varphi_X(\xi) = \varphi_Y(\xi) \text{ for all } \xi \in \mathbb{R} \implies X \stackrel{\text{Law}}{=} Y.$$

2.3 Modelling random events: two examples

2.3.1 Modelling with probability space

We want to model the following experiment: flip a fair coin until you get “head”. We will model this event using an appropriate probability space. We start by finding a set of realisations. We can take the possible sequences of coin flips:

$$\Omega = \{H, TH, TTH, TTTH, \dots\}.$$

We then need to describe the probability measure on that set of realisations. To get the sequence

$$\underbrace{TT \dots TT}_n H,$$

we need to flip the coin $n + 1$ times and to get “tail” on the first n flips and “head” on the last flip which has probability $(\frac{1}{2})^{n+1}$, so we can take the probability mass function $p : \Omega \rightarrow [0, 1]$ given by

$$p(\underbrace{TT \dots TT}_n H) = \frac{1}{2^{n+1}}.$$

Our probability measure is then P_p . We can check that p is indeed a probability mass function:

$$\sum_{\omega \in \Omega} p(\omega) = \sum_{n \geq 0} \frac{1}{2^{n+1}} = \frac{1}{2} \frac{1}{1 - \frac{1}{2}} = 1.$$

Equipped with our probability space, we can start to make measurements/observations. For example, we can ask about the number of flips we need to perform in our experiment. This can be modelled using a random variable X :

$$X : \Omega \rightarrow \mathbb{R}, \quad X(\underbrace{TT \dots TT}_n H) = n + 1.$$

This is a discrete random variable as

$$\text{Image}(X) = \mathbb{N}^*.$$

We can then study its law. For any $n \in \mathbb{N}^*$, we have

$$P_p(X = n) = P_p(\underbrace{TT \dots TT}_{n-1 \text{ times}} H) = \frac{1}{2^n}.$$

We can then compute its CDF²:

$$F_X(t) = P_p(X \leq t) = \begin{cases} 0 & \text{if } t < 1, \\ \sum_{n=1}^{\lfloor t \rfloor} \frac{1}{2^n} = \frac{\frac{1}{2} - (\frac{1}{2})^{\lfloor t \rfloor + 1}}{1 - \frac{1}{2}} = 1 - 2^{-\lfloor t \rfloor} & \text{if } t \geq 1. \end{cases}$$

Finally, we can compute its expectation.

$$\begin{aligned} E_p(X) &= \sum_{n \in \text{Image}(X)} P_p(X = n)n = \sum_{n=1}^{\infty} n\left(\frac{1}{2}\right)^n = \frac{1}{2} \sum_{n=1}^{\infty} nx^{n-1} \Big|_{x=\frac{1}{2}} \\ &= \frac{1}{2} \frac{d}{dx} \sum_{n=0}^{\infty} x^n \Big|_{x=\frac{1}{2}} = \frac{1}{2} \frac{d}{dx} \frac{1}{1-x} \Big|_{x=\frac{1}{2}} = \frac{1}{2} \frac{1}{(1-x)^2} \Big|_{x=\frac{1}{2}} = 2. \end{aligned}$$

where we used the geometric series.

2.3.2 Modelling with random variables

We can take the same example as in the previous subsection: flip a fair coin until you get “head”. Note that the only thing needed to study, for example, the mean number of coin flips needed to get “head” was the random variable X . We could therefore have picked a different probability space; for example:

$$\Omega = \mathbb{R}, \quad P = P_f, \quad \text{with } f(x) = \mathbb{1}_{[0,1]}(x);$$

and defined a random variable on this space; for example:

$$Y : [0, 1] \rightarrow \mathbb{R}, \quad Y(x) = \sum_{n=0}^{\infty} \mathbb{1}_{[0, 2^{-n}]};$$

which has the same law as the variable X of the previous section. In our example: $P(Y = 0) = P_f([0, 1]^c) = 0$, and for $n \geq 1$,

$$P(Y = n) = P(Y \geq n) - P(Y \geq n+1) = P_f([0, 2^{1-n}]) - P_f([0, 2^{-n}]) = 2^{1-n} - 2^{-n} = 2^{-n}.$$

2.3.3 A general working principle

When modelling random events using random variables, it is often the case in probability theory (and in statistics) that we do not really care about the *particular probability space* we work with (i.e. the choice of Ω and P), as long as we can define on it all the *random variables/vectors* that we want to study. The consequence of this is that we will often omit the index P in the notation for expectation, or not mention precisely the realisation space Ω , and only specify the properties of the random variables/vectors that we want to study. **We will apply this principle most of the time.**

²We can use the truncated geometric series identity: $\sum_{n=i}^j q^n = \frac{q^i - q^{j+1}}{1-q}$.

2.4 Conditional probability, independence

2.4.1 Conditional probability

Let us first discuss what should conditional probability model. We saw in Chapter 1 that the probability of an event in an random experiment represents the frequency with which the event occurs if we perform several repetitions of the same experiment. Conditional probability will represent the *relative frequency*: for example, if we play Texas hold 'em poker and our hand is 7 and 10 of diamond. If we have no other information on the rest of the game (start of the round), we have a certain probability of win. Now, after the flop, we get some new information on the situation: if we see 2, 4 and ace of club, our chances of having a good hand are not so nice. On the other hand, if we see 6, queen and king of diamond, we are guaranteed to have at least a flush. Conditional probability is designed to model this effect of “additional information”. From the “frequency of event” point of view we would like to ask the following questions. “If I play 100 billions billions games...”

- before the dealing of hands: “... what fraction of the games do I win?”,
- after I received 7 and 10 of diamond: “... amongst the games where I had a 7 and 10 of diamond hand, what fraction of them do I win?”,
- and after I see the 2, 4 and ace of club flop: “... amongst the games where I had a 7 and 10 of diamond hand and a 2, 4 and ace of club flop, what fraction of them do I win?”.

Let us formalize this with events: if I have events $\text{Win} = \text{“I win the poker game”}$, $\text{hand} = \text{“I received a 7 and 10 of diamond hand”}$, and $\text{flop} = \text{“the flop is 2, 4 and ace of club”}$, the three previous questions correspond to the quantities (read $P(A|B)$ as “the probability of A given that B happened”)

- before the dealing of hands:

$$P(\text{Win}) \approx \frac{\text{number of wins}}{\text{number of games}},$$

- after I received 7 and 10 of diamond:

$$\begin{aligned} P(\text{Win} | \text{hand}) &\approx \frac{\# \text{ of wins with 7, 10 dia. hand}}{\# \text{ of games with 7, 10 dia. hand}} \\ &= \frac{\# \text{ of wins with 7, 10 dia. hand}}{\# \text{ of games}} \frac{\# \text{ of games}}{\# \text{ of games with 7, 10 dia. hand}} \\ &\approx \frac{P(\text{Win} \cap \text{hand})}{P(\text{hand})}, \end{aligned}$$

- and after I see the 2, 4 and ace of club flop:

$$\begin{aligned} P(\text{Win} | \text{hand} \cap \text{flop}) &\approx \\ &\frac{\# \text{ of wins with 7, 10 dia. hand and 2, 4 and ace of club flop}}{\# \text{ of games with 7, 10 dia. hand and 2, 4 and ace of club flop}} \\ &\approx \frac{P(\text{Win} \cap \text{hand} \cap \text{flop})}{P(\text{hand})} \frac{P(\text{hand})}{P(\text{hand} \cap \text{flop})} = \frac{P(\text{Win} \cap \text{flop} | \text{hand})}{P(\text{flop} | \text{hand})}. \end{aligned}$$

The second point illustrates what conditional probability represents, whilst the third illustrated that further information corresponds to conditioning the conditional measure or, in other words, to look at relative frequencies *in the conditional experiment* (i.e. the realisations of the experiment where the first event occurs). With this intuition in mind, we can define conditional probability in a formal manner.

Definition 2.4.1 (Conditional measure). Let Ω be a set of realisations, and P a probability measure on Ω . Let $A \subset \Omega$ be an event such that $P(A) > 0$. Define^a then the *probability measure P conditioned on A* , denoted $P(\cdot | A)$, by

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \quad \forall \text{ event } B.$$

We can then define the *conditional expectation* of a random variable $X : \Omega \rightarrow \mathbb{R}$ by

$$E_P(X | A) = E_{P_A}(X) = \frac{E_P(X \mathbf{1}_A)}{P(A)},$$

where P_A stands for $P_A = P(\cdot | A)$.

^aIt is easy to check that we indeed have a probability measure: $P(\Omega | A) = P(A)/P(A) = 1$, and $P(\sqcup_{i \geq 1} A_i | A) = \frac{1}{P(A)} P(\sqcup_{i \geq 1} (A_i \cap A)) = \frac{1}{P(A)} \sum_{i \geq 1} P(A_i \cap A) = \sum_{i \geq 1} P(A_i | A)$.

Conditional probability is a very helpful notion to understand un-intuitive facts. Let us look at a few examples.

Example 2.4.1. We make the following simplifying assumptions: at every birth, we suppose that the child is either of XX-type or of XY-type, each with probability 1/2. Consider the following problems.

1. Pick uniformly at random a family amongst the families with exactly two children, at least one of whom is of XX-type. What is the probability that both children are of XX-type?
2. Pick uniformly at random a family amongst the families with exactly two children, and with the eldest child being of XX-type. What is the probability that both children are of XX-type?

We start by modelling the situation: the realisation space is

$$\Omega = \{(X, X), (Y, X), (X, Y), (Y, Y)\},$$

with the first coordinate representing the type of the eldest child, and the second the type of the youngest. Consider P the uniform probability measure on Ω , which amounts to pick uniformly at random a family amongst the families with exactly two children. In both situations, we are looking at the measure P conditioned on some event, and we wonder about the probability of seeing the realisation (X, X) , which is looking at the probability of the set $\{(X, X)\}$.

In the first situation, we are conditioning on the event “at least one for the two child is of XX-type” which, in math, is the set $\{(X, X), (Y, X), (X, Y)\}$. We therefore search the probability of

$$P(\{(X, X)\} | \{(X, X), (Y, X), (X, Y)\}) = \frac{P(\{(X, X)\})}{P(\{(X, X), (Y, X), (X, Y)\})} = \frac{1}{3}.$$

In the second situation, we are conditioning on the event “the eldest child is of XX-type” which, in math, is the set $\{(X, X), (X, Y)\}$. We therefore search the probability of

$$P(\{(X, X)\} \mid \{(X, X), (X, Y)\}) = \frac{P(\{(X, X)\})}{P(\{(X, X), (X, Y)\})} = \frac{1}{2}.$$

Example 2.4.2. Let’s now look at a variation of the previous example. Alice is visiting her friend Bob. Alice knows that Bob has two kids. She rings the bell, and a child of XX-type opens. What is the probability that Bob has two children of XX-type?

We again formalize the situation. The realisation set should encompass the type of each kid and which of the two opens the door. We can for example take

$$\Omega = \{(X_*, X), (Y_*, X), (X_*, Y), (Y_*, Y), (X, X_*), (Y, X_*), (X, Y_*), (Y, Y_*)\},$$

with the first coordinate representing the type of the eldest child, the second the type of the youngest, and the star indicates who opens the door. We suppose that all realisations have the same probability, we thus consider P the uniform probability measure on Ω . We wonder about the event the both kids are of XX-type, which is $\{(X_*, X), (X, X_*)\}$, and we work under the condition that “a child of XX-type opens”, which is the event $\{(X_*, X), (X_*, Y), (X, X_*), (Y, X_*)\}$. We then can compute

$$\begin{aligned} P(\{(X_*, X), (X, X_*)\} \mid \{(X_*, X), (X_*, Y), (X, X_*), (Y, X_*)\}) \\ = \frac{P(\{(X_*, X), (X, X_*)\})}{P(\{(X_*, X), (X_*, Y), (X, X_*), (Y, X_*)\})} = \frac{2}{4} = \frac{1}{2}. \end{aligned}$$

This is to be contrasted with the first case of the previous example: in both cases, we have the information that at least one of the kids is of XX-type, but the way the information is obtained can bias the final probability!

Remark 2.4.1. When calculating with conditional probabilities, it is easy to make mistakes when resorting to our intuition. So, it is much safer to systematically use the rules and to perform formal computations rather than trying to “guess” the answer.

Exercise 2.25. 1. Let Ω be a finite set. Let P be the uniform probability measure on Ω , and let $A \subset \Omega$ be non-empty. Show that $P(\cdot \mid A)$ can be seen as the uniform probability measure on A (i.e.: for every $B \subset A$, $P(B \mid A) = |B|/|A|$).

2. Let P be the law of a fair (6 faces) dice roll. Let even be the event that we obtain a face with an even number of points. What is the probability mass function associated to $P(\cdot \mid \text{even})$?

Solution is given in [A.2](#).

Exercise 2.26. Nicole would like to go to the hairdresser, but she can't decide between hairdresser A and hairdresser B. So she roll a fair 6-faces dice: if she obtains a 5 or a 6, she goes to hairdresser A, and if she obtains a 1,2,3, or 4 she goes to hairdresser B. Suppose that the waiting time (in minutes) is random and is a uniform random variable on $[0, 30]$ for hairdresser A, and that it is a uniform random variable on $[0, 20]$ for hairdresser B. *The following questions are independent of each others.*

1. Nicole obtains a 5. What is the probability that she waits at least 25 minutes?
2. What is the probability that she waits at least 15 minutes?
3. Nicole waited for at least 15 minutes. What is the probability that she rolled a 4?

Solution is given in [A.2](#).

***Exercise 2.27** (The two envelopes problem). The problem is a famous seemingly paradoxical situation. It goes as follows.

Imagine you are given two identical envelopes, each containing money. One contains twice as much as the other. You may pick one envelope and keep the money it contains. Having chosen an envelope at will, but before inspecting it, you are given the chance to switch envelopes. Should you switch?

(I)

The two contradictory conclusions are

1. The situation is obviously symmetric, so it does not matter whether you switch or not. Formally, let $x, 2x$ be the money in the two envelopes, and S be the money in the first envelope you pick. Then, on the one hand, if you keep your envelope, you have an expected gain of

$$2xP(S = 2x) + xP(S = x) = \frac{3}{2}x,$$

as, by symmetry, $P(S = 2x) = P(S = x) = 1/2$. On the other hand, if you switch, you have an expected gain of

$$xP(S = 2x) + 2xP(S = x) = \frac{3}{2}x.$$

2. You should always switch as, given the money in the envelope you have, you can double the sum with probability $1/2$ or divide it by two with probability $1/2$, which has higher average gain than the number you see. More formally, let A be the money in the first envelope you pick. Then, the other envelope contains either $2A$ or $A/2$ each with probability $1/2$, so you gain A if you keep your envelope, but on average you gain

$$\frac{1}{2} \frac{A}{2} + \frac{1}{2} 2A = \frac{5}{4}A > A$$

if you switch.

Which one of the two reasoning is incorrect? can you spot why?

Solution is given in [A.2](#).

2.4.2 Independence

Intuitively, independence is the property that “one thing does not depend on the other”. From an experimental point of view, it is that “the result of experiment 1 does not influence the result of experiment 2”. This can also be said as follows: if A is an event depending only on experiment 1, and B is an event depending only on experiment 2, the frequency with which we see A and B simultaneously when performing the two experiments many times is the frequency with which we see A in experiment 1 times the frequency with which we see B in experiment 2. We formalize this idea for several different “things”.

Definition 2.4.2. • Two events A, B are said to be *independent* if

$$P(A \cap B) = P(A)P(B).$$

- A family of events $(A_i)_{i \in I}$ is said to be *two-by-two independent* if for any $i \neq j$, A_i and A_j are independent.
- A family of events $(A_i)_{i \in I}$ is said to be *an independent family* if for any $J \subset I$ finite,

$$P\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} P(A_i).$$

Note that if A, B are two events with $P(A) > 0$, $P(B) > 0$, the fact that they are independent is equivalent to saying that the conditional probability of A given B is equal to the probability of A :

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A).$$

In other words: A and B being independent is equivalent to saying that the fact that B is realized has no influence on the realisation of A (and vice-versa). From independence of event, one can define independent random variables. The spirit is that two random variables are independent if *all* pair of events of the form $\{X \in A\}$, $\{Y \in B\}$ are independent events.

Definition 2.4.3. • Two random variables X, Y are said to be *independent* if for any events $A, B \subset \mathbb{R}$,

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B).$$

Equivalently, X, Y are independent if for any $f, g : \mathbb{R} \rightarrow \mathbb{R}$,

$$E(f(X)g(Y)) = E(f(X))E(g(Y))$$

- A family of random variables $(X_i)_{i \in I}$ is said to be *two-by-two independent* if for any $i \neq j$, X_i and X_j are independent.
- A family of random variables $(X_i)_{i \in I}$ is said to be *an independent family* if for any $J \subset I$ finite, and any events $A_i \subset \mathbb{R}$, $i \in J$,

$$P\left(\bigcap_{i \in J} \{X_i \in A_i\}\right) = \prod_{i \in J} P(X_i \in A_i).$$

Equivalently, $(X_i)_{i \in I}$ is an independent family if for any $J \subset I$ finite, and any functions $f_i : \mathbb{R} \rightarrow \mathbb{R}$, $i \in J$,

$$E\left(\prod_{i \in J} f_i(X_i)\right) = \prod_{i \in J} E(f_i(X_i)).$$

The same definition holds with “random vectors” replacing “random variables”.

One has to be careful: two-by-two independent is strictly weaker than being an independent family. See Exercise 2.30

Remark 2.4.2. The “function” definition of independence is not completely rigorous: one should restrict to f, g such that $f(X), g(Y)$ are themselves random variables... we will happily ignore this issue!

From the point of view of mathematics, independence is a *factorisation property*. Note that it fits what we want from the experimental viewpoint: if we are running two experiments which have set of realisations Ω_1 for experiment 1, and Ω_2 for experiment 2, our probability space for the two experiments done together is the set $\Omega_1 \times \Omega_2$ (a realisation of experiment 1 and a realisation of experiment 2), an event depending only on the result of experiment 1 is an event of the form

$$E_1 = A \times \Omega_2 \text{ with } A \subset \Omega_1,$$

and an event depending only on the result of experiment 2 is an event of the form

$$E_2 = \Omega_1 \times B \text{ with } B \subset \Omega_2.$$

The event for the joint system corresponding to “ A occurs in experiment 1, and B occurs in experiment 2” is then the event

$$E_1 \cap E_2 = (A \times \Omega_2) \cap (\Omega_1 \times B) = A \times B.$$

The independence between experiment 1 and experiment 2 is then the property that for any A, B as above,

$$P(A \times B) = P(E_1 \cap E_2) = P(E_1)P(E_2),$$

i.e.: events depending only on experiment 1 are independent from events depending only on experiment 2.

Example 2.4.3 (Guess the larger number). A showman offers the following challenge: he choses two distinct numbers $A < B \in \mathbb{N}$ and write them on two different pieces of paper that are put face down on a table. Now, everyone in the (supposed as large as wanted) crowd, can come and try to guess the larger number by: 1) looking at the number on one of the papers, 2) chose whether to keep this number or take the number below the other piece of paper. The challenger wins if his number is the largest of the two. We suppose that the people who did not yet attempt the challenge are not able to gain any information from the ones that have tried or are trying the challenge.

Note that if everyone picks an envelope uniformly are random and simply keep it, there is a probability $1/2$ of winning for each individual, so there is a strategy such that half of the challengers win. The question is “can the crowd do better?”.

The answer is... yes! And the strategy involves resorting to additional randomness. Let p be a probability mass function on $\mathbb{N}$ such that $p(n) > 0$ for all $n \in \mathbb{N}$. Now, the strategy of each individual is the following: sample X a random variable with law $P(X = n) = p(n)$ for all $n \in \mathbb{N}$ (and thus $P(X \in \mathbb{N}) = 1$). Pick an envelope uniformly at random (independently of X), denote N_1 the number in that envelope. If $N_1 < X + \frac{1}{2} =: Y$, change number. Else, keep the number. The challenger wins in any of the two mutually exclusive situations

1. $N_1 = A$ and $A < Y$,
2. $N_1 = B$ and $B < Y$.

So, as Y and N_1 are independent,

$$\begin{aligned}
P(\text{win}) &= P(N_1 = A, A < Y) + P(N_1 = B, B < Y) \\
&= \frac{1}{2}P(A < Y) + \frac{1}{2}P(B < Y) \\
&= \frac{1}{2}(P(Y > A) + P(Y < A) + P(A < Y < B)) \\
&= \frac{1}{2}(1 + P(A < Y < B))
\end{aligned}$$

as $P(Y < B) = P(Y < A) + P(A < Y < B)$, and $P(Y > A) + P(Y < A) = 1$ as $A, B \in \mathbb{N}$ and $Y \in \frac{1}{2} + \mathbb{N}$. Now, $P(A < Y < B) > 0$ as p is strictly positive and $A < B$. So, $P(\text{win}) > \frac{1}{2}$.

We will often encounter a particular case of independent family. It is so present in probability that it is worthwhile to give it a particular name.

Definition 2.4.4 (i.i.d. sequence/family). A family $(X_i)_{i \in I}$ of random variables is called an *independent identically distributed* family (or sequence when $I \subset \mathbb{N}$), abbreviated *i.i.d. family (sequence)*, if the family $(X_i)_{i \in I}$ is an independent family, and for any $i, j \in I$, $X_i \stackrel{\text{Law}}{=} X_j$.

In the case of discrete and continuous random variables, there is another useful equivalent definition of independence: the factorisation property carries over to the probability mass functions/density functions.

Theorem 2.4.1. Let $d, d' \geq 1$. Let $X : \Omega \rightarrow \mathbb{R}^d$, $Y : \Omega \rightarrow \mathbb{R}^{d'}$ be random vector.

- **If X, Y are continuous random vectors:** X and Y are independent if and only if the random vector $(X, Y) : \Omega \rightarrow \mathbb{R}^{d+d'}$ has density

$$f_{(X,Y)}(x, y) = f_X(x)f_Y(y),$$

where $f_X : \mathbb{R}^d \rightarrow \mathbb{R}$ is a density for X , and $f_Y : \mathbb{R}^{d'} \rightarrow \mathbb{R}$ is a density for Y .

- **If X, Y are discrete random vectors:** X and Y are independent if and only if for any $x \in \mathbb{R}^d$, $y \in \mathbb{R}^{d'}$,

$$P(X = x, Y = y) = P(X = x)P(Y = y).$$

- **If X is discrete and Y is continuous:** X and Y are independent if and only if for any $x \in \mathbb{R}^d$, $A \subset \mathbb{R}^{d'}$,

$$P(X = x, Y \in A) = P(X = x) \int_A f_Y(y) dy$$

where $f_Y : \mathbb{R}^{d'} \rightarrow \mathbb{R}$ is a density for Y .

The above theorem generalises to more than two random vectors, as long as the random vectors form an independent family.

Exercise 2.28. A communication system is composed of n components. Each component works with probability p , independently of the others.

1. What is the law of the random variable X which gives the number of working components? (i.e.: give $P(X = x)$ for all x).
2. For the system to work properly, we need that at least half of the components are working. For which values of p does a system with 5 components has bigger chances of working than a system with 3 components?

Solution is given in [A.2](#).

Exercise 2.29. Bob has an appointment with the dentist at 10:00. The waiting time between the appointment time and the actual moment Bob will be able to see the dentist is random and assumed to be a continuous random variable with density $f(x) = c\mathbb{1}_{[0,+\infty)}(x)e^{-2x}$ for some $c > 0$. The time the meeting takes is independent of the waiting time, and is uniformly distributed on $[5, 15]$.

1. What is the value of c ?
2. What is the density of the meeting duration?
3. On average, at what time will Bob get out of the dentist office?

At 10:20, a gas leak in the building of the dentist office kills everyone in the office.

4. What is the probability that Bob survives?

Solution is given in [A.2](#).

Exercise 2.30. Let P be the uniform probability measure on $\{1, 2, 3, 4\}$. Define the events

$$A = \{1, 2\}, \quad B = \{1, 3\}, \quad C = \{2, 3\}.$$

1. Show that A and B are independent, that A and C are independent, and that B and C are independent.
2. Show that $P(A \cap B \cap C) \neq P(A)P(B)P(C)$. Are A, B, C forming an independent family?
3. Based on the two previous points, construct three random variables X_1, X_2, X_3 that are two-by-two independent but that do not form an independent family.

Solution is given in [A.2](#).

Exercise 2.31. 1. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be two independent random variables. Show that

$$M_{X+Y}(t) = M_X(t)M_Y(t).$$

2. Let $n \geq 2$, let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be an independent family of random variables. Show that

$$M_{\sum_{i=1}^n X_i}(t) = \prod_{i=1}^n M_{X_i}(t).$$

Solution is given in [A.2](#).

Exercise 2.32. 1. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be two independent random variables. Show that

$$P(\max(X, Y) \leq t) = F_X(t)F_Y(t).$$

2. Let $n \geq 2$, let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be an independent family of random variables. Show that

$$P\left(\max_{i=1, \dots, n} X_i \leq t\right) = \prod_{i=1}^n F_{X_i}(t).$$

Solution is given in [A.2](#).

2.4.3 Bayes law, formula of total probability

Bayes law is a way to “reverse the computation” of conditional probabilities.

Lemma 2.4.2 (Bayes law). *Let Ω be a set of realisations, and let P be a probability measure on Ω . Let $A, B \subset \Omega$ be two events such that $P(A), P(B) > 0$. Then,*

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

Proof. Using the definition, we have

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}.$$

□

It is often paired with the formula of total probability, which is a way to “split the space into components”, and study each components separately.

Lemma 2.4.3 (Formula of total probability). *Let Ω be a set of realisations, and let P be a probability measure on Ω . Let I be a finite or countable set. Let A_i , $i \in I$ be a collection of events such that*

- *if $i \neq j$, then $A_i \cap A_j = \emptyset$;*
- *$\cup_{i \in I} A_i = \Omega$ ^a.*

Suppose moreover that $P(A_i) > 0$ for all $i \in I$. Then for any event B ,

$$P(B) = \sum_{i \in I} P(B \cap A_i) = \sum_{i \in I} P(B | A_i) P(A_i).$$

In the same fashion, for every random variable X ,

$$E(X) = \sum_{i \in I} E(X | A_i) P(A_i).$$

^aSuch a collection is called a *partition* of Ω .

Remark 2.4.3. In the case $|I| = 2$, we have the very useful formula

$$P(B) = P(B | A)P(A) + P(B | A^c)P(A^c).$$

Remark 2.4.4. The decomposition of the expected value is a nice way to compute random variables that are neither continuous nor discrete, but some combination of the two. See Exercise 2.34.

To gain some intuition, let us look at a simple example.

Example 2.4.4. Consider two identical jars containing marbles (all of the same shape, size, and mass). Suppose that one of the two, call it J_1 , contains 10 blue marbles and 5 red ones; and that the other jar, call it J_2 , contains 8 blue marbles and 7 red ones.

1. Bob picks a jar (uniformly at random as he has no information on what they contain). What is now the probability that he picked J_1 ?
2. Bob now draws a marble from the urn he picked. The marble is blue. What is now the probability that he picked J_1 ?

We can compute the probabilities using Bayes rule and the formula of total probability. Denote

$$\text{Blue} = \{\text{Bob draws a blue marble}\}, \quad J_i = \{\text{Bob picked the jar } i\}, \quad i = 1, 2.$$

1. Here, there is no further information, the choice of the jar is done uniformly at random, so $P(J_1) = 1/2$.

2. We have new information: using Bayes rule, and the law of total probability,

$$\begin{aligned}
 P(J_1 | \text{Blue}) &= \frac{P(\text{Blue} | J_1)P(J_1)}{P(\text{Blue})} \\
 &= \frac{P(\text{Blue} | J_1)P(J_1)}{P(\text{Blue} | J_1)P(J_1) + P(\text{Blue} | J_2)P(J_2)} \\
 &= \frac{\frac{10}{15} \cdot \frac{1}{2}}{\frac{10}{15} \cdot \frac{1}{2} + \frac{8}{15} \cdot \frac{1}{2}} = \frac{10}{18} = \frac{5}{9}.
 \end{aligned}$$

This is an example of *Bayesian inference*: we start with a *prior estimation*, the probability of picking the jar 1 before drawing a marble, and update the probability after picking the marble according to the drawn colour.

Example 2.4.5 (Monthy Hall problem). We can now discuss the Monthy Hall problem presented in Section 1.1.3. Let us formally model the situation. Suppose that the doors are numbered 1,2,3. Let $C \in \{1, 2, 3\}$ be the number of the door behind which the car is hidden, let $D \in \{1, 2, 3\}$ be the door first picked by the candidate (this is *not* a random number), and let $H \in \{1, 2, 3\}$ be the door opened by the host. From the rules of the game, we have $P(D = H) = 0$, $P(C = H) = 0$, and $P(C = 1) = P(C = 2) = P(C = 3) = \frac{1}{3}$. We want to compare two strategies.

1. Let us first look at what happens if the candidate keeps his initial choice. We have

$$\begin{aligned}
 P(\text{win}) &= P(\text{win} | D = C)P(D = C) + P(\text{win} | D \neq C)P(D \neq C) \\
 &= P(D = C) = \frac{1}{3},
 \end{aligned}$$

as $P(\text{win} | D = C) = 1$ and $P(\text{win} | D \neq C) = 0$ as the candidate keeps his initial choice.

2. Now, let us see what happens if he swaps doors.

$$\begin{aligned}
 P(\text{win}) &= P(\text{win} | D = C)P(D = C) + P(\text{win} | D \neq C)P(D \neq C) \\
 &= 0 \cdot P(D = C) + 1 \cdot P(D \neq C) = \frac{2}{3},
 \end{aligned}$$

as here we have that under the event $D = C$, the candidates always ends up with a goat door as he is changing away from the door with the car (regardless of what the host is doing), and, under the event $D \neq C$ the host can only open one door: the goat one that was not picked by the candidate, and thus the candidate swaps to the only available door: the car one!

Bayes law is “trivial” from a mathematical viewpoint, yet it is the cornerstone of many methods in statistics, and it tremendously helps with the understanding of seemingly paradoxical facts. Let us look at a classical example.

Example 2.4.6. We consider the following situation: consider a population in which a virus is spreading. We evaluate that a proportion $\rho \in (0, 1)$ of the population is infected by the virus. There is a simple test to check whether someone is infected or not, but the test can sometimes fail. It can fail in two ways:

- false positive: the test says “infected” (write this T_+) whilst the individual is not (which we will denote I_-). We estimate that 0.9% of healthy people are tested positive;
- false negative: the test says “healthy” (denote this T_-) whilst the individual was infected (which we will write I_+). We estimate that 0.1% of the infected population are tested negative.

We now wonder about what can be inferred from the result of the test for a fixed individual.

As usual, we start by modelling the situation using probabilist language. As realisation space, we can take

$$\Omega = \{(I_-, T_-), (I_+, T_-), (I_-, T_+), (I_+, T_+)\},$$

for “healthy individual and negative test”, “infected individual and negative test”,... Let us now look at the probability measure P . As we do not have further information about the individual, we can suppose that he is picked uniformly at random in the population, so he has a probability ρ of being infected and $1 - \rho$ of being healthy. This amounts to say

$$P(\{(I_+, T_-), (I_+, T_+)\}) = \rho, \quad P(\{(I_-, T_-), (I_-, T_+)\}) = 1 - \rho.$$

We also know that the probability that the test is negative knowing that the individual is infected is 0.1% and that the probability of the test being positive knowing that the individual is healthy is 0.9%. Formally, this amounts to

$$\begin{aligned} P(\{(I_-, T_-), (I_+, T_-)\} \mid \{(I_+, T_-), (I_+, T_+)\}) &= 0.1\% \\ P(\{(I_-, T_+), (I_+, T_+)\} \mid \{(I_-, T_-), (I_-, T_+)\}) &= 0.9\% \end{aligned}$$

To make the reading easier, introduce the notation

$$\begin{aligned} \text{infec} &= \{(I_+, T_-), (I_+, T_+)\}, & \text{health} &= \{(I_-, T_-), (I_-, T_+)\}, \\ \text{test}_+ &= \{(I_-, T_+), (I_+, T_+)\}, & \text{test}_- &= \{(I_-, T_-), (I_+, T_-)\} \\ \text{fail} &= \{(I_-, T_+), (I_+, T_-)\}. \end{aligned}$$

We can start by wondering if the test is reliable: i.e., what is the probability that the test gives the wrong result. Formally, we wonder about $P(\text{fail})$. By the total probability formula,

$$\begin{aligned} P(\text{fail}) &= P(\text{fail} \mid \text{infec})P(\text{infec}) + P(\text{fail} \mid \text{health})P(\text{health}) \\ &= P(\text{fail} \cap \text{infec} \mid \text{infec})\rho + P(\text{fail} \cap \text{health} \mid \text{health})(1 - \rho) \\ &= P(\text{test}_- \cap \text{infec} \mid \text{infec})\rho + P(\text{test}_+ \cap \text{health} \mid \text{health})(1 - \rho) \\ &= P(\text{test}_- \mid \text{infec})\rho + P(\text{test}_+ \mid \text{health})(1 - \rho) \\ &= \frac{0.9}{100}(1 - \rho) + \frac{0.1}{100}\rho \leq 0.9\%, \end{aligned}$$

as

$$\begin{aligned} \text{fail} \cap \text{infec} &= \{(I_+, T_-)\} = \text{test}_- \cap \text{infec} \\ \text{fail} \cap \text{health} &= \{(I_-, T_+)\} = \text{test}_+ \cap \text{health}. \end{aligned}$$

So, our test is 99.1% reliable *independently of the proportion of infected people*. We can consider this test as very reliable.

We can now ask a different question: suppose that the individual is tested positive, what is the probability that he is indeed infected by the virus? Formally, this asks about the value of $P(\text{infec} | \text{test}_+)$. We can use Bayes law and the formula of total probability to obtain

$$\begin{aligned} P(\text{infec} | \text{test}_+) &= \frac{P(\text{test}_+ | \text{infec})P(\text{infec})}{P(\text{test}_+)} \\ &= \frac{P(\text{test}_+ | \text{infec})P(\text{infec})}{P(\text{test}_+ | \text{infec})P(\text{infec}) + P(\text{test}_+ | \text{health})P(\text{health})} \\ &= \frac{\frac{99.9}{100}\rho}{\frac{99.9}{100}\rho + \frac{0.9}{100}(1 - \rho)} = \frac{99.9\rho}{99\rho + 0.9}. \end{aligned}$$

where we used

$$P(\text{test}_+ | \text{infec}) = 1 - P(\text{test}_+ | \text{infec}) = 1 - 0.1\% = 99.9\%.$$

Looking at the obtained expression

$$P(\text{infec} | \text{test}_+) = \frac{99.9\rho}{99\rho + 0.9},$$

we observe that if a large proportion of the population is infected (ρ close to 1), this probability is close to 1, and thus a positive test result often reflects the fact that the individual is indeed infected. On the other side, if ρ is very small (rare virus), the result of a positive test is often wrong!

This observation that a highly reliable test seems to “often fail” comes from a cognitive bias : *neglecting the sample size* (or *forgetting about the base frequency*). In the example, we work under the hypotheses that the very rare event “the individual is tested positive” occurs. It is clear when doing calculations formally, but it is often less obvious when facing real life situations. For example, hearing many stories about people being tested positive whilst not being infected can make you forget about all the people that where correctly tested negative...

Example 2.4.7 (Simpson’s paradox³). Simpson’s paradox is a phenomenon in probability and statistics in which a trend appears in several groups of data but disappears or reverses when the groups are combined. We will look here at the best known example of this paradox, and how to resolve it.

The example is a study of gender bias among graduate school admissions to University of California, Berkeley, performed on the admissions on the fall of 1973. The numbers for the whole university are the following.

	All		Women		Men	
	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
Total	12763	41%	4321	35%	8442	44%

³Edward Hugh Simpson (1922 – 2019), British statistician. The paradox, discussed by Simpson in 1951, had already been discussed in 1899 by Karl Pearson and his collaborators, and in 1903 by George Udny Yule.

These numbers show that substantially more male applicants than female applicants were admitted. The early conclusion, leading to the study, was that men were favoured during the admission process. The study team then went to check the different departments to see where was the problem in the admission process. They had the following numbers for the six largest departments.

	All		Women		Men	
Dep.	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
A	933	64%	108	82%	825	62%
B	585	63%	25	68%	560	63%
C	918	35%	593	34%	325	37%
D	792	34%	375	35%	417	33%
E	584	25%	393	24%	191	28%
F	714	6%	341	7%	373	6%
Total	4526	39%	1835	30%	2691	45%

Figure 2.3: Difference of strictly more than 3% are highlighted in red/blue.

Looking at the table, we notice that the bias per department is not really in favour of male applicants... and even slightly in favour of female applicants! But the general admission rate still indicates a bias in favour of male applicants! This is Simpson paradox. The way to “resolve” this apparent contradiction, is to look more precisely at how are the numbers combined to go from the “admissions per department” to “total admissions”. To simplify the computations, we will look only at the case of departments A and C.

	All		Women		Men	
Dep.	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
A	933	64%	108	82%	825	62%
C	918	35%	593	34%	325	37%
Total	1851	39%	701	38%	1150	55%

Figure 2.4: Difference of strictly more than 3% are highlighted in red/blue.

We see the same phenomenon as previously. Let us model the situation with conditional probabilities. Let us consider the probability space with realisation set

$$\Omega = \{M, W\} \times \{A, C\} \times \{\text{Ad}, \text{Rej}\}.$$

The first entry is the gender of the candidate, the second is the department to which the candidate applies, and the third is the result of the application. Consider the events:

$$\begin{aligned} \text{Admitted} &= \{M, W\} \times \{A, C\} \times \{\text{Ad}\}, & \text{Rejected} &= \{M, W\} \times \{A, C\} \times \{\text{Rej}\}, \\ \text{Women} &= \{W\} \times \{A, C\} \times \{\text{Ad}, \text{Rej}\}, & \text{Men} &= \{M\} \times \{A, C\} \times \{\text{Ad}, \text{Rej}\}, \\ \text{depA} &= \{M, W\} \times \{A\} \times \{\text{Ad}, \text{Rej}\}, & \text{depC} &= \{M, W\} \times \{C\} \times \{\text{Ad}, \text{Rej}\}. \end{aligned}$$

We then look at the total probability to be admitted given that the candidate is a women: using the formula of total probability,

$$\begin{aligned} P(\text{Admitted} | \text{Women}) &= P(\text{Admitted} | \text{Women, depA})P(\text{depA} | \text{Women}) \\ &\quad + P(\text{Admitted} | \text{Women, depC})P(\text{depC} | \text{Women}) \\ &= \frac{82}{100} \cdot \frac{108}{701} + \frac{34}{100} \cdot \frac{593}{701}. \end{aligned}$$

Doing the same computation for a men, we obtain

$$P(\text{Admitted} | \text{Men}) = \frac{62}{100} \cdot \frac{825}{1150} + \frac{37}{100} \cdot \frac{325}{1150}.$$

We can then see where is the issue: more men are applying to department A, which has “large” acceptance rates (82% and 62%), whilst the largest proportion of women are applying to department C, which has “small” acceptance rates (34% and 37%). Thus the apparent bias in favour of men in the total numbers is not a sign of a selection bias in favour of men, but rather a reflection of a bias *in the choice of department on the side of the candidates*. The candidates choices is a particular example of *confounding variable*. The apparent paradox is caused by our tendency to see causality where there is only correlation: in the example, being a men and being admitted are positively correlated. But that does not necessarily mean that there is a direct causality link between the two.

For references to the original studies and further examples of Simpson’s paradox, see the Wikipedia page devoted to it: https://en.wikipedia.org/wiki/Simpson%27s_paradox.

Exercise 2.33. Three teams, numbered 1, 2, 3 are handling customer service. They all have different efficiency. We suppose that the treatment time (in hours) of team i to a given customer request is a continuous random variable T_i with density

$$f_{T_1}(x) = \mathbb{1}_{[0,+\infty)}(x)e^{-x}, \quad f_{T_2}(x) = 2\mathbb{1}_{[0,+\infty)}(x)e^{-2x}, \quad f_{T_3}(x) = 3\mathbb{1}_{[0,+\infty)}(x)e^{-3x}.$$

When a customer submits a request, it is assigned uniformly at random to one of the three teams. You submit a request to customer service. Denote X the random variable corresponding to the time that your request takes to be treated.

1. What is the expected value of X ?
2. What is the probability that X is less than 1?

You now have waited for an hour and your request has not yet been treated.

3. What is the probability that the team assigned to your request is team 1? team 2? team 3?

Solution is given in A.2.

Exercise 2.34. Let X be a discrete random variable with $P(X = -1) = P(X = 3) = \frac{1}{4}$, and $P(X = 0) = \frac{1}{2}$. Let Y be a continuous random variable with density $f_Y(x) = \mathbb{1}_{[0,+\infty)}(x) \frac{3}{(x+1)^4}$. Finally, let Z be a discrete random variable with $P(Z = 0) = P(Z = 1) = \frac{1}{2}$. Suppose that X, Y, Z form an independent family. Define the random variable

$$W = ZX + (1 - Z)Y.$$

Compute $E(W)$.

Solution is given in [A.2](#).

Exercise 2.35. Suppose that we have two identical bags contain each 30 marbles. In one of the bag (bag 1), there are 10 red, 10 green, and 10 blue marbles. In the other one (bag 2), there are 3 red, 8 green, and 19 blue. You chose one of the bags uniformly at random and start drawing marbles out of the bag, one by one without putting them back in the bag. The first marble you pick is blue.

1. What is the probability that you chose bag 1?
2. What is the probability that the next marble you pick is red?

You draw a second marble. It is again blue.

3. What is now the probability that you chose bag 1?
4. What is now the probability that the next marble you pick is red?

Based on the first two draws, you evaluate that the bag you picked is bag 2.

5. What is the probability that your evaluation of picking bag 2 is correct?

You draw a third marble. It is red.

6. What is now the probability that your evaluation of picking bag 2 is correct?

Solution is given in [A.2](#).

2.4.4 Almost sure properties

In probability, we often encounter statements such as “Property XXX is true *almost surely*”. This kind of statement is a reflect of the following fact: *in probability, only the events which occur with strictly positive probability matter*.

Definition 2.4.5. An event A is said to occur *almost-surely* if

$$P(A) = 1.$$

Note that events that occur almost surely are independent of any other event: if $P(A) = 1$ and B is another event,

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 1 + P(B) - 1 = 1 \cdot P(B) = P(A)P(B)$$

as $1 \geq P(A \cup B) \geq P(A) = 1$. In particular, P and $P(\cdot | A)$ give the same probability to every events, and are therefore the same probability measures.

The reason why we talk about “almost sure properties” is that they are way more permissive than “sure properties”. Indeed, let us look at a simple example. Consider the following setup:

- Look at the probability space $\Omega = \mathbb{Z}$. Look at the probability mass function $p(1) = p(0) = \frac{1}{2}$, and $p(n) = 0$ for $n \notin \{0, 1\}$.
- Look at the function (random variable) $X : \mathbb{Z} \rightarrow \mathbb{R}$, $X(n) = n^2$.

On the one hand, we have that $P_p(X \in \{0, 1\})$ is one:

$$P_p(X \in \{0, 1\}) = P_p(X = 0) + P_p(X = 1) = p(0) + p(1) = 1.$$

In particular, the random variable X is *almost-surely bounded* (it is upper bounded by 1 with probability 1 and lower bounded by 0 with probability 1). But the function $X : \mathbb{Z} \rightarrow \mathbb{R}$ is obviously *not* bounded (it is the square function).

The advantage of working with almost-sure properties is that they are the *only* ones that matter from a probabilistic viewpoint as events of probability 0 occur with frequency... zero! So we can use this less restrictive notion rather than asking that things are true “surely”.

Remark 2.4.5. It is to be noted that “not happening almost surely” does not mean “impossible”: if you sample a uniform random number in the interval $[0, 1]$, you obtain a fixed number. But that particular fixed number has probability 0 of being drawn... “Not happening almost surely” has more the meaning of “statistically irrelevant”.

Exercise 2.36. Let X be a continuous random variable with density function $f_X(x) = \frac{1}{2}\mathbb{1}_{[-1,1]}$. Which of the following events are occurring almost-surely?

1. $|X| \leq 1$
2. X is non-negative
3. X is strictly greater than -1
4. X is not a rational number
5. $|X| > 1000^{-2}$

Solution is given in [A.2](#).

2.5 Correlation

2.5.1 Variance, Covariance

We saw that the expectation of a random variable represents the quantity that we observe *on average*. Variance is a way to quantify “how far from its mean is typically my variable”?

Definition 2.5.1 (Variance). Let Ω be a realisation set and P a probability measure on Ω . Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. The *variance* of X is given by

$$\text{Var}_P(X) := E_P((X - E_P(X))^2).$$

Alternatively, $\text{Var}_P(X) = E_P(X^2) - E_P(X)^2$, see Exercise 2.38.

If the variance of X is small, it means that $(X - E_P(X))^2$ is typically small, which in turns means that X is typically close to its expected value. The “how far is typically X from it expected value” is encoded in the *standard deviation*.

Definition 2.5.2 (Standard deviation). The *standard deviation* of a random variable X , often denoted σ_X , is the square-root of its variance:

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

As the variance can be understood at the “typical square distance to the mean”, the standard deviation can be thought of as “the typical distance to the mean”, in the sense that the variable X takes value in an interval centred around $E(X)$ having length of the order of σ_X .

Remark 2.5.1. The variance of a random variable can equivalently be defined as

$$\text{Var}(X) = \inf_{\mu \in \mathbb{R}} E((X - \mu)^2).$$

Indeed, expanding the square, we have

$$E((X - \mu)^2) = E(X^2) - 2\mu E(X) + \mu^2.$$

Differentiating w.r.t. μ , and equating to 0, we find that the minima of the above is attained at $\mu = E(X)$, so

$$\inf_{\mu \in \mathbb{R}} E((X - \mu)^2) = \min_{\mu \in \mathbb{R}} E((X - \mu)^2) = E((X - E(X))^2).$$

We also can observe that $E(X)$ is the “best” approximation of X in the sense that it is the number which minimizes the expected square distance to X .

Example 2.5.1. To get some grasp at what standard deviation represents geometrically. It will also be a good occasion to discuss what “typical values” can mean (there is not a unique notion of what “typical” means). Let $p \in [0, 1]$, and $K > 0$ be numbers. Let X be a discrete random variable with

$$P(X = K) = P(X = -K) = \frac{p}{2}, \quad P(X = 0) = 1 - p.$$

By symmetry, $E(X) = 0$, so

$$\text{Var}(X) = E(X^2) - E(X)^2 = E(X^2) = pK^2 + (1 - p)0^2 = pK^2.$$

Its standard deviation is then $\sigma_X = \sqrt{p}K$. Often in statistics, we say that the *typical values* of the variable X is the interval $[E(X) - \sigma_X, E(X) + \sigma_X]$. On the other hand,

the intuitive meaning of “typical values of X ” is “the values that X takes with very high probability”. To see what is going on, let us consider the case of p very small: in this case, X typically is equal to 0 (with probability $1 - p$) and rarely takes on the extreme values $\pm K$. We would like to say that the “typical values” of X is some small interval around $E(X) = 0$, for example $[-1, 1]$ or $[-0.1, 0.1]$. Let us see in what conditions do standard deviation gives us a good match for this. We want to see when is the interval $[E(X) - \sigma_X, E(X) + \sigma_X]$ close to the “expected” typical values interval $[-1, 1]$. We need that σ_X is of the same order of magnitude as 1, which says that K should be (at most) of order $1/\sqrt{p}$. So the “typical values” notion given by standard deviation matches the “intuitive” notion of typical values when the frequency of the extreme value K is at most the inverse square of K .

As we shall see when studying limit theorems, many very strong results are available about *independent* random variables. But in the real world, nothing is *truly* independent: for example, successive coin tosses are only *very close to be independent*. It is therefore useful to search for quantities that can measure a form of “closeness to independence”. Covariance is a “first order” (if we think in terms of “Taylor expanding around independence”, which means nothing well defined) notion of such closeness.

Definition 2.5.3. Let Ω be a realisation set and P a probability measure on Ω . Let $X, Y : \Omega \rightarrow \mathbb{R}$ be two random variables. The *covariance between X and Y* is given by

$$\text{Cov}_P(X, Y) := E_P(XY) - E_P(X)E_P(Y).$$

When $\text{Cov}_P(X, Y) = 0$, we say that X and Y are *uncorrelated*.

The covariance between X and Y is a measure of how much “typical large values of X ” and “typical large values of Y ” are influencing each other. A motivation for introducing this quantity is the following identity:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).$$

So, the covariance is a “correction to additivity” for the variance. It is a good exercise to verify the above identity.

Remark 2.5.2. Be careful: whilst independent random variables are clearly uncorrelated, the reverse is not true. **Uncorrelated does not imply independent.** It is instructive to look at the following counter-example: let X, Z be two independent random variables and suppose that X is a discrete random variable with law $P(X = 10) = P(X = 0) = 1/2$, and that Z is a discrete random variable with law $P(Z = -1) = P(Z = 1) = 1/2$. Define then $Y = X \cdot Z$. On the one hand, we have

$$\begin{aligned}\text{Cov}(X, Y) &= E(XY) - E(X)E(Y) = E(X^2Z) - E(X)E(XZ) \\ &= E(X^2)E(Z) - E(X)E(X)E(Z) = 0,\end{aligned}$$

where we used that X, Z are independent, and that $E(Z) = 0$. On the other hand, we have

$$\begin{aligned}P(X = 0, Y = 0) &= P(X = 0, XZ = 0) = P(X = 0) = \frac{1}{2} \\ &\neq \frac{1}{4} = P(X = 0)P(X = 0) = P(X = 0)P(Y = 0)\end{aligned}$$

as $P(Z = 0) = 0$, so $Y = 0$ iff $X = 0$. So, the events $\{X = 0\}$ and $\{Y = 0\}$ are not independent and thus X, Y are not independent.

There is one particular case where uncorrelated implies independent: it is when one is looking at *Bernoulli random variables*.

Lemma 2.5.1. *Let X, Y be two random variables such that*

$$P(X \in \{0, 1\}) = P(Y \in \{0, 1\}) = 1.$$

Such variable are called Bernoulli random variables. Then, X and Y are independent if and only if $\text{Cov}(X, Y) = 0$.

Proof. See Exercise 2.37. □

Lemma 2.5.2. *Let X, Y, Y_1, Y_2 be random variables and $a, b \in \mathbb{R}$. Then,*

$$\begin{aligned}\text{Cov}(X, Y) &= \text{Cov}(Y, X), \quad \text{Cov}(aX, bY) = ab\text{Cov}(X, Y), \\ \text{Cov}(X, Y_1 + Y_2) &= \text{Cov}(X, Y_1) + \text{Cov}(X, Y_2).\end{aligned}$$

In words: Cov is symmetric, and linear in each of its arguments.

Proof. See Exercise 2.38. □

Exercise 2.37. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be two random variables such that $P(X \in \{0, 1\}) = P(Y \in \{0, 1\}) = 1$. Suppose that

$$\text{Cov}(X, Y) = 0.$$

1. Show that for any $a, b \in \{0, 1\}$, $P(X = a, Y = b) = P(X = a)P(Y = b)$.
2. Using the first point, show that for any events $A, B \subset \mathbb{R}$, $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$.

Hint: what is $P(X \in (A \cap \{0, 1\}))$?

Solution is given in [A.2](#).

Exercise 2.38. Let (X, Y) and (X', Y') be two independent vector such that $(X, Y) \stackrel{\text{Law}}{=} (X', Y')$. Show the following points.

1. $\text{Var}(X) = E(X^2) - E(X)^2 = \frac{1}{2}E((X - X')^2) = \text{Cov}(X, X)$;
2. $\text{Cov}(X, Y) = E((X - E(X))(Y - E(Y)))$;
3. $2\text{Cov}(X, Y) = E((X - X')(Y - Y'))$;
4. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$;
5. $\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$;
6. $\text{Cov}(X, Y + Z) = \text{Cov}(X, Y) + \text{Cov}(X, Z)$.

Solution is given in [A.2](#).

Exercise 2.39. In each of the following cases, compute the covariance between X and Y .

1. The random vector (X, Y) is a discrete random vector with

$$P((X, Y) = (2, 1)) = \frac{1}{4}, \quad P((X, Y) = (1, 0)) = \frac{1}{2}, \quad P((X, Y) = (0, 0)) = \frac{1}{4}.$$

2. The random vector (X, Y) is a discrete random vector with

$$P((X, Y) = (0, 1)) = \frac{1}{15}, \quad P((X, Y) = (2, 3)) = \frac{7}{15}, \quad P((X, Y) = (5, 5)) = \frac{4}{15}, \\ P((X, Y) = (1, 0)) = \frac{2}{15}, \quad P((X, Y) = (0, 0)) = \frac{1}{15}.$$

3. The random vector (X, Y) is a continuous random vector with density

$$f_{(X, Y)}(x, y) = 96\mathbf{1}_{[1, +\infty)}(x)\mathbf{1}_{[1, +\infty)}(y)\frac{1}{(x + y)^5}.$$

Solution is given in [A.2](#).

Exercise 2.40. We do a study on the processors produced by a small firm. We study the price and durability of each models. The firm proposes three models, and the observed data are given in the following table.

Model	Price	Average lifetime	Number produced
A	537	5.6	135k
B	425	6	208k
C	368	3.7	67k

We pick a processor produced by this firm uniformly at random, what is the covariance between its price and average lifetime?

Solution is given in [A.2](#).

2.5.2 Pearson correlation coefficient

In statistics, a coefficient obtained from the covariance and standard deviations is frequently used: the *Pearson⁴ correlation coefficient*.

Definition 2.5.4 (Pearson correlation coefficient). For two random variables X, Y define their *Pearson correlation coefficient*^a:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

$|\rho_{XY}| = 1$ is and only if X and Y are related by an affine transformation (i.e.: there are $a, b \in \mathbb{R}$ such that $Y = aX + b$).

^aThe fact that $|\rho_{XY}| \leq 1$ follows from the Cauchy-Schwartz inequality that we will see later (see Lemma [2.8.4](#)).

The reason for the normalization by the product of the standard deviations is as follows: suppose that X, Y measure two lengths, for example, the heights of two individuals. Saying that “the height of Bob influences the height of Alice” should not depend on the unit we chose to measure height, but the covariance does: a length measured in centimetres is a 100 times bigger than a length measured in metres, so the covariance between two lengths measure in cm. is 10000 times bigger than the covariance between the same lengths measured in m.. Normalizing by the standard deviation makes the coefficient ρ_{XY} unit-independent.

We saw that uncorrelated does not imply independent. We will see another type of common mistake: being correlated does not imply the presence of a causality relation.

⁴Karl Pearson, 27 March 1857 – 27 April 1936. English biostatistician and mathematician.

Remark 2.5.3 (Correlation does not imply causality). Let $X \sim \text{Uni}(\{0, 1, \dots, 30\})$ a random variable taking values $\{0, 1, \dots, 30\}$, each with equal probability (which can represent, for example, the number of rainy days in Lausanne during the month of November). Define then two new random variables:

$$Y = aX, \quad Z = bX, \quad a, b \in (0, +\infty).$$

We can imagine that Y represents the number of umbrellas sold in Lausanne during the month, and Z the number of hours that a typical Lausanne resident will spend watching Netflix during the month. Computing the correlation between Y and Z :

$$\begin{aligned} \text{Var}(Y) &= a^2 \text{Var}(X), & \text{Var}(Z) &= b^2 \text{Var}(X), \\ \text{Cov}(Y, Z) &= ab \text{Cov}(X, X) = ab \text{Var}(X), \\ \rho_{XY} &= \frac{\text{Cov}(Y, Z)}{\sqrt{\text{Var}(Y)\text{Var}(Z)}} = \frac{ab \text{Var}(X)}{\sqrt{a^2 \text{Var}(X) b^2 \text{Var}(X)}} = \frac{ab}{|ab|} = 1. \end{aligned}$$

We have that these two variables are strongly correlated, but it is clear from the way we defined the variables that this correlation comes from a *common causality* rather than a *mutual causality*. For example, it would be strange to infer from the correlation measurement that an increased time spend on Netflix is due to the recent purchase of an umbrella or the other way around...

2.5.3 A first linear regression

Linear regression is an “approximation by affine function”. To be more precise, if X, Y are two random variables and we believe them to be very dependent, we can try to see if the two are functions of each other (which is a very strong form of dependence, as knowing the value of X would also give the value of Y). The simplest type of function we can try is an affine function. In mathematical terms, we try to model the random vector (X, Y) by a random vector of the form $(X, aX + b)$ with $a, b \in \mathbb{R}$ to be chosen. Linear regression is about finding the “best” such a, b .

Pearson correlation coefficient naturally appears when trying to find the best values of a, b , as we shall now see. Here, by “best” we mean “that minimizes the expected square distance”. The choice of the square distance rather than the distance is due to its mathematical niceness.

Lemma 2.5.3. *Let X, Y be random variables. Then,*

$$\min_{a, b \in \mathbb{R}} E((Y - aX - b)^2) = (1 - \rho_{XY}^2) \text{Var}(Y).$$

Moreover, the minimum is attained for

$$a = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}, \quad b = E(Y - aX).$$

**Proof.* Write $\bar{X} = X - E(X)$, $\bar{Y} = Y - E(Y)$. We have

$$E((Y - aX - b)^2) = E((\bar{Y} - a\bar{X} + \tilde{b})^2)$$

where we introduced $\tilde{b} = E(Y) - aE(X) - b$. Then, expanding the square,

$$\begin{aligned} E((\bar{Y} - a\bar{X} + \tilde{b})^2) &= E(\bar{Y}^2) + a^2 E(\bar{X}^2) + \tilde{b}^2 - 2aE(\bar{X}\bar{Y}) - 2\tilde{b}aE(\bar{X}) + 2\tilde{b}E(\bar{Y}) \\ &= E(\bar{Y}^2) + a^2 E(\bar{X}^2) + \tilde{b}^2 - 2aE(\bar{X}\bar{Y}) \\ &= \text{Var}(Y) + a^2 \text{Var}(X) + \tilde{b}^2 - 2a \text{Cov}(X, Y) \end{aligned}$$

as $E(\bar{X}) = E(\bar{Y}) = 0$. To minimize the above, we want to take $\tilde{b} = 0$, and a such that $a^2 \text{Var}(X) - 2a \text{Cov}(X, Y) + \text{Var}(Y)$ is minimized. Differentiating w.r.t. a and equating to 0, we find that $a_* = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$ is the minimal value, and so $\tilde{b} = 0$ gives

$$b_* = E(Y) - a_* E(X) = E(Y - a_* X).$$

Plugging these values in the wanted quantity, we get

$$\begin{aligned} E((Y - a_* X - b_*)^2) &= E\left(\left(Y - E(Y) - \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E(X))\right)^2\right) \\ &= \text{Var}(Y) + \frac{\text{Cov}(X, Y)^2}{\text{Var}(X)^2} \text{Var}(X) - 2 \frac{\text{Cov}(X, Y)}{\text{Var}(X)} \text{Cov}(X, Y) \\ &= (1 - \rho_{XY}^2) \text{Var}(Y). \end{aligned}$$

□

Remark 2.5.4. The above result is theoretical. In practice, we suppose that we are given a sequence of (independently sampled) realisations of the the random vector (X, Y) . We then plot these realisations in a coordinate frame and draw the line that minimizes the sum over the realisations of the square distances to the line. As the size of the sequence augments, the line converges to the affine function given in Lemma 2.5.3. See Figure 2.5. This partially explains the name “linear regression” (which would probably have been better named “affine regression”, but who are we to judge?).

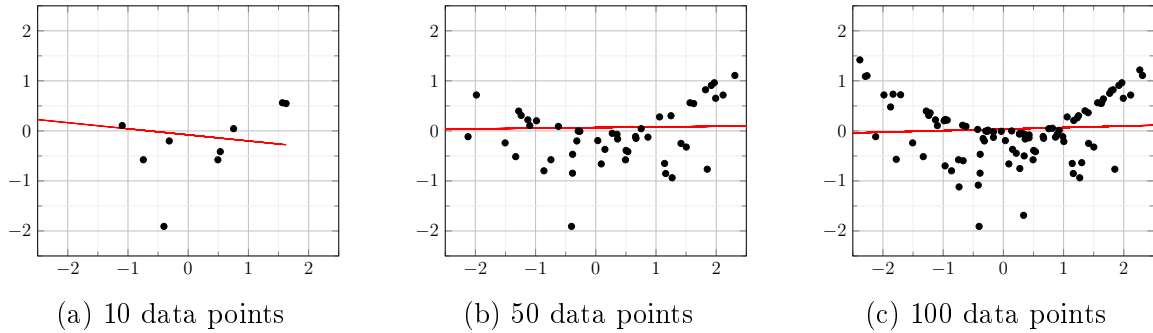
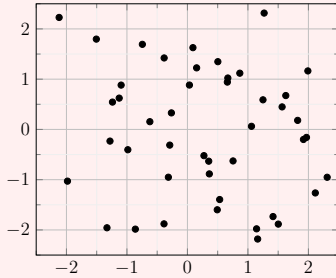


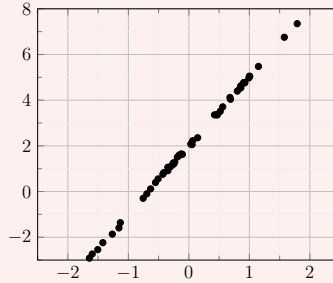
Figure 2.5: Regression line for the sum and product of two independent standard Gaussian.

Exercise 2.41. Associate each of the values of ρ_{XY} to a graph of samples from (X, Y) .

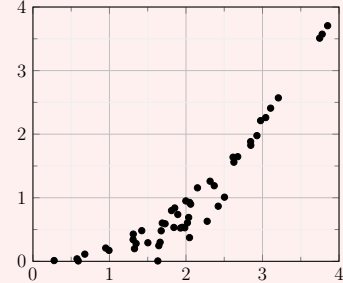
(A) $\rho_{XY} = 0.66$, (B) $\rho_{XY} = 0$, (C) $\rho_{XY} = 1$.



(a)



(b)



(c)

Solution is given in [A.2](#).

Exercise 2.42. Compute ρ_{XY} in each of the following situations.

1. $Y = aX + b$ for some $a, b \in \mathbb{R}$ with $a \neq 0$.
2. X is a continuous random variable with density $f_X(x) = \mathbb{1}_{[0,1]}(x)$, and $Y = \mathbb{1}_{[0,1/2]}(X)$.
3. U, V are two independent continuous random variables with density $f(x) = \mathbb{1}_{[0,1]}(x)$. X, Y are given by $X = U + V$, $Y = UV$.
4. X is a continuous random variable with density $f_X(x) = \frac{1}{3}\mathbb{1}_{[0,3]}(x)$, and $Y = X^2$.
5. (X, Y) is a continuous random vector with density $f_{(X,Y)}(x, y) = \frac{1}{2\pi}e^{-\sqrt{x^2+y^2}}$.

Solution is given in [A.2](#).

2.5.4 Covariance matrix of random vectors

When dealing with a random vector $X = (X_1, \dots, X_n)$, one can look at its expected value:

$$E(X) = (E(X_1), \dots, E(X_n)).$$

The next thing we can study is its *covariance matrix*: it is the symmetric matrix with entry ij given by the covariance between X_i and X_j .

$$\begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \cdots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \cdots & \text{Var}(X_n) \end{pmatrix}$$

It is simply a convenient way to store the covariance structure of the random variables constituting the random vector.

2.6 Classical examples of random variables

2.6.1 Discrete random variables

Constant random variable, Dirac mass

The simplest random variable is a constant one: for $c \in \mathbb{R}$,

$$X : \Omega \rightarrow \mathbb{R}, \quad X(\omega) = c \quad \forall \omega \in \Omega.$$

Its law is called *the Dirac mass at c* , denoted δ_c : it is the probability measure on $\mathbb{R}$ given by

$$\delta_c(A) = \begin{cases} 1 & \text{if } c \in A, \\ 0 & \text{else.} \end{cases}$$

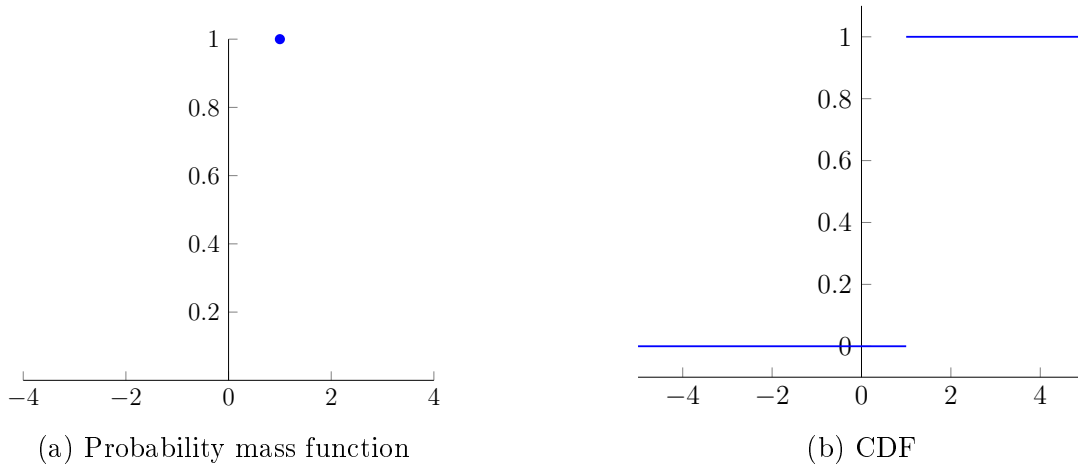


Figure 2.7: δ_1 .

Bernoulli random variable

The Bernoulli random variable corresponds to a “yes/no” measurement: it can take only values 0 or 1. It depends on a parameter $p \in [0, 1]$ which gives the probability of the variable being 1. More formally, a Bernoulli random variable of parameter $p \in [0, 1]$ is a random variable $X : \Omega \rightarrow \mathbb{R}$ such that

$$P(X = 1) = p = 1 - P(X = 0).$$

We will denote $X \sim \text{Bern}(p)$ for “ X is a Bernoulli random variable of parameter p ”.

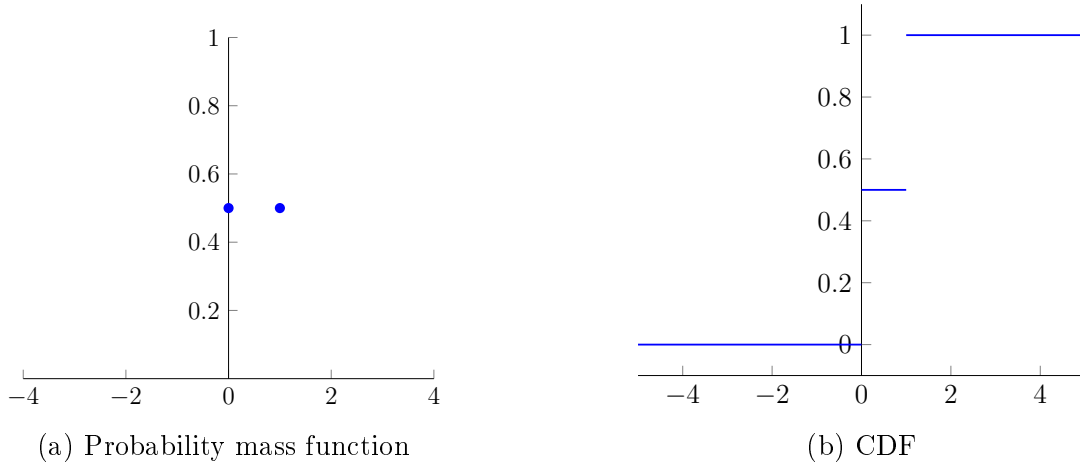


Figure 2.8: Bern(1/2).

We can compute its expectation

$$E(X) = 1 \cdot P(X = 1) + 0 \cdot P(X = 0) = P(X = 1) = p.$$

Its variance

$$\text{Var}(X) = E(X^2) - E(X)^2 = 1^2 \cdot P(X = 1) + 0^2 \cdot P(X = 0) - p^2 = p - p^2 = p(1 - p).$$

And its moment generating function: for $t \in \mathbb{R}$,

$$E(e^{tX}) = e^{t \cdot 1} P(X = 1) + e^{t \cdot 0} P(X = 0) = e^t p + 1 - p = 1 + p(e^t - 1).$$

Example 2.6.1. A simple example, corresponding precisely to the “yes-no” question motivation is the following. Let A be an event. Then, $\mathbb{1}_A : \Omega \rightarrow \mathbb{R}$ is a Bernoulli random variable of parameter $P(A)$.

Binomial random variable

A random variable $X : \Omega \rightarrow \mathbb{R}$ is a binomial random variable with parameter $n \in \mathbb{N}, p \in [0, 1]$, denoted $X \sim \text{Bin}(n, p)$ if

$$P(X = k) = \mathbb{1}_{\{0, \dots, n\}}(k) \binom{n}{k} p^k (1 - p)^{n-k}$$

In particular $P(X \in \{0, \dots, n\}) = 1$. The binomial random variable corresponds to the law of the number of 1's obtained when repeating (independently) n times a Bernoulli measurement.

Lemma 2.6.1. Let $n \in \mathbb{N}$, $p \in [0, 1]$. Let $X_1, \dots, X_n$ be an independent family of Bernoulli random variables of parameter p . Define

$$Y = \sum_{k=1}^n X_k.$$

Then, $Y \sim \text{Bin}(n, p)$.

Proof. See Exercise 2.43. □

Exercise 2.43. 1. Prove Lemma 2.6.1.

2. Let $X \sim \text{Bin}(n, p)$. Show that

$$E(X) = np, \quad \text{Var}(X) = np(1 - p), \quad E(e^{tX}) = (1 + p(e^t - 1))^n$$

for any $t \in \mathbb{R}$.

Solution is given in A.2.

Geometric random variable

A random variable $X : \Omega \rightarrow \mathbb{R}$ is a geometric random variable with parameter $p \in (0, 1]$, denoted $X \sim \text{Geo}(p)$, if

$$P(X = k) = \mathbf{1}_{k \in \mathbb{N}^*} (1 - p)^{k-1} p.$$

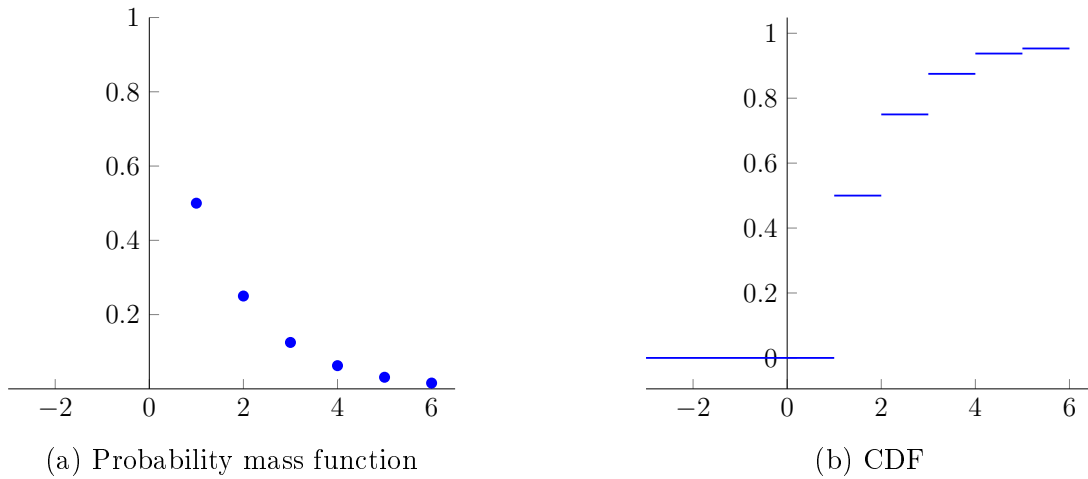


Figure 2.9: $\text{Geo}(1/2)$.

A geometric random variable is the number of independent Bernoulli random variables one has to sample before getting a 1.

Lemma 2.6.2. Let $X_1, X_2, \dots$ be an i.i.d. sequence of Bernoulli random variables with parameter p . Define

$$Y = 1 + \sum_{n \geq 1} \prod_{i=1}^n (1 - X_i),$$

the number of trials before getting a 1 in the sequence^a. Then, $Y \sim \text{Geo}(p)$.

^aNote that if $\epsilon_1, \epsilon_2, \dots \in \{0, 1\}$, then $\prod_{i=1}^n (1 - \epsilon_i) = \mathbf{1}_{\epsilon_1 = \dots = \epsilon_n = 0}$

Proof. See Exercise 2.44. □

Remark 2.6.1. Some people define a geometric random variable to be given by

$$P(X = k) = \mathbb{1}_{\mathbb{N}}(k)(1 - p)^k p.$$

In that case, it is the number of 0's before getting a 1 in an i.i.d. sequence of Bernoulli random variables.

Lemma 2.6.3 (Memory loss). *Let $X \sim \text{Geo}(p)$ be a geometric random variable. Then, for any $n > k \in \mathbb{N}$,*

$$P(X = n \mid X > k) = P(X = n - k).$$

In particular, under the law $P(\cdot \mid X > k)$, $X - k$ follows a geometric law of parameter p .

Proof. Start by computing $P(X > a)$ pour $a \geq 0$ (integer):

$$P(X > a) = \sum_{k > a} P(X = k) = \sum_{k > a} p(1 - p)^{k-1} = p \sum_{k \geq 0} (1 - p)^{k+a} = (1 - p)^a.$$

We thus find that for $n > k$,

$$P(X = n \mid X > k) = \frac{P(X = n)}{P(X > k)} = \frac{p(1 - p)^{n-1}}{(1 - p)^k} = p(1 - p)^{n-k-1} = P(X = n - k).$$

□

We can see this “loss of memory” property as follows: a geometric random variable is the number of independent coin toss needed to make a 1. If we pause after k tosses and that the first k coins all gave 0, the following coins being independent of the first k , we end up with simply a sequence of independent coins tosses, exactly as we started.

Exercise 2.44. Let $X \sim \text{Geo}(p)$.

1. Show that

$$E(X) = \frac{1}{p}, \quad \text{Var}(X) = \frac{1 - p}{p^2}.$$

2. Show that

$$E(e^{tX}) = \begin{cases} \frac{pe^t}{1 - (1 - p)e^t} & \text{for } t < |\ln(1 - p)|, \\ +\infty & \text{for } t \geq |\ln(1 - p)|. \end{cases}$$

3. Show Lemma 2.6.2.

Solution is given in A.2.

Poisson random variable

Let $\lambda \geq 0$. A random variable $X : \Omega \rightarrow \mathbb{R}$ is a *Poisson random variable of parameter λ* , denoted $X \sim \text{Poi}(\lambda)$, if

$$P(X = k) = \mathbb{1}_{k \in \mathbb{N}} e^{-\lambda} \frac{\lambda^k}{k!}.$$

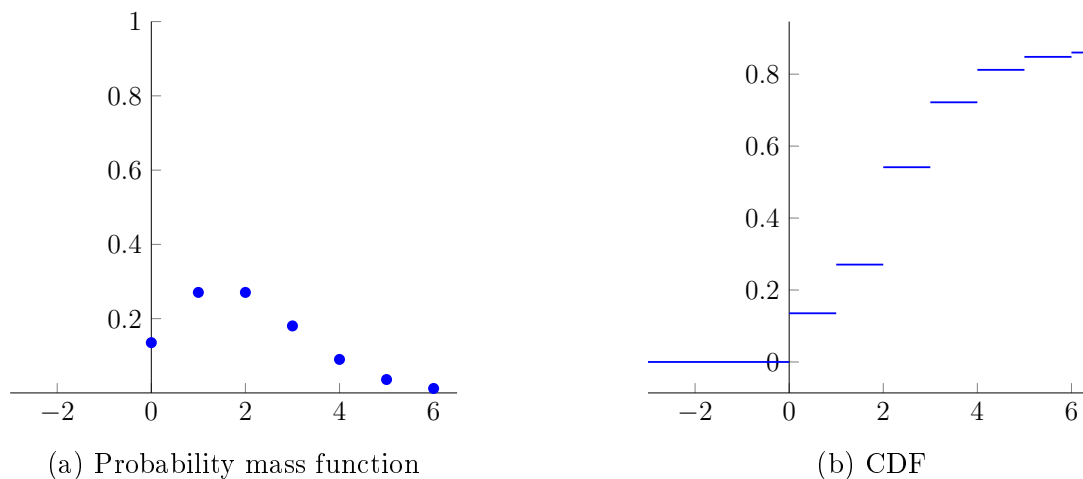


Figure 2.10: Poi(2).

Poisson random variables have a useful characterisation.

Lemma 2.6.4. *Let X be a random variable. Then, the two following points are equivalent:*

1. $X \sim \text{Poi}(\lambda)$;
2. $P(X = 0) = e^{-\lambda}$ and for all $k \in \mathbb{N}$,

$$\frac{P(X = k + 1)}{P(X = k)} = \frac{\lambda}{k + 1}.$$

Proof. See Exercise 2.45. □

Exercise 2.45. Let $\lambda \in [0, +\infty)$, and $X \sim \text{Poi}(\lambda)$.

1. Show that

$$E(X) = \lambda, \quad \text{Var}(X) = \lambda, \quad E(e^{tX}) = \exp(\lambda(e^t - 1)) \quad \forall t \in \mathbb{R}.$$

2. Show Lemma 2.6.4.

Solution is given in A.2.

Uniform random variable (finite case)

Let $J \subset \mathbb{R}$ be finite. A random variable $X : \Omega \rightarrow \mathbb{R}$ is a *uniform random variable on J* denoted $X \sim \text{Uni}(J)$ if

$$P(X = x) = \frac{1}{|J|} \quad \forall x \in J.$$

In particular, $P(X \in J) = 1$. We will often look at $J = \{0, 1, \dots, n\}$ or $J = \{1, \dots, n\}$ for some $n \geq 1$.

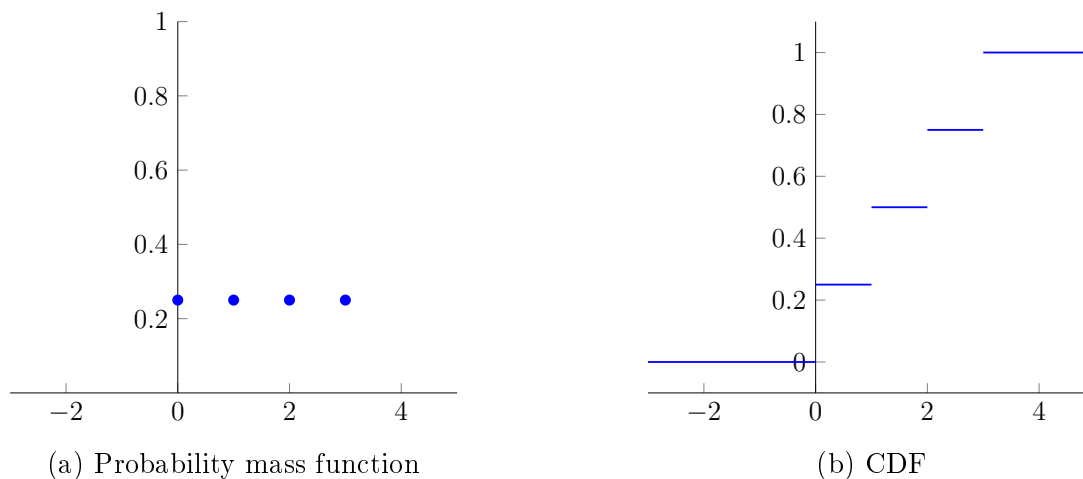


Figure 2.11: $\text{Uni}(\{0, 1, 2, 3\})$.

A property of uniform random variables is that conditioning on $\{X \in I\}$ for some $I \subset J$ yields a uniform distribution on I .

Lemma 2.6.5. *Let $J \subset \mathbb{R}$ be finite, and let $X \sim \text{Uni}(J)$. Let $I \subset J$. Then, for any $A \subset I$,*

$$P(X \in A | X \in I) = \frac{|A|}{|I|}.$$

Proof. This was part of Exercise 2.25. □

Remark 2.6.2. Uniform random variables on a finite set J are, in some sense, the “most random” random variable on J : they have the smallest predictive power. Therefore, it is common use in modelization to use uniform random variables when the only information we have about a measurement is the set of values it can take.

Exercise 2.46. Let $n \geq 1$.

1. Let $X \sim \text{Uni}(\{0, 1, \dots, n\})$. Show that^a

$$E(X) = \frac{n}{2}, \quad \text{Var}(X) = \frac{n^2 + 2n}{12}, \quad E(e^{tX}) = \frac{e^{(n+1)t} - 1}{(n+1)(e^t - 1)} \quad \forall t \in \mathbb{R}.$$

2. Let $Y \sim \text{Uni}(\{1, \dots, n\})$. Show that

$$E(Y) = \frac{n+1}{2}, \quad \text{Var}(Y) = \frac{n^2 - 1}{12}, \quad E(e^{tY}) = \frac{e^{nt} - 1}{n(1 - e^{-t})} \quad \forall t \in \mathbb{R}.$$

Solution is given in A.2.

^aHint: first show that $\sum_{k=1}^n k = \frac{n(n+1)}{2}$, that $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$, and recall the truncated geometric series: $\sum_{k=m}^n z^k = \frac{z^m - z^{n+1}}{1-z}$.

2.6.2 Continuous random variables

Uniform random variable on an interval

Let $a < b \in \mathbb{R}$. A random variable $X : \Omega \rightarrow \mathbb{R}$ is a *uniform random variable on $[a, b]$* , denoted $X \sim \text{Uni}([a, b])$, if it is a continuous random variable with probability density given by

$$f_X(x) = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x).$$

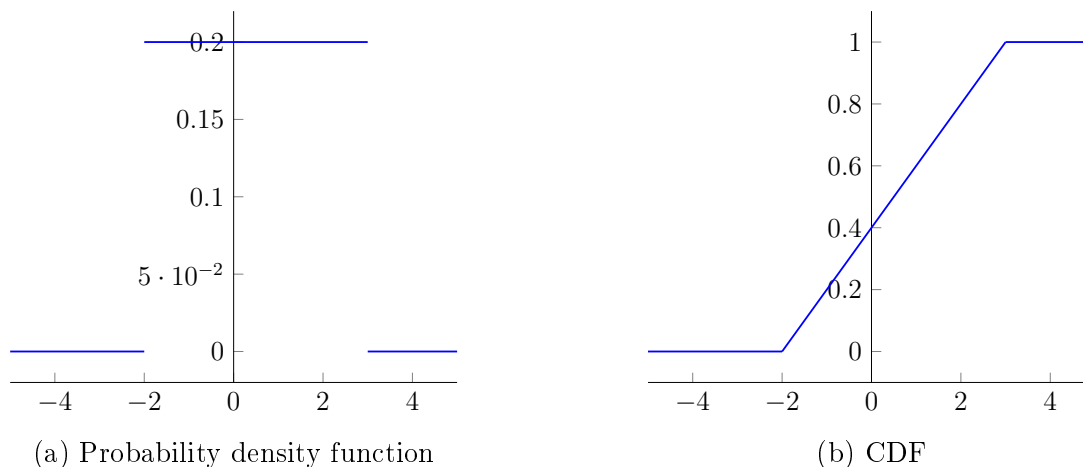


Figure 2.12: $\text{Uni}([-2, 3])$.

As in the uniform case on finite sets, continuous uniforms react well to conditioning.

Lemma 2.6.6. *Let $a < b < c < d \in \mathbb{R}$. Then, if $X \in \text{Uni}([a, d])$,*

$$P(t_1 \leq X \leq t_2 \mid b \leq X \leq c) = \frac{t_2 - t_1}{c - b}, \quad \forall b \leq t_1 < t_2 \leq c,$$

which is equivalent to say that under the conditioning $\{X \in [b, c]\}$, X is a uniform random variable on $[b, c]$.

Proof. See Exercise 2.47. □

Exercise 2.47. Let $a < b$ and $X \sim \text{Uni}([a, b])$.

1. Show that

$$E(X) = \frac{b+a}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}, \quad E(e^{tX}) = \begin{cases} 1 & \text{if } t = 0, \\ \frac{e^{bt} - e^{at}}{t(b-a)} & \text{if } t \neq 0. \end{cases}$$

2. Prove Lemma 2.6.6.

Solution is given in A.2.

Gaussian random variables

Let $\mu \in \mathbb{R}$, $\sigma \geq 0$. A random variable $X : \Omega \rightarrow \mathbb{R}$ is a *Gaussian random variable with mean μ and variance σ^2* , denoted $X \sim \mathcal{N}(\mu, \sigma^2)$, if it is a continuous random variable with density

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \quad (2.3)$$

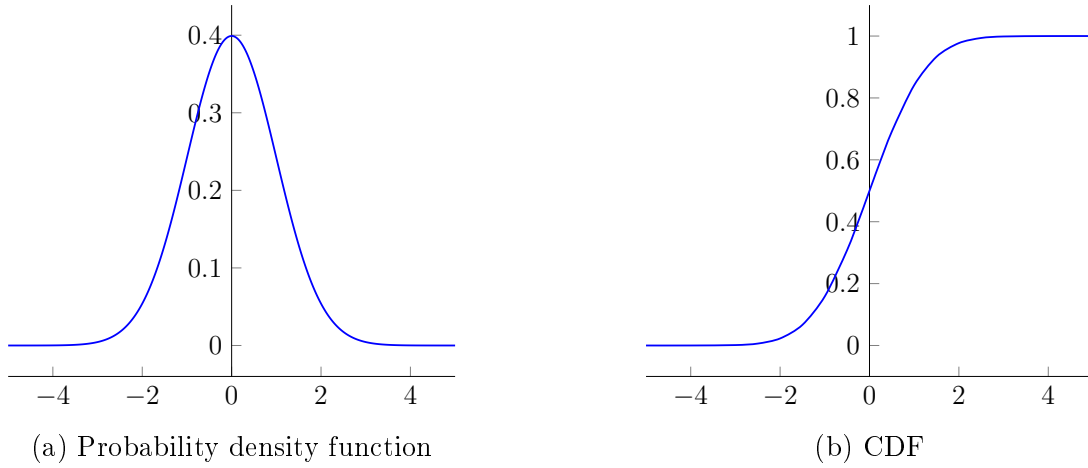


Figure 2.13: $\mathcal{N}(0, 1)$.

Exercise 2.48. Let $\mu \in \mathbb{R}$, $\sigma \geq 0$.

1. Show that $\int_{-\infty}^{+\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \sqrt{2\pi\sigma^2}$, so that f_X defined in (2.3) is indeed a probability density.

Hint: change variables to reduce to $\mu = 0$, $\sigma = 1$. Then, try to compute the square of this integral as an integral over $\mathbb{R}^2$, and use polar coordinates.

2. Show that if $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$E(X) = \mu, \quad \text{Var}(X) = \sigma^2, \quad E(e^{tX}) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right) \quad \forall t \in \mathbb{R}.$$

Hint: for the last point, try completing the square.

3. Show that if $X \sim \mathcal{N}(0, 1)$, and $m \in \mathbb{N}^*$,

$$E(X^m) = \begin{cases} 0 & \text{if } m \text{ is odd,} \\ \frac{m!}{2^{m/2}(m/2)!} & \text{if } m \text{ is even.} \end{cases}$$

Hint: remember what is an induction.

Solution is given in A.2.

Part of the interest and the omnipresence of Gaussian random variable is the following “stability under sums and scaling” Lemma.

Lemma 2.6.7. Let $X, Y : \Omega \rightarrow \mathbb{R}$ be two independent Gaussian random variables. Suppose that $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$. Then,

1. the random variable $\tilde{X} = (X - \mu_1)/\sigma_1$ is a centred and reduced Gaussian random variable: $\tilde{X} \sim \mathcal{N}(0, 1)$;
2. the random variable $Z = X + Y$ is a Gaussian random variable with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$: $Z \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof. We will use two different tools (we could use twice any of the two tools).

1. In this case, we use the cumulative distribution function. We have that for any $t \in \mathbb{R}$,

$$\begin{aligned} P(\tilde{X} \leq t) &= P(X \leq \sigma_1 t + \mu_1) \\ &= \int_{-\infty}^{\sigma_1 t + \mu_1} \frac{1}{\sqrt{2\pi\sigma_1^2}} e^{-(x-\mu_1)^2/(2\sigma_1^2)} dx \\ &= \int_{-\infty}^t \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy \end{aligned}$$

where we used the change of variable $y = (x - \mu_1)/\sigma_1$. But this is the cumulative distribution function of a $\mathcal{N}(0, 1)$ evaluated at t .

2. For the second point, we use the moment generating function. For $t \in \mathbb{R}$, we have that the moment generating function of Z is

$$\begin{aligned} E(e^{tZ}) &= E(e^{t(X+Y)}) = E(e^{tX})E(e^{tY}) = e^{\mu_1 t + \sigma_1^2 t^2/2} e^{\mu_2 t + \sigma_2^2 t^2/2} \\ &= \exp((\mu_1 + \mu_2)t + (\sigma_1^2 + \sigma_2^2)t^2/2) \end{aligned}$$

as X, Y are independent, and Gaussian. The last expression is the moment generating function of a $\mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$ evaluated at t as wanted.

□

Iterating this Lemma, we obtain that if $X_1, \dots, X_n$ are an i.i.d. sequence of $\mathcal{N}(0, 1)$ random variables, then their sum normalized by $\sqrt{n}$ also is a $\mathcal{N}(0, 1)$:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i \sim \mathcal{N}(0, 1).$$

The “invariance property” of Gaussian random variables is the main reason behind the Central Limit Theorem (see Section 3.2.3).

Exponential random variable

Let $\lambda > 0$. A random variable $X : \Omega \rightarrow \mathbb{R}$ is an *exponential random variable of parameter λ* , denoted $X \sim \text{Exp}(\lambda)$, if X is a continuous random variable with density

$$f_X(x) = \mathbb{1}_{[0, \infty)}(x) \lambda e^{-\lambda x}. \quad (2.4)$$

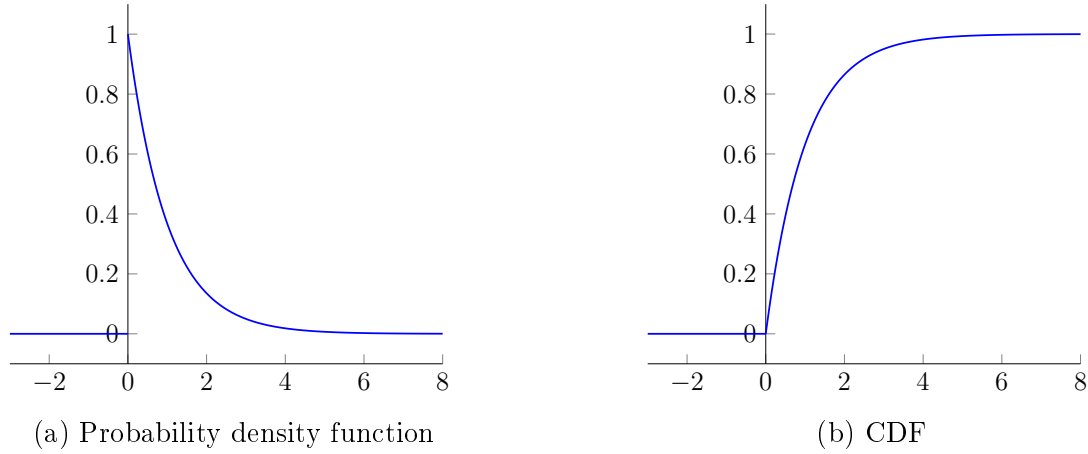


Figure 2.14: $\text{Exp}(1)$.

The exponential random variable is the continuous version of the geometric random variable, it is therefore not a surprise that they share the “memory loss” property.

Lemma 2.6.8. *Let $\lambda > 0$, and $X \sim \text{Exp}(\lambda)$. Then for any $0 < a < b$,*

$$P(X \geq b \mid X \geq a) = P(X \geq b - a).$$

In particular, under the conditioning $\{X \geq a\}$, the variable $X - a$ is an exponential random variable with parameter λ .

Proof. See Exercise 2.49 □

Remark 2.6.3. The “loss of memory” property makes exponential random variables very well suited to model waiting times.

Exercise 2.49. Let $\lambda > 0$ and $X \sim \text{Exp}(\lambda)$.

1. Show that

$$\int_0^{+\infty} e^{-\lambda x} dx = \frac{1}{\lambda},$$

so that f_X defined in (2.4) is indeed a probability density function.

2. Show that

$$E(X) = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

3. Show that

$$E(e^{tX}) = \begin{cases} \frac{\lambda}{\lambda - t} & \text{if } t < \lambda, \\ +\infty & \text{if } t \geq \lambda. \end{cases}$$

4. Prove Lemma 2.6.8.

Solution is given in A.2.

Cauchy random variable

Let $x_0 \in \mathbb{R}$ and $\alpha > 0$. A random variable $X : \Omega \rightarrow \mathbb{R}$ is a *Cauchy random variable*, denoted $X \sim \text{Cauchy}(x_0, \alpha)$, if it is a continuous random variable with density

$$f_X(x) = \frac{\alpha}{\pi((x - x_0)^2 + \alpha^2)}. \quad (2.5)$$

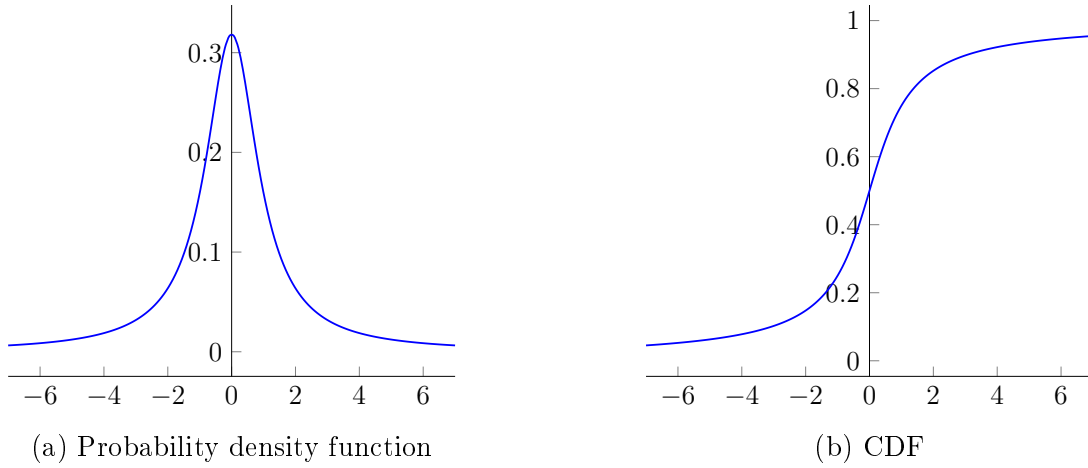


Figure 2.15: Cauchy(0, 1).

Cauchy random variables are often useful to study what goes wrong if we forget the assumption $E(|X|) < \infty$ when trying to compute expectations: indeed, Cauchy random variables *do not admit a well defined expected value*. As example, let us look at the case $x_0 = 0$ and $\alpha = 1$. We have

$$\int_{-\infty}^{+\infty} dx \frac{|x|}{\pi(x^2 + 1)} \geq \frac{1}{\pi} \int_1^{+\infty} dx \frac{1}{x + 1} = \infty.$$

2.6.3 Summary of usual random variables

Variable	Expectation	Variance	Mom. Gen. Fct.
δ_c	c	0	e^{tc}
Bern(p)	p	$p(1-p)$	$1 + p(e^t - 1)$
Bin(n, p)	np	$np(1-p)$	$(1 + p(e^t - 1))^n$
Geo(p)	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
Poi(λ)	λ	λ	$\exp(\lambda(e^t - 1))$
Uni($\{0, 1, \dots, n\}$)	$\frac{n}{2}$	$\frac{n^2+1}{12}$	$\frac{e^{(n+1)t}-1}{(n+1)(e^t-1)}$
Uni($\{1, \dots, n\}$)	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$	$\frac{e^{nt}-1}{n(1-e^{-t})}$
Uni($[a, b]$)	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{bt}-e^{at}}{t(b-a)}$
$\mathcal{N}(\mu, \sigma^2)$	μ	σ^2	$\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$
Exp(λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}$
Cauchy(x_0, α)	N.D.	N.D.	N.D.

Remark 2.6.4. The properties of the different random variables (being integer-valued or real valued, memory loss, etc.) are useful when choosing what variable to use when modelling a real situation: we can use the known properties of the experiment we want to model to select what properties we want for the random variable we will use to perform the modelling.

2.6.4 Further relations between random variables

The next exercise shows that Poisson random variables typically appear in situations where we are looking at “the number of occurrences of an event over a fixed time interval”: we can see the X_i ’s as the times between two occurrences, and M as the number of occurrences before time A .

Exercise 2.50. Let $X_1, X_2, \dots$ be an independent family of exponential random variables with rate 1 ($X_i \sim \text{Exp}(1)$ for all i 's). Let $S_0 = 0$, and for $n \geq 1$, define

$$S_n = \sum_{i=1}^n X_i.$$

Now, let $A > 0$ and define

$$M = \max\{n \geq 0 : S_n \leq A\}.$$

1. What is the value of $P(M = k)$ for $k \in \mathbb{N}$?
2. Deduce the law of M .

Solution is given in [A.2](#).

The next exercise is a particular case of a very useful sampling method allowing to sample from any given distribution using the CDF and a uniform random variable on $[0, 1]$. It is used in many programming languages to sample random variables with “complicated” distributions.

Exercise 2.51. Let $Z \sim \mathcal{N}(0, 1)$ and $U \sim \text{Uni}([0, 1])$.

1. Show that F_Z is continuous and strictly increasing.
2. Show that $F_Z : \mathbb{R} \rightarrow (0, 1)$ is invertible. Denote its inverse $F_Z^{-1} : (0, 1) \rightarrow \mathbb{R}$.
3. Show that the random variable $F_Z^{-1}(U)$ has the same law as Z .

Solution is given in [A.2](#).

Exercise 2.52. Let $\lambda > 0$. Let $n \geq 1$. Let $X_1, X_2, \dots, X_n$ be an i.i.d. family of random variables with common distribution $\text{Exp}(\lambda)$. Define the random variables $T_- = \min(X_1, \dots, X_n)$, and $T_+ = \max(X_1, \dots, X_n)$.

1. Show that $T_- \sim \text{Exp}(\lambda n)$.
2. Compute F_{T_+} , the CDF of T_+ .
3. *Difficult.* Based on the previous point, show that $E(T_+) = \frac{1}{\lambda} \sum_{k=1}^n \frac{1}{k}$. In particular, as $n \rightarrow \infty$, $\frac{E(T_+)}{\ln(n)}$ converges towards $\frac{1}{\lambda}$.

Hint: draw inspiration from Exercise 2.32. For the last point: remember that, when defined, F'_X is a density for the variable X .

Solution is given in [A.2](#).

2.7 Gaussian vectors

We saw in the last section several examples of random variables. There is an extremely important class of examples of random vectors that are closely related to the Gaussian

random variable: *Gaussian vectors*. They play a central role in Statistics, Physics, Theoretical Computer Science, Machine Learning, Image Processing, and many others.

2.7.1 Definition

Let $X = (X_1, X_2, \dots, X_n) : \Omega \rightarrow \mathbb{R}^n$ be a random vector. Let Σ be a symmetric positive semi-definite matrix⁵, and $\mu \in \mathbb{R}^n$. X is a *Gaussian vector* (or a *multivariate normal*) with mean μ and covariance matrix Σ if any of the following equivalent conditions holds.

- X has moment generating function $E(e^{t \cdot X}) = e^{\mu \cdot t + \frac{1}{2} t \cdot \Sigma t}$.
- $X \stackrel{\text{Law}}{=} \Sigma Z + \mu$, where $Z = (Z_1, \dots, Z_n)^T$, and $Z_1, \dots, Z_n$ form an i.i.d. sequence of $\mathcal{N}(0, 1)$.
- If Σ is positive definite, X is a continuous random vector with density

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} e^{-\frac{1}{2}(x-\mu) \cdot \Sigma^{-1}(x-\mu)}. \quad (2.6)$$

The law of X is entirely determined by the mean of X , $E(X) = \mu \in \mathbb{R}^n$, and its covariance matrix Σ ,

$$\Sigma_{ij} = \text{Cov}(X_i, X_j).$$

When $\mu = 0$, we say that X is a *centred* Gaussian vector.

Part of the power of Gaussian vectors lie in the next characterisation of Gaussian vectors.

Lemma 2.7.1. *Let $X = (X_1, X_2, \dots, X_n) : \Omega \rightarrow \mathbb{R}^n$ be a random vector. X is a Gaussian vector if and only if for any $t_1, \dots, t_n \in \mathbb{R}$, the random variable $t \cdot X = \sum_{i=1}^n t_i X_i$ is a Gaussian random variable (not necessarily centred or reduced).*

Remark 2.7.1. In particular, Lemma 2.7.1 implies that linear functions of Gaussian vectors (like taking a subset of the variables) also are Gaussian vectors.

2.7.2 Properties

Gaussian vectors enjoy a wealth of properties that makes working with them very convenient. A first property is that independence is implied by absence of correlations.

Lemma 2.7.2. *Let $X = (X_1, X_2, \dots, X_n) : \Omega \rightarrow \mathbb{R}^n$ be a Gaussian vector with covariance matrix Σ . Let $I \subset \{1, \dots, n\}$. Then, the vectors $(X_i)_{i \in I}$ and $(X_j)_{j \in I^c}$ are independent if and only if*

$$\text{Cov}(X_i, X_j) = 0 \quad \forall i \in I, \forall j \in I^c.$$

In particular, $(X_i)_{i=1, \dots, n}$ is an independent family if and only if Σ is a diagonal matrix.

⁵Recall that a matrix A is positive semi-definite if $x^T \cdot Ax \geq 0$ for all $x \in \mathbb{R}^n$ non-zero, or, equivalently, if all the eigenvalues of A are non-negative. Positive definite asks for strict inequalities.

We already saw that independent implies un-correlated. The other direction follows from the expression (2.6) for the density of Gaussian vectors: a diagonal covariance matrix gives a product density function.

Remark 2.7.2. Note however that one has *first* to show that the vector is a Gaussian vector in order to apply Lemma 2.7.2. When true, proving Gaussianity is in general a highly non-trivial task if the law of the vector is not given explicitly.

Another properties, fundamental for computations, is the *Gaussian integration by parts*. We already encountered such a principle in Exercise 2.48, and it allowed us to compute the moment of a standard Gaussian random variable.

Lemma 2.7.3. Let $X = (X_1, X_2, \dots, X_n) : \Omega \rightarrow \mathbb{R}^n$ be a centred ($E(X_i) = 0$ for all i 's) Gaussian random vector with covariance matrix Σ . Then, for any $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable, growing at most like $e^{\|x\|^{2-\epsilon}}$ at infinity for some $\epsilon > 0$, and any $i \in \{1, \dots, n\}$,

$$E(X_i f(X)) = \sum_{j=1}^n \Sigma_{ij} E(\partial_j f(X))$$

where $\partial_j f(x) = \frac{\partial}{\partial x_j} f(x)$.

Exercise 2.53. Show Lemma 2.7.3 when $n = 1$.

Solution is given in A.2.

Exercise 2.54. Let $X = (X_1, X_2)$ be a centred Gaussian vector with covariance matrix Σ .

1. In the case $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, find a 2×2 matrix A such that AX is a centred Gaussian vector with identity covariance matrix (X is seen as a column vector).
2. In the case $\Sigma = \begin{pmatrix} a+1 & -1 \\ -1 & a^2 \end{pmatrix}$, find the values of a such that $(X_1 - X_2, X_1 + X_2)$ is a centred Gaussian vector with diagonal covariance matrix.

Solution is given in A.2.

Exercise 2.55. Let $n \geq 2$. Let $(X_1, \dots, X_n)$ be a centred Gaussian vector with covariance matrix Σ . Define

$$R_n = \sum_{i=1}^n X_i^2.$$

Compute $\text{Var}(R_n)$ in terms of the entries of $\Sigma = (\Sigma_{ij})_{i,j=1}^n$.

Solution is given in A.2.

2.8 Probabilistic inequalities and applications

2.8.1 Markov's inequality

Theorem 2.8.1. *Let X be a non-negative random variable ($X : \Omega \rightarrow [0, +\infty)$). Then, for any $a > 0$,*

$$P(X \geq a) \leq \frac{E(X)}{a}.$$

Proof. For any $a > 0$, we have that for every realisation $\omega \in \Omega$,

$$a\mathbb{1}_{X(\omega) \geq a} \leq X(\omega)\mathbb{1}_{X(\omega) \geq a} \leq X(\omega)$$

as $X(\omega) \geq 0$. So,

$$E(X) \geq E(X\mathbb{1}_{X \geq a}) \geq aE(\mathbb{1}_{X \geq a}) = aP(X \geq a).$$

Dividing everything by a gives the wanted inequality. $\square$

2.8.2 First moment method

The first moment method is a simple observation: if we have a random variable X taking values in the non-negative integers, $P(X \in \mathbb{N}) = 1$, we can upper bound the probability that X is non-zero by using its mean:

$$P(X \neq 0) = P(X > 0) = E(\mathbb{1}_{X > 0}) \leq E(\mathbb{1}_{X > 0}X) \leq E(X).$$

As an application, we will study the following problem. We are given a uniform sequence of n bits, and we want to understand what is the longest run of 1's in the sequence as n becomes large (one can think that the bits represent the colours in roulette, and the run sequence of 1's corresponds to the largest number of time “red” was drawn in a row). Formally, we have an i.i.d. sequence $X_1, \dots, X_n$ of Bernoulli random variables of parameter $1/2$, and we wonder about the typical value of

$$M_n = \max \{k \geq 0 : \exists i \in \{1, \dots, n-k\} X_{i+j} = 1, j = 0, \dots, k-1\}.$$

To study M_n , it will be convenient to introduce

$$Y_k = \sum_{i=1}^{n-k+1} \prod_{j=0}^{k-1} \mathbb{1}_{X_{i+j}=1} = \sum_{i=1}^{n-k+1} \prod_{j=0}^{k-1} X_{i+j},$$

the number of runs of 1's with length k in the whole sequence⁶. We can now upper bound the probability to see a run of length k using the first moment method:

$$\begin{aligned} P(\exists \text{ a 1-run of length } k) &\leq E(Y_k) = \sum_{i=1}^{n-k+1} E\left(\prod_{j=0}^{k-1} X_{i+j}\right) \\ &= \sum_{i=1}^{n-k+1} \prod_{j=0}^{k-1} E(X_{i+j}) = \sum_{i=1}^{n-k+1} 2^{-k} = (n-k+1)2^{-k} \end{aligned}$$

⁶Indeed: $\prod_{j=0}^{k-1} \mathbb{1}_{X_{i+j}=1}$ is 1 if and only if $X_i, X_{i+1}, \dots, X_{i+k-1}$ are all 1, which is equivalent to say that there is a 1-run of length k starting at i . Summing over i accounts for all the possible starting point of a 1-run of length k .

where we used that the X'_i 's are independent in the last equality. Now, to have $M_n \geq k$, one needs to have at least one 1-run of length k (the two are equivalent), so

$$P(M_n \geq k) = P(\exists \text{ a 1-run of length } k) \leq (n - k + 1)2^{-k} \leq n2^{-k}.$$

In particular, we can see that if $k > \log_2(n)$, this probability starts to be much smaller than 1. We obtained that for any $\epsilon > 0$,

$$P(M_n \geq \lceil (1 + \epsilon) \log_2(n) \rceil) \leq n2^{-\lceil (1 + \epsilon) \log_2(n) \rceil} \leq n2^{-(1 + \epsilon) \log_2(n)} = n^{-\epsilon} \xrightarrow{n \rightarrow \infty} 0.$$

In particular, we obtain that the longest run of 1's is at most of order $\log_2(n)$.

2.8.3 Chebychev's inequality

Chebychev's inequality can be seen as a re-phrasing of Markov's inequality.

Lemma 2.8.2. *Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Let $g : \mathbb{R} \rightarrow \mathbb{R}_+$ be an increasing function such that $g(X)$ is a random variable. Then,*

$$P(X \geq a) \leq \frac{E(g(X))}{g(a)}, \quad \forall a \in \mathbb{R}. \quad (2.7)$$

The following cases are of particular interest.

- *p-th moment version: for all $p \in (0, +\infty)$,*

$$P(X \geq a) \leq \frac{E(|X|^p)}{a^p}, \quad \forall a > 0. \quad (2.8)$$

- *Exponential version: pour tout $\delta \in \mathbb{R}_+$,*

$$P(X \geq a) \leq e^{-\delta a} E(e^{\delta X}), \quad \forall a > 0. \quad (2.9)$$

Proof. Apply the Markov inequality to the non-negative random variable $g(X)$ to get

$$P(g(X) \geq g(a)) \leq \frac{E(g(X))}{g(a)}, \quad \forall a \in \mathbb{R}.$$

Note then that as g is increasing, $g(X) \geq g(a)$ if and only if $X \geq a$. The two particular cases correspond to the choices $g(x) = |x|^p$ and $g(x) = e^{\delta x}$. $\square$

Remark 2.8.1. Taking the function $g(x) = x^2$ applied to the random variable $|X - E(X)|$, we obtain the extremely useful particular case

$$P(|X - E(X)| \geq a) \leq \frac{\text{Var}(X)}{a^2}, \quad \forall a > 0.$$

Exercise 2.56. A professor knows from experience that students grades at a final exam are random variables with expectation 4.3.

1. Give an upper bound on the probability that a given student grade is higher than 5.5.

Suppose now that the professor also knows that the variance of the grade is 0.4.

2. What can we say about the probability that a given student obtains a grade between 3.3 and 5.3?
3. How large should a class of students be in order to have a probability at least 0.9 to have that the average grade of the class is in $[4, 4.6]$?

Solution is given in [A.2](#).

Exercise 2.57. X is a Pareto random variable of parameter $(\alpha, \theta) \in (0, +\infty)^2$ if its CDF is

$$F_X(t) = \begin{cases} 1 - \left(\frac{\alpha}{t}\right)^\theta & \text{if } t > \alpha, \\ 0 & \text{else.} \end{cases}$$

1. Check that F_X is indeed a repartition function (right-continuous, non-decreasing, and tends to 0 at $-\infty$ and to 1 at $+\infty$).
2. Compute the first and second moments of X . Under which conditions are they finite?
3. Pareto random variables are often used to model rare events (big stock market drop, heavy snowfall,...). For $c > \alpha$, and $t \in \mathbb{R}$, compute

$$P(X \leq t | X > c), \quad E(X | X > c).$$

Solution is given in [A.2](#).

2.8.4 A weak Law of Large Numbers

An application of Chebychev's inequality is a version of the *weak Law of Large Numbers*. In words, this law states that *with high probability, the empirical average of i.i.d. random variables is close to its expectation*. This is the kind of principle we used in [Section 1.1.1](#) to approximate the distribution of a dice roll.

Theorem 2.8.3 (Weak LLN, second moment version). *Let $X_1, X_2, \dots$ be a sequence of **identically distributed** random variables. Suppose that*

1. *they admit a second moment: $E(X_1^2) < \infty$,*
2. *they are **uncorrelated**: $\text{Cov}(X_i, X_j) = 0$ if $i \neq j$.*

Denote

$$E(X_1) = \mu, \quad \text{Var}(X_1) = \sigma^2, \quad \bar{S}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, for all $\epsilon > 0$,

$$P(|\bar{S}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2 n}.$$

In particular, $\bar{S}_n$ converges in probability^a towards μ .

^aWe will see what this means later in Chapter 3.

Proof. Use the second moment Chebychev's inequality (the Chebychev's inequality with $g(x) = x^2$) on the random variable $|\bar{S}_n - \mu|$. We get

$$P(|\bar{S}_n - \mu| \geq \epsilon) \leq \frac{E((\bar{S}_n - \mu)^2)}{\epsilon^2}.$$

Now,

$$\begin{aligned} E((\bar{S}_n - \mu)^2) &= \frac{1}{n^2} E\left(\left(\sum_{i=1}^n X_i - E(X_i)\right)^2\right) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n E((X_i - E(X_i))(X_j - E(X_j))) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\sigma^2}{n}, \end{aligned}$$

where we used that the variables are un-correlated in the fourth equality. Plugging this in the first display of the proof, we get the claim. $\square$

We will see another version of the LLN: the *strong Law of Large Numbers*. The main difference will lie in the following subtlety: the weak law says that

Once you fixed an allowed error window for the approximation, ϵ , **and** an allowed probability of failure, α , you can find n such that having a sample of size n will give you an approximation inside your ϵ -window with probability at least $1 - \alpha$.

(wLLN)

The strong law will say that **with probability 1**, your approximation will be within any fixed ϵ -error window provided n is large enough. We will discuss this subtlety in Chapter 3.

Exercise 2.58. Let $X_1, X_2, \dots$ be an independent family of random variables with $X_i \sim \text{Bern}(p)$ for all i 's, $p \in (0, 1)$. Let $S_0 = 0$, and for $n \geq 1$,

$$S_n = \sum_{i=1}^n X_i, \quad \bar{S}_n = \frac{1}{n} S_n.$$

1. What are the values of $E(S_n)$ and $E(\bar{S}_n)$?
2. What is the value of $\text{Cov}(S_5, S_{10})$?
3. Give a value of n (as small as you can) such that

$$P(|\bar{S}_n - p| \geq 0.01) \leq 0.01.$$

Solution is given in [A.2](#).

2.8.5 Cauchy-Schwartz and Hölder's inequalities

Lemma 2.8.4 (Cauchy-Schwarz inequality). *For any random variables X, Y ,*

$$E(XY)^2 \leq E(X^2)E(Y^2).$$

Proof. Let X, Y be two random variables (we can assume $E(X^2), E(Y^2) < \infty$, otherwise the inequality is trivial as the R.H.S. is $+\infty$). Define $h(t) = E((tX + Y)^2) \geq 0$. Expanding the square and using linearity, we get

$$h(t) = t^2 E(X^2) + 2t E(XY) + E(Y^2),$$

which is a polynomial of degree 2 in t . As this polynomial does not change sign, it has at most one real zero, and thus the discriminant $4E(XY)^2 - 4E(X^2)E(Y^2)$ is non-positive. This is precisely the wanted inequality. $\square$

***Remark.** Cauchy-Schwarz inequality is a general property of Hilbert spaces^a. The Hilbert space structure here is obtained by looking at the (infinite dimensional) vector space $L_P^2(\Omega)$, the set of functions from Ω to $\mathbb{R}$ with a finite second moment under P , and to endow it with the scalar product

$$(X, Y) \mapsto E_P(XY).$$

^aHilbert spaces are a generalisation of Euclidean spaces.

A direct application is that Pearson correlation coefficient defined in Definition 2.5.4 takes values in $[-1, 1]$ as stated.

Lemma 2.8.5. *Let X, Y be two random variables. Then,*

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}.$$

In particular, $\rho_{XY} \in [-1, 1]$.

Proof. We use

$$\begin{aligned}\text{Cov}(X, Y)^2 &= E((X - E(X))(Y - E(Y)))^2 \\ &\leq E((X - E(X))^2)E((Y - E(Y))^2) = \text{Var}(X)\text{Var}(Y),\end{aligned}$$

by Cauchy-Schwarz applied to the variables $X - E(X)$ et $Y - E(Y)$. Taking the square-root on both sides, we get the wanted result. $\square$

There is a generalisation of the Cauchy-Schwarz inequality, called Hölder's inequality, of which Cauchy-Schwarz is the special case $p = q = 2$.

Lemma 2.8.6 (*Hölder's inequality). *Let $p, q \in (1, +\infty)$ be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then, for all random variables X, Y ,*

$$E(|XY|) \leq E(|X|^p)^{1/p} E(|Y|^q)^{1/q}.$$

2.8.6 Second moment method

The second moment method is the “complement” of the first moment method. The first moment method was the observation that for $\mathbb{N}$ -valued random variable, having a small expectation implies having a large probability to be 0. The second moment method is about finding a quantity which implies that the probability to be 0 is small. It relies on the following: for X a $\mathbb{N}$ -valued random variable,

$$P(X > 0) \geq \frac{E(X)^2}{E(X^2)}. \quad (2.10)$$

It follows from the Cauchy-Schwarz inequality (Lemma 2.8.4): as $X \geq 0$,

$$E(X) = E(X\mathbf{1}_{X>0}) \leq E(X^2)^{1/2} E(\mathbf{1}_{X>0})^{1/2} = E(X^2)^{1/2} P(X > 0)^{1/2}.$$

Squaring both sides, and dividing by $E(X^2)$ gives the wanted bound. As application, we come back to our example from Section 2.8.2. There, we found that the longest run of 1's in a uniform sequence of n bits was *at most* of order $\log_2(n)$. We will use the second moment method to show that it is *at least* $\log_2(n)$. Use the same notations as in Section 2.8.2. We computed the first moment of Y_k , and found

$$E(Y_k) = (n - k + 1)2^{-k}.$$

We now want to use the second moment method:

$$P(Y_k > 0) \geq \frac{E(Y_k)^2}{E(Y_k^2)} = \frac{E(Y_k)^2}{\text{Var}(Y_k) + E(Y_k)^2} = \left(1 + \frac{\text{Var}(Y_k)}{E(Y_k)^2}\right)^{-1}$$

We then need to compute the variance of Y_k .

$$\text{Var}(Y_k) = \text{Cov}(Y_k, Y_k) = \sum_{i=1}^{n-k+1} \sum_{j=1}^{n-k+1} \text{Cov}(B_i, B_j),$$

where we used the linearity of the covariance, and we introduced

$$B_i = \prod_{l=0}^{k-1} X_{i+l}.$$

Now, notice that when $|i - j| \geq k$, we have that $(X_i, X_{i+1}, \dots, X_{i+k-1})$ and $(X_j, X_{j+1}, \dots, X_{j+k-1})$ are independent, and so $\text{Cov}(B_i, B_j) = 0$ in this case. Now, when $|i - j| < k$ (without loss of generality, assume $j \geq i$), we have

$$\begin{aligned} \text{Cov}(B_i, B_j) &= E(B_i B_j) - E(B_i)E(B_j) \\ &= E\left(\prod_{u=i}^{i+k-1} X_u \prod_{v=j}^{j+k-1} X_v\right) - 2^{-2k} = E\left(\prod_{u=i}^{j+k-1} X_u\right) - 2^{-2k} = 2^{-j+i-k} - 2^{-2k} \end{aligned}$$

as $E(B_i) = \prod_{l=0}^{k-1} E(X_{i+l}) = 2^{-k}$, and $X_i^2 = X_i$ for any i as the X_i 's are $\{0, 1\}$ -valued. Using all these observations,

$$\begin{aligned} \sum_{i=1}^{n-k+1} \sum_{j=1}^{n-k+1} \text{Cov}(B_i, B_j) &= \sum_{i=1}^{n-k+1} \text{Cov}(B_i, B_i) + 2 \sum_{1 \leq i < j \leq n-k+1} \text{Cov}(B_i, B_j) \\ &= (n-k+1)(2^{-k} - 2^{-2k}) + 2 \sum_{i=1}^{n-k+1} \sum_{r=1}^{k-1} \mathbb{1}_{i+r \leq n-k+1} (2^{-r-k} - 2^{-2k}) \\ &\leq (n-k+1)2^{-k} (1 - 2^{-k}) + 2 \sum_{r=1}^{k-1} 2^{-r} - 2(k-1)2^{-k} \\ &\leq (n-k+1)2^{-k} \left(1 + \sum_{r=0}^{\infty} 2^{-r}\right) \\ &= 3(n-k+1)2^{-k} \end{aligned}$$

where we used the change of variable $r = j - i$ in the second line, and the geometric series⁷ in the fifth. In particular, we obtained that

$$P(Y_k > 0) \geq \left(1 + \frac{\text{Var}(Y_k)}{E(Y_k)^2}\right)^{-1} \geq \left(1 + \frac{3(n-k+1)2^{-k}}{(n-k+1)^2 2^{-2k}}\right)^{-1} = \left(1 + \frac{3}{(n-k+1)2^{-k}}\right)^{-1}.$$

So, for any $\epsilon > 0$, and n large enough,

$$P(M_n \geq (1 - \epsilon) \log_2(n)) = P(Y_{\lceil (1-\epsilon) \log_2(n) \rceil} > 0) \geq \left(1 + \frac{6}{n^\epsilon}\right)^{-1} \xrightarrow{n \rightarrow \infty} 1,$$

where we used that $(1 - \epsilon) \log_2(n) + 1$ is less than $n/2$ for n large.

2.8.7 Jensen's inequality

A last very useful inequality is deeply linked with convex analysis.

⁷Recall that for $|z| < 1$, $\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$, and for any z , $\sum_{n=0}^m z^n = \frac{1-z^{m+1}}{1-z}$.

Lemma 2.8.7 (Jensen's inequality). *Let $I \subset \mathbb{R}$ be an interval. Let $g : I \rightarrow \mathbb{R}$ be a convex function. Then, for any random variable X taking values in I ($P(X \in I) = 1$),*

$$E(g(X)) \geq g(E(X)).$$

Moreover, for any $h : I \rightarrow \mathbb{R}$ concave,

$$E(h(X)) \leq h(E(X)).$$

Proof. As g is convex, there is at least one $a \in \mathbb{R}$ (slope of g at $E(X)$) such that for any $y \in I$,

$$g(y) \geq g(E(X)) + a(y - E(X)).$$

Applying this, we obtain

$$E(g(X)) \geq E(g(E(X)) + a(X - E(X))) = g(E(X)) + a(E(X) - E(X)) = g(E(X)).$$

The concave case follows: h is concave if and only if $-h$ is convex. □

Exercise 2.59. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable.

1. Show that if $E(X^2) < \infty$, then $E(|X|) < \infty$.
2. Show that if $E(X) = 0$, $M_X(t) \geq 1$ for any $t \in \mathbb{R}$.

Solution is given in [A.2](#).

Chapter 3

Convergence of random variables

3.1 What does “converge” mean?

3.1.1 Modes of convergences

The goal of this section is to make sense of having a sequence of random variables which “looks more and more like” a fixed random variable. This is a notion of *convergence*, which was already encountered in Analysis... the question is then “what do we mean by convergence of random variables?”. There are several ways to formalize this convergence, which are more or less useful in extracting properties of the limit from the properties of the sequence or the other way around. We will see here the most commonly used notions of convergence for random variables.

The weakest notion of convergence is the convergence in law. In words, convergence in law means that the statistical properties of the sequence approach the statistical properties of the limiting random variable as we go further and further in the sequence.

Definition 3.1.1 (Convergence in law). Let $X, X_1, X_2, \dots : \Omega \rightarrow \mathbb{R}$ be random variables. Say that $(X_i)_{i \geq 1}$ *converges in law* towards X , denoted $X_n \xrightarrow{\text{Law}} X$, if one of the two following equivalent conditions is fulfilled.

1. $F_{X_n}(t) \rightarrow F_X(t)$ for all $t \in \mathbb{R}$ such that F_X is continuous at t .
2. $E_P(\varphi(X_n)) \xrightarrow{n \rightarrow \infty} E_P(\varphi(X))$ for all $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ continuous and bounded.

Note that the second condition makes sense also for random vectors, and gives the definition of convergence in law for random vectors. The fact that these points are equivalent is non-trivial, and its proof goes way beyond the scope of these notes, so we will not discuss it and simply accept the result.

Remark 3.1.1. The condition “for all $t \in \mathbb{R}$ such that F_X is continuous at t ” is natural for the following reason: if F_X has a jump at t , the random variables X_n could have a behaviour not fitting the one of X at t for every n , but if the “problem” gets localised *closer and closer* to t , the variables X_n will still be more and more similar to X . A very simple example is to take $X_n = \frac{1}{n}$ to be constant random variables, and $X = 0$ to also be a constant random variable. We then “clearly see” that $X_n \xrightarrow{n \rightarrow \infty} X$, yet

$$F_{X_n}(t) = \begin{cases} 1 & \text{if } t \geq 1/n \\ 0 & \text{if } t < 1/n \end{cases}, \quad F_X(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}.$$

So $F_{X_n}(0) = 0 \not\rightarrow 1 = F_X(0)$.

Remark 3.1.2. Note that in the particular case of X being a continuous random variable, F_X is continuous (primitive of the density function). Moreover, still for X continuous, as $P(X = x) = 0$ for all x , we have that $P(X \in (a, b)) = P(X \in [a, b])$ for all $a < b \in \mathbb{R}$. In particular, if X is a continuous random variable, convergence in law of X_n towards X is equivalent to: for any interval $I \subset \mathbb{R}$,

$$P(X_n \in I) \xrightarrow{n \rightarrow \infty} P(X \in I).$$

There is a very useful characterization of the convergence in law using point-wise convergence of moment generating functions.

Theorem 3.1.1. Let $X, X_1, X_2, \dots : \Omega \rightarrow \mathbb{R}$ be random variables. Suppose that there is $\delta > 0$ such that

$$\sup_{|t| < \delta} E(e^{tX}) < +\infty, \quad \sup_{|t| < \delta} E(e^{tX_i}) < +\infty \quad \forall i \geq 1,$$

$$E(e^{tX_n}) \xrightarrow{n \rightarrow \infty} E(e^{tX}) \quad \forall |t| < \delta.$$

Then,

$$X_n \xrightarrow{\text{Law}} X \text{ as } n \rightarrow \infty,$$

and for any $p \geq 1$,

$$E(X_n^p) \xrightarrow{n \rightarrow \infty} E(X^p).$$

Example 3.1.1. 1. Let $X_n \sim \text{Bern}(p + n^{-1})$, and $X \sim \text{Bern}(p)$. Then, $X_n \xrightarrow{\text{Law}} X$. Indeed, we have $F_X(t) = (1 - p)\mathbb{1}_{t \geq 0} + p\mathbb{1}_{t \geq 1}$ is continuous everywhere except at 0 and 1. For $t \in \mathbb{R} \setminus \{0, 1\}$,

$$P(X_n \leq t) = (1 - p + n^{-1})\mathbb{1}_{t \geq 0} + (p + n^{-1})\mathbb{1}_{t \geq 1} \xrightarrow{n \rightarrow \infty} F_X(t).$$

2. Let $X_n \sim \mathcal{N}(0, n^{-1})$, and $X \sim \delta_0$. Then, $X_n \xrightarrow{\text{Law}} X$. Indeed, we have $E(e^{tX}) = 1$, and for any $t \in \mathbb{R}$,

$$E(e^{tX_n}) = e^{t^2/(2n)} \xrightarrow{n \rightarrow \infty} 1.$$

So Theorem 3.1.1 implies $X_n \xrightarrow{\text{Law}} X$. This is a particular instance of “Gaussian random variables with 0 variance are Dirac random variables”.

Another useful result concerns random variables with values in $\mathbb{Z}$.

Lemma 3.1.2. *Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable with $\text{Image}(X) \subset \mathbb{Z}$. Let $X_1, X_2, \dots : \Omega \rightarrow \mathbb{R}$ be random variables. Then, the two following conditions are equivalent.*

1. $X_n \xrightarrow{\text{Law}} X$.
2. For every $k \in \mathbb{Z}$,

$$\lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} P(X_n \in (k - \epsilon, k + \epsilon)) = P(X = k).$$

**Proof.* See Exercise 3.1. □

***Exercise 3.1.** Show Lemma 3.1.2.

Hint: where are the discontinuities of F_X ? How does F_X behaves between these discontinuities?

Solution in given in A.3.

The second mode of convergence is the one we encountered in Theorem 2.8.3. It is a version saying “if I allow a small probability of failure ϵ , and a small error of approximation ϵ' , I can find a rank in the sequence which approximate the limit object up to an error ϵ' with probability of success at least $1 - \epsilon$ ”. It is much stronger than the convergence in law, as it asks for some form of “uniform” convergence of the statistical properties of the variables. This mode of convergence is the one that we will encounter when studying estimators in statistics.

Definition 3.1.2 (Convergence in probability). Let $X, X_1, X_2, \dots : \Omega \rightarrow \mathbb{R}$ be random variables. Say that $(X_n)_{n \geq 1}$ converges in probability towards X , denoted $X_n \xrightarrow{\text{Proba}} X$, if for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X - X_n| \geq \epsilon) = 0.$$

The third mode of convergence is the almost sure convergence. It states that the sequence of random variables converge *with probability one*, meaning: if you sample a realisation $\omega \in \Omega$ under the probability measure P , with probability one, the sequence of measurements $X_n(\omega)$ will converge to the measurement $X(\omega)$.

Definition 3.1.3 (Almost sure convergence). Let $X, X_1, X_2, \dots : \Omega \rightarrow \mathbb{R}$ be random variables. We say that the sequence $(X_n)_{n \geq 1}$ converges almost-surely towards X , denoted $X_n \xrightarrow{\text{a.s.}} X$, if

$$P(\lim_{n \rightarrow \infty} X_n = X) = 1.$$

Lemma 3.1.3. $P(\lim_{n \rightarrow \infty} X_n = X) = 1$ is equivalent to

$$\forall \epsilon > 0, \lim_{n \rightarrow \infty} P(\sup_{m \geq n} |X_m - X| \geq \epsilon) = 0.$$

**Proof.* The almost sure convergence of X_n towards X is the fact that the set

$$\{\omega \in \Omega : \exists 1 > \epsilon > 0, \limsup_{n \rightarrow \infty} |X_n(\omega) - X(\omega)| \geq \epsilon\}$$

has probability 0. But we can re-write this set as

$$\begin{aligned} \bigcup_{\epsilon > 0} \{\omega \in \Omega : \inf_{m \geq 1} \sup_{n \geq m} |X_n(\omega) - X(\omega)| \geq \epsilon\} \\ = \bigcup_{l \geq 1} \bigcap_{m \geq 1} \{\omega \in \Omega : \sup_{n \geq m} |X_n(\omega) - X(\omega)| \geq \frac{1}{l}\}. \end{aligned}$$

where we used that if $1 > \epsilon > 0$, there exists $l \in \mathbb{N}$ such that $1/(l+1) \leq \epsilon \leq 1/l$. The union over l is an increasing limit, we can thus use Theorem 2.1.1 to get

$$\begin{aligned} P\left(\bigcup_{l \geq 1} \bigcap_{m \geq 1} \{\omega \in \Omega : \sup_{n \geq m} |X_n(\omega) - X(\omega)| \geq \frac{1}{l}\}\right) = \\ = \sup_{l \geq 1} P\left(\bigcap_{m \geq 1} \{\omega \in \Omega : \sup_{n \geq m} |X_n(\omega) - X(\omega)| \geq \frac{1}{l}\}\right). \end{aligned}$$

So almost-sure convergence is equivalent to: for all $l \geq 1$,

$$P\left(\bigcap_{m \geq 1} \{\omega \in \Omega : \sup_{n \geq m} |X_n(\omega) - X(\omega)| \geq \frac{1}{l}\}\right) = 0.$$

Using that the intersection over m is a decreasing limit and that for any $1 > \epsilon > 0$ there is $l \in \mathbb{N}$ such that $1/(l+1) \leq \epsilon \leq 1/l$, almost-sure convergence is then equivalent to

$$\forall \epsilon > 0, \lim_{m \rightarrow \infty} P\left(\{\omega \in \Omega : \sup_{n \geq m} |X_n(\omega) - X(\omega)| \geq \epsilon\}\right) = 0,$$

which is the wanted result. $\square$

Convergence in probability and almost-sure are both linked to the limiting probability of suitable events. Convergence in law has both a version linked to the convergence of probabilities, and a version linked with convergence of expected values of suitable functions. The next convergence modes are about refining this second point of view.

Definition 3.1.4 (Convergence in L^p). Let $p \in (0, +\infty)$. Let $X, X_1, X_2, \dots : \Omega \rightarrow \mathbb{R}$ be random variables admitting a moment of order p (remember Definition 2.2.12). We say that the sequence $(X_n)_{n \geq 1}$ converges in L^p towards X , denoted $X_n \xrightarrow{L^p} X$, if

$$\lim_{n \rightarrow \infty} E(|X_n - X|^p) = 0.$$

This notion of convergence controls the *expected distance* between X_n and X (with different weight put on small/large distances depending on the value of p). The last mode of convergence is a $p = \infty$ version of the previous one.

Definition 3.1.5 (*Essentially uniform convergence). Let $X, X_1, X_2, \dots : \Omega \rightarrow \mathbb{R}$ be essentially bounded random variables (remember Definition 2.2.12). We say that $(X_n)_{n \geq 1}$ converges essentially uniformly towards X , denoted $X_n \xrightarrow{L^\infty} X$, if

$$\lim_{n \rightarrow \infty} \|X - X_n\|_\infty = 0,$$

where $\|Y\|_\infty = \inf\{M \geq 0 : P(|Y| \leq M) = 1\}$.

- Exercise 3.2.** 1. Let $X_n \sim \text{Uni}([-1, \frac{1}{n}])$. Show that X_n converges in law towards a $\text{Uni}([-1, 0])$ random variable.
2. For $n \geq 1$, let $Y_n \sim \text{Bern}(1/n)$. Let X be a random variable with $E(X^2) < \infty$. Define $X_n = X + Y_n$. Show that X_n converges in probability towards X .
3. Let $X \sim \text{Bern}(1/2)$ and $X_n = 1 - X$ for $n \geq 1$. We have that $X_n \sim \text{Bern}(1/2)$ for all $n \geq 1$ (in particular, $X_n \xrightarrow{\text{Law}} X$ as $X_n \stackrel{\text{Law}}{=} X$). Show that X_n does not converge in probability towards X .

Solution in given in [A.3](#).

Exercise 3.3. This exercise is a prequel to the CLT, more precisely, to Theorem [3.2.6](#). Let $p \in (0, 1)$. Let $X_1, \dots, X_n$ be an i.i.d. sequence of Bernoulli random variables with parameter p . Introduce

$$\tilde{S}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - p).$$

Show that $\tilde{S}_n \xrightarrow{\text{Law}} Z$, where $Z \sim \mathcal{N}(0, p(1-p))$.

Hint: remember Theorem [3.1.1](#).

Solution in given in [A.3](#).

Exercise 3.4. This exercise studies a different regime than Exercise [3.3](#). Here we will let the parameter of the Bernoulli's tend to 0 as n goes to ∞ and show that it changes the natural re-scaling (from $\sqrt{n}$ to 1), as well as the nature of the limiting random variable. Let $\lambda > 0$. For $n \geq 1$, let $p_n \in (0, 1)$ be a sequence of numbers such that

$$\lim_{n \rightarrow \infty} np_n = \lambda,$$

and let $X_{n,1}, X_{n,2}, \dots, X_{n,n} : \Omega \rightarrow \mathbb{R}$ be an i.i.d. family of random variable with law $\text{Bern}(p_n)$. Define

$$S_n = \sum_{i=1}^n X_{n,i}.$$

Show that $S_n \xrightarrow{\text{Law}} N$, where $N \sim \text{Poi}(\lambda)$.

Hint: recall Lemma [2.6.1](#), and look first at $\lim_{n \rightarrow \infty} P(S_n = k)$ for k fixed.

Solution in given in [A.3](#).

3.1.2 Relations between convergence modes

We shall not prove all relations, a few are done in Exercise [3.5](#). All the implications are summarized in the following diagram.

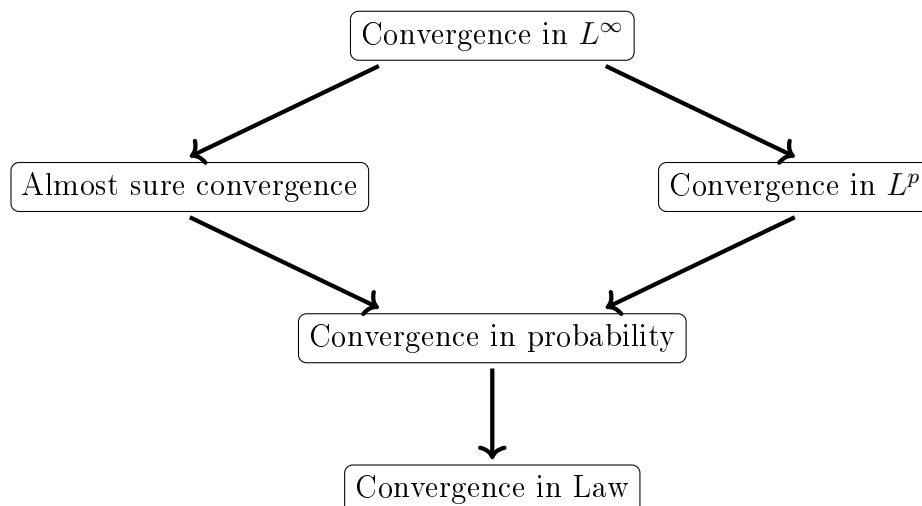


Figure 3.1: Each arrow represents an implication.

Exercise 3.5. Show that

1. Convergence in probability implies convergence in law.
2. Convergence in L^p implies convergence in probability.

Solution is given in [A.3](#).

3.2 Limit theorems

We now will study the two fundamental results on which most of statistics are based: the *law of large number* which formalizes the fact that “the probability of an event represents its occurrence frequency”; and the *central limit theorem* which quantifies the *typical deviations of frequency in a long sequence of experiments*. Both will be about *independent sequences of identically random variables* (i.i.d. sequences for short): the setup will be as follows.

- Take a random variable X . For example: some measurement in a random experiment.
- Take a sequence $X_1, X_2, \dots$ of random variables such that
 1. $(X_i)_{i=1,2,\dots}$ is an independent family,
 2. X_i has the same law as X for every $i \geq 1$.

This will be the repeated measurement in a sequence of repetitions of our random experiment.

- The goal: for n large, study
 1. whether $\frac{1}{n} \sum_{i=1}^n X_i$ gets typically close or not to $E(X)$ as $n \rightarrow \infty$,
 2. quantify how far from $E(X)$ $\frac{1}{n} \sum_{i=1}^n X_i$ typically is.

3.2.1 Weak Law of Large Numbers

One of the central results in parametric estimation is the *weak law of large numbers*.

Theorem 3.2.1 (wLLN). *Let $X, X_1, X_2, \dots$ be an independent family of identically distributed random variables. Then, if $E(|X|) < \infty$, for any $\epsilon > 0$,*

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - E(X)\right| \geq \epsilon\right) = 0.$$

In other words, $\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{Proba}} E(X)$ as $n \rightarrow \infty$.

In fact, this law admits a *strong* version: Theorem 3.2.3. It will have the advantage that it guarantees that we can work with convergence of empirical means “as if” their were functions, equipped with pointwise convergence of functions. The disadvantage of the strong version is that it is much harder to obtain quantitative versions of it. On the other hand, we already obtained a quantitative version of Theorem 3.2.1: Theorem 2.8.3. In our context of i.i.d. sequences, it gives the following statement.

Theorem 3.2.2 (Quantitative wLLN). *Let $X, X_1, X_2, \dots$ be an independent family of identically distributed random variables. Then, if $E(X^2) < \infty$, for any $\epsilon > 0$, and any $n \geq 1$,*

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - E(X)\right| \geq \epsilon\right) \leq \frac{\text{Var}(X)}{\epsilon^2 n}.$$

3.2.2 Strong Law of Large numbers

Theorem 3.2.3 (strong Law of Large Numbers). *Let $X, X_1, X_2, \dots$ be an independent family of identically distributed random variables. Then, if $E(|X|) < \infty$,*

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} E(X)$$

as $n \rightarrow \infty$.

The condition $E(|X|) < \infty$ is optimal, but the independence is not. There are multiple extensions of the sLLN to dependent sequences, but this goes beyond the scope of this course.

3.2.3 Central Limit Theorem

The LLN study the convergence of $\frac{1}{n} \sum_{i=1}^n X_i$ towards $E(X)$. In particular, it says that typically, $\sum_{i=1}^n X_i \approx nE(X)$, i.e.: $\sum_{i=1}^n X_i$ is close to its expectation. The Central Limit Theorem studies deviations of $\sum_{i=1}^n X_i$ away from $nE(X)$. We can get a first idea of what those typical deviations should be by studying the variance of $\sum_{i=1}^n X_i$:

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \text{Cov}\left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i\right) = \sum_{i,j=1}^n \text{Cov}(X_i, X_j) = \sum_{i=1}^n \text{Var}(X_i) = n\text{Var}(X)$$

as $\text{Cov}(X_i, X_j) = 0$ for $i \neq j$ by independence. In particular, we obtain that the standard deviation of $\sum_{i=1}^n X_i$ is of order $\sqrt{n}$. This is the scaling that pops out in the CLT.

Theorem 3.2.4 (Central Limit Theorem). *Let $X, X_1, X_2, \dots$ be an independent family of identically distributed random variables. Then, if $E(X^2) < \infty$,*

$$\frac{1}{\sqrt{\sigma_X^2 n}} \sum_{i=1}^n (X_i - \mu) \xrightarrow{\text{Law}} \mathcal{N}(0, 1)$$

as $n \rightarrow \infty$, where

$$\mu = E(X), \quad \sigma_X^2 = \text{Var}(X).$$

As in the case of the weak LLN, one can obtain a quantitative version of the CLT provided we are ready to sacrifice more moments. The more quantitative statement is of paramount importance in rigorous statistics.

Theorem 3.2.5 (Berry–Esseen CLT). *Let $X, X_1, X_2, \dots$ be an independent family of identically distributed random variables. Then, if $E(|X|^3) < \infty$, for any $n \geq 1$,*

$$\sup_{t \in \mathbb{R}} \left| P\left(\frac{1}{\sqrt{\sigma_X^2 n}} \sum_{i=1}^n (X_i - \mu) \leq t\right) - P(Z \leq t) \right| \leq \frac{0.5E(|X|^3)}{\sqrt{n}},$$

where $Z \sim \mathcal{N}(0, 1)$, and

$$\mu = E(X), \quad \sigma_X^2 = \text{Var}(X).$$

Remark 3.2.1. The number 0.5 is a refinement of the original theorem: 7.59 is the number found by Esseen in 1942. Since then, it has been improved to some number slightly below 0.5. It is also known that one cannot do better than some number slightly above 0.4.

We will prove a more restrictive version of the CLT using moment generating functions.

Theorem 3.2.6. *Let $X, X_1, X_2, \dots$ be an independent family of identically distributed random variables. Then, if for some $\delta > 0$, $E(e^{\delta|X|}) < \infty$,*

$$\frac{1}{\sqrt{\sigma_X^2 n}} \sum_{i=1}^n (X_i - \mu) \xrightarrow{\text{Law}} \mathcal{N}(0, 1)$$

as $n \rightarrow \infty$, where

$$\mu = E(X), \quad \sigma_X^2 = \text{Var}(X).$$

**Proof.* We use Theorem 3.1.1: to prove the convergence in law, it is sufficient to check the pointwise convergence of the moment generating functions in a neighbourhood of 0. For $|t| < \delta$,

$$E\left(\exp\left(t \frac{1}{\sqrt{\sigma_X^2 n}} \sum_{i=1}^n (X_i - \mu)\right)\right) = \prod_{i=1}^n E\left(\exp\left(\frac{t}{\sqrt{\sigma_X^2 n}} (X_i - \mu)\right)\right) = \left(M_{X-\mu}\left(\frac{t}{\sqrt{\sigma_X^2 n}}\right)\right)^n,$$

where we used independence in the first equality, and that the variables are identically distributed in the second. But now, a second order Taylor expansion gives

$$\begin{aligned} M_{X-\mu}\left(\frac{t}{\sqrt{\sigma_X^2 n}}\right) &= M_{X-\mu}(0) + M'_{X-\mu}(0) \frac{t}{\sqrt{\sigma_X^2 n}} + M''_{X-\mu}(0) \frac{t^2}{2\sigma_X^2 n} + o(n^{-1}) \\ &= 1 + \frac{t^2}{2n} + o(n^{-1}) \end{aligned}$$

as $M_{X-\mu}(0) = 1$, $M'_{X-\mu}(0) = E(X - \mu) = 0$, and $M''_{X-\mu}(0) = E((X - \mu)^2) = \text{Var}(X) = \sigma_X^2$. Taking the limit $n \rightarrow \infty$, we get¹

$$\lim_{n \rightarrow \infty} E\left(\exp\left(t \frac{1}{\sqrt{\sigma_X^2 n}} \sum_{i=1}^n (X_i - \mu)\right)\right) = \lim_{n \rightarrow \infty} \left(1 + \frac{t^2}{2n} + o(n^{-1})\right)^n = e^{t^2/2}$$

which is the moment generating function of a $\mathcal{N}(0, 1)$. □

3.3 Exercises on Limit Theorems

Exercise 3.6. Let $X_1, X_2, \dots$ be a sequence of measurements on repetition of the same experiment (independent sequence of identically distributed random variables). What can we say on

$$S_n = \sum_{i=1}^n X_i,$$

as $n \rightarrow \infty$ in each of the following cases.

1. The experiment is picking an individual in the population uniformly at random, and X_i is the indicator function that the i th individual knows how to fix a leaking pipe.
2. The experiment is flipping a fair coin until getting “head”. X_i is 10 to the power the number of tails before getting head in the i th repetition of the experiment (for example, you play a game where your lose is multiplied by 10 every times you get “tail”).
3. We can assume that the X_i ’s are continuous random variables with common density

$$f(x) = 2\mathbb{1}_{[1, +\infty)}(x) \frac{1}{x^3}.$$

Solution in given in [A.3](#).

¹Recall that for any $x \in \mathbb{R}$, $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$

Exercise 3.7. We perform the following experiment: we pick graphic cards uniformly at random in the NVIDIA production and test them. Let X_i be the random variable giving 0 if the i th card tested functions well, and 1 if there is a malfunction. Denote ρ the proportion of cards that have malfunction in the production. ρ is supposed to be unknown, so you cannot use it directly in computations.

In the following, you can use that if $Z \sim \mathcal{N}(0, 1)$,

$$P(|Z| > 1.96) = 0.05, \quad P(|Z| > 2.31) = 0.04.$$

1. Propose (and justify your proposition) a method to “guess” ρ from the sequence $X_1, X_2, X_3, \dots$
2. Using Theorem 3.2.4 and some non-rigorous approximation, can you give a minimal number of tests to be performed if we want to have an estimation of ρ that is precise up to an error 0.01, with a risk a failing the estimation at most 0.05?
3. Using Theorem 3.2.5, can you **rigorously** give a minimal number of tests to be performed if we want to have an estimation of ρ that is precise up to an error 0.01?

Solution is given in A.3.

3.4 *Delta-method

The delta-method is a way to determine the limit of certain sequences of random variables: variables that are image by a smooth function of a sequence of random variables converging to a Gaussian one. It is useful in statistic when computing confidence intervals using asymptotic normality, see Section 4.3.2.

Lemma 3.4.1. *Let $\theta \in \mathbb{R}$. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function in a neighbourhood around θ with $g'(\theta) \neq 0$. Then, if $X_1, X_2, \dots$ is a sequence of random variables such that*

$$\sqrt{n}(X_n - \theta) \xrightarrow{\text{Law}} \mathcal{N}(0, \sigma^2),$$

then,

$$\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{\text{Law}} \mathcal{N}(0, \sigma^2(g'(\theta))^2)$$

The idea of the proof is to Taylor expand g around θ and to use the convergence of the X_n 's.

Example 3.4.1. The main example of this method is when the sequence $X_1, X_2, \dots$ is a sequence of *empirical means*: if $Y_1, Y_2, \dots$ is an i.i.d. sequence of random variables (with at least two moments), and we set

$$\theta = E(Y_1), \quad \sigma^2 = \text{Var}(Y_1), \quad X_n = \frac{1}{n} \sum_{i=1}^n Y_i,$$

the CLT (Theorem 3.2.4) says that

$$\sqrt{n}(X_n - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n (Y_i - \theta) \xrightarrow{\text{Law}} \mathcal{N}(0, \sigma^2).$$

So, we can apply Lemma 3.4.1 to this case! For example, if we take the function $g(x) = x^4$, we obtain

$$\sqrt{n} \left(\left(\frac{1}{n} \sum_{i=1}^n Y_i \right)^4 - \theta^4 \right) \xrightarrow{\text{Law}} \mathcal{N}(0, 16\sigma^2\theta^6).$$

3.5 *Random Number Generator

In the light of the LLN, let us discuss briefly how is randomness handled in a computer. There are several instances where it is desirable to generate a random number on a computer, one can think of computer games, of landscape generation in movies, but many algorithms rely on randomness (Quicksort, Karger's algorithm for the min-cut problem, noised gradient descent to name a few). But one runs into a problem: how to “roll a dice” on demand for our computer? At some point, “farms” of coin flips/dice rolls existed (for example for online casinos), but they cannot accommodate for the needs of everyone. Nowadays two main methods exist.

- **Hardware Random Number Generator:** the idea is to construct a device that generates numbers using a un-predictable physical phenomena. The observation is that many signals are “naturally noised” due to interferences, for example, jitter of electronic circuit. The point is thus to generate numbers using that noise.
- **PseudoRandom Number Generators:** there the idea is particularly useful when the numbers only need to “look” random. PRNGs are algorithms that produce sequences of numbers in an inductive fashion, in such a way that *the sequence produced has the same statistical properties as an i.i.d. sequence*. In other words, PRNGs generate sequences of numbers so that one has a sLLN holding for that sequence. The upside is that one can ask for many numbers in a very short time without increasing much computation time (making them very useful for Monte-Carlo methods), the downside is that these sequences not being truly random, one can theoretically predict them (making them not very well suited for online casinos)...

3.6 Chi-square distribution

The χ^2 (chi-square) distribution and the related convergence results are widely used in statistics to construct hypotheses tests. The results concerning convergence that we will use in Chapter 4 are gathered here and will be recalled in the relevant section.

Definition 3.6.1 (Chi-square distribution). Let $k \in \mathbb{N}^*$ be a positive integer. A random variable X follows the χ^2 *distribution with k degrees of freedom*, denoted $X \sim \chi_k^2$, if it is a continuous random variable with density given by^a

$$f_X(x) = \mathbb{1}_{[0,+\infty)}(x) \frac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{k}{2}} \Gamma(\frac{k}{2})}. \quad (3.1)$$

^aRecall the Gamma function $\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$. For $n \in \mathbb{N}^*$, we have $\Gamma(n) = (n-1)!$.

These variables can also be defined through their MGF: $X \sim \chi_k^2$ if and only if

$$M_X(t) = E(e^{tX}) = (1 - 2t)^{-k/2}, \quad \text{for } t < \frac{1}{2}. \quad (3.2)$$

This law arises in statistics through the following links with Gaussian random variables.

- First, if $k \geq 1$ is an integer, and $X_1, \dots, X_k$ are independent $\mathcal{N}(0, 1)$ random variables, then

$$\sum_{i=1}^k X_i^2 \sim \chi_k^2. \quad (3.3)$$

For the motivated reader, one can prove this directly by computing the density of $\sum_{i=1}^k X_i^2$ using a change of variable in the Gaussian density. Another way is to use the MGFs, see Exercise 3.8. Note that this implies that if $X_1, \dots, X_k$ are independent $\mathcal{N}(0, \sigma^2)$, then $\frac{1}{\sigma^2} \sum_{i=1}^k X_i^2 \sim \chi_k^2$.

- Second, if $k \geq 1$ is an integer, $\mu \in \mathbb{R}$, $\sigma^2 > 0$, and $X_1, \dots, X_k$ are independent $\mathcal{N}(\mu, \sigma^2)$ random variables, then the following rescaling of the empirical variance follows a χ_{k-1}^2 law:

$$\frac{1}{\sigma^2} \sum_{i=1}^k \left(X_i - \frac{1}{k} \sum_{i=1}^k X_i \right)^2 \sim \chi_{k-1}^2. \quad (3.4)$$

The $k-1$ appears from the fact that we “lost” one degree of freedom by estimating the expectation using the empirical mean. See Exercise 3.9.

Remark 3.6.1. A crucial property of Gaussian making the second point (i.e.: (3.4)) valid is the following observation: if $(X_1, \dots, X_n)$ is a family of i.i.d. $\mathcal{N}(\mu, \sigma^2)$, then the empirical means $\bar{X}_n$ is *independent of* the deviations from the mean $(X_1 - \bar{X}_n, \dots, X_n - \bar{X}_n)$. This is deduced in the first point of Exercise 3.9.

Definition 3.6.2 (Multinomial distribution). Let $k \geq 2$ be a positive integer. Let $n \geq 1$. Let $p_1, \dots, p_k \in [0, 1]$ be such that $\sum_{i=1}^k p_i = 1$. A random vector $X = (X_1, \dots, X_k)$ follows the *multinomial distribution with parameters* $(k, n; p_1, \dots, p_k)$ if it is a discrete random vector with mass function given by

$$P(X = (x_1, \dots, x_k)) = \begin{cases} \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k} & \text{if } x_1, \dots, x_k \in \mathbb{N}, \sum_{i=1}^k x_i = n, \\ 0 & \text{else.} \end{cases}$$

When $k = 2$, it reduces to the binomial distribution.

The convergence result we will use is the following.

Lemma 3.6.1. Let $k \geq 2$, $p_1, \dots, p_k \in (0, 1)$ such that $p_1 + \dots + p_k = 1$. For $n \geq 1$, let $(N_{n,1}, \dots, N_{n,k})$ be a multinomial random vector with parameters $(n, k; p_1, \dots, p_k)$ (see (4.4)). Then, the random variable

$$\sum_{i=1}^k \frac{(N_{n,i} - np_i)^2}{p_i}$$

converges in law to a χ^2 law with $k - 1$ degrees of freedom:

$$\sum_{i=1}^k \frac{(N_{n,i} - np_i)^2}{p_i} \xrightarrow{\text{Law}} \chi_{k-1}^2, \text{ as } n \rightarrow \infty.$$

The way this result is usually used is as follows. Let $X_1, X_2, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be some independent sequence of identically distributed random variables (the results of a sequence of repeated experiments). Let $A_1, \dots, A_k \subset \mathbb{R}$ be a partition of $\mathbb{R}$ (some way to classify the result of the measurement we make on our experiment):

$$A_i \cap A_j = \emptyset \text{ if } i \neq j, \quad \cup_{i=1}^k A_i = \mathbb{R}.$$

Then, the random vector

$$\left(\sum_{i=1}^n \mathbb{1}_{A_1}(X_i), \sum_{i=1}^n \mathbb{1}_{A_2}(X_i), \dots, \sum_{i=1}^n \mathbb{1}_{A_k}(X_i) \right), \quad (3.5)$$

which simply counts the number of measurements which fell into each class, follows a multinomial distribution with parameters $(n, k; P(X_1 \in A_1), \dots, P(X_1 \in A_k))$.

***Exercise 3.8.** Let $n \geq 1$, $\sigma > 0$. Let $X, X_1, \dots, X_n \sim \mathcal{N}(0, \sigma^2)$ be an i.i.d. sequence. Define

$$S = \frac{1}{\sigma^2} \sum_{i=1}^n X_i^2.$$

Let $R \sim \chi_n^2$.

1. Show that for all $t < \frac{1}{2}$, $M_R(t) = (1 - 2t)^{-\frac{n}{2}}$. *Hint: you can proceed by induction over n , and use the property $\Gamma(z + 1) = z\Gamma(z)$ of Γ function (this is obtained by a direct integration by parts). Using as base cases $n = 1$ and $n = 2$ helps!*
2. Show that for all $t < \frac{1}{2}$, $M_{\frac{X^2}{\sigma^2}}(t) = E(e^{tX^2/\sigma^2}) = (1 - 2t)^{-\frac{1}{2}}$.
3. Deduce from the previous point and independence that $M_S(t) = (1 - 2t)^{-\frac{n}{2}}$ for $t < \frac{1}{2}$.
4. Conclude that $S \sim \chi_n^2$.

Solution in given in [A.3](#).

Exercise 3.9. Let $\mu \in \mathbb{R}, \sigma > 0$. Let $X_1, \dots, X_n$ be an i.i.d. sequence of $\mathcal{N}(0, \sigma^2)$. Define

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Define $Y_i = X_i - \bar{X}_n$.

1. Show that $(Y_1, \dots, Y_n, \bar{X}_n)$ is a Gaussian vector. Compute its covariance matrix. What do you deduce about the relation between $\bar{X}_n$ and $(Y_1, \dots, Y_n)$ from that?
2. Show that $\sum_{i=1}^n (X_i - \bar{X}_n)^2 = \sum_{i=1}^n X_i^2 - n\bar{X}_n^2$.
3. Using the first point, show that $R = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ and $T = \frac{n}{\sigma^2} \bar{X}_n^2$ are independent. Letting $S = \frac{1}{\sigma^2} \sum_{i=1}^n X_i^2$, what do you deduce about the relation between the MGFs of R , S , and T ?
4. Using MGF, show that $R \sim \chi_{n-1}^2$, (i.e.: that $\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \sim \chi_{n-1}^2$).
5. Deduce the same result in the case $X_1, \dots, X_n$ is an i.i.d. sequence of $\mathcal{N}(\mu, \sigma^2)$ with $\mu \in \mathbb{R}$.

Solution is given in [A.3](#).

Chapter 4

Statistics: starter guide

As mentioned in the introduction, statistics have a very different purpose than probability theory (whilst using the same language). In probability we *are given the laws of random variables and we wish to study their properties*, in statistics we *are given realisations of random variables and we wish to guess their laws*.

4.1 Basic objects

4.1.1 Statistics

Definition 4.1.1. Let P be a probability measure on $\mathbb{R}^d$. Let $n \geq 1$. A *size n sample* of law P is an i.i.d. sequence $X_1, \dots, X_n$ of random variables/vectors having law P . A *realisation* of a size n sample is a sequence of values of the random variables: that is a sequence $(x_1, \dots, x_n) \in (\mathbb{R}^d)^n$.

Definition 4.1.2. A *statistic* on a size n sample $X_1, \dots, X_n$ with values in $\mathbb{R}^d$ is a random variable $Y : (\mathbb{R}^d)^n \rightarrow \mathbb{R}$.

Example 4.1.1. We study a sequence of 10 coin tosses, and we define the random variables

$$X_i = \begin{cases} 1 & \text{if the coin } i \text{ is head,} \\ 0 & \text{if the coin } i \text{ is tail.} \end{cases}$$

We toss our coins and find the sequence

$$(0, 0, 0, 1, 0, 1, 0, 1, 1, 0).$$

We then measure the number of heads:

$$N(X_1, \dots, X_{10}) = \sum_{i=1}^{10} X_i.$$

Then, the law that we want to study is the law of a coin toss (i.e.: the law of X_1). The sample size is 10, the sample is $(X_1, \dots, X_{10})$, the realisation is $(0, 0, 0, 1, 0, 1, 0, 1, 1, 0)$, and N is a possible statistic on our sample.

We will be interested in *parametric estimation*: we will suppose that the objects which we have to guess/approximate are numbers or sequences of numbers and not more complicated quantities such as functions or probability measures (which is also within the capabilities of statistics).

Let us start with an example. We have a coin, potentially biased. We want to find what is the “law of the coin”, i.e.: model the experiment “tossing the coin” by a random variable with some explicit law.

1. We start by, arbitrarily, restricting the (too vast!) space of “all random variables”: we can assume that each coin toss gives either head or tail (which is not a priori entirely true: the coin could fall on the side or get disintegrated by a meteor before falling, but for modelling purposes, we can simplify the situation). So, writing a 1 for head and a 0 for tail, our “candidate law” is a Bernoulli random variable, which has a parameter $p \in [0, 1]$ which value we have to estimate. We therefore make the assumption “there is $p \in [0, 1]$ such that the coin flip has law $\text{Bern}(p)$ ”.
2. To estimate p , we start by performing the experiment (tossing the coin) a large number of times, n , say $n = 1000$. Denote $X_1, X_2, \dots, X_{1000}$ the sequence of $\{0, 1\}$ -valued coins tosses. This is our *size 1000 sample* of law $\text{Bern}(p)$. We record the results as a sequence of numbers $x_1, \dots, x_{1000}$, this is our *realisation of a size 1000 sample*¹. For the sake of concreteness, here is a realisation of 1000 (biased) coin flips:

```
0101000000000010100000100000100001000000101000001000111001100
0011011000100001000100001000011100000000010000100000001000000
0110101010000011001010000000001000010011001010011010000000001
0111010110010000000000101000001100110100001010000000010101000
1001100000100001000101000100000000000011010010000010010010001
1100000000000000100001100011100000100000000000001000100000000
0000001001101110000010110001010000010000010011000011000001000
0000010001000001000000010010000111000011011000000110000010100
0000101100100011101000000001100100100000001101001100100010001
00000000000010101100101000000110001011100000000000000100100010
0010100101000010000111000101100010100000100010100100000001000
001100001100101100001001001011000000110101000000000010000001
10000001000110110000001010010100100000000001000000000010011000
1010000011000010000000001100010010000010010110110001001011001
0010100111010111011010110110000010100001011100100000000001001
001110100101010110001000010000000011100100001000010010001
100001001101100100111000
```

(PRNG based sample)

3. We now want to find a suitable *statistic* that we can compute on that sample that would approximate the wanted parameter p . As we did in the introduction

¹Note the subtle difference: the sequence of coin tosses gives a sequence of random variables, the sample, whilst the sequence of *results* of the coins tosses gives a sequence of numbers, the realisation of the sample.

of the notes, we can use the “frequency of occurrence” of 1 to estimate p : we define

$$\hat{p}_n = \hat{p}_n(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{X_i=1} = \frac{1}{n} \sum_{i=1}^n X_i.$$

$\hat{p}_n$ is our *estimator of p for the size n sample*. In our example, we have

$$\hat{p}_{1000}(x_1, \dots, x_{1000}) = \frac{289}{1000}.$$

We could wonder if our choice of $\hat{p}_n$ is a “good one”... which leads to the question “what makes a good estimator?” and there is no universal answer, as it depends on what you want to achieve with your model. Let us see a few options.

1. We can for example test if our estimator give the correct value “on average”: $\hat{p}_n$ is a function of the size n sample, so we can wonder if its expected value is p or at least close to p .
2. We can also wonder whether increasing the sample size n creates a better estimator.
3. We can ask what is the probability that $\hat{p}_n$ is far from the value p .
4. And many others...

In our example, we have

1. The expected value of $\hat{p}_n$ is given by

$$E(\hat{p}_n) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n p = p,$$

so, *on average*, our estimator of p gives the correct value.

2. For this point, we can use the strong Law of Large Numbers: if we have a realisation $x_1, x_2, \dots$ of an infinite sequence of coin flips, the sLLN says that, with probability one,

$$\hat{p}_n(x_1, \dots, x_n) \xrightarrow{n \rightarrow \infty} E(X_1) = p.$$

Note that here the sLLN really says that *performing more times the experiment will eventually bring us as close as wanted to the real value*.

3. Finally, for this point, we can use the Central Limit Theorem. From Theorem 3.2.4,

$$\begin{aligned} P(\hat{p}_n \notin (p - \epsilon, p + \epsilon)) &= P(\hat{p}_n - p \notin (-\epsilon, \epsilon)) \\ &= P\left(\frac{1}{\sqrt{np(1-p)}} \sum_{i=1}^n (X_i - p) \notin \left(-\frac{\epsilon\sqrt{n}}{\sqrt{p(1-p)}}, \frac{\epsilon\sqrt{n}}{\sqrt{p(1-p)}}\right)\right) \\ &\approx P\left(Z \notin \left[-\frac{\epsilon\sqrt{n}}{\sqrt{p(1-p)}}, \frac{\epsilon\sqrt{n}}{\sqrt{p(1-p)}}\right]\right) \\ &= 2P\left(Z > \frac{\epsilon\sqrt{n}}{\sqrt{p(1-p)}}\right) \end{aligned}$$

where $Z \sim \mathcal{N}(0, 1)$. Note that we are doing an approximation: we say that $\frac{1}{\sqrt{np(1-p)}} \sum_{i=1}^n (X_i - p)$ is *approximately* a Gaussian random variable. This is true for n very large by Theorem 3.2.4, but having an explicit control about how large should one take n requires a more quantitative result, such as Theorem 3.2.5. We can then compute

$$\begin{aligned} P\left(Z > \frac{\epsilon\sqrt{n}}{\sqrt{p(1-p)}}\right) &= \frac{1}{\sqrt{2\pi}} \int_{\frac{\epsilon\sqrt{n}}{\sqrt{p(1-p)}}}^{\infty} e^{-x^2/2} dx \leq \frac{\sqrt{p(1-p)}}{\sqrt{2\pi}} \frac{e^{-\epsilon^2 n/(2p(1-p))}}{\epsilon\sqrt{n}} \\ &\leq \frac{1}{\epsilon\sqrt{2\pi n}} e^{-\epsilon^2 n/(2p(1-p))} \end{aligned}$$

see Exercise 4.2. In particular, we obtained that our estimator $\hat{p}_n$ is ϵ -close to the correct value with probability at least $1 - \epsilon$ provided n is taken large enough.

Remark 4.1.1. In practical situations, we do not bother finding rigorous estimations on $P\left(Z > \frac{\epsilon\sqrt{n}}{\sqrt{p(1-p)}}\right)$, instead we use *quantile tables* (see Section 4.4).

Exercise 4.1. Consider the following situation. We play roulette 30 times (with possible results $\{0, 1, 2, \dots, 36\}$). Denote $X_1, X_2, \dots, X_{30}$ the sequence of random values representing the results of each game. Suppose that after playing, we obtained the numbers

12, 15, 19, 21, 13, 6, 34, 9, 18, 4, 34, 5, 5, 27, 14, 34, 8,
26, 7, 25, 2, 36, 28, 21, 27, 2, 6, 14, 35, 12.

The total sum is 519. Define the empirical average and the average number of results in $[10, 26]$:

$$\bar{X}_{30} = \frac{1}{30} \sum_{i=1}^{30} X_i, \quad Y = \frac{1}{30} \sum_{i=1}^{30} \mathbb{1}_{[10,26]}(X_i).$$

1. What are the sample and the sample size in this experiment?
2. Are $\bar{X}_{30}, Y$ statistics?
3. What is the realization of the sample?
4. Compare the values of $\bar{X}_{30}, Y$ in the obtained sample realization to their expected values in a fair game.

We run the same experiment 3 times in different casinos and find the numbers

- Casino 1:

14, 10, 11, 34, 22, 24, 32, 18, 20, 31, 12, 10, 24, 24, 13, 22, 32,
31, 35, 19, 18, 24, 32, 15, 28, 32, 20, 35, 27, 24,

total sum: 693.

- Casino 2:

7, 19, 20, 0, 20, 11, 12, 3, 25, 11, 0, 4, 9, 8, 19, 22, 20,
1, 19, 16, 4, 23, 25, 8, 14, 8, 6, 17, 15, 16,

total sum: 382.

- Casino 3:

13, 17, 11, 10, 24, 15, 22, 19, 20, 13, 18, 19, 25, 23, 10, 14, 21,
13, 10, 17, 23, 22, 25, 12, 16, 26, 17, 24, 13, 25,

total sum: 537.

5. What can be said about the game being fair or not in each cases?

Solution is given in [A.4](#).

Exercise 4.2. Show that for any $a > 0$,

$$\frac{1}{1+a^{-2}} \frac{e^{-a^2/2}}{a} \leq \int_a^\infty e^{-x^2/2} dx \leq \frac{e^{-a^2/2}}{a}$$

Hint: write $1 = \frac{x}{x}$ and integrate by part.

Solution is given in A.4.

4.1.2 Estimators

Definition 4.1.3 (Parametric estimation). We are interested in *parametric estimation*. We will denote $\mathbb{P}_\theta$ the law that we are trying to approximate, $\mathbb{P}_\theta$ is thus a probability measure on $\mathbb{R}^d$. We will suppose that it is entirely determined by a *parameter* $\theta \in \Theta$, where $\Theta \subset \mathbb{R}^m$ for some $m \geq 1$ is the *parameter space*.

Example 4.1.2. We could for example look at

1. a sequence of height measurements in the Swiss population,
2. a sequence of life duration for computers in a company,
3. the number of blue cars exiting the highway at Ecublens during a day.

We can now choose an appropriate family of probability measures as “candidates” for approximating each of these situations.

1. One can use a Gaussian law $\mathbb{P}_\theta = \mathcal{N}(\mu, \sigma^2)$. The parameter defining the law is $\theta = (\mu, \sigma)$. It leads to the parameter space $\Theta = \mathbb{R} \times [0, +\infty)$.
2. One can use an exponential law $\mathbb{P}_\theta = \text{Exp}(\lambda)$. The parameter is then $\theta = \lambda$, and the parameter space is $\Theta = (0, +\infty)$.
3. One can use a Poisson law $\mathbb{P}_\theta = \text{Poi}(\lambda)$. The parameter is then $\theta = \lambda$, and the parameter space is $\Theta = [0, +\infty)$.

Definition 4.1.4. Let $X_1, \dots, X_n$ be an n -sample of law $\mathbb{P}_\theta$. For $f : \Theta \rightarrow \mathbb{R}$, an *estimator* of $f(\theta)$ is a statistic $\hat{f} : (\mathbb{R}^d)^n \rightarrow \mathbb{R}$, which **does not depend on the parameter** θ . In words, an estimator is a way to probe the parameter space *without prior knowledge of the parameter value* to approximate the parameter defining our probability law $\mathbb{P}_\theta$.

Remark 4.1.2. The interest of looking at functions f of the parameter rather than at the parameter itself is to allow for more flexibility (for example, it is often more convenient to estimate σ^2 rather than σ in the case of the Gaussian law). It also permits to estimate parameters one at the time when the dimension of Θ is strictly bigger than 1: one can then take f to be the first/second/... coordinate of the parameter. It is also often of interest to estimate the mean of $\mathbb{P}_\theta$, or its variance, or higher moments. These are all functions of θ (as they are functions of $\mathbb{P}_\theta$ which is a function of θ).

The first notion makes precise the idea that as we increase a sample size, we can find better and better approximations of our parameter.

Definition 4.1.5. For $n \geq 1$, let $X_1, \dots, X_n$ be an n -sample of law $\mathbb{P}_\theta$, and let $\hat{f}_n$ be an estimator of $f(\theta)$. We say that the sequence of estimators $(\hat{f}_n)_{n \geq 1}$ is *convergent* (or *consistent*) if $\hat{f}_n$ converges in probability towards $f(\theta)$: for any $\epsilon > 0$ and for any $\theta \in \Theta$,

$$\lim_{n \rightarrow \infty} P(|\hat{f}_n(X_1, \dots, X_n) - f(\theta)| \geq \epsilon) = 0.$$

The next notions are formalizations of “the estimator gives the correct value on average”.

Definition 4.1.6. Let $X_1, \dots, X_n$ be an n -sample of law $\mathbb{P}_\theta$. Let $\hat{f}$ be an estimator of $f(\theta)$. The *bias* of $\hat{f}$ is the difference between the expected value of $\hat{f}$ and $f(\theta)$:

$$\text{Bias}_\theta(\hat{f}) = E(\hat{f}(X_1, \dots, X_n)) - f(\theta).$$

$\hat{f}$ is called *unbiased* if for any $\theta \in \Theta$, $\text{Bias}_\theta(\hat{f}) = 0$. Otherwise, it is called *biased*.

Definition 4.1.7. For $n \geq 1$, let $X_1, \dots, X_n$ be an n -sample of law $\mathbb{P}_\theta$, and let $\hat{f}_n$ be an estimator of $f(\theta)$. We say that the sequence of estimators $(\hat{f}_n)_{n \geq 1}$ is *asymptotically unbiased* if for any $\theta \in \Theta$,

$$\lim_{n \rightarrow \infty} \text{Bias}_\theta(\hat{f}_n) = 0.$$

Definition 4.1.8. For $n \geq 1$, let $X_1, \dots, X_n$ be an n -sample of law $\mathbb{P}_\theta$, and let $\hat{f}_n$ be an estimator of $f(\theta)$. The *variance* of $\hat{f}_n$, denoted $\text{Var}_\theta(\hat{f}_n)$, is simply the variance of the random variable $\hat{f}_n(X_1, \dots, X_n)$:

$$\text{Var}_\theta(\hat{f}_n) = \text{Var}(\hat{f}_n(X_1, \dots, X_n)).$$

Remark 4.1.3. When working with Definitions 4.1.5, 4.1.6, and 4.1.7, it is important to remember that there is an implicit dependency on θ of the quantities involving $\hat{f}(X_1, \dots, X_n)$: the law of the X_i 's is $\mathbb{P}_\theta$, so changing the value of θ changes this law!

We can give a first example of criterion guaranteeing that a sequence of estimators converges.

Lemma 4.1.1. For $n \geq 1$, let $X_1, \dots, X_n$ be an n -sample of law $\mathbb{P}_\theta$, and let $\hat{f}_n$ be an estimator of $f(\theta)$. Suppose that

1. $\hat{f}_n$ is asymptotically unbiased,
2. $\text{Var}(\hat{f}_n) \xrightarrow{n \rightarrow \infty} 0$.

Then, $\hat{f}_n$ is a convergent sequence of estimators.

Proof. This will follow from a direct application of Chebychev's inequality. Indeed, by Lemma 2.8.2 (the second moment version), for any $\epsilon > 0$,

$$\begin{aligned} P(|\hat{f}_n(X_1, \dots, X_n) - f(\theta)| \geq \epsilon) &\leq \frac{E((\hat{f}_n(X_1, \dots, X_n) - f(\theta))^2)}{\epsilon^2} \\ &= \epsilon^{-2} (\text{Var}(\hat{f}_n) + (\text{Bias}_\theta(\hat{f}_n))^2) \xrightarrow{n \rightarrow \infty} 0, \end{aligned}$$

as $E(\hat{f}_n(X_1, \dots, X_n)) \xrightarrow{n \rightarrow \infty} f(\theta)$ (recall that we assumed that $\hat{f}_n$ is asymptotically unbiased), and we assumed that $\text{Var}(\hat{f}_n) \xrightarrow{n \rightarrow \infty} 0$. $\square$

Exercise 4.3. In each of the following examples, give the parameter space Θ and the family of probability laws $(\mathbb{P}_\theta)_{\theta \in \Theta}$ that one is led to consider.

1. We make a sequence of (potentially biased) coin flips.
2. We roll a (potentially biased) 6-faces dice many times.
3. We measure the sequence of inter-arrival times between clients in a candy store.

Solution is given in [A.4](#).

Exercise 4.4. Let $X_1, \dots, X_n$ be a sample of law $\mathbb{P}_\theta = \text{Bern}(p)$ (so that the parameter θ is p). In which of the following cases is $\hat{f}$ an estimator of $f(\theta)$? When it is the case, is $\hat{f}$ biased or not?

1. Consider $f(\theta) = p$, and

$$\hat{f}(X_1, \dots, X_n) = X_1.$$

2. Consider $f(\theta) = p$, and

$$\hat{f}(X_1, \dots, X_n) = X_1 + X_3.$$

3. Consider $f(\theta) = \text{Var}(X_1)$, and

$$\hat{f}(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n (X_i - p)^2.$$

4. Consider $f(\theta) = p^2$, and

$$\hat{f}(X_1, \dots, X_n) = \frac{p}{n} \sum_{i=1}^n X_i.$$

5. Consider $f(\theta) = p^2$, and

$$\hat{f}(X_1, \dots, X_n) = \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2.$$

Solution is given in [A.4](#).

4.1.3 Some examples of estimators

Empirical mean

The empirical mean is the estimator we already encountered several times. It is an estimator for the expected value and it is given by

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i. \tag{4.1}$$

It is so widely used that it got its own “standard” notation.

It is direct to see that this estimator is unbiased:

$$E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = E(X_1),$$

as the X_i 's are identically distributed. Moreover, the (weak) Law of Large Numbers implies that it is a convergent estimator as soon as $E(X_1)$ is well defined.

Empirical median

A median for a random variable X is a number M such that

$$P(X \leq M) = P(X \geq M) = \frac{1}{2}.$$

The empirical median is obtained as follows.

1. Order the sample $X_1, \dots, X_n$ in a non-decreasing fashion: let $(\tilde{X}_1, \dots, \tilde{X}_n)$ be a permutation of $(X_1, \dots, X_n)$ such that $\tilde{X}_{i+1} \geq \tilde{X}_i$ for all i 's.
2. Define the empirical median of $X_1, \dots, X_n$ to be

$$\begin{cases} \tilde{X}_{\frac{n+1}{2}} & \text{if } n \text{ is odd,} \\ \frac{1}{2}(\tilde{X}_{\frac{n}{2}} + \tilde{X}_{1+\frac{n}{2}}) & \text{if } n \text{ is even.} \end{cases}$$

One can show that when the law $\mathbb{P}_\theta$ is symmetric around its mean (if it is a continuous law with density f_θ , this means $f_\theta(\mu + x) = f_\theta(\mu - x)$ with $\mu = \int_{-\infty}^{+\infty} x f_\theta(x) dx$), the median is an unbiased estimator of the mean. See Exercise 4.5.

Exercise 4.5. 1. What is the median for $X \sim \mathcal{N}(\mu, \sigma^2)$?

2. What is the median for $X \sim \text{Uni}([0, 1])$?

3. Give two different medians for $X \sim \text{Uni}(\{-1, 34, 45, 55\})$.

4. Suppose $X \stackrel{\text{Law}}{=} -X$. Give a median for X .

5. Let $X_1, X_2, \dots, X_n$ be an n -sample of a continuous law with density f . Suppose that $f(x) = f(-x) > 0$ for all $x \in \mathbb{R}$. Show that the median for X_1 is 0 and that the empirical median of $X_1, \dots, X_n$ is an un-biased estimator of the median of X_1 and of the mean of X_1 .

Solution is given in A.4

Empirical variance

Recall that the variance of a random variable X is given by

$$\text{Var}(X) = E((X - E(X))^2) = E(X^2) - E(X)^2.$$

We already have an estimator for the expectation: $\bar{X}_n$. A natural thing to do is the to define the empirical variance as

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}_n^2 = \overline{X^2}_n - \bar{X}_n^2.$$

I.e.: take the difference between the empirical mean of $X_1^2, \dots, X_n^2$ and the square of the empirical mean of $X_1, \dots, X_n$.

Note that the (weak)LLN implies that as soon as $E(X_1^2) < \infty$, this estimator is convergent: $\overline{X^2}_n$ converges to its mean which is $E(X_1^2)$, and we already saw that $\bar{X}_n$ converges to $E(X_1)$.

Exercise 4.6. Show that the empirical variance

$$\hat{\sigma}_n^2 = \overline{X^2}_n - \bar{X}_n^2$$

is a biased estimator. Show that one can remedy this by considering the estimator

$$\frac{n}{n-1} \hat{\sigma}_n^2$$

instead. Show that this estimator is convergent when $E(X_1^2) < \infty$ (i.e.: $\mathbb{P}_\theta$ has a second moment).

Solution is given in [A.4](#)

Empirical covariance

Consider an n -sample $(X_1, Y_1), \dots, (X_n, Y_n)$ of random vectors of law $\mathbb{P}_\theta$. The goal is to estimate the covariance $\text{Cov}(X, Y)$ where $(X, Y) \sim \mathbb{P}_\theta$. Define the estimator

$$\hat{\tau}_n((X_1, Y_1), \dots, (X_n, Y_n)) = \frac{1}{n} \sum_{i=1}^n X_i Y_i - \frac{1}{n} \left(\sum_{i=1}^n X_i \right) \frac{1}{n} \left(\sum_{i=1}^n Y_i \right) = \overline{XY}_n - \bar{X}_n \bar{Y}_n.$$

We have a phenomenon similar than in the case of the empirical variance.

Exercise 4.7. Show that $\hat{\tau}_n$ is a biased estimator of the covariance. Show that $\frac{n}{n-1} \hat{\tau}_n$ is unbiased.

Solution is given in [A.4](#)

*Empirical distribution

This last estimator leaves the realm of parametric estimation. One can go further and directly ask “given a sample of law $\mathbb{P}$, can I approximate the probability measure $\mathbb{P}$?”. This is what the empirical distribution attempts to do. The idea is to create a histogram of the obtained data, and to attribute “mass” to each realisation proportional to its frequency of occurrence. For a the realisation $x_1, \dots, x_n$ of a sample $X_1, \dots, X_n$ of law $\mathbb{P}$, the *empirical distribution* (or *empirical measure*) is given by

$$(\hat{P}_n(x_1, \dots, x_n))(A) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_A(x_i).$$

In words: the empirical measure gives to an event A a probability equal to the proportion of realisations that fell into A . The LLN gives that this number will converge towards $\mathbb{P}(A)$, as $(\hat{P}_n(x_1, \dots, x_n))(A)$ is the empirical mean of the Bernoulli random variable $\mathbb{1}_A(X)$ with $X \sim \mathbb{P}$, which has mean $\mathbb{P}(A)$.

Exercise 4.8. Suppose that we do the following experiment: pick 10 first year students uniformly at random in EPFL, and ask them their coffee consumption in litres per week, and their grade in Analysis I. Suppose that we obtain the numbers

Quest. / Student	1	2	3	4	5	6	7	8	9	10
Coffee	1.12	0.3	0.95	0.76	0.04	0.88	1.1	1.1	0.1	0.73
Analysis I	5.5	3.5	4	5	3	4.5	5	5.5	4.5	4

Based on these numbers, answer the following using a suitable convergent estimator in each cases.

1. Estimate the average weekly coffee consumption (in litres) of a first year student.
2. Estimate the average Analysis I grade of a first year student.
3. Estimate the variance of the weekly coffee consumption amongst first year students.
4. Estimate the covariance between coffee consumption and Analysis I grades.

Solution is given in [A.4](#)

4.1.4 *Computing integrals using the Monte-Carlo method

Suppose that we want to evaluate the integral of a function over some compact interval: for real numbers $a < b$, and $f : [a, b] \rightarrow \mathbb{R}$, we want to evaluate

$$\int_a^b f(x)dx.$$

The idea is to view this as the expectation of a suitable random variable and to use the law of large numbers. First, let $X \sim \text{Uni}([a, b])$. By definition of a uniform random variable, can re-write (using Theorem [2.2.2](#))

$$\int_a^b f(x)dx = (b-a) \int_{-\infty}^{+\infty} \frac{\mathbb{1}_{[a,b]}(x)}{b-a} f(x)dx = (b-a)E(f(X)).$$

The question is then reduced to evaluate $E(f(X))$. But we know that we can do it using the empirical mean: if $X_1, \dots, X_n$ is a sample of law $\text{Uni}([a, b])$,

$$\frac{1}{n} \sum_{i=1}^n f(X_i)$$

is a unbiased, convergent estimator of $E(f(X))$ (provided f is not too horribly behaved, for example: f bounded works). This is the general idea behind the Monte-Carlo

method: to evaluate some quantity, write it as the expected value of some random variable and use the empirical mean to obtain an estimation of the quantity from a sample of the correct law. The question “evaluate the quantity” is then reduced to “sample random variables with a given law”.

4.2 Constructing estimators

4.2.1 Moments method

Let us start with an example. Suppose that we have an n -sample, $X_1, \dots, X_n$, of exponential random variables with a parameter to be determined (i.e.: $X_i \sim \text{Exp}(\lambda)$ with λ the parameter to be estimated). We know from the computation of the mean of the exponential law that

$$E(X_1) = \frac{1}{\lambda}.$$

We thus know from the convergence of the empirical mean that $\overline{X}_n$ is a convergent estimator of $E(X_1) = \frac{1}{\lambda}$. We could thus use the estimator of λ given by

$$(\overline{X}_n)^{-1}.$$

From this example, we can extract a general method. Suppose one wants to estimate a parameter θ from a sample $X_1, \dots, X_n$ of law $\mathbb{P}_\theta$. The general idea is

1. find functions h, g such that we have the relation

$$\theta = h(E(g(X)))$$

with $X \sim \mathbb{P}_\theta$;

2. use the empirical mean $\frac{1}{n} \sum_{i=1}^n g(X_i)$ to estimate $E(g(X))$;
3. use the estimator of θ given by

$$\hat{\theta}_n = h\left(\frac{1}{n} \sum_{i=1}^n g(X_i)\right).$$

In the example seen before, we had $g(x) = x$, and $h(x) = x^{-1}$.

Remark 4.2.1. The name “method of moments” is due to the fact that in practice, one often use $g(x) = x^r$ for some $r > 0$.

Example 4.2.1. We can look at the case of $\mathbb{P}_\theta = \text{Geo}(p)$ (so that $\theta = p$, $\Theta = [0, 1]$). We know that if $X \sim \text{Geo}(p)$,

$$E(X) = \frac{1}{p}, \quad E(X^2) = \frac{2-p}{p^2}.$$

Each of these give rise to an estimator via the moment method.

- First consider the first moment. We can use $h(x) = x^{-1}$ and $g(x) = x$. This gives the estimator of p

$$\hat{p}_n(X_1, \dots, X_n) = \frac{n}{\sum_{i=1}^n X_i}.$$

- Then consider the second moment. We can use $h(x) = \frac{\sqrt{1+8x}-1}{2x}$ and $g(x) = x^2$. This gives the estimator of p

$$\hat{p}_n(X_1, \dots, X_n) = \frac{n}{2 \sum_{i=1}^n X_i^2} \left(\left(1 + \frac{8}{n} \sum_{i=1}^n X_i^2 \right)^{1/2} - 1 \right).$$

Exercise 4.9. We look at $\mathbb{P}_\theta = \text{Uni}([0, \theta])$, $\theta \in \Theta = (0, +\infty)$. Let $a \in \mathbb{N}^*$. Find an estimator of θ using the moment method with $g(x) = x^a$.

Solution is given in [A.4](#)

Exercise 4.10. A cow looks at the cars passing a mountain pass. As she is a modern cow, she has a watch, a computer, and an excel sheet to take notes. During a day, she observes the following times (in minutes) between each cars:

1.12, 5.67, 42.93, 5.56, 9.98, 2.13, 14.21, 19.51, 9.38, 59.44, 4.75, 1.07, 30.67, 6.37

Supposing that the inter-arrival times are independent, identically distributed, and follow an exponential distribution $\text{Exp}(\lambda)$ for some $\lambda > 0$, use the moment's method to estimate λ .

Solution is given in [A.4](#)

Exercise 4.11. Let $\mu \in \mathbb{R}$, $\sigma > 0$, and $X \sim \mathcal{N}(\mu, \sigma^2)$. Define the random variable $Y = e^X$. The law of Y is called the *log-normal distribution*.

1. Using the moment generating function of X , show that

$$E(Y) = e^{\mu + \frac{\sigma^2}{2}}, \quad \text{Var}(Y) = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1).$$

2. If $Y_1, Y_2, \dots, Y_n$ is a n -sample from the log-normal distribution, find estimators of $\psi = \mu + \frac{\sigma^2}{2}$, and for σ^2 using the method of moments.

Solution is given in [A.4](#)

4.2.2 Maximum likelihood estimator

The idea behind this estimator is deeply linked to the Bayes formula: imagine that you have a realisation $x_1, \dots, x_n$ of a sample $X_1, \dots, X_n$ of law $\mathbb{P}_\theta$. The idea is to look for the value of θ which *maximises the probability to observe the realisation* $x_1, \dots, x_n$.

Definition 4.2.1. Let $n \geq 1$, and a sample $X_1, \dots, X_n$ of law $\mathbb{P}_\theta$. For a realisation $x_1, \dots, x_n$ of $X_1, \dots, X_n$, the *likelihood of $x_1, \dots, x_n$ given θ* , $\mathcal{L}(x_1, \dots, x_n | \theta)$, is defined by

- if $\mathbb{P}_\theta$ is a discrete probability measure (i.e.: the X_i 's are discrete random variables),

$$\mathcal{L}(x_1, \dots, x_n | \theta) = P(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n P(X_i = x_i) = \prod_{i=1}^n \mathbb{P}_\theta(\{x_i\})$$

as the X_i 's are independent and of law $\mathbb{P}_\theta$;

- if $\mathbb{P}_\theta$ is a continuous probability measure (i.e.: the X_i 's are continuous random variables),

$$\mathcal{L}(x_1, \dots, x_n | \theta) = \prod_{i=1}^n f_{X_i}(x_i) = \prod_{i=1}^n f_\theta(x_i)$$

where f_θ is the density function associated to $\mathbb{P}_\theta$.

From the likelihood function, we can define the *maximum likelihood estimator*.

Definition 4.2.2. The *maximum likelihood estimator* of θ is given by

$$\text{MLE}(X_1, \dots, X_n) = \arg \max_{\theta} \mathcal{L}(X_1, \dots, X_n | \theta),$$

the point θ at which $\mathcal{L}(X_1, \dots, X_n | \theta)$ is maximized.

Note that it is a priori not obvious (and true) at all that the maxima exists and is unique. We will apply our general principle of happily ignoring this issue as it is a subtle one...

Example 4.2.2. Consider the case $X_1, \dots, X_n$ is a n -sample of law $\text{Geo}(p)$. The likelihood function of a realisation $x_1, \dots, x_n \in \mathbb{N}^*$ is given by

$$\mathcal{L}(x_1, \dots, x_n | \lambda) = \prod_{i=1}^n ((1-p)^{x_i-1} p) = \left(\frac{p}{1-p}\right)^n (1-p)^{x_1+x_2+\dots+x_n}.$$

Maximizing the likelihood function is the same as maximizing the logarithm of the likelihood function. We thus try to find the value p_* of p which maximizes

$$n \ln \left(\frac{p}{1-p}\right) + (x_1 + x_2 + \dots + x_n) \ln(1-p).$$

To simplify the computations, we can set $\frac{p}{1-p} = e^\beta$ (i.e.: $p = \frac{e^\beta}{1+e^\beta}$ and $1-p = (1+e^\beta)^{-1}$) with $\beta \in \mathbb{R}$. Then, we maximize

$$n\beta - (x_1 + x_2 + \dots + x_n) \ln(1 + e^\beta)$$

by looking at the critical point. We find that the maxima is attained at the value of β such that $\frac{e^\beta + 1}{e^\beta} = \frac{1}{n} \sum_{i=1}^n x_i$, which is equivalent to

$$p_* = \frac{1}{\frac{1}{n} \sum_{i=1}^n x_i}.$$

We thus obtained that the maximum likelihood estimator for Geometric random variables is given by the inverse of the empirical mean (the same as given by the method of moments for the first moment).

$$\text{MLE}(X_1, \dots, X_n) = \frac{1}{\bar{X}_n}.$$

Example 4.2.3. Consider the case $X_1, \dots, X_n$ is a n -sample of law $\mathcal{N}(\mu, \sigma^2)$ (and take $\theta = (\mu, \sigma^2)$). The likelihood function of a realisation $x_1, \dots, x_n \in \mathbb{N}^*$ is given by

$$\mathcal{L}(x_1, \dots, x_n | \mu, \sigma^2) = \prod_{i=1}^n \frac{e^{-(x_i - \mu)^2 / 2\sigma^2}}{\sqrt{2\pi\sigma^2}}.$$

Maximizing $\mathcal{L}$ is equivalent to minimizing $-\ln(\mathcal{L})$.

$$-\mathcal{L}(x_1, \dots, x_n | \mu, \sigma^2) = \frac{n}{2} \ln(2\pi\sigma^2) + \sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2}.$$

For any fixed σ^2 , the minima of this expression is reached at the empirical mean: $\mu_* = \frac{1}{n} \sum_{i=1}^n x_i$ (which is found by finding the critical points as a function of μ). Then, the minima is reached at the empirical variance: $\sigma_*^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_*)^2$ (which is again found by finding the critical points as a function of σ). The maximum likelihood estimator of (μ, σ^2) is thus given by

$$\text{MLE}(X_1, \dots, X_n) = (\bar{X}_n, \hat{\sigma}_n^2).$$

Exercise 4.12. Find the maximum likelihood estimator in the following cases.

1. $X_1, \dots, X_n$ is a sample of law $\text{Exp}(\lambda)$ (so $\theta = \lambda$).
2. $X_1, \dots, X_n$ is a sample of law $\text{Bern}(p)$ (so $\theta = p$).
3. $X_1, \dots, X_n$ is a sample of law $\text{Uni}([0, \theta])$, $\theta > 0$ (so $\theta = \theta$).

You observe cars passing on a small country road and note the time that passes between two cars (in minutes). You find the numbers

8.15, 2.77, 2.94, 19.20, 13.65, 4.99, 5.48, 17.37, 11.65, 4.54

Which law would you chose to approximate the times between two cars? Use the maximum likelihood estimator to estimate the parameter of the chosen law using the given data set.

Solution is given in [A.4](#)

4.3 Comparing estimators

4.3.1 Mean square error

Definition 4.3.1. Let $n \geq 1$, $X_1, \dots, X_n$ be an n -sample of law $\mathbb{P}_\theta$, and let $\hat{f}$ be an estimator of $f(\theta)$. The *mean square error* of $\hat{f}$ is

$$\text{MSE}_\theta(\hat{f}) = E((\hat{f}(X_1, \dots, X_n) - f(\theta))^2).$$

The mean square error measure the typical square distance between the value of the estimator and the “true value” of the quantity we attempt to estimate. An estimator is usually considered better than another if it has a smaller mean square error. From a direct computation, we obtain

$$\text{MSE}_\theta(\hat{f}) = \text{Var}_\theta(\hat{f}) + (\text{Bias}_\theta(\hat{f}))^2.$$

We can notice that this decomposition suggests that it might be worth introducing a small bias in our estimator if it reduces its variance. This is illustrated in the next exercise.

Exercise 4.13. Let $X_1, \dots, X_n$ be an n -sample of law $\text{Uni}([0, \theta])$, $\theta > 0$ (so $\Theta = (0, +\infty)$).

1. Show that the estimator of θ $\text{Est}_1(X_1, \dots, X_n) = 2\bar{X}_n$ is an unbiased estimator of θ .
2. Show that the estimator $\text{Est}_2(X_1, \dots, X_n) = \text{MLE}(X_1, \dots, X_n)$ is a continuous random variable with density $f(x) = \frac{n}{\theta^n} x^{n-1} \mathbf{1}_{(0, \theta)}(x)$.
3. Show that Est_2 is a biased estimator of θ , but that it is asymptotically unbiased.
4. Compute $\text{MSE}_\theta(\text{Est}_1)$ and $\text{MSE}_\theta(\text{Est}_2)$.
5. What do you observe?

Solution is given in [A.4](#)

Minimizing the mean square error is yet another way to construct an estimator.

Definition 4.3.2. Let $n \geq 1$, $X_1, \dots, X_n$ be an n -sample of law $\mathbb{P}_\theta$, and $f(\theta)$ be a quantity to estimate. The *minimal mean square error estimator* of $f(\theta)$ is

$$\text{MMSE}_{f(\theta)}(X_1, \dots, X_n) = \arg \min_{\hat{f}} E((\hat{f}(X_1, \dots, X_n) - f(\theta))^2).$$

where the minimum is over all estimators $\hat{f}(X_1, \dots, X_n)$ of $f(\theta)$. The existence of this estimator is a non-trivial fact of probability theory. It is in general not so easy to compute.

4.3.2 Asymptotic normality

Another desirable property of estimator is asymptotic normality.

Definition 4.3.3. For $n \geq 1$, let $X_1, \dots, X_n$ be an n -sample of law $\mathbb{P}_\theta$, and let $\hat{f}_n$ be an estimator of $f(\theta)$. We say that $\hat{f}_n$ is an *asymptotically normal* sequence of estimators if

1. $\hat{f}_n$ is convergent;
2. there is $C \geq 0$ such that $\sqrt{n}(\hat{f}_n - f(\theta)) \xrightarrow{\text{Law}} \mathcal{N}(0, C)$ as $n \rightarrow \infty$.

Asymptotically normal estimators have the nice property that we can usually do the same trick as will be used in Example 4.5.1: when computing confidence intervals or rejection regions using $\hat{f}_n$, we can replace $\sqrt{n}(\hat{f}_n - f(\theta))$ by a $\mathcal{N}(0, C)$ random variable, which simplifies greatly the computations. The asymptotic normality says that doing so will produce a small error if n is taken large enough (without saying how large). To obtain a quantitative version (i.e.: which says how large should n be, one can ask that a stronger version of the CLT holds for $\sqrt{n}(\hat{f}_n - f(\theta))$), for example an equivalent of Theorem 3.2.5.

Remark 4.3.1. The CLT says that the empirical mean is an asymptotically normal estimator of the expected value. In the same spirit, asymptotic normality can often be obtained using the delta-method (Lemma 3.4.1) combined with the CLT. This is one of the main interest of the delta-method in applications.

4.4 Quantile tables

In finding confidence intervals, which is the topic of the next section, we are often confronted to the problem of finding the value of t for equations of the form

$$P(X \leq t) = 0.95,$$

with X some explicit random variable (typically, $X \sim \mathcal{N}(0, 1)$). To avoid doing computations every time all over again, people have computed these values to very high precision for many classical random variables, and many other values than 0.95. These are called *quantile tables*.

Definition 4.4.1. Let X be a random variable, $q > 0$ be an integer. For integer $k \geq 1$, $t \in \mathbb{R}$ is a k th q -quantile of X if

$$P(X < t) \leq \frac{k}{q} \text{ AND } P(X \leq t) \geq \frac{k}{q}.$$

If X is a continuous random variable with strictly positive density, there is only one k th q -quantile of X . Sometimes, one speaks about ν -quantiles with $\nu \in (0, 1)$. In, this case, a k th ν -quantile of X is a number $t \in \mathbb{R}$ such that

$$P(X < t) \leq k\nu \text{ AND } P(X \leq t) \geq k\nu.$$

The value q corresponds to a discretization of $\mathbb{R}$ in intervals such that for each of them X has probability $\frac{1}{q}$ of falling into that particular interval. The q -quantiles then correspond to the values at which one passes from one interval to the other.

Example 4.4.1. The 10-quantiles of the $\mathcal{N}(0, 1)$ distribution are given in the next table. See the graph after for an illustration of the concept. In the table, t_k is the number such that $P(X \leq t) = \frac{k}{10}$ where $X \sim \mathcal{N}(0, 1)$.

k	1	2	3	4	5	6	7	8	9	10
t_k	-1.282	-0.841	-0.524	-0.253	0	0.253	0.524	0.842	1.282	$+\infty$

Figure 4.1: Table of the 10-quantiles of the $\mathcal{N}(0, 1)$ distribution (rounded to 3 digits).

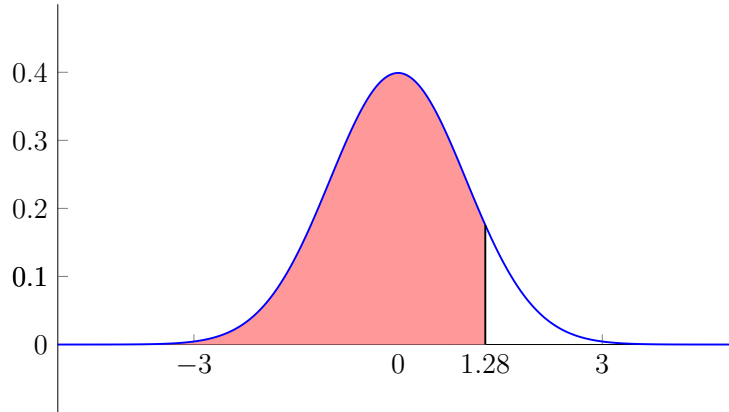


Figure 4.2: 1.28 is (approximately) the 9th 10-quantile (or 9th 0.1-quantile) of the $\mathcal{N}(0, 1)$ distribution as $P(X \leq 1.28) \approx \frac{9}{10}$ for $X \sim \mathcal{N}(0, 1)$.

Tables of quantiles for some classical distributions are given in Appendix C. In R (<https://cran.r-project.org/>), the command to find the value t such that $P(X \leq t) = p$ with $X \sim \mathcal{N}(0, 1)$ is “`qnorm(p)`”. For finding the value t such that $P(X \leq t) = p$ with $X \sim \mathcal{N}(\mu, \sigma^2)$, the command is “`qnorm(p, mean =  $\mu$ , sd =  $\sigma$ )`”.

Exercise 4.14. Let $X \sim \text{Uni}([0, 1])$, and $Y \sim \text{Exp}(1)$.

1. Find the values of the 5-quantiles of X .
2. Same question for Y .

Solution is given in A.4

Exercise 4.15. Let $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(2, 4)$. Answer the following questions using the table of Figure C.2.

1. For what value of t do we have $P(X \leq t) = 0.9$?
2. For what value of t do we have $P(X > t) = 0.05$?
3. For what value of t do we have $P(|X| > t) = 0.05$?
4. For what value of t do we have $P(Y \leq t) = 0.8$?

Solution is given in A.4

Exercise 4.16. For $k \geq 1$ integer, the χ^2 square law with k degrees of freedom is the continuous probability measure on $\mathbb{R}$ with density

$$f_{\chi_k^2}(x) = \mathbb{1}_{[0,+\infty)}(x) \frac{x^{\frac{k}{2}-1}}{2^{\frac{k}{2}}\Gamma(\frac{k}{2})} e^{-x/2},$$

where $\Gamma : \mathbb{R}_+ \rightarrow \mathbb{R}$ is the Gamma function. We denote $X \sim \chi_1^2$ for “the random variable X follows a χ_1^2 law”.

1. Using $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, show that if $Z \sim \mathcal{N}(0, 1)$, then $Z^2 \sim \chi_1^2$.
2. Let $X_1, X_2, \dots$ be an independent family of uniform random variables on $\{-1, 1\}$. Using the CLT, show that

$$\frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2 \xrightarrow{\text{Law}} \chi_1^2.$$

Then, using Table C.5, find the value of

$$\lim_{n \rightarrow \infty} P \left(\left(\sum_{i=1}^n X_i \right)^2 \in [2.26n, 2.5n] \right).$$

3. Let $X, Y \sim \mathcal{N}(0, 1)$ be two independent Gaussian random variables. Using $\Gamma(1) = 1$, show that $X^2 + Y^2 \sim \chi_2^2$.
4. For X, Y as in the previous point, use Table C.6 to find $C \in \mathbb{R}$ such that $P(X^2 + Y^2 \leq C) = 0.893$. Also, find the value of $P(X^2 + Y^2 \in [3, 4])$.

Solution is given in A.4

4.5 Confidence intervals

The goal here is to find ways to make formal (and quantitative) statements corresponding to sentences of the kind “I trust that my estimation is good”, or “I think that my estimation is good”, or “My estimation is probably very bad”... More precisely, suppose that we want to estimate some parameter θ out of a sequence of measurements $X_1, X_2, \dots, X_n$. Confidence intervals will allow to state things such as “given the measurements $X_1, X_2, \dots, X_n$, the parameter θ belongs to the interval $I(X_1, \dots, X_n)$ with probability $p(X_1, \dots, X_n)$ ”, where I, p are to be specified. One is typically interested in finding intervals with very short length (precise estimation) with p close to 1 (small probability of error).

Definition 4.5.1. Let $X_1, \dots, X_n$ be a sample of law $\mathbb{P}_\theta$. Let $\alpha \in (0, 1)$. A random interval $I = I(X_1, \dots, X_n)$ not depending on θ is called a *level $1 - \alpha$ confidence interval* for $f(\theta)$ if for every $\theta \in \Theta$,

$$P(f(\theta) \in I(X_1, \dots, X_n)) = 1 - \alpha.$$

$1 - \alpha$ is called the *confidence level* of the estimation.

Example 4.5.1. After performing a survey asking the number of hours spend watching TV per day (on average) on 100 US citizens and 100 British citizen, we are given the following data:

- US citizen answers:

$(x_1, \dots, x_{100}) = (99.3, 94.6, 88.9, 70.8, 76.2, 59.9, 78.5, 75.4, 58.8, 71.2, 109.4,$
 $109.6, 100, 114.6, 63, 69.4, 86.2, 117.1, 84.8, 71.2, 72.3, 65.7,$
 $89.7, 111, 95.6, 74.5, 114.1, 115.9, 82.7, 70.6, 83.5, 101.3, 75.5,$
 $104.2, 73.1, 99.4, 80.6, 74.1, 84, 102.4, 112.9, 85.4, 91.9, 80.6,$
 $97.7, 78.4, 105.6, 79.4, 71.9, 91, 99.2, 81.7, 92.7, 76.6, 97.5,$
 $113.1, 97.1, 97.5, 70.4, 95.9, 82, 72.7, 65.1, 93.1, 74.7, 90.9,$
 $83.3, 70.3, 87.5, 78.1, 81.8, 78.8, 112.5, 100.5, 108.3, 78.8, 123.3,$
 $108, 68, 82, 95.6, 98.9, 117.4, 71, 93.5, 94.4, 100.2, 75.6,$
 $88.7, 107.4, 94.9, 83.9, 64.4, 97.1, 90.3, 89.9, 96.6, 93.9, 91.7, 83.1)$

- British citizen answers:

$(y_1, \dots, y_{100}) = (79.8, 85.8, 54.9, 37, 21.7, 165.2, 18.5, 62, 4.5, 33.6, 77.4,$
 $67.1, 9.6, 130, 72.3, 28.8, 87.5, 124.2, 109.9, 85.3, 79.6, 137.7,$
 $105.3, 142.6, 57.5, 147.4, 145.5, 100.2, 75.4, 138.3, 34.3, 99.8, 60,$
 $43, 54.5, 64.2, 36.5, 131.5, 109.8, 147.9, 26.4, 59.2, 102.2, 111.9,$
 $123, 110.3, 135.2, 23.9, 116.4, 145.9, 134.7, 96.1, 38.6, 45.7, 90.8,$
 $86, 121.2, 75.1, 125.4, 73.6, 91.1, 126.1, 101.8, 48.6, 117.6, 115.9,$
 $22.2, 104, 186.3, 29.6, 11.8, 136.9, 185.6, 94.7, 159.9, 124.8, 112.2,$
 $63.7, 31.4, 102.2, 45.1, 80.1, 90, 179.2, 86.5, 55.8, 75.9, 93.5,$
 $6.8, 114.3, 156.7, 145.7, 107.8, 28.3, 101, 144.5, 0.6, 35.3, 142.2, 59)$

We write $X_1, \dots, X_{100}$ for the US sample, and $Y_1, \dots, Y_{100}$ for the British sample. Our goal is to estimate the mean TV time of both these countries, $E(X_1)$, and $E(Y_1)$, with 90% confidence. We can then compute the mean TV times per day in each cases using the empirical mean: this gives 88.538 minutes for the US, and 88.224 minutes for the British. We now want to establish 90% confidence intervals for both our estimations. We can for example (this is an arbitrary but convenient decision) decide to use an interval that is centred around the empirical mean in each case. We therefore need to find $\delta, \delta' \geq 0$ such that

$$P(|\bar{X}_{100} - E(X_1)| \geq \delta) = \frac{10}{100}, \quad P(|\bar{Y}_{100} - E(Y_1)| \geq \delta') = \frac{10}{100}.$$

Let us look at the histograms of the collected data to see what to expect before computing δ, δ' .

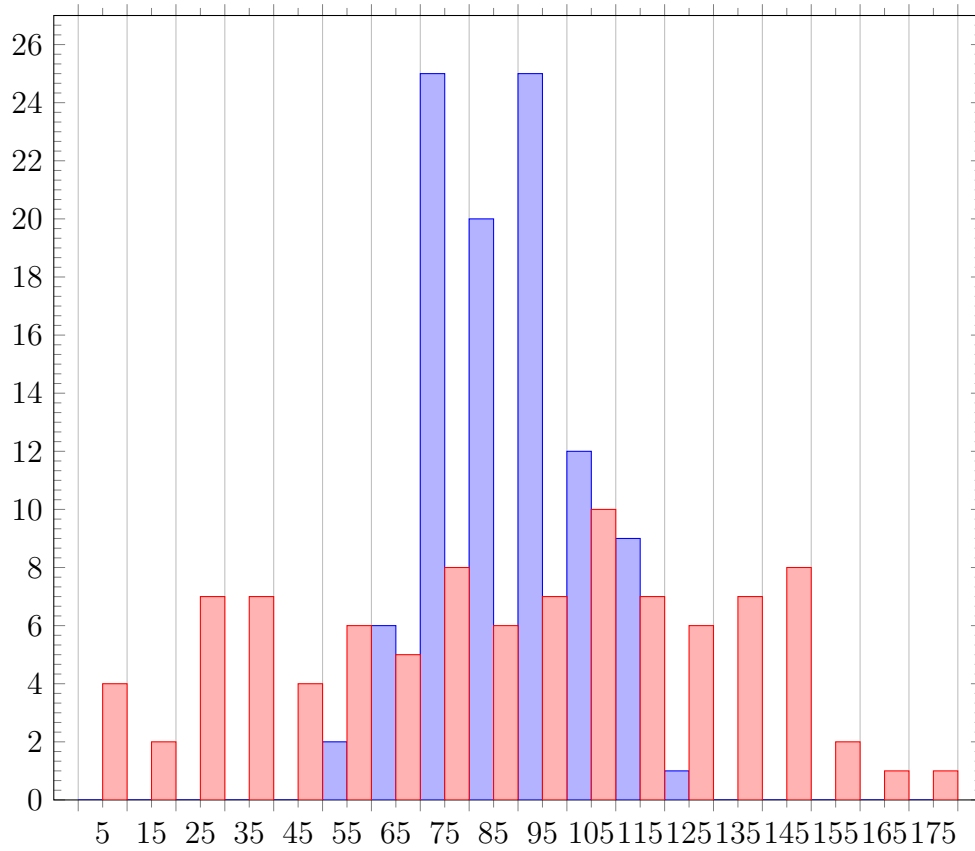


Figure 4.3: Minutes spend watching TV per day, by stretches of 10mn. Blue for US, Red for British.

We see that the data for the US is much more concentrated than for Britain. We should therefore expect a larger (less precise) confidence in the second case. Let us compute δ, δ' . We want to use that (assuming 100 is a large enough number), $\sqrt{100/\text{Var}(X_1)}(\bar{X}_{100} - E(X_1))$ follows approximately a $\mathcal{N}(0, 1)$ law. Then, we could use that if $Z \sim \mathcal{N}(0, 1)$, $P(|Z| > 1.65) = 0.1$ to get δ, δ' using

$$\begin{aligned} P(|\bar{X}_{100} - E(X_1)| \geq \delta) &= P(\sqrt{100/\text{Var}(X_1)}|\bar{X}_{100} - E(X_1)| \geq \sqrt{100/\text{Var}(X_1)}\delta) \\ &\approx P(|Z| \geq 10\delta/\sqrt{\text{Var}(X_1)}), \end{aligned}$$

and similarly for Y . This would give $\delta = \frac{1.65}{10}\sqrt{\text{Var}(X_1)}$, $\delta' = \frac{1.65}{10}\sqrt{\text{Var}(Y_1)}$. The only remaining problem is... that we do not know $\text{Var}(X_1), \text{Var}(Y_1)$! We therefore need to estimate them. We can do so using the empirical variance: in our case, this gives

$$\begin{aligned} \text{Var}(X_1) &\approx \frac{1}{100} \sum_{i=1}^{100} (x_i - 88.538)^2 = 223.4214, \\ \text{Var}(Y_1) &\approx \frac{1}{100} \sum_{i=1}^{100} (y_i - 88.538)^2 = 2027.724, \end{aligned}$$

which gives the values

$$\delta = 0.165 \cdot \sqrt{223.4214} \approx 2.47, \quad \delta' = 0.165 \cdot \sqrt{2027.724} \approx 7.43. \quad (4.2)$$

and the confidence intervals

$$\begin{aligned} I(x_1, \dots, x_{100}) &= [88.538 - 2.47, 88.538 + 2.47] \approx [86, 91], \\ I(y_1, \dots, y_{100}) &= [88.224 - 7.43, 88.224 + 7.43] \approx [80.8, 95.7]. \end{aligned}$$

Note that if we really wanted to be more rigorous in our derivation, we would need to find some confidence intervals for our estimations of $\text{Var}(X_1), \text{Var}(Y_1) \dots$. But this goes beyond the scope of this course, so we will leave it as is.

Example 4.5.2. Consider a n -sample of law $\mathbb{P}_\theta = \mathcal{N}(\mu, 1)$ (so $\theta = \mu$). We already saw that the empirical mean $\bar{X}_n$ is an unbiased estimator of μ . The goal is here to construct a confidence interval of the form $[\bar{X}_n - a, \bar{X}_n + a]$. By the properties of Gaussian random variables (Lemma 2.6.7), we have that $\bar{X}_n \sim \mathcal{N}(\mu, 1/n)$. In particular, $Z = \sqrt{n}(\bar{X}_n - \mu)$ is a $\mathcal{N}(0, 1)$ random variable (still using Lemma 2.6.7). We obtained that

$$P([\bar{X}_n - a, \bar{X}_n + a] \ni \mu) = 1 - \alpha \iff P(|\bar{X}_n - \mu| \leq a) = P(|Z| \leq \sqrt{n}a) = 1 - \alpha.$$

For example, to have a level 90% confidence interval for μ (i.e.: $\alpha = 10\%$) we need to find $a > 0$ such that $P(|Z| > a\sqrt{n}) = 10\%$. Consulting tables for the standard Gaussian law, we find that $a\sqrt{n} \approx 1.64$ does the job. In particular, we obtained that the (random) interval

$$I(X_1, \dots, X_n) = [\bar{X}_n - \frac{1.64}{\sqrt{n}}, \bar{X}_n + \frac{1.64}{\sqrt{n}}]$$

is a level 90% confidence interval for μ .

Example 4.5.3. Consider a n -sample of law $\mathbb{P}_\theta = \text{Uni}([0, \theta])$ (so $\theta = \theta \in [0, +\infty)$). We have that the maximum likelihood estimator is given by

$$\hat{\theta}(X_1, \dots, X_n) = \max\{X_1, \dots, X_n\},$$

see Exercise 4.12. As under $\mathbb{P}_\theta$, we always have $X_i \leq \theta$ for every i 's, we obtain that $\hat{\theta}(X_1, \dots, X_n) \leq \theta$. So we can look for a confidence interval of the form $[\hat{\theta}, C\hat{\theta}]$, with $C \geq 1$ to be determined. If we want to find a confidence interval at level $1 - \alpha$, we need to find C such that

$$P(C\hat{\theta} \geq \theta) = 1 - \alpha.$$

We then have

$$P(C\hat{\theta} \geq \theta) = 1 - P(\max\{X_1, \dots, X_n\} < \theta/C) = 1 - \prod_{i=1}^n P(X_i < \theta/C) = 1 - \prod_{i=1}^n \frac{\theta/C}{\theta}.$$

as the X_i are independent (see Exercise 2.32). We need to take C such that $C^{-n} = \alpha$, which is $C = \alpha^{-1/n}$. To have a confidence interval at level α , we can thus take

$$I(X_1, \dots, X_n) = [\hat{\theta}, \alpha^{-1/n}\hat{\theta}].$$

In applications, it is usually sufficient to have a lower bound on the probability to be in a given interval (rather than an equality), this leads to the notion of *excess confidence intervals*.

Definition 4.5.2. A confidence interval $I = I(X_1, \dots, X_n)$ is an *excess confidence interval* for $f(\theta)$ at level $1 - \alpha$ if

$$P(f(\theta) \in I(X_1, \dots, X_n)) \geq 1 - \alpha.$$

To get numeric answers to the following exercises, read Section 4.4 before.

Exercise 4.17. During a test, we asked 190 people who was Niki Lauda. 19 of them answered “a songwriter” (wrong), and the others had the correct answer. Based on this experiment, answer the following.

1. Give an estimation of the percentage of the population who do not know who Niki Lauda is.
2. Given a 95% confidence interval for this proportion.

Solution is given in [A.4](#)

Exercise 4.18. We pick randomly 10 apricots from a tree. Their weight (in grams) are

30.2 40.7 35.1 36.2 38.8 29.9 42.1 45.2 39.5 37.1

Assume that the weight of an apricot is Gaussian with positive variance that is given by the empirical variance of our realisation of the sample.

1. Obtain a 90% confidence interval for the mean weight μ of an apricot.
2. Try to explain what “90% confidence interval” means in this example.

Solution is given in [A.4](#)

Exercise 4.19. A physical constant μ is measured 30 times in a lab. We make the assumption that the measurements follow a normal $\mathcal{N}(\mu, \sigma^2)$ with some known variance σ^2 . Based on the measurements, the 90% confidence interval for μ obtained using an interval centred around the empirical mean is $[6.04, 6.18]$. How many more measurements must be taken if we want:

1. to divide the length of the confidence interval for μ by 2?
2. to obtain a 95% confidence interval of the same length?

Solution is given in [A.4](#)

4.6 Hypotheses testing: General principle

As often, we start with an example. Imagine that you are working for a construction firm who wants to buy a large amount of plastic pipes. They have two offers: one

from company 1, which they do not know but is cheaper, and one from company 2, which they know well but is more expensive. They are worried about the solidity of the pipes from company 1, so they mandate you to check if the pipes of company 1 are as solid as the pipes of company 2. You are given a sample of 500 pipes from each makers on which you can perform tests.

Let us phrase the situation in probabilistic language:

- Let p_1 be the frequency with which pipes from company 1 break, and p_2 the frequency with which pipes from company 2 break.
- Denote $X_1, \dots, X_{500}$ the Bernoulli random variables corresponding to the test result for the pipes of company 1 ($X_i = 1$ if pipe number i from company 1 breaks during the tests). Denote $Y_1, \dots, Y_{500}$ the Bernoulli random variables corresponding to the test result for the pipes of company 2.

Our task is then to determine whether $p_1 \leq p_2$ or the reverse. Our sample is then $(X_1, Y_1), \dots, (X_n, Y_n)$, the parameter is $\theta = (p_1, p_2)$, the parameter space $\Theta = [0, 1]^2$, and the law is $\mathbb{P}_\theta = \text{Bern}(p_1) \times \text{Bern}(p_2)$ (the law of two independent Bernoulli random variables). We do not need to determine the values of θ , rather *we need to determine to which of two regions of the parameter space it belongs*. Denote

$$\Theta_0 = \{(x, y) \in [0, 1]^2 : x > y\}, \quad \Theta_1 = \Theta \setminus \Theta_0 = \{(x, y) \in [0, 1]^2 : x \leq y\}.$$

$\theta \in \Theta_1$ corresponds to “company 1 is at least as safe as company 2”. The idea is to test the *null hypotheses* H_0 , given by $\theta \in \Theta_0$, against the *alternative hypotheses* H_1 , given by $\theta \in \Theta \setminus \Theta_0 = \Theta_1$. In our example, the null hypotheses is “company 1 produces less safe pipes”.

We now use an estimator $\hat{\theta}_{500}$ (for example, the empirical mean $\hat{\theta}_{500} = (\bar{X}_{500}, \bar{Y}_{500})$) to estimate the value of the parameter θ . We can then directly check whether $\hat{\theta}_{500} \in \Theta_0$ or not. Doing so, we take two risks:

- **Type-I error:** we reject H_0 whereas it was correct.
- **Type-II error:** we do not reject H_0 whereas it was false.

We see that the type of error committed does not have the same consequences: if H_0 is wrong and we do not reject it, we “only” take the risk that our firm misses an opportunity of saving a bit of money. On the other hand, if we reject H_0 whereas it was correct, our firm is likely to buy and use pipes that are not safe enough, risking a lot of future damages. We did not take that into account when we simply checked $\hat{\theta}_{500} \in \Theta_0$ or not. A smarter thing to do is to find $D \subset \Theta_1$ and reject H_0 if $\hat{\theta}_{500} \notin D$. We can then optimize over D to bring down the risk of type-I error to a reasonable level. Let us formally define what we just discussed.

Remark 4.6.1. We could have exchanged the roles of Θ_0 and Θ_1 , but the general convention is to set things so that type-I errors are the worst and are therefore to be minimized in priority.

The error that a practising statistician would consider the more important to avoid (which is a subjective judgment) is called the error of the first kind. The first demand of the mathematical theory is to deduce such test criteria as would ensure that the probability of committing an error of the first kind would equal (or approximately equal, or not exceed) a preassigned number α , such as $\alpha = 0.05$ or 0.01 , etc. This number is called the level of significance.

(Jerzy Neyman^a)

^aJerzy Sława-Neyman (April 16, 1894 – August 5, 1981) was a Polish mathematician and statistician who first introduced the modern concept of a confidence interval into statistical hypothesis testing.

General framework: we are trying to decide whether a parameter θ belongs to a region $\Theta_0 \subset \Theta$ or not, based on a sample $X_1, \dots, X_n$ of law $\mathbb{P}_\theta$.

Definition 4.6.1. The hypotheses “ $\theta \in \Theta_0$ ”, usually denoted H_0 , is called the *null hypotheses*. Its complement, the hypotheses “ $\theta \in \Theta \setminus \Theta_0$ ”, usually denoted H_1 , is called the *alternative hypotheses*.

Definition 4.6.2. A *rejection region* D is an event for the random variables $X_1, \dots, X_n$. I.e.: if the X_i take values in $\mathbb{R}^d$, $D \subset (\mathbb{R}^d)^n$. In practice, one usually takes

$$D = \{(x_1, \dots, x_n) : T(x_1, \dots, x_n) \in [a, b]\} \equiv \{T(X_1, \dots, X_n) \in [a, b]\}$$

for some statistic T and real numbers $a \leq b$. In this case, T is called the *test statistic*.

Definition 4.6.3. Given D a rejection region, and H_0, H_1 two hypotheses that are tested one against the other, a *test procedure* corresponds to

1. reject H_0 if $(X_1, \dots, X_n) \in D$;
2. do not reject (accept) H_0 if $(X_1, \dots, X_n) \notin D$.

Failure of prediction are divided into two classes:

- *Type-I error:* we reject H_0 whereas it was correct.
- *Type-II error:* we do not reject H_0 whereas it was false.

Definition 4.6.4. Let $\alpha \in [0, 1]$. We say that the test procedure has a *risk level* α , or a *confidence level* $1 - \alpha$ if^a

$$\sup_{\theta \in \Theta_0} P((X_1, \dots, X_n) \in D) = \alpha.$$

^aRecall that the law of the sample $X_1, \dots, X_n$ *does* depend on θ .

Definition 4.6.5. The *power* of a test is given by^a

$$\inf_{\theta \in \Theta_1} P((X_1, \dots, X_n) \in D) = 1 - \beta.$$

In particular,

$$\beta = \sup_{\theta \in \Theta_1} P((X_1, \dots, X_n) \notin D).$$

^aRecall again that the law of the sample $X_1, \dots, X_n$ *does* depend on θ .

In words: the risk level α is the “worst case scenario” of the probability to fall in the rejection region whilst having a value of the parameter satisfying H_0 (producing a type-I error), and the β in the power of a test is the “worst case scenario” of the probability to fall outside of the rejection region whilst having a value of the parameter not satisfying H_0 (producing a type-II error). The game is then to fix a value of α which is acceptable in a given situation, and to find a rejection region D which minimizes the value β for fixed value of α .

Example 4.6.1. We can try to perform a Hypotheses test on the example of Example 4.5.1. We could test the Hypotheses “Americans watch TV more than British do”. Let us set up the situation highlighting the Hypotheses made along the way. Recall that our samples were $X_1, \dots, X_{100}$ for the times of the US, and $Y_1, \dots, Y_{100}$ for the times of the British. We do parametric estimation, so we need to specify the parameter space. We can assume that the X_i, Y_i ’s are Gaussian (this is a very standard assumption in applied statistics). Now, in Example 4.5.1, we decided to use the empirical variances as proxy for the true variance without including this in the computations of confidence intervals. This would translate in assuming that the X_i ’s follow a $\mathcal{N}(\mu_X, 223)$ with parameter $\mu_X \in \mathbb{R}$ to be determined, and the Y_i ’s follow a $\mathcal{N}(\mu_Y, 2028)$ with parameter $\mu_Y \in \mathbb{R}$ to be determined (223 and 2028 are (rounding of) the empirical variances computed in Example 4.5.1). Our parameter space is thus $\Theta = \mathbb{R}^2$, and our parameter is $\theta = (\mu_X, \mu_Y)$. Let us say that in our estimation, we really want to avoid saying that British and US citizen have very different TV watch time, so we can chose H_0 to be “ $\mu_X = \mu_Y$ ”, so $\Theta_0 = \{(\mu, \mu) : \mu \in \mathbb{R}\}$. We now need to chose a test. We could take as rejection region

$$D = \{|\bar{X}_{100} - \bar{Y}_{100}| \geq \delta\}$$

for some $\delta > 0$ to be chosen according to the wanted confidence level. If we want a 95% confidence level, we need to find δ such that,

$$\sup_{(\mu_X, \mu_Y) \in \Theta_0} P(|\bar{X}_{100} - \bar{Y}_{100}| \geq \delta) = 5\%.$$

under our assumptions, $\bar{X}_{100} \sim \mathcal{N}(\mu_X, 22.3)$, $\bar{Y}_{100} \sim \mathcal{N}(\mu_Y, 202.8)$, and they are independent, so by the properties of Gaussian (Lemma 2.6.7), $\bar{X}_{100} - \bar{Y}_{100} \sim \mathcal{N}(\mu_X - \mu_Y, 225.1)$. Thus,

$$\sup_{(\mu_X, \mu_Y) \in \Theta_0} P(|\bar{X}_{100} - \bar{Y}_{100}| \geq \delta) = P(|Z| \geq \delta) = 2P(\mathcal{N}(0, 1) \geq \delta/\sqrt{225.1})$$

where $Z \sim \mathcal{N}(0, 225.1)$, as $\mu_X = \mu_Y$ for $(\mu_X, \mu_Y) \in \Theta_0$. We thus look for δ such that

$$P(\mathcal{N}(0, 1) \geq \delta/\sqrt{225.1}) = 0.025 \iff P(0 \leq \mathcal{N}(0, 1) \leq \delta/\sqrt{225.1}) = 0.475$$

which is found to be $\delta = 1.96 \cdot \sqrt{225.1} \approx 29.4$ by looking at Table C.2. We therefore reject H_0 and accept H_1 if

$$|\bar{X}_{100} - \bar{Y}_{100}| \geq 29.4.$$

Computing in our situation, we have that the difference of empirical means is $|88.5 - 88.2| = 0.3$ so we do not reject H_0 . We can finally wonder about the power of our test. We compute the value β ($1 - \beta$ is the power of the test)

$$\begin{aligned} \beta &= \sup_{(\mu_X, \mu_Y) \in \Theta_1} P(|\bar{X}_{100} - \bar{Y}_{100}| < 29.4) = \sup_{(\mu_X, \mu_Y) \in \Theta_1} P(|\mathcal{N}(\mu_X - \mu_Y, 225.1)| < 29.4) \\ &= \sup_{\mu \in \mathbb{R}^*} P(|\mathcal{N}(\mu, 1)| < \frac{29.4}{\sqrt{225.1}}) = \sup_{\mu \in \mathbb{R}^*} P(|\mathcal{N}(0, 1) - \mu| < 1.96) \\ &= \sup_{\mu \in \mathbb{R}^*} \frac{1}{\sqrt{2\pi}} \int_{\mu-1.96}^{\mu+1.96} e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}} \int_{-1.96}^{+1.96} e^{-x^2/2} dx \\ &= P(|\mathcal{N}(0, 1)| \leq 1.96) = 0.95. \end{aligned}$$

So the power of this test is only 0.05.

Remark 4.6.2. Note that we used a variance estimated empirically, which is not a very satisfying way to proceed: we should also test for the failure of our variance estimation. This, in part, motivates the more sophisticated test procedures that we shall study later. Some deal precisely with testing the mean without knowledge of the variance.

Exercise 4.20. We will try Hypotheses testing on a size-1 sample. Let $X \sim \text{Uni}([0, \theta])$ (the parameter space is $\Theta = (0, +\infty)$). We want to test the null Hypotheses H_0 “the mean of X is at least 10”. Let $C \geq 0$. Consider the rejection region

$$D_C = \{x \in \mathbb{R} : x < C\}.$$

1. What is the parameter space region Θ_0 corresponding to the null Hypotheses H_0 ? and the region Θ_1 corresponding to the alternative hypotheses H_1 ?
2. If we sample X and find 23.45, for which values of C do we reject H_0 in the test procedure associated with the rejection region D_C ?
3. Compute the risk level and the power of the test associated with the rejection region D_C .

Solution is given in A.4

The next exercise justifies (a bit) the i.i.d. assumption we have on samples picked out of a population: in substance, it shows that if we pick n individuals uniformly at random in a population of size N *with replacement* (so that we have a risk of drawing twice the same individual which induces correlations, but we are really in the framework of repeated experiments with the same setup each time), the probability of drawing twice the same individual is tiny as long as N is much bigger than n .

Exercise 4.21. Let $1 \leq n \leq N$ be integers. For $i_1, \dots, i_n \in \{1, \dots, N\}$, define

$$p_{N,n}(i_1, \dots, i_n) = \frac{(N-n)!}{N!} \prod_{j=1}^n \prod_{l \in \{1, \dots, n\} \setminus j} \mathbb{1}_{i_j \neq i_l},$$

$$q_{N,n}(i_1, \dots, i_n) = \frac{1}{N^n},$$

$p_{N,n}$ is the mass function of the probability measure associated with putting N numbered balls in a bag, and drawing *without replacement* n of them. $q_{N,n}$ is the mass function of the probability measure associated with putting N numbered balls in a bag, and drawing *with replacement* n of them. Let $P_{N,n}$ be the probability measure on $\{1, \dots, N\}^n$ with mass function $p_{N,n}$, and let $Q_{N,n}$ be the probability measure on $\{1, \dots, N\}^n$ with mass function $q_{N,n}$.

Introduce the event

$$A_n = \{(i_1, \dots, i_n) \in \{1, \dots, N\}^n : \forall j \neq l, i_j \neq i_l\},$$

that all drawn numbers are different.

1. Show that for any event $B \subset \{1, \dots, N\}^n$, $P_{N,n} = Q_{N,n}(B | A_n)$.
2. Show that if $n \leq N/2$, $Q_{N,n}(A_n) \geq 1 - \frac{n^2}{N}$. *Hint: show and use that for any $x \in [0, 0.5]$, $1 - x \geq e^{-2x}$, and that for any $x \geq 0$, $e^{-x} \geq 1 - x$.*

Solution is given in [A.4](#)

4.7 t-tests

4.7.1 Student t-distribution

This family of tests is based on finding statistics that follow a *Student t-distribution*.

Definition 4.7.1 (Student's t-distribution). Let $\nu \in (0, +\infty)$ be a positive real. A random variable X follows the *Student's t-distribution with parameter ν* , denoted $X \sim \text{Student}_t(\nu)$, if it is a continuous random variable with density given by^a

$$f_X(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}.$$

Often, one takes $\nu \in \mathbb{N}^*$. In this case, we call $\text{Student}_t(\nu)$ the *Student's t-distribution with ν degrees of freedom*.

^aRecall the Gamma function $\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$. $\Gamma(n+1) = n!$ for n integer.

Note that this law is symmetric. In particular, for any $\delta > 0$, if $X \sim \text{Student}_t(\nu)$,

$$P(|X| > \delta) = 2P(X > \delta), \quad P(|X| \leq \delta) = 2P(0 \leq X \leq \delta).$$

It occurs in statistics (in the context of hypotheses testing) through the following property.

Lemma 4.7.1. Let Z, V be two independent random variables with

$$Z \sim \mathcal{N}(0, 1), \quad V \sim \chi_k^2, \quad k \in \{2, 3, 4, \dots\}.$$

Then, the random variable $Z\sqrt{\frac{k}{V}}$ follows a Student's t -distribution with parameter k :

$$Z\sqrt{\frac{k}{V}} \sim \text{Student}_t(k).$$

We will only see the simplest examples of this family of tests: the *one-sample t -test* and *two-samples t -test*.

4.7.2 One-sample t -test

It is used to test the null Hypotheses that the mean of a sample is given by a fixed value.

Setup: We have a sample $X_1, \dots, X_n$ of a law $\mathbb{P}$. We want to test the Hypotheses H_0 “ $E(X_1) = \mu_0$ ” for a fixed $\mu_0 \in \mathbb{R}$.

Assumptions: We assume that there are $\mu \in \mathbb{R}$, $\sigma^2 > 0$ such that $\mathbb{P} = \mathcal{N}(\mu, \sigma^2)$.

Test statistic: $T(X_1, \dots, X_n) = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2}} = \frac{\sqrt{n}}{\sigma}(\bar{X}_n - \mu_0) \frac{\sqrt{n-1}}{\sqrt{\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2}}.$

Running the test: Under the null Hypotheses ($\mu = \mu_0$), our assumptions and (3.4) imply that $\frac{\sqrt{n}}{\sigma}(\bar{X}_n - \mu_0) \sim \mathcal{N}(0, 1)$, that $\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \sim \chi_{n-1}^2$, and that they are independent (see Remark 3.6.1). Thus, *under the null Hypotheses*, Lemma 4.7.1 implies that $T \sim \text{Student}_t(n-1)$. The rejection region is then taken to be of the form $\{|T| \geq C\}$ with $C > 0$ depending on the wanted confidence level.

Remark 4.7.1. Note the remarkable feature that the law of T is independent of the variance of the sample. In particular, the test can be used without having to estimate $\text{Var}(X_1)$ beforehand, making it a great improvement over our attempt of Example 4.6.1.

Example 4.7.1. We want to test whether eggs bought in a supermarket have the weight that is announced on the packaging: 53 grammes. We buy 36 eggs and weight them, obtaining the following numbers:

48.85, 52.01, 55.87, 55.25, 53.73, 50.44, 52.89, 52.50, 57.15, 48.75, 56.7, 49.49,
51.63, 56.34, 50.08, 51.94, 55.27, 54.49, 49.68, 56.36, 51.42, 51.43, 54.76, 55.56,
50.61, 54.55, 51.92, 49.86, 51.39, 50.50, 49.65, 52.03, 49.81, 55.95, 50.33, 50.43.

We want to run the test for the null Hypotheses “ $E(X_1) = 53$ ” (i.e.: $\mu_0 = 53$). From the data we compute the needed statistics:

$$\bar{x}_{36} = 52.48944, \quad T(x_1, \dots, x_{36}) = \frac{6(\bar{x}_{36} - 53)}{\sqrt{\frac{1}{35} \sum_{i=1}^{36} (x_i - \bar{x}_{36})^2}} = \frac{-3.06 \dots}{2.56 \dots} = -1.19 \dots$$

Under the null Hypotheses, $T \sim \text{Student}_t(35)$. So, if we want to run a test with confidence level 95%, we take the rejection region $D = \{|T| > C\}$ with C such that $P(|Z| > C) = 0.05$ for $Z \sim \text{Student}_t(35)$. C is obtained using the R command “qt(.95, 35)” and found to be 1.689572. For a 95% confidence level test, our rejection region is therefore $\{|T| > 1.689572\}$. We have $|-1.19| \leq 1.689572$, so we do not reject (accept) the null Hypotheses.

4.7.3 Two-samples t-test

It is used to check whether two independent samples have the same mean or not.

Setup: We have two samples of the same size $X_1, \dots, X_n$ of law $\mathbb{P}$, $Y_1, \dots, Y_n$ of a law $\mathbb{Q}$. We want to test the Hypotheses H_0 “ $E(X_1) = E(Y_1)$ ”.

Assumptions: We assume that $\mathbb{P} = \mathcal{N}(\mu_X, \sigma^2)$, and that $\mathbb{Q} = \mathcal{N}(\mu_Y, \sigma^2)$ for some $\mu_X, \mu_Y \in \mathbb{R}$, $\sigma^2 > 0$. Note that we in particular assume that the variance is the same in both samples (σ^2). We moreover assume that the samples are independent of one-another.

Test statistic: We use the statistic

$$T(X_1, \dots, X_n, Y_1, \dots, Y_n) = \frac{\sqrt{n}(\bar{X}_n - \bar{Y}_n)}{\sqrt{s_X^2 + s_Y^2}} = \frac{\sqrt{n}}{\sigma}(\bar{X}_n - \bar{Y}_n) \frac{\sqrt{n-1}}{\sqrt{s_X^2 + s_Y^2}}$$

where

$$s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2, \quad s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y}_n)^2,$$

are the non-biased versions of the empirical variances.

Running the test: Under the null Hypotheses ($\mu_X = \mu_Y$), we have that, by our assumptions and (3.4),

$$\frac{\sqrt{n}}{\sigma}(\bar{X}_n - \bar{Y}_n) \sim \mathcal{N}(0, 1), \quad \frac{n-1}{\sigma^2}(s_X^2 + s_Y^2) \sim \chi_{2n-2}^2,$$

and that the two variables above are independent (see Remark 3.6.1). Thus, by Lemma 4.7.1 $T \sim \text{Student}_t(2n-2)$ under the null Hypotheses. On the other hand, if H_0 does not hold, $\bar{X}_n - \bar{Y}_n \xrightarrow{\text{a.s.}} \delta \neq 0$ as $n \rightarrow \infty$, which implies that T takes extremely large values for large n . We thus take a rejection region of the form $\{|T| \geq C\}$ with C to be chosen depending on the wanted confidence level.

Example 4.7.2. We want to test a new drug, so we split $2n$ test subjects into two groups: one to whom we give the real drug, and another one, the control group, to whom we give a placebo. This gives our two independent, equal sizes, samples. This test then allows to test whether the drug has an effect or not. This will be the object of Exercise 4.22.

Exercise 4.22. We have a group of 60 students who have to pass an exam. We want to test a new concentration enhancing drug on them. We assume that the variance of the test result is the same for all student, independently of whether they take the drug or not. Set a test procedure to test the null Hypotheses “the drug has an effect on the average performance of students” using a two-samples t-test. Explicitly state the assumptions you make along the way. Find a 0.9 confidence level test given your assumptions.

You now obtain the test results of 30 students who were given the drug: $x_1, \dots, x_{30}$ is equal to

63.9, 72.8, 78.1, 76.7, 50.7, 77.7, 77.9, 60.2, 57.9, 47.3, 94.6, 37.1, 66.7, 43.8, 50.4, 56.0, 51.5, 51.6, 49.5, 55.4, 37.6, 43.1, 83.0, 48.3, 49.8, 63.7, 57.5, 69.7, 61.8, 93.9

and the test results of 30 students who were not given the drug: $y_1, \dots, y_{30}$ is equal to

54.0, 22.3, 39.4, 43.7, 41.2, 61.1, 47.7, 26.0, 41.2, 38.4, 38.3, 34.3, 37.4, 33.6, 48.4, 22.0, 44.3, 28.8, 70.0, 22.1, 34.7, 47.6, 37.8, 58.6, 58.8, 57.5, 47.2, 31.7, 15.7, 16.6

Do you accept the null Hypotheses in this case? You can use that $P(0 \leq \text{Student}_t(58) \leq 1.67) \approx 0.45$, and that the above data satisfy

$$\begin{aligned} \sum_{i=1}^{30} x_i &= 1828.2, & \sum_{i=1}^{30} y_i &= 1200.4, \\ \sum_{i=1}^{30} (x_i - \bar{x}_n)^2 &= 6880.032, & \sum_{i=1}^{30} (y_i - \bar{y}_n)^2 &= 5426.395 \end{aligned}$$

Solution is given in [A.4](#)

4.7.4 Relaxing the Gaussian sample assumption

The assumption that the sample is a Gaussian one can be relaxed at the cost of not having an exact test, rather an *asymptotically exact* one: the test statistic does not follow a Student law under the null Hypotheses, but converges in law to one as the sample size grows. In practice, one can use this test as if it was an exact one as soon as the sample size is large enough (30 is usually considered a good realistic threshold for “large enough”).

4.8 F-tests

4.8.1 Fisher distribution

F-tests are designed to compare variances. As for χ^2 -tests, the name comes from the limit law that the statistics under consideration converge to: the Fisher² law (or

²Sir Ronald Aylmer Fisher, February 17 1890 – July 29 1962, British biologist and statistician. One of the most influential statistician of the 20th century.

Fisher-Snedecor³ law).

Definition 4.8.1 (Fisher distribution). Let $k_1, k_2 > 0$ be two positive integers. A random variable X follows the *Fisher's law with degrees of freedom* (k_1, k_2) , denoted $X \sim F(k_1, k_2)$, if it is a continuous random variable with density given by^a

$$f_X(x) = \mathbb{1}_{[0,+\infty)}(x) \frac{1}{x\mathrm{B}(k_1/2, k_2/2)} \left(\frac{k_1 x}{k_1 x + k_2} \right)^{k_1/2} \left(1 - \frac{k_1 x}{k_1 x + k_2} \right)^{k_2/2}.$$

^aRecall the Beta function $\mathrm{B}(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$.

The interest of these laws lies in the following link with the χ^2 laws: if $X \sim \chi_{k_1}^2$ and $Y \sim \chi_{k_2}^2$ are independent, then,

$$\frac{X/k_1}{Y/k_2} \sim F(k_1, k_2). \quad (4.3)$$

One can check this by computing the density of $\frac{X/k_1}{Y/k_2}$ using the density of the χ^2 distributions, and suitable change of variables. We shall not do it.

As this link is the key to all F-tests, they are particularly well suited for cases where we are interested in hypotheses testing under the assumption that the observations we make are Gaussian. We will have the use of the following properties of F random variables.

Exercise 4.23. Let $n, m \geq 1$ be integers. Let $X \sim F(n, m)$. Then,

1. Show that if $m > 2$,

$$E(X) = \frac{m}{m-2}.$$

2. Show that if $m > 4$,

$$\mathrm{Var}(X) = \frac{2m^2(m+n-2)}{n(m-2)^2(m-4)}.$$

3. Show that if $n = m > 4$,

$$P(|X - E(X)| \geq \epsilon) \leq \frac{12}{(n-4)\epsilon^2}.$$

Solution is given in [A.4](#)

We will see two particular instances of F-tests in the setup of variance analysis (ANOVA⁴) for Gaussian estimation.

4.8.2 F-test for equality of variances

The test is about verifying whether two samples have the same variance or not. We decompose the setup and procedure.

³George Waddel Snedecor, October 20 1881 – 15 February 1974, American statistician.

⁴ANalysis Of VAriance

Setup: Two *independent* samples $X_1, \dots, X_n$ of law $\mathbb{P}$, and $Y_1, \dots, Y_m$ of law $\mathbb{Q}$, $n, m \geq 1$. We want to test the null Hypotheses “ $\text{Var}(X_1) = \text{Var}(Y_1)$ ”.

Assumptions: Assume that the variables are Gaussian: $\mathbb{P} = \mathcal{N}(\mu_X, \sigma_X^2)$, and $\mathbb{Q} = \mathcal{N}(\mu_Y, \sigma_Y^2)$, $\mu_X, \mu_Y \in \mathbb{R}$, $\sigma_X^2, \sigma_Y^2 > 0$. The parameter space is thus $\Theta = \mathbb{R} \times \mathbb{R} \times (0, +\infty) \times (0, +\infty)$, and $\theta = (\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2)$. The null Hypotheses is then “ $\sigma_X^2 = \sigma_Y^2$ ”.

Test statistic: Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, $\bar{Y}_m = \frac{1}{m} \sum_{i=1}^m Y_i$, and

$$s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2, \quad s_Y^2 = \frac{1}{m-1} \sum_{i=1}^m (Y_i - \bar{Y}_m)^2.$$

The test statistic is then

$$T(X_1, \dots, X_n, Y_1, \dots, Y_m) = \frac{S_X^2}{S_Y^2}.$$

Running the test: By our assumption that the X_i 's and Y_i 's are Gaussian and independent, (3.4), and (4.3), we have that,

$$T(X_1, \dots, X_n, Y_1, \dots, Y_m) \stackrel{\text{Law}}{=} \frac{\sigma_X^2}{\sigma_Y^2} \cdot Z,$$

where $Z \sim F(n-1, m-1)$. Under the null Hypotheses, we have that $\frac{S_X^2}{S_Y^2} \sim F(n-1, m-1)$, whilst under the alternative hypotheses, $\frac{S_X^2}{S_Y^2}$ is a scaled version of a Fisher random variable by a positive number different than one. From Exercise 4.23, we get that for n, m large, the typical values of a random variable following a $F(n-1, m-1)$ are close to 1. We take a rejection region that corresponds to T being either too large (which is likely to happen if $\frac{\sigma_X^2}{\sigma_Y^2} > 1$) or too small (which is likely to happen if $\frac{\sigma_X^2}{\sigma_Y^2} < 1$) compare to the value it should take under the null Hypotheses: for a confidence level $1 - \alpha$ we can for example take

$$D(X_1, \dots, X_n, Y_1, \dots, Y_m) = \left\{ T \in \mathbb{R} \setminus [1 - c_\alpha^-, 1 + c_\alpha^+] \right\}$$

with c_α^-, c_α^+ such that for $Z \sim F(n-1, m-1)$,

$$P(Z \leq 1 + c_\alpha^+) = 1 - \frac{\alpha}{2}, P(Z \leq 1 - c_\alpha^-) = \frac{\alpha}{2}.$$

Let us check that the proposed rejection region at the last point indeed gives a confidence level $1 - \alpha$ for our test. Recall that under the Hypotheses $H_0 = \{\theta \in \Theta_0\}$, we have that $\frac{S_X^2}{S_Y^2} \sim F(n-1, m-1)$, so our risk level is

$$\begin{aligned} \sup_{\theta \in \Theta_0} P\left(\frac{S_X^2}{S_Y^2} \in \mathbb{R} \setminus [1 - c_\alpha^-, 1 + c_\alpha^+]\right) &= P(\{Z < 1 - c_\alpha^-\} \cup \{Z > 1 + c_\alpha^+\}) \\ &= P(Z \leq 1 - c_\alpha^-) + 1 - P(Z \leq 1 + c_\alpha^+) = \frac{\alpha}{2} + 1 - 1 + \frac{\alpha}{2} = \alpha \end{aligned}$$

where $Z \sim F(n-1, m-1)$. If we try to look at the particular case $n = m = 100$, and we want to achieve a confidence level 0.95 (i.e.: $\alpha = 0.05$), we can ask R to give us the values of c_α^+, c_α^- corresponding via the commands

- $1 + c_{\alpha}^{+} = \text{qf}(1 - \alpha/2, n - 1, m - 1)$, so $1 + c_{0.05}^{+} = \text{qf}(0.975, 99, 99) \approx 1.486$ in our particular case,
- $1 - c_{\alpha}^{-} = \text{qf}(\alpha/2, n - 1, m - 1)$, so $1 - c_{0.05}^{-} = \text{qf}(0.025, 99, 99) \approx 0.673$ in our particular case.

This gives the rejection region

$$D(X_1, \dots, X_{100}, Y_1, \dots, Y_{100}) = \left\{ \frac{S_X^2}{S_Y^2} \in \mathbb{R} \setminus [0.673, 1.486] \right\}.$$

Example 4.8.1. We look at two distinct populations (for example: latest AMD graphic cards, and latest NVIDIA graphic cards), and look at a certain quantity on the individuals of these populations (for example: largest frame-rate achievable whilst playing *Clair Obscur: Expedition 33* with maximal graphics setup without frying the card). The assumptions correspond then to the frame-rates of different cards to be independent and Gaussian.

The test goal is then to test whether the variance of the frame-rate is the same for both NVIDIA and AMD cards, *without knowing the mean frame-rates of either*.

4.8.3 One way ANOVA

The test is about verifying whether belonging to a certain sub-group of a population has an influence of a given characteristic or not. We again decompose the setup and procedure.

Setup: We look at a fixed population, Pop, that we divide into k distinct groups ($k \geq 2$), $\text{Pop} = G_1 \cup G_2 \cup \dots \cup G_k$, $G_i \cap G_j = \emptyset$ if $i \neq j$. We take k samples with sizes $n_1, \dots, n_k \geq 1$: $X_{(1)} = (X_{1,1}, \dots, X_{1,n_1})$, $\dots$, $X_{(k)} = (X_{k,1}, \dots, X_{k,n_k})$, and assume that they are all independent. The sample $X_{(l)}$ is a sequence of measures made on individual in the group G_l , and has a law $\mathbb{P}_l$. We want to test the null Hypotheses “for every $l = 1, \dots, k$, $E(X_{l,1}) = \sum_{j=1}^k \frac{|G_j|}{|\text{Pop}|} E(X_{j,1})$ ”.

Assumptions: We assume that all variables are Gaussian, and that *they all have the same variance*: $\mathbb{P}_l = \mathcal{N}(\mu_l, \sigma^2)$, $l = 1, \dots, k$, $\sigma^2 > 0$, $\mu_1, \dots, \mu_k \in \mathbb{R}$. The parameter space is thus $\Theta = (0, +\infty) \times \mathbb{R}^k$, and the parameter $\theta = (\sigma^2, \mu_1, \dots, \mu_k)$. The null Hypotheses is then “ $\{\exists \bar{\mu} \in \mathbb{R} : \mu_1 = \mu_2 = \dots = \mu_k = \bar{\mu}\}$ ”.

Test statistic: Letting $N = n_1 + n_2 + \dots + n_k$,

$$\begin{aligned} \overline{X_{(l)}} &= \frac{1}{n_l} \sum_{i=1}^{n_l} X_{l,i}, & \bar{X} &= \frac{1}{N} \sum_{l=1}^k \sum_{i=1}^{n_l} X_{l,i} = \sum_{l=1}^k \frac{n_l}{N} \overline{X_{(l)}}, \\ Y_{\text{betGrpVar}} &= \frac{1}{k-1} \sum_{l=1}^k n_l (\overline{X_{(l)}} - \bar{X})^2, & Y_{\text{inGrpVar}} &= \frac{1}{N-k} \sum_{l=1}^k \sum_{i=1}^{n_l} (X_{l,i} - \overline{X_{(l)}})^2, \\ T(X_{1,1}, \dots, X_{k,n_k}) &= \frac{Y_{\text{betGrpVar}}}{Y_{\text{inGrpVar}}}. \end{aligned}$$

Running the test: Under the null Hypotheses, $X_{l,i} \sim \mathcal{N}(\bar{\mu}, \sigma^2)$ independently of i, l , and so $T \sim F(k-1, N-k)$ by our Gaussian assumption, (4.3), and (3.4) (the needed independence follows from an argument similar to Exercise 3.9). Under the alternative Hypotheses, there is (at least one) l_* such that $(\mu_{l_*} - \bar{\mu})^2 > 0$. Thus, on the one hand, Y_{inGrpVar} takes value close to σ^2 when N is large by the LLN: $X_{l,i} - \bar{X}_{(l)} \approx X_{l,i} - \mu_l \sim \mathcal{N}(0, \sigma^2)$, whilst on the other hand (still by LLN) $Y_{\text{betGrpVar}} \approx \frac{1}{k-1} \sum_{l=1}^k n_l (\mu_l - \bar{\mu})^2 \geq \frac{1}{k-1} n_{l_*} (\mu_{l_*} - \bar{\mu})^2$ for $n_1, \dots, n_k$ large, so $T \geq \frac{(\mu_{l_*} - \bar{\mu})^2}{(k-1)\sigma^2} n_{l_*}$ with high probability. If all groups sample sizes are very large, T takes some value close to 1 with high probability under the null Hypotheses, whilst it takes very large values under the alternative Hypotheses (about N/k if all group sample sizes are equal). We thus take a rejection region of the form

$$D(X_{l,i}, l = 1, \dots, k, i = 1, \dots, n_l) = \{T > c\}$$

with c to be chosen as function of the wanted confidence level. For a confidence level $1 - \alpha$, we want to take c such that

$$P(F(k-1, N-k) > c) = \alpha \iff P(F(k-1, N-k) \leq c) = 1 - \alpha.$$

Remark 4.8.1. Note that *the test does not say anything about the value $\bar{\mu}$ and does not need an estimation of this value.*

The statistic $\bar{X}_{(l)}$ is simply the empirical mean in the group l (estimates the mean value of the quantity of interest inside the group l). The statistic $\bar{X}$ is the *overall* empirical mean (estimates the mean value of the quantity of interest *in the whole population*). The statistic $Y_{\text{betGrpVar}}$ is the *between-groups variability*: $(\bar{X}_{(l)} - \bar{X})^2$ gives the square deviation between the “group l mean” and the “global population mean”, it is then multiplied by n_l to account for the number of times we will see this square deviation in our sample. The statistic Y_{inGrpVar} is the *within-group variability*: $\sum_{i=1}^{n_l} (X_{l,i} - \bar{X}_{(l)})^2$ measures the total square deviation away from the mean *inside the group l* . The normalization $k-1, N-k$ are here to make Fischer law appear in the end.

Example 4.8.2. To get some intuition about what one way ANOVA is about, we follow the example of looking at the influence of linguistic region on television use⁵. We can look at Pop being all Swiss residents older than 3, and divide it into the 3 linguistic groups G_F = “French speaking”, G_G = “German speaking”, G_I = “Italian speaking” (we omit Romansh for reasons of absence of available data, and consider only geographic location, not actual spoken language). From available data, we have

$$|\text{Pop}| = 8'800'000, |G_F| = 2'200'000, |G_G| = 6'072'000, |G_I| = 528'000.$$

We then draw representative samples of sizes $n_F = n_G = n_I = 30$ from each groups (so $N = 90$). For example, $X_{G,i}$, $i = 1, \dots, n_G$ represents then the numbers of minutes

⁵Samples are computer generated. Means of TV times and population numbers are rounded versions of actual data, source: <https://www.mediapulse.ch/wp-content/uploads/2025/07/Media-release-Semester-Publication-TV-1-25.pdf>, <https://www.bfs.admin.ch/bfs/en/home/statistics/population/languages-religions/languages.html>. Standard deviation is set (arbitrarily) to 10min to generate the sample.

n_G individuals picked uniformly at random amongst German-speaking region Swiss residents spend watching TV per day. A realisation of our samples is then a family of real numbers

$$x_{F,1}, \dots, x_{F,n_F}, x_{G,1}, \dots, x_{G,n_G}, x_{I,1}, \dots, x_{I,n_I}.$$

Here is an example of sample: for the $x_{F,i}$'s:

$$\begin{aligned} &79.67, 79.29, 101.03, 115.64, 117.53, 124.98, 97.73, 94.06, 98.37, 116.24, \\ &104.17, 103.28, 126.77, 106.08, 104.93, 100.05, 105.49, 106.46, 110.51, 105.87, \\ &115.03, 98.68, 122.14, 103.75, 123.28, 119.38, 100.05, 98.58, 129.21, 107.86, \end{aligned}$$

for the $x_{G,i}$'s:

$$\begin{aligned} &119.02, 105.33, 97.5, 94.46, 104.71, 109.08, 96.94, 99.78, 92.38, 94.08, \\ &94.9, 94.89, 111.95, 112.68, 97.28, 115.05, 104.52, 116.94, 110.54, 103.66, \\ &89.04, 90.76, 86.92, 105.56, 94.44, 94.26, 91.95, 110.15, 84.6, 91.11, \end{aligned}$$

for the $x_{I,i}$'s:

$$\begin{aligned} &159.9, 147.07, 154.13, 134.23, 150.08, 165.26, 151.92, 146.39, 150.65, 148.27, \\ &161.16, 159.52, 167.49, 158.51, 154.21, 156.36, 143.95, 137.4, 157.77, 142.49, \\ &156.06, 154.76, 140.48, 140.8, 147.34, 156.2, 143.2, 146.97, 149.59, 150.03. \end{aligned}$$

The values of our different statistics are:

$$\begin{aligned} \overline{x_{(F)}} &= 107.2037, \quad \overline{x_{(G)}} = 100.4827, \quad \overline{x_{(I)}} = 151.073, \quad \bar{x} = 119.5864, \\ y_{\text{betGrpVar}} &= \frac{1}{2} \left((\overline{x_{(F)}} - \bar{x})^2 + (\overline{x_{(G)}} - \bar{x})^2 + (\overline{x_{(I)}} - \bar{x})^2 \right) = 754.8453, \\ y_{\text{inGrpVar}} &= \frac{1}{87} \sum_{l \in \{F, G, I\}} \sum_{i=1}^{30} (x_{l,i} - \overline{x_{(l)}})^2 = 101.7591, \\ T(x_{F,1}, \dots, x_{I,n_I}) &= \frac{y_{\text{betGrpVar}}}{y_{\text{inGrpVar}}} = 7.417964. \end{aligned}$$

We then determine our rejection region. Say that we want a confidence level $1 - \alpha = 95\%$. We thus find a c such that

$$P(F(2, 87) \leq c) = 0.95.$$

We find it in R using the command “`qf(.95, df1=2, df2=87)`”, which gives $c = 3.101296$. For a test at confidence level 0.95, we therefore use the rejection region $D = \{T > 3.101296\}$. In our case, $7.417964 > 3.101296$ and we therefore reject the null Hypotheses (i.e.: at least one group has a mean different from the others).

Exercise 4.24. We measured the wine consumption of 74 former Swiss politicians (in litres per week). We found the following data.

For 12 former members of the “conseil fédéral”: $x_1, \dots, x_{12}$ are equal to

5.4, 5.2, 5.2, 4.7, 5.6, 5.6, 4.8, 5.4, 6.1, 4.9, 3.2, 4.8

For 26 former members of the “assemblée fédérale”: $y_1, \dots, y_{26}$ are equal to

2.9, 2.6, 3.4, 3.8, 5.5, 2.1, 4.3, 3.2, 2.6, 3.4, 1.5, 2.0, 2.4, 5.0, 4.9, 2.3,
3.0, 3.2, 2.3, 3.2, 4.9, 4.5, 3.0, 3.5, 4.7, 4.2

For 36 former members of “conseils d’états”: $z_1, \dots, z_{36}$ are equal to

3.4, 1.3, 3.8, 3.2, 2.2, 4.4, 0.9, 4.4, 2.8, 0.4, 0.8, 1.5, 2.8, 1.2, 2.6, 2.9, 3.7, 2.6, 3.6,
0.3, 1.9, 1.7, 3.4, 3.3, 3.2, 1.5, 2.4, 1.7, 2.7, 1.9, 1.3, 2.7, 3.9, 2.1, 2.0, 1.8

We want to test whether belonging to different political levels has an influence on the amount of wine drank per week using a one-way ANOVA. Setup the test procedure stating the assumptions made at each step. Set the parameters so that the test has a confidence level of 0.9. Do you reject the Hypotheses H_0 “all political levels have the same average wine consumption” based on the test procedure you created and the given data?

You can use

$$\sum_{i=1}^{12} x_i = 60.9, \quad \sum_{i=1}^{26} y_i = 88.4, \quad \sum_{i=1}^{36} z_i = 86.3,$$

$$\sum_{i=1}^{12} (x_i - \bar{x}_{12})^2 \approx 5.7, \quad \sum_{i=1}^{26} (y_i - \bar{y}_{26})^2 \approx 28.6, \quad \sum_{i=1}^{36} (z_i - \bar{z}_{36})^2 \approx 41.1,$$

and the fact that

$$P(F(2, 71) \leq 2.379) = 0.9.$$

Solution is given in [A.4](#)

4.9 χ^2 tests

The χ^2 tests are an example of *non-parametric tests*. So far, we always considered that the goal was to estimate a parameter. χ^2 tests do not assume prior knowledge of the family of laws to which the law of the sample belongs. The tests are build around the χ^2 -laws which are presented in Section 3.6. The central idea of each test is to find a suitable *parametric estimation problem* that we can study in place of the *non-parametric hypotheses*. In short, we reduce the problem to the comfortable realm of parametric testing.

4.9.1 Adequacy tests

Adequacy tests are a way to check whether the observed realisation of the sample was obtained from a given law or not. The χ^2 -adequacy test is amongst the most used in this category. Consider an experiment with a certain realisation space Ω . Suppose that you can partition Ω into disjoint classes $\Omega_1, \dots, \Omega_k$. If we let $\mathbb{P}$ be the law of the sample (which we do not know), we then have that the probability of falling in each of the classes is a number $p_i = \mathbb{P}(\Omega_i) \in [0, 1]$, and that $\sum_{i=1}^k p_i = 1$. Observing from a n -sample $X_1, \dots, X_n$, we can associate the numbers of times we observed a realisation in the class Ω_i :

$$N_{n,i}(X_1, \dots, X_n) = \sum_{j=1}^n \mathbb{1}_{\Omega_i}(X_j).$$

If we assume that the law of the sample is $\mathbb{P}$, we have that the vector $(N_{n,1}, \dots, N_{n,k})$ follows a *multinomial distribution* of parameters $(n, k; p_1, \dots, p_k)$: recall (see Section 3.6) that it is a discrete random vector with, for $m_1, \dots, m_k \in \mathbb{N}$,

$$P((N_{n,1}, \dots, N_{n,k}) = (m_1, \dots, m_k)) = \mathbb{1}_{m_1 + \dots + m_k = n} \frac{n!}{m_1! \dots m_k!} p_1^{m_1} \dots p_k^{m_k}. \quad (4.4)$$

We can now describe the idea of the test. Our goal is to test whether $\mathbb{P}$, the law of the sample, is a certain *fixed* law $\mathbb{Q}$ or not. We thus introduce

$$q_i = \mathbb{Q}(\Omega_i), \quad i = 1, \dots, k.$$

We then have that if $\mathbb{P} = \mathbb{Q}$, $p_i = q_i$ for every $i = 1, \dots, k$... but if $p_i \neq q_i$ for some i , then we know for sure that $\mathbb{P} \neq \mathbb{Q}$! We will therefore test the null hypotheses H_0 : “ $p_i = q_i$ for $i = 1, \dots, k$ ” against the alternative hypotheses H_1 “there is $i \in \{1, \dots, k\}$ such that $p_i \neq q_i$ ”.

Under H_0 , the q_i 's should be well approximated by the empirical frequencies: $q_i \approx N_{n,i}/n$. We just transformed a non-parametric question, “is $\mathbb{P}$ equal to $\mathbb{Q}$?” into a parametric one about the parameters of a multinomial law!

We now need to construct our rejection region. Introduce the statistic

$$T_n(X_1, \dots, X_n) = \sum_{i=1}^k \frac{(N_{n,i}(X_1, \dots, X_n) - nq_i)^2}{nq_i} = n \sum_{i=1}^k \frac{\left(\frac{N_{n,i}(X_1, \dots, X_n)}{n} - q_i\right)^2}{q_i}.$$

It is some measure of the difference between the observed frequencies $\frac{N_{n,i}(X_1, \dots, X_n)}{n}$ and the ones we should see under the null hypotheses. The reason to chose this particular statistic is Lemma 3.6.1: in our setup, it says that, *under the null Hypotheses H_0* ,

$$T_n(X_1, \dots, X_n) \xrightarrow{\text{Law}} \chi_{k-1}^2$$

Remark 4.9.1. The $k - 1$ degrees of freedom come from the fact that there are $k - 1$ parameters being effectively estimated and not k as the p_i 's are related by $p_1 + \dots + p_k = 1$.

From this Lemma, we have on the one hand that under H_0 , T_n follows asymptotically a χ^2 law with $k - 1$ degrees of freedom (note that the limit does not depend on the

particular values of the q_i 's!). On the other hand, under H_1 , there is $i_* \in \{1, \dots, k\}$ such that

$$\left(\frac{N_{n,i_*}}{n} - q_{i_*}\right)^2 \xrightarrow{\text{a.s.}} (p_{i_*} - q_{i_*})^2 > 0,$$

as $n \rightarrow \infty$, which implies that $T_n \xrightarrow{\text{a.s.}} +\infty$. If we want a risk level α for our test, we can thus take a rejection region of the form $D = \{T_n > C\}$ with C such that, with $Z \sim \chi_{k-1}^2$,

$$P(Z > C) = \alpha.$$

This gives a region with risk level α asymptotically as, under H_0 ,

$$P(T_n > C) \xrightarrow{n \rightarrow \infty} P(Z > C).$$

Let us summarize this before providing an example of application.

Setup: We have a sample $X_1, \dots, X_n$ taking value in some realisation space Ω , and having *unknown* law $\mathbb{P}$. We fix a law $\mathbb{Q}$ on Ω , and want to test if $\mathbb{P} = \mathbb{Q}$ or not.

Test construction: We fix some partition $\Omega_1, \dots, \Omega_k$ of Ω , and for $i = 1, \dots, k$, let $p_i = \mathbb{P}(\Omega_i)$ (which is unknown), and $q_i = \mathbb{Q}(\Omega_i)$ (which is known). We introduce the random variables

$$N_{n,i}(X_1, \dots, X_n) = \sum_{j=1}^n \mathbb{1}_{\Omega_i}(X_j), \quad i = 1, \dots, k,$$

and note that $(N_{n,1}, \dots, N_{n,k})$ is a multinomial random variable with parameters $(k, n; p_1, \dots, p_k)$.

Null Hypotheses: If $\mathbb{P} = \mathbb{Q}$, then $(N_{n,1}, \dots, N_{n,k})$ follows a multinomial law with parameters $(k, n; q_1, \dots, q_k)$. We will test the null Hypotheses $H_0 = "p_i = q_i \text{ for } i = 1, \dots, k"$.

Test statistic: We will construct the rejection region using the test statistic

$$T_n(X_1, \dots, X_n) = \sum_{i=1}^k \frac{(N_{n,i}(X_1, \dots, X_n) - nq_i)^2}{nq_i}.$$

Running the test: Under the null Hypotheses, $T_n \xrightarrow{\text{Law}} \chi_{k-1}^2$, whilst under the alternative Hypotheses, $T_n \xrightarrow{n \rightarrow \infty} \infty$ almost surely. To get (asymptotically as $n \rightarrow \infty$) a test with confidence level $1 - \alpha$, we take a rejection region of the form $D = \{T > C\}$, with C such that $P(Z > C) = \alpha$ for $Z \sim \chi_{k-1}^2$.

Example 4.9.1. On February 8, 1865, Gregor Mendel⁶ publishes his “Versuche über Pflanzenhybriden” (“Experiments on Plant Hybridization”), where he first describes the laws of heredity called after him. He discovered those laws by studying the transmission of biological characteristic in peas. He studied in particular the characteristics “colour” and “shape”. Both of them are coded by a gene with two alleles:

⁶20 July 1822 – 6 January 1884, Austrian biologist, meteorologist, mathematician, Augustinian friar and abbot of St. Thomas' Abbey in Brno (Brünn), Margraviate of Moravia.

- “colour” is either c (green) and recessive, or C (yellow) and dominant;
- “shape” is either s (wrinkled) and recessive, or S (round) and dominant;

By crossing two individuals with genotype $CcSs$, there are 16 equiprobable genotypes for the descendants, and the genes expression should be distributed as follows: a proportion 9/16 of yellow and round peas, 3/16 of yellow and wrinkled peas, 3/16 of green and round peas, 1/16 of green and wrinkled peas. Mendel results are summarized in the next table.

	yel. + round	yel. + wrink.	green + round	green + wrink.
Number of peas	315	101	108	32
Empirical freq.	315/556	101/556	108/556	32/556
Theoretical freq.	9/16	3/16	3/16	1/16

We can then use a χ^2 -test to test the hypotheses H_0 “the frequencies are the ones predicted by Mendel’s theory”, against the alternative hypotheses. Using the statistic Z_n from before, we obtain

$$Z_{556} = \frac{(315 - 556 \cdot \frac{9}{16})^2}{556 \cdot \frac{9}{16}} + \frac{(101 - 556 \cdot \frac{3}{16})^2}{556 \cdot \frac{3}{16}} + \frac{(108 - 556 \cdot \frac{3}{16})^2}{556 \cdot \frac{3}{16}} + \frac{(32 - 556 \cdot \frac{1}{16})^2}{556 \cdot \frac{1}{16}} \approx 0.47.$$

Let us find the rejection region for a 5% risk. For $Z \sim \chi_3^2$, we obtain that $P(Z > C) = 5\%$ pour $C \approx 7.82$, using the R command “qchisq(.95, 3)”. So we use the rejection region $D = \{Z_{556} > 7.82\}$, and therefore do not reject H_0 (as $0.47 < 7.82$).

Nowadays, the results are considered “a bit too good”, and it is strongly believed that Mendel might have “optimized” his data a little...

Exercise 4.25. We have a sample $X_1, \dots, X_{50}$ from a law $\mathbb{P}$ on $\mathbb{R}$, and we wonder if this law is a centred Gaussian. More precisely, we want to test the Hypotheses H_0 “ $\mathbb{P} = \mathcal{N}(0, \sigma^2)$ for some $\sigma^2 > 0$ ”. We want to do so with an χ^2 adequacy test. We partition $\mathbb{R}$ (which is the realisation space of our experiment) as the disjoint union of $\mathbb{R}_- = (-\infty, 0)$ and $\mathbb{R}_+ = [0, +\infty)$.

1. Under the null Hypotheses, what is the probability for X_1 to fall into $\mathbb{R}_-$ and $\mathbb{R}_+$?
2. Give the test statistic and the degrees of freedom in this case if we want to perform a χ^2 adequacy test.
3. Give the rejection region corresponding to a level 0.9 confidence test ($\alpha = 0.1$). Use the approximation that 50 is close enough to infinity to be able to replace the test statistic by its limit in doing the computation of the rejection region.

We find the realisation

1.10, -0.33, -0.03, 0.90, 2.75, 1.55, -0.46, -1.12, 1.21, 1.65, -1.11, 0.04,
 3.40, 3.37, 1.47, 0.69, 0.13, 1.07, 2.72, 0.63, -1.57, 2.72, 1.11, 0.30,
 -0.57, 6.37, 0.08, 3.25, 1.30, 1.09, 3.20, 2.57, 2.56, -3.30, 2.16, 1.99,
 0.98, 3.39, 3.09, 2.16, 0.84, -0.05, -1.64, -0.57, -0.30, -0.70, 4.84, -0.54,
 0.91, -2.12

4. Using the test setted up in the first points, do you reject the null Hypotheses?

Solution is given in [A.4](#)

4.9.2 Independence test

The goal of the test is to test the hypotheses “are property A and property B independent?”. More precisely, we have a sample $(X_1, Y_1), \dots, (X_n, Y_n)$ of random vectors (i.e.: sequence of independent observations) taking value in some realisation space $\Omega_X \times \Omega_Y$. The law of the sample (i.e.: the law of (X_1, Y_1)) is an *unknown* probability measure $\mathbb{P}$ on $\Omega = \Omega_X \times \Omega_Y$. The goal is to test whether X_1 and Y_1 are independent or not. We will consider the particular case where Ω_X and Ω_Y are finite.

Example 4.9.2. Here are two examples.

1. We take n individuals uniformly at random in a population, and look at their eyes and hair colour. Here, we would have X_i is the eye colour of individual number i , and Y_i is the hair colour of individual number i . We could take $\Omega_X = \{\text{Blue, Green, Brown}\}$, $\Omega_Y = \{\text{Blond, Brown, Black, Ginger}\}$.
2. We take n individuals uniformly at random in a population, and perform two tests on them. We take X_i the result of an antibody test for chickenpox on individual number i , so $\Omega_X = \{\text{positive, negative}\}$. We take Y_i the result of a vision test on whether individual i needs glasses to drive or not, so $\Omega_Y = \{\text{positive, negative}\}$.

Let $\Omega_X = \{\omega_1, \dots, \omega_m\}$, $\Omega_Y = \{\omega'_1, \dots, \omega'_{m'}\}$. $\mathbb{P}$ is specified by the numbers

$$p_{ij} = \mathbb{P}((\omega_i, \omega'_j)), \quad i = 1, \dots, m, j = 1, \dots, m'. \quad (4.5)$$

We can then introduce

$$p_{i*} = \mathbb{P}(\{\omega_i\} \times \Omega_Y) = \sum_{j=1}^{m'} p_{ij}, \quad i = 1, \dots, m, \quad (4.6)$$

and

$$p_{*j} = \mathbb{P}(\Omega_X \times \{\omega'_j\}) = \sum_{i=1}^m p_{ij}, \quad j = 1, \dots, m'. \quad (4.7)$$

Now, X_1 and Y_1 are independent if and only if

$$p_{ij} = p_{i*}p_{*j}, \quad i = 1, \dots, m, j = 1, \dots, m'. \quad (4.8)$$

The idea is precisely to test all the above equalities simultaneously, which is in the realm of parametric testing. The null Hypotheses is therefore $H_0 = "p_{ij} = p_{i*}p_{*j} \text{ for all } i, j\text{'s}"$. As for the adequacy test, introduce the “counting statistics”:

$$\begin{aligned} N_{n;ij} &\equiv N_{n;ij}((X_1, Y_1), \dots, (X_n, Y_n)) = \sum_{l=1}^n \mathbb{1}_{(X_l, Y_l) = (\omega_i, \omega'_j)}, \\ N_{n;i*} &\equiv N_{n;i*}((X_1, Y_1), \dots, (X_n, Y_n)) = \sum_{l=1}^n \mathbb{1}_{X_l = \omega_i} = \sum_{j=1}^{m'} N_{n;ij}, \\ N_{n;*j} &\equiv N_{n;*j}((X_1, Y_1), \dots, (X_n, Y_n)) = \sum_{l=1}^n \mathbb{1}_{Y_l = \omega'_j} = \sum_{i=1}^m N_{n;ij}. \end{aligned} \quad (4.9)$$

By the law of large numbers, as $n \rightarrow \infty$, $\frac{1}{n}N_{n;ij} \xrightarrow{\text{a.s.}} p_{ij}$, $\frac{1}{n}N_{n;i*} \xrightarrow{\text{a.s.}} p_{i*}$, $\frac{1}{n}N_{n;*j} \xrightarrow{\text{a.s.}} p_{*j}$.

The test statistic is then

$$\begin{aligned} T_n &\equiv T_n((X_1, Y_1), \dots, (X_n, Y_n)) \\ &= \sum_{i=1}^m \sum_{j=1}^{m'} \frac{(N_{ij} - n \frac{N_{i*}N_{*j}}{n^2})^2}{n \frac{N_{i*}N_{*j}}{n^2}} = n \sum_{i=1}^m \sum_{j=1}^{m'} \frac{(\frac{N_{ij}}{n} - \frac{N_{i*}N_{*j}}{n^2})^2}{\frac{N_{i*}N_{*j}}{n^2}}. \end{aligned} \quad (4.10)$$

Now, as in the case of the adequacy test, one can extend Lemma 3.6.1 to prove that, under the null Hypotheses, T_n converges in law to a $\chi_{mm'-m-m'+1}^2$ law. The number of degrees of freedom comes from the following count: we *as priori* test mm' quantities (the p_{ij} 's). But they are not independent: one has the linear relation $\sum_{ij} p_{ij} = 1$ which removes one degree of freedom. Moreover, we used empirical means to centred the counting variables: the empirical evaluations of the p_{i*} cost $m - 1$ degrees of freedom (m quantities being estimated, but they have one linear relation connecting them: $\sum_i p_{i*} = 1$), and the empirical evaluations of the p_{*j} cost $m' - 1$ degrees of freedom (same reason). This gives $mm' - 1 - (m - 1) - (m' - 1) = mm' - m - m' + 1$.

Moreover, under the alternative Hypotheses, there are i_0, j_0 such that $(p_{i_0 j_0} - p_{i_0*}p_{*j_0})^2 > 0$. In particular, this implies

$$T_n = n \sum_{i=1}^m \sum_{j=1}^{m'} \frac{(\frac{N_{ij}}{n} - \frac{N_{i*}N_{*j}}{n^2})^2}{\frac{N_{i*}N_{*j}}{n^2}} \geq n \frac{(\frac{N_{i_0 j_0}}{n} - \frac{N_{i_0*}N_{*j_0}}{n^2})^2}{\frac{N_{i_0*}N_{*j_0}}{n^2}} \xrightarrow{n \rightarrow \infty} \infty$$

almost surely, as by the sLLN, $\frac{\left(\frac{N_{i_0 j_0}}{n} - \frac{N_{i_0*} N_{*j_0}}{n^2}\right)^2}{\frac{N_{i_0*} N_{*j_0}}{n^2}} \xrightarrow{\text{a.s.}} \frac{(p_{i_0 j_0} - p_{i_0*} p_{*j_0})^2}{p_{i_0*} p_{*j_0}}$. To get a test with confidence level (asymptotically) $1 - \alpha$, we take a rejection region of the form $D = \{T_n > C\}$ with C such that $P(Z > C) = \alpha$ for $Z \sim \chi_{mm'-m-m'+1}^2$.

Let us summarize this before providing an example of application.

Setup: We make two measurements on individual of a population the first one with values in $\Omega_X = \{\omega_1, \dots, \omega_m\}$, and the second one in $\Omega_Y = \{\omega'_1, \dots, \omega'_{m'}\}$. $(X_1, Y_1), \dots, (X_n, Y_n)$ is a sample of those measurements. The *unknown* law of the sample is $\mathbb{P}$, a probability measure on $\Omega_X \times \Omega_Y$. We want to test whether the first measurement, X_1 , is independent form the second, Y_1 .

Test construction: As $\Omega_X \times \Omega_Y$ is finite, $\mathbb{P}$ is entirely determined by the p_{ij} given in (4.5), and independence between the two measurements is equivalent to (4.8). We construct approximation of the p_{ij} 's using empirical mean $\frac{1}{n}N_{n;ij}$ given in (4.9).

Null Hypotheses: Our parameter space is then

$$\Theta = \left\{ \theta \in [0, 1]^{\Omega_X \times \Omega_Y} : \sum_{(\omega, \omega') \in \Omega_X \times \Omega_Y} \theta_{\omega, \omega'} = 1 \right\}.$$

The null Hypotheses is $H_0 =$ “for all $i = 1, \dots, m$ and all $j = 1, \dots, m'$, $p_{ij} = p_{i*} p_{*j}$ ”, which corresponds to the parameter region $\Theta_0 = \{\theta \in \Theta : \theta_{ij} = \theta_{i*} \theta_{*j} \forall i, j\}$.

Test statistic: We will construct the rejection region using the test statistic

$$T_n((X_1, Y_1), \dots, (X_n, Y_n)) = n \sum_{i=1}^m \sum_{j=1}^{m'} \frac{\left(\frac{N_{ij}}{n} - \frac{N_{i*} N_{*j}}{n^2}\right)^2}{\frac{N_{i*} N_{*j}}{n^2}}.$$

Running the test: Under the null Hypotheses, $T_n \xrightarrow{\text{Law}} \chi_{mm'-m-m'+1}^2$, whilst under the alternative Hypotheses, $T_n \xrightarrow{n \rightarrow \infty} \infty$ almost surely. To get (asymptotically as $n \rightarrow \infty$) a test with confidence level $1 - \alpha$, we take a rejection region of the form $D = \{T > C\}$, with C such that $P(Z > C) = \alpha$ for $Z \sim \chi_{mm'-m-m'+1}^2$.

We now look at an example of implementation.

Example 4.9.3. Here are the measures of hairs and eyes colour in a group of people.

eyes \ hairs	Blond	Brown	Black	Ginger	Total	Freq.
Blue	25	9	3	7	44	44/124
Green	13	17	10	7	47	47/124
Brown	7	13	8	5	33	33/124
Total	45	39	21	19	124	
Freq	45/124	39/124	21/124	19/124		

We then want to test whether the colour of the eyes is independent of the colour of the hairs. Phrased in a “statistic testing” framework, we want to test the null hypotheses “the eyes and hairs colours are independent” against the alternative hypotheses. Denote the eyes colours by

$$1 \equiv \text{blue}, \quad 2 \equiv \text{green}, \quad 3 \equiv \text{brown},$$

and the hair colours by

$$1 \equiv \text{blond}, \quad 2 \equiv \text{brown}, \quad 3 \equiv \text{black}, \quad 4 \equiv \text{ginger}.$$

Denote then p_{ij} the probability that an individual has eyes colour i and hair colour j , and set

$$p_{i*} = \sum_{k=1}^4 p_{ik}, \quad p_{*j} = \sum_{k=1}^3 p_{kj},$$

the probability that an individual has eye colour i , hair colour j . Under the null hypotheses, we have that hair and eyes colours are independent, which translates into

$$p_{ij} = p_{i*}p_{*j}, \quad i = 1, 2, 3, \quad j = 1, 2, 3, 4.$$

We therefore have that the null Hypotheses is “ $p_{ij} = p_{i*}p_{*j}$ for every $i \in \{1, 2, 3\}$, and every $j \in \{1, 2, 3, 4\}$ ”. Under the null Hypotheses, we have that the number of individual with eye colour i and hair colour j in a sample of n people is on average

$$np_{ij} = np_{i*}p_{*j}.$$

Also, we know from the LLN that p_{i*} is well approximated by the empirical frequency. We can then test this Hypotheses by introducing the statistic

$$T_n = \sum_{i=1}^3 \sum_{j=1}^4 \frac{(N_{ij} - n \frac{N_{i*}N_{*j}}{n^2})^2}{n \frac{N_{i*}N_{*j}}{n^2}} = n \sum_{i=1}^3 \sum_{j=1}^4 \frac{(\frac{N_{ij}}{n} - \frac{N_{i*}N_{*j}}{n^2})^2}{\frac{N_{i*}N_{*j}}{n^2}}$$

where N_{ij} counts the number of individual with eye colour i and hair colour j , N_{i*} counts the number of individual with eye colour i , and N_{*j} counts the number of individual with hair colour j . Now, under the null hypotheses, T_n converges (in law) to a χ_6^2 law as $n \rightarrow \infty$, whilst $T_n \xrightarrow{\text{a.s.}} +\infty$ under the alternative Hypotheses. To achieve an (asymptotic) confidence level $1 - \alpha$, we take a rejection region of the form $D = \{T_n > C\}$ with C such that $P(Z > C) = \alpha$ for $Z \sim \chi_6^2$.

To get a 95% confidence region, we find $C = 12.59$ using the R command “qchisq(.95, 6)”. Using our data in T_{124} , we find the value $\approx 15.06 > 12.59$, we therefore should reject the null Hypotheses (i.e.: no independence) in the context of a 95% confidence level test.

4.10 *Neyman–Pearson test

The Neyman–Pearson test is a test where it is assume that the parameter space contains only two elements: $\Theta = \{\theta_0, \theta_1\}$. The test is a hypotheses test where we take the null hypotheses to be $H_0: “\theta = \theta_0$ (i.e.: $\Theta_0 = \{\theta_0\}$), and the alternative hypotheses to be

H_1 : “ $\theta = \theta_1$ ” (i.e.: $\Theta_1 = \{\theta_1\}$). The test will use the likelihood function defined in Definition 4.2.1. The key object in the test procedure is the *likelihood ratio*:

$$R_{x_1, \dots, x_n}(\theta_0, \theta_1) = \frac{\mathcal{L}(x_1, \dots, x_n | \theta_1)}{\mathcal{L}(x_1, \dots, x_n | \theta_0)}.$$

In words: this ratio is large if the probability to witness the realisation $x_1, \dots, x_n$ is much larger if the parameter θ is equal to θ_1 than if it is equal to θ_0 , and this ratio is tiny if the contrary holds.

One then takes the rejection region

$$D = \{(x_1, \dots, x_n) : R_{x_1, \dots, x_n}(\theta_0, \theta_1) > C\}$$

where C is some constant that we have to chose. If we want to obtain a test with risk level α , we have to chose C such that, if we take $X_1, \dots, X_n$ a sample of $\mathbb{P}_{\theta_0}$,

$$P(R_{X_1, \dots, X_n}(\theta_0, \theta_1) > C) = \alpha.$$

Before continuing, let us look at an example.

Example 4.10.1. Suppose that we are given two coins: a fair one, and a biased one which falls on Head with probability $1/3$ and on Tail with probability $2/3$. We pick one of the two without knowing which one. We then toss it 200 times and record the numbers of Heads, denoted $\#H = 71$, and the number of Tails, denoted $\#T = 129$.

Writing 1 for head and 0 for tail, our model is: the parameter is $\theta \in \Theta = \{1/2, 1/3\}$, the law which parameter we want to determine is $\mathbb{P}_\theta = \text{Bern}(\theta)$, and we have a 200-sample of law $\mathbb{P}_\theta$ with a realisation giving $\#H$ 1's. We want to test the null hypotheses $\theta = \theta_0 = \frac{1}{2}$, against the alternative hypotheses $\theta = \theta_1 = \frac{1}{3}$. The likelihood function of an n -sample of a $\text{Bern}(\theta)$ is given by, for $x_1, \dots, x_n \in \{0, 1\}$,

$$\mathcal{L}(x_1, \dots, x_n | \theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \left(\frac{\theta}{1-\theta}\right)^{\sum_{i=1}^n x_i} (1 - \theta)^n.$$

As $\sum_{i=1}^n x_i$ is the number of 1's in the realisation, the likelihood ratio in our example is

$$R = \frac{\left(\frac{\theta_1}{1-\theta_1}\right)^{\#H} (1 - \theta_1)^n}{\left(\frac{\theta_0}{1-\theta_0}\right)^{\#H} (1 - \theta_0)^n} = \left(\frac{1-\theta_1}{1-\theta_0}\right)^n \left(\frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)}\right)^{\#H} = \left(\frac{4}{3}\right)^n 2^{-\#H}.$$

This ratio is monotone (decreasing) in $\#H$, so the event $\{R \geq C\}$ is equivalent to the event

$$\{\#H \geq \log_2\left(\frac{1}{C} \left(\frac{4}{3}\right)^n\right)\}.$$

We could then compute exactly the value C for which we obtain a rejection region with, say, $\alpha = 5\%$. Instead, we will use the CLT to obtain an approximate value of C (which is often useful in practical cases where exact computations can be very heavy or even impossible!). The idea is as follows:

- We know that if $\theta = 1/2$, $\#H$ has mean $\frac{200}{2} = 100$, and variance $\frac{1}{2}(1-\frac{1}{2})200 = 50$.
- The CLT then says that $\frac{\#H-100}{\sqrt{50}}$ is close to be a Gaussian random variable $\mathcal{N}(0, 1)$.

- We therefore simply do calculations with $Z \sim \mathcal{N}(0, 1)$ replacing $\frac{\#H-100}{\sqrt{50}}!$

We thus need to find C such that (recall $n = 200$)

$$P\left(Z \geq \frac{200 \log_2\left(\frac{4}{3}\right) - \log_2(C) - 100}{\sqrt{50}}\right) \approx 10\%.$$

Many tables of these type of probabilities are available for $\mathcal{N}(0, 1)$ random variables as this type of “replacement” is common use! Browsing them allows to find the wanted C .

Note that the above is justified “philosophically” by the CLT, and becomes reasonably accurate when the sample size (here 200) is large enough, but it is *purely heuristic* and more precise/rigorous derivations usually involve 1) more/heavier computations, and 2) an explicit control over the error in the CLT, for example the Berry-Esseen version, Theorem 3.2.5.

The reason this test is so interesting is that it is optimal in the sense that it minimizes type-II error probability at fixed type-I error risk. This is the content of the Neyman-Pearson Lemma.

Lemma 4.10.1 (*Neyman-Pearson Lemma). *Work in the setup described at the beginning of this section. Suppose moreover that $\mathbb{P}_{\theta_0}$ and $\mathbb{P}_{\theta_1}$ are continuous probability measures with densities $f_{\theta_0}, f_{\theta_1}$. Let $\alpha \in (0, 1)$, and set*

$$D_C = \left\{ (x_1, \dots, x_n) : \prod_{i=1}^n f_{\theta_1}(x_i) > C \prod_{i=1}^n f_{\theta_0}(x_i) \right\}.$$

Then, there is $C_\alpha > 0$ such that testing $\theta = \theta_0$ against $\theta = \theta_1$ using the rejection region D_{C_α} has risk level α , and D_{C_α} maximizes the power of the test amongst tests with risk level α .

A similar result is available in the discrete case, but it is a bit harder to phrase. This is in part due to the fact that it is not always possible to find C such that risk level α is attainable (this is due to the CDF being discontinuous in the discrete case).

Appendix A

Solution to the exercises

A.1 Solutions to Chapter 1

Solution to Exercise 1.1. 1. $68 + 20 + 15 + 5 + 119 + 84 + 54 + 29 + 26 + 17 + 14 + 14 + 7 + 94 + 10 + 16 = 592$.

2. We need to normalize each entry in the table by the total number of participants:

Hair \ Eyes	Brown	Blue	Hazel	Green
Black	68/592	20/592	15/592	5/592
Brown	119/592	84/592	54/592	29/592
Ginger	26/592	17/592	14/592	14/592
Blond	7/592	94/592	10/592	16/592

3. The proportion of these students with black hair is: $(68 + 20 + 15 + 5)/592 = 108/592 \approx 0.18$. With blue eyes: $(20 + 84 + 17 + 94)/592 = 215/592 \approx 0.36$.

4. There are $68 + 20 + 15 + 5 + 119 + 84 + 54 + 29 = 394$ students with dark hair and $68 + 119 = 187$ students with dark hair and dark eyes. So, the proportion of the students with dark hair that have dark eyes is $187/394 \approx 0.47$, whilst the proportion of the students with dark hair that have light eyes is $1 - \frac{187}{394} = \frac{207}{394} \approx 0.53$. There are more light than dark eyes amongst the students with dark hair.

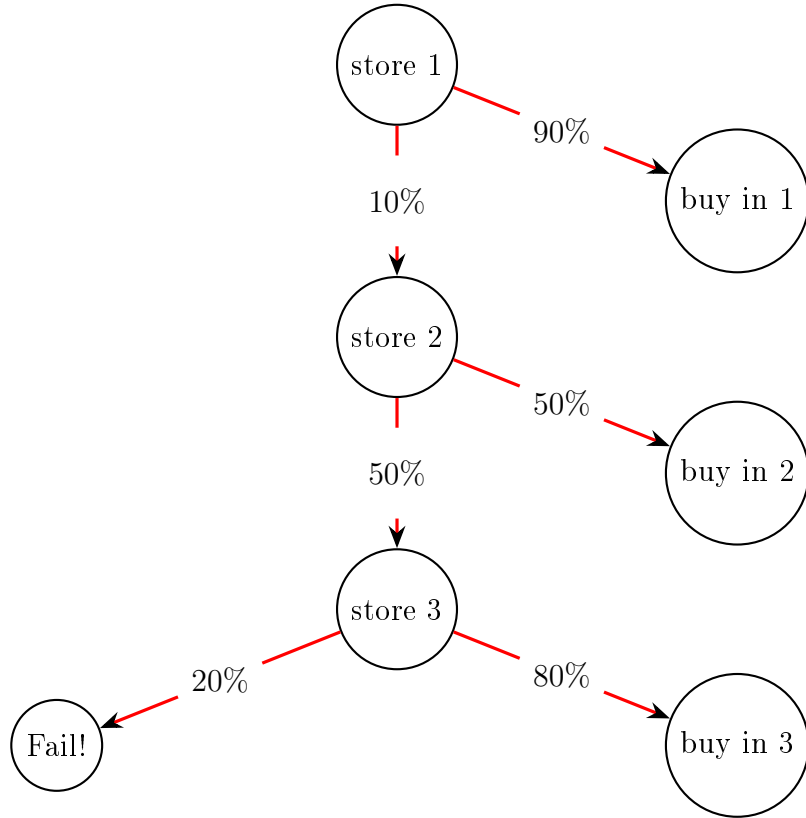


Figure A.1: The probability tree.

Solution to Exercise 1.2. The probability of not finding toothpaste is thus $\frac{10}{100} \cdot \frac{50}{100} \cdot \frac{20}{100} = \frac{1}{100}$.

Solution to Exercise 1.3. 1. No: the proportions add up to $1 = 100\%$.

2. $0.071 + 0.353 = 0.424$.

3. It is 1 minus the proportion that are, which gives $1 - 0.574 = 0.426$.

Solution to Exercise 1.4. First, there are $\binom{52}{5}$ possible hands in total (pick 5 cards amongst 52 different cards).

1. The values being fixed, we only have 4 choices for the colour. We obtain a probability of $4/\binom{52}{5} \approx 0,000154\%$.
2. There are 4 choices for the colour and 9 choices for the smallest value (1 to 9). We obtain a probability of $4 \cdot 9/\binom{52}{5} \approx 0,00139\%$.
3. We need to choose the value of the Four of a Kind (13 choices), and the value (12 choices) as well as the colour (4 choices) of the 5th card. We obtain a probability of $13 \cdot 12 \cdot 4/\binom{52}{5} \approx 0,0240\%$.
4. For the 3 cards with the same value, we have 13 choices for the value, and 4 choices for the colour (3 colours to pick amongst 4). For the 2 cards of the same value, we have 12 choices for the value (has to be different than the three cards), and 6 choices for the colour (2 colours amongst 4). We obtain a probability of $13 \cdot 4 \cdot 12 \cdot 6/\binom{52}{5} \approx 0,144\%$.

5. There are 4 choices for the colour. For the values, we can consider the $\binom{13}{5} = 1287$ possible arrangements, and remove the 10 giving rise to a straight or a straight flush. We obtain a probability of $1277 \cdot 4 / \binom{52}{5} \approx 0,197\%$.
6. There are 10 choices for the smallest value (1 to 10) and 4^5 choices for the colours. Amongst these, we need to remove the 4 arrangements giving a royal straight flush, and the 36 giving a straight flush. We obtain a probability of $(10 \cdot 4^5 - 40) / \binom{52}{5} \approx 0,393\%$.
7. There are 13 choices for the value of the Three of a Kind, and 4 choices for the colour of these 3 cards (3 colours amongst 4). There are $\binom{12}{2} = 66$ choices for the values of the 2 other cards, and 4^2 choices for their colours. We obtain a probability of $13 \cdot 4 \cdot 66 \cdot 16 / \binom{52}{5} \approx 2,113\%$.
8. There are $\binom{13}{2} = 78$ choices for the values of the two pairs, and $\binom{4}{2}^2 = 36$ choices for their colours. Then, there are 11 choices for the value of the 5th card (has to be different from the two pairs) and 4 choices for its colour. We obtain a probability of $78 \cdot 36 \cdot 11 \cdot 4 / \binom{52}{5} \approx 4,754\%$.
9. There are 13 choices for the value of the pair and $\binom{4}{2} = 6$ choices for the colours. Then, there are $\binom{12}{3} = 220$ choices for the values of the 3 other cards, and 4^3 choices for their colours. We obtain a probability of $13 \cdot 6 \cdot 220 \cdot 64 / \binom{52}{5} \approx 42,26\%$.
10. There are $\binom{13}{5} = 1287$ choices for the values of the cards, and $4^5 = 1024$ choices for their colours. To this, we have to remove the 10200 straights, the 36 straight flush, the 4 royal straight flush, and the 5108 flush. We obtain a probability of $(1287 \cdot 1024 - 15348) / \binom{52}{5} \approx 50,12\%$.

Solution to Exercise 1.5. The probability that two people are born the same day is 1 minus the probability that they are all born different days. The total number of possible birthdays for each of the 23 people is 365^{23} (chose a day per person). The number of birthdays so that all 23 people are born different days is

$$365 \cdot 364 \cdot \dots \cdot 343 = \frac{365!}{342!},$$

as there are 365 possibilities for the first person, 364 possibilities for the second (has to be different from the first), 363 for the third (has to be different from the first and second), and so on. The wanted probability is thus

$$1 - \frac{365 \cdot 364 \cdot \dots \cdot 343}{365^{23}} \approx 0.507.$$

More than 50% chances!

Solution to Exercise 1.6. 1. *First approach:* The first member sits anywhere. The second ends up at the same table as the first with probability $5/59$ (5 seats left at the table over 59 remaining seats); the third with probability $4/58$ (same reasoning); the fourth with probability $3/57$ (same reasoning); the last with probability $2/56$ (same reasoning). The probability of the event is therefore

$$5!55!/59! = 5/455126 \approx 0.001\%.$$

Second approach: Number of favourable cases: $\binom{10}{1}$ choices for the table, $\binom{6}{5}$ choices for the seats occupied at that table by the members of family A.. Number of seating options for the 5 members of family A.: $\binom{60}{5}$ (pick 5 seats amongst 60). The probability is thus $10 \cdot 6 / \binom{60}{5} = 5/455126$.

2. *First approach:* We chose the first seat arbitrarily; the probability that the second seat chosen is not at the same table is $54/59$ (54 seats not at the table over 59 remaining seats); the probability that the third is not at one of the two already chosen table is $48/58$ (same reasoning); and so on. We find a probability of $54 \cdot 48 \cdot 42 \cdot 36 \cdot 30 \cdot 24 / (59! / 53!) = 31104 / 357599 \approx 8,7\%$.

Second approach: Number of favourable cases: $\binom{10}{7}$ for the table with one missing person; $\binom{6}{1}^7$ for the empty seat at each of the tables with a missing person. Total number of cases: $\binom{60}{7}$ (7 empty seats to be picked in the 60 seats). We find a probability of $31104 / 357599$.

3. *First approach:* Mr. C. sits on an arbitrary seat. The probability that his wife is seated next to him is $2/59 \approx 3.4\%$ (two seats are next to Mr. C. over the 59 seats that are not Mr. C. seat).

Second approach: Number of favourable cases: $\binom{10}{1}$ for the choice of the table; $\binom{6}{1}$ for the first seat at the table (the other one being on the right of the first seat). Total number of cases: $\binom{60}{2}$ (two seats to be picked in the 60 possible ones). we find a probability of $10 \cdot 6 / (60 \cdot 59 / 2) = 2/59$.

4. *First approach:* The first man sits anywhere at the table; the second has a probability $2/5$ to chose one of the two seats permitting the wanted arrangement; the third has a probability $1/4$ of choosing the last compatible seat. We obtain a probability of $1/10 = 10\%$.

Second approach: Number of favourable cases: there are 2 choices for the 3 seats attributed to men. Total number of cases: there are $\binom{6}{3}$ ways to seat the 3 men at the table. We find a probability of $2 / (6! / 3!3!) = 1/10$.

A.2 Solutions to Chapter 2

Solution to Exercise 2.1. 2.

Solution to Exercise 2.2. 1. One can take as realisation space

$$\Omega = \{H, T\} \times \{H, T\} \times \{H, T\},$$

the first coordinate representing the result of the 1CHF coin, the second coordinate the result of the 2CHF coin, and the last coordinate gives the result of the 5CHF coin. The probability mass function is the uniform probability measure on Ω as all coins are fair:

$$p(\omega) = \frac{1}{|\Omega|} = \frac{1}{2^3} = \frac{1}{8}, \quad \forall \omega \in \Omega.$$

2. The result of our experiment is a pair “result of the coin” and “number of points on the face of a dice”. We have several options, we illustrate two here.

- We could take $\Omega = \{H, T\} \times \{1, 2, 3, 4, 5, 6\}$, and p given by

$$\begin{aligned} p(T, i) &= \frac{1}{3} \cdot \frac{1}{6}, \quad i = 1, \dots, 6, \\ p(H, i) &= \frac{2}{3} \cdot \frac{1}{4}, \quad i = 1, \dots, 4, \quad p(H, 5) = p(H, 6) = 0. \end{aligned}$$

- We could take

$$\Omega = \{(H, 1), (H, 2), (H, 3), (H, 4), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\},$$

and p given by

$$\begin{aligned} p(T, i) &= \frac{1}{2} \cdot \frac{1}{6}, \quad i = 1, \dots, 6, \\ p(H, i) &= \frac{1}{2} \cdot \frac{1}{4}, \quad i = 1, \dots, 4. \end{aligned}$$

3. We only look at the total number of points, which ranges from $2 = 1 + 1$ to $12 = 6 + 6$, so we can take $\Omega = \{2, 3, \dots, 12\}$ and p given by

$$\begin{aligned} p(2) = p(12) &= \frac{1}{36}, \quad p(3) = p(11) = \frac{2}{36}, \quad p(4) = p(10) = \frac{3}{36}, \\ p(5) = p(9) &= \frac{4}{36}, \quad p(6) = p(8) = \frac{5}{36}, \quad p(7) = \frac{6}{36}, \end{aligned}$$

where the 36 is the total number of cases for the result of the two dices, and the numerators correspond to the number of cases which give the wanted number of points.

Solution to Exercise 2.3. 1. We have

$$P_p(\Omega) = \sum_{\omega \in \Omega} p(\omega) = 1,$$

by definition of a mass function. On the other hand,

$$\sum_{\omega \in \emptyset} p(\omega) = 0$$

as it is an empty sum.

2. As $A \cap B = \emptyset$, for any $\omega \in \Omega$, $\mathbb{1}_{A \cup B}(\omega) = \mathbb{1}_A(\omega) + \mathbb{1}_B(\omega)$. Thus,

$$\begin{aligned} P_p(A \cup B) &= \sum_{\omega \in A \cup B} p(\omega) = \sum_{\omega \in \Omega} \mathbb{1}_{A \cup B}(\omega) p(\omega) = \sum_{\omega \in \Omega} (\mathbb{1}_A(\omega) + \mathbb{1}_B(\omega)) p(\omega) \\ &= \sum_{\omega \in \Omega} \mathbb{1}_A(\omega) p(\omega) + \sum_{\omega \in \Omega} \mathbb{1}_B(\omega) p(\omega) = \sum_{\omega \in A} p(\omega) + \sum_{\omega \in B} p(\omega) = P_p(A) + P_p(B) \end{aligned}$$

where we used that sums are linear.

3. The previous point show the claim when $n = 2$. We suppose that the equality is true for $n \geq 2$, and let us show that is it true for $n + 1$. By assumption, $B = A_1 \cup \dots \cup A_n$ and A_{n+1} are such that $B \cap A_{n+1} = \emptyset$. By the previous point, we have

$$\begin{aligned} P_p(B \cup A_{n+1}) &= P_p(B) + P_p(A_{n+1}) = P_p(A_1 \cup \dots \cup A_n) + P_p(A_{n+1}) \\ &= P_p(A_1) + \dots + P_p(A_n) + P_p(A_{n+1}) \end{aligned}$$

as we supposed the equality true for n .

4. For any $\omega \in \Omega$, $\mathbb{1}_A(\omega) = 1 - \mathbb{1}_{\Omega \setminus A}$. So,

$$\begin{aligned} P_p(A) &= \sum_{\omega \in \Omega} \mathbb{1}_A(\omega) p(\omega) = \sum_{\omega \in \Omega} (1 - \mathbb{1}_{\Omega \setminus A}(\omega)) p(\omega) \\ &= 1 - \sum_{\omega \in \Omega} \mathbb{1}_{\Omega \setminus A}(\omega) p(\omega) = 1 - P_p(\Omega \setminus A). \end{aligned}$$

5. Proceed as in points 2. and 4. using that for any $\omega \in \Omega$, we have that

$$\mathbb{1}_{A \cup B}(\omega) = \mathbb{1}_A(\omega) + \mathbb{1}_B(\omega) - \mathbb{1}_{A \cap B}(\omega).$$

Solution to Exercise 2.4. 1. This is not a probability mass function as it does not sum to 1:

$$\sum_{k=0}^L \frac{1}{L} = \frac{L+1}{L} > 1.$$

2. This is a probability mass function: $p(n) \geq 0$ for ever $n \in \Omega$, and

$$\sum_{n \in \Omega} p(n) = \frac{6}{\pi^2} \sum_{k=1}^{\infty} k^{-2} = 1.$$

3. This is not a probability mass function as $p(-1) = -1 < 0$.

4. This is a probability mass function: $p(n) \geq 0$ and

$$\sum_{n \in \Omega} p(n) = \frac{1}{2} \sum_{k=0}^{\infty} 2^{-k} = \frac{1}{2} \frac{1}{1 - \frac{1}{2}} = 1.$$

Solution to Exercise 2.5. We start by computing the total number of possible draws. We can suppose that all the balls are numbered from 1 to 25 with the blue going from 1 to 10 and the red from 11 to 25. In that case, there are $\binom{25}{4}$ possible draws.

1. We need to pick two balls out of the 10 blues, which gives $\binom{10}{2}$ possibilities, and two out of the 15 red, which gives $\binom{15}{2}$. The probability is thus

$$\frac{\binom{10}{2} \cdot \binom{15}{2}}{\binom{25}{4}} = \frac{10!15!2!14!}{2!8!2!13!25!} = \frac{10 \cdot 9 \cdot 15 \cdot 14 \cdot 6}{25 \cdot 24 \cdot 23 \cdot 22} = \frac{189}{506} \approx 0.37.$$

2. We need to pick either 0 red ball or 1 red ball. The first case has $\binom{10}{4}$ favourable cases (pick 4 balls amongst the 10 blue), the second has $\binom{10}{3} \cdot \binom{15}{1}$ favourable cases. This gives a probability of

$$\frac{\binom{10}{4} + \binom{10}{3} \cdot \binom{15}{1}}{\binom{25}{4}} = \frac{201}{1265} \approx 0.16.$$

3. The wanted probability is 1 minus the probability of picking 0 blue ball. The later has $\binom{15}{4}$ favourable cases (4 red balls amongst 15), so the wanted probability is

$$1 - \frac{\binom{15}{4}}{\binom{25}{4}} = \frac{2257}{2530} \approx 0.89.$$

Solution to Exercise 2.6. 1. This is not a probability density function as $\int_{-\infty}^{+\infty} f(x)dx = \int_0^3 dx = 3 \neq 1$.

2. This is not a probability density function as $f(-1/2) = \frac{-1}{8} < 0$.
3. This is a probability density function: $f(x) \geq 0$ for all x , and

$$\begin{aligned} \int_{-\infty}^{+\infty} f(x)dx &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx \\ &= \frac{1}{\pi} \lim_{R_-, R_+ \rightarrow \infty} (\arctan(R_+) - \arctan(-R_-)) = \frac{1}{\pi} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = 1. \end{aligned}$$

4. This is not a probability density function: as it does not integrate to 1:

$$\begin{aligned} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-\sqrt{x^2+y^2}} dx dy = \int_0^{2\pi} \int_0^{+\infty} r e^{-r} dr d\theta \\ &= 2\pi \int_0^{+\infty} r e^{-r} dr = 2\pi \int_0^{+\infty} e^{-r} dr = 2\pi \end{aligned}$$

where we changed to polar coordinates: $x = r \cos(\theta)$, $y = r \sin(\theta)$, $dx dy = r d\theta dr$, and integrated by parts in the fourth equality.

5. This is a probability density function: $f(x, y) \geq 0$ for all x, y , and

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = \int_0^{+\infty} \int_0^{+\infty} e^{-x-y} dx dy = \left(\int_0^{+\infty} e^{-x} dx \right)^2 = 1.$$

Solution to Exercise 2.7.

$$f(x, y) = \frac{1}{2\pi} \frac{1}{1+x^2} e^{-|y|}.$$

1. We have that $f(x, y) \geq 0$ for all x, y , so we need to check that it integrates to 1. We have

$$\begin{aligned} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{1}{1+x^2} e^{-|y|} dx dy \\ &= \frac{1}{2\pi} \left(\int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx \right) \left(\int_{-\infty}^{+\infty} e^{-|y|} dy \right) \\ &= \frac{1}{2\pi} \left(\lim_{L, R \rightarrow \infty} (\arctan(R) - \arctan(-L)) \right) \left(2 \int_0^{+\infty} e^{-y} dy \right) = \frac{1}{2\pi} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) \cdot 2 = 1. \end{aligned}$$

2. For any events $A, B \subset \mathbb{R}$, we have

$$\begin{aligned} P_f(A \times B) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \mathbf{1}_A(x) \mathbf{1}_B(y) f(x, y) dx dy \\ &= \left(\int_{-\infty}^{+\infty} \mathbf{1}_A(x) \frac{dx}{\pi(1+x^2)} \right) \left(\int_{-\infty}^{+\infty} \mathbf{1}_B(y) \frac{e^{-|y|} dy}{2} \right). \end{aligned}$$

But now, as (see the first point) $\int_{-\infty}^{+\infty} \frac{e^{-|y|} dy}{2} = 1$, we have

$$\begin{aligned} \int_{-\infty}^{+\infty} \mathbf{1}_A(x) \frac{dx}{\pi(1+x^2)} &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \mathbf{1}_A(x) \frac{dx}{\pi(1+x^2)} \frac{e^{-|y|} dy}{2} \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \mathbf{1}_{A \times \mathbb{R}}(x, y) f(x, y) dx dy = P_f(A \times \mathbb{R}), \end{aligned}$$

and proceed similarly to obtain $\int_{-\infty}^{+\infty} \mathbf{1}_B(y) \frac{e^{-|y|} dy}{2} = P_f(\mathbb{R} \times B)$.

Solution to Exercise 2.8. 1. We can represent a permutation by a list of n elements: the first is the image of 1, the second is the image of 2 and so on and so forth. We have

$$\mathcal{S}_3 = \{(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 2, 1), (3, 1, 2)\}.$$

We then get

$$P_3(D_{3,0}) = \frac{2}{6} = \frac{1}{3}, \quad P_3(D_{3,1}) = \frac{3}{6} = \frac{1}{2}, \quad P_3(D_{3,2}) = \frac{0}{6} = 0, \quad P_3(D_{3,3}) = \frac{1}{6}.$$

2. $n!$: there are n possible images for 1, $n-1$ for 2 (n minus the one that was picked for 1), $n-2$ for 3, and so on and so forth.
3. For $1 \leq i_1 < \dots < i_k \leq n$, we have that permutations fixing $I = \{i_1, \dots, i_k\}$ are in bijection with permutations of $I^c = \{1, \dots, n\} \setminus I$: one simply restricts the first sort of permutation to I^c to obtain a permutation of the second sort, and complete a permutation of the second sort by setting it to be the identity on I to obtain a permutation of the first sort. As there are $|I^c|! = (n-k)!$ permutations of the elements of I^c , we obtain

$$P_n(A_{n;i_1, \dots, i_k}) = \frac{(n-k)!}{n!}.$$

4. Every permutation that is not a derangement possess at least one fixed point, so $\mathcal{S}_n \setminus D_{n,0} \subset \cup_{i=1}^n A_{n,i}$. Moreover, if $\phi \in \cup_{i=1}^n A_{n,i}$, then there is i_* such that $\phi \in A_{n,i_*}$, so ϕ fixes i_* and is therefore not a derangement, so $\phi \in \mathcal{S}_n \setminus D_{n,0}$, which gives $\mathcal{S}_n \setminus D_{n,0} \supset \cup_{i=1}^n A_{n,i}$.
5. By the complement formula and the previous point,

$$P_n(D_{n,0}) = 1 - P(\mathcal{S}_n \setminus D_{n,0}) = 1 - P\left(\cup_{i=1}^n A_{n,i}\right).$$

Now, from the Inclusion-Exclusion principle (Lemma 2.1.3), we obtain

$$\begin{aligned} P\left(\cup_{i=1}^n A_{n,i}\right) &= \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P\left(\cap_{j=1}^k A_{n,i_j}\right) \\ &= \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P(A_{n,i_1, \dots, i_k}) \\ &= \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \frac{(n-k)!}{n!} \\ &= \sum_{k=1}^n (-1)^{k+1} \frac{(n-k)!}{n!} \binom{n}{k} = - \sum_{k=1}^n (-1)^k \frac{1}{k!} \end{aligned}$$

where we used the third point and the fact that $\sum_{1 \leq i_1 < \dots < i_k \leq n} 1 = \binom{n}{k}$ (the sum counts the number of ways to pick k elements in $\{1, \dots, n\}$). Plugging this in the previous display, we obtain

$$P_n(D_{n,0}) = 1 + \sum_{k=1}^n (-1)^k \frac{1}{k!}.$$

Notice that this is the Taylor series of the exponential function truncated at rank n and evaluated at -1 . By convergence of the Taylor series, we obtain the wanted limit.

Solution to Exercise 2.9. Suppose that $X : \Omega \rightarrow \mathbb{R}$ is a random variable such that $\text{Image}(X) = \{2, 5, 7\}$. Let $p_2 = P(X = 2)$, $p_5 = P(X = 5)$, $p_7 = P(X = 7)$.

1. We can take the set $\mathcal{D}_X = \{2, 5, 7\}$ as $P(X \in \text{Image}(X)) = 1$.
2. We have

$$p_2 + p_5 + p_7 = P(X = 2) + P(X = 5) + P(X = 7) = P(X \in \{2, 5, 7\}) = 1.$$

3. We have that $P_X(\emptyset) = P(X \in \emptyset) = 0$ and $P_X(\mathbb{R}) = P(X \in \mathbb{R}) = 1$. We only have to check additivity. Let $A_1, A_2, \dots \subset \mathbb{R}$ be disjoint. Then, the events $\{X \in A_1\}, \{X \in A_2\}, \dots$ are disjoint events (as $X : \Omega \rightarrow \mathbb{R}$ is a function). Thus, as P is a probability measure,

$$\begin{aligned} P_X\left(\cup_{i \geq 1} A_i\right) &= P\left(X \in \cup_{i \geq 1} A_i\right) = P\left(\cup_{i \geq 1} \{X \in A_i\}\right) \\ &= \sum_{i \geq 1} P(X \in A_i) = \sum_{i \geq 1} P_X(A_i) \end{aligned}$$

where we used that an element belong to the union of disjoint sets if and only if it belongs to exactly one of them.

4. $P(X \in [0, 6]) = P(X \in [0, 6] \cap \{2, 5, 7\})$ as $P(X \in \{2, 5, 7\}) = 1$. So,

$$P(X \in [0, 6]) = P(X \in \{2, 5\}) = p_2 + p_5.$$

5. $P(X \leq 3) = P(X \in (-\infty, 3]) = P(X \in (-\infty, 3] \cap \{2, 5, 7\})$ as in the previous point, so $P(X \leq 3) = p_2$.

Solution to Exercise 2.10. From the definitions, we have that X takes only the values $\{0, 1, 3, 5\}$, and we have

$$P(X = 0) = P(X^{-1}(\{0\})) = P(\{2, 4, 6\}) = \frac{3}{6} = \frac{1}{2},$$

and, for $i = 1, 3, 5$,

$$P(X = i) = P(X^{-1}(\{i\})) = P(\{i\}) = \frac{1}{6}.$$

In the same fashion, Y only takes the values $\{0, 2 - 1, 4 - 1, 6 - 1\} = \{0, 1, 3, 5\}$ and we have

$$P(Y = 0) = P(Y^{-1}(\{0\})) = P(\{1, 3, 5\}) = \frac{3}{6} = \frac{1}{2}$$

and for $i = 1, 3, 5$,

$$P(Y = i) = P(Y^{-1}(\{i\})) = P(\{i + 1\}) = \frac{1}{6}.$$

We note that X and Y have the same law!

Now, let us study Z . We have

$$\begin{aligned} Z(1) &= X(1) - Y(1) = 1 - 0 = 1, & Z(2) &= X(2) - Y(2) = 0 - 1 = -1, \\ Z(3) &= X(3) - Y(1) = 3 - 0 = 3, & Z(4) &= X(4) - Y(4) = 0 - 3 = -3, \\ Z(5) &= X(5) - Y(5) = 5 - 0 = 5, & Z(6) &= X(6) - Y(6) = 0 - 5 = -5. \end{aligned}$$

Thus, the law of Z is given by

$$P(Z = -5) = P(Z = -3) = P(Z = -1) = P(Z = 1) = P(Z = 3) = P(Z = 5) = \frac{1}{6}.$$

In particular, $P(X = Y) = P(|X - Y| = 0) = P(Z = 0) = 0$.

Solution to Exercise 2.11. 1. We use the definition:

$$P_f(X \leq 4) = P_f(\{x : X(x) \leq 4\}) = P_f(\{x : x^2 \leq 4\}) = P_f([-2, 2]) = \int_{-2}^2 f(x)dx = \frac{1}{2}$$

as $x^2 \leq 4$ if and only if $x \in [-2, 2]$. We proceed similarly to obtain $P_f(X \leq 9) = \frac{3}{4}$.

2. Writing the definitions, we want to find f_X such that for any set A :

$$\begin{aligned} P_f(X \in A) &= P_f(\{x : x^2 \in A\}) = \int_{\mathbb{R}} \mathbb{1}_A(x^2) f(x) dx \\ &= \frac{1}{4} \int_0^4 \mathbb{1}_A(x^2) dx = \int_{\mathbb{R}} \mathbb{1}_A(x) f_X(x) dx. \end{aligned}$$

We can then make the change of variable $y = x^2$ to obtain

$$\frac{1}{4} \int_0^4 \mathbb{1}_A(x^2) dx = \frac{1}{4} \int_0^{16} \mathbb{1}_A(y) \frac{1}{2\sqrt{y}} dy = \frac{1}{4} \int_{\mathbb{R}} \mathbb{1}_A(y) \mathbb{1}_{[0,16]}(y) \frac{1}{2\sqrt{y}} dy.$$

We thus find that we can take $f_X(x) = \mathbb{1}_{[0,16]}(x) \frac{1}{8\sqrt{x}}$.

Solution to Exercise 2.12. 1. A number of days is an integer, in particular, the variable takes value in (a subset of) a countable set, it is therefore discrete.

2. The volume of rain is a continuous non-negative quantity, it thus takes value in the whole of $[0, +\infty)$ and is thus a continuous random variable.
3. This is again an integer, so discrete RV as in the first point.
4. This is again an integer, so discrete RV as in the first point.
5. This is a continuous random variable (you can take anywhere between 0 and the max duration of the exam), so continuous random variable.

Solution to Exercise 2.13. 1. We have that $P(X \in \{-1, 1\}) = 1$, so,

- for $t < -1$, $P(X \leq t) = 0$,
- for $-1 \leq t < 1$, $P(X \leq t) = P(X = -1) = \frac{1}{2}$,
- for $t \geq 1$, $P(X \leq t) = P(X \in \{-1, 1\}) = 1$.

The associated graph is the (b).

2. This is a constant variable, so
 - for $t < 3$, $P(X \leq t) = 0$,
 - for $t \geq 3$, $P(X \leq t) = P(X = 3) = 1$.

The associated graph is (d).

3. We can compute:

$$P(X \leq t) = \int_{-\infty}^t f_X(x) dx = \frac{1}{4} \int_{-\infty}^t \mathbb{1}_{[0,4]}(x) dx = \begin{cases} 0 & \text{if } t < 0, \\ \frac{t}{4} & \text{if } t \in [0, 4], \\ 1 & \text{if } t > 4. \end{cases}$$

The associated graph is (a).

4. We can compute:

$$\begin{aligned} P(X \leq t) &= \int_{-\infty}^t f_X(x) dx = \int_{-\infty}^t \mathbb{1}_{[0,+\infty)}(x) e^{-x} dx \\ &= \begin{cases} 0 & \text{if } t < 0, \\ \int_0^t e^{-x} dx = 1 - e^{-t} & \text{if } t \geq 0 \end{cases} \end{aligned}$$

The associated graph is (c).

Solution to Exercise 2.14. 1. For $x, y \in \text{Image}(X)$, consider the distance between them $|x - y|$. If $x \neq y$, $|x - y| > 0$. We can take

$$\epsilon = \frac{1}{3} \min\{|x - y| : x, y \in \text{Image}(X), x \neq y\} > 0$$

as it is the minimum of finitely many positive numbers. Indeed, for any $x \in \text{Image}(X)$, we have that $(x - \epsilon, x + \epsilon) \cap \text{Image}(X) = \{x\}$ as the distance between x and any other points in $\text{Image}(X)$ is at least 3ϵ . So,

$$P(X \in (x - \epsilon, x + \epsilon)) = P(X \in (x - \epsilon, x + \epsilon) \cap \text{Image}(X)) = P(X \in \{x\}) = P(X = x).$$

2. They are precisely $\text{Image}(X)$ as for any $t \notin \text{Image}(X)$, $F_X(t) = P(X \leq x(t))$ where $x(t)$ is the smallest point in $\text{Image}(X)$ which is larger than t . In particular, F_X is constant between two successive points in $\text{Image}(X)$, if is therefore continuous on $\mathbb{R} \setminus \text{Image}(X)$. On the other side, if $t \in \text{Image}(X)$, we have that for any $\delta > 0$,

$$\begin{aligned} |F_X(t) - F_X(t - \delta)| &= F_X(t) - F_X(t - \delta) = P(X \leq t) - P(X \leq t - \delta) \\ &= P(X \in (t - \delta, t]) \geq P(X = t) > 0 \end{aligned}$$

as we assume the X takes any of its values with strictly positive probability. In particular, this shows that F_X is not left continuous at t .

3. Using the same ϵ as in the first point, we have that

$$P(X = x) = P(X \in (x - \epsilon, x + \epsilon]) = F_X(x + \epsilon) - F_X(x - \epsilon).$$

4. Use the same ϵ as in the first point. For any $\epsilon' \in (0, \epsilon]$, we have (recall that by assumption, $F_Y = F_X$)

$$\begin{aligned} P(Y \in (x - \epsilon', x + \epsilon']) &= F_Y(x + \epsilon') - F_Y(x - \epsilon') = F_X(x + \epsilon') - F_X(x - \epsilon') \\ &= P(X \in (x - \epsilon', x + \epsilon']). \end{aligned}$$

Letting $\epsilon' \rightarrow 0$, we obtain that for any $x \in \text{Image}(X)$, $P(Y = x) = P(X = x)$. In particular,

$$P(Y \in \text{Image}(X)) = \sum_{x \in \text{Image}(X)} P(Y = x) = \sum_{x \in \text{Image}(X)} P(X = x) = 1.$$

So, Y is a discrete random variable (we can take $\mathcal{D}_Y = \text{Image}(X)$), and for every event $A \subset \mathbb{R}$,

$$P(Y \in A) = \sum_{x \in A \cap \text{Image}(X)} P(Y = x) = \sum_{x \in A \cap \text{Image}(X)} P(X = x) = P(X \in A).$$

Solution to Exercise 2.15. 1. We can take $\Omega = \mathbb{R}^{1000}$, each coordinate representing the temperature reached by the corresponding card. The probability measure can be taken to be a continuous probability measure

$$P(A) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \mathbb{1}_A(x_1, \dots, x_{1000}) \prod_{i=1}^{1000} f(x_i) dx_i$$

where f is some density function on $\mathbb{R}$ which is 0 on the negative numbers. The product structure reflects the fact that the results of the tests are independent, and the fact that we use the same f for different cards reflects the fact that they are all of the same model (and so their temperatures share the same law). We can then take the random variable “average number of cards reaching 100 degrees”:

$$X : \Omega \rightarrow \mathbb{R}, \quad X(x_1, \dots, x_{1000}) = \frac{1}{1000} \sum_{i=1}^{1000} \mathbb{1}_{[100, +\infty)}(x_i)$$

2. Denote by Pop the world population (elements of Pop are living individuals). For $i \in \text{Pop}$, let IQ_i denote the IQ of i . Let then $\text{IQPop} = \{(i, IQ_i) : i \in \text{Pop}\}$. We can take Ω to be the set of subsets of IQPop with size 100:

$$\Omega = \{A : A \subset \text{IQPop}, |A| = 100\}.$$

The probability measure can then be taken to be the uniform probability measure on Ω : $P(A) = |A|/|\Omega|$ for any $A \subset \Omega$. The random variables of interest are then

- the maximal IQ of an individual in our sample: for $\omega \in \Omega$,

$$\text{MaxIQ}(\omega) = \max\{IQ_i : (i, IQ_i) \in \omega\}.$$

- the minimal IQ of an individual in our sample: for $\omega \in \Omega$,

$$\text{MinIQ}(\omega) = \min\{IQ_i : (i, IQ_i) \in \omega\}.$$

Solution to Exercise 2.16. Recall that in the present case,

$$E_{P_p}(X) = \sum_{\omega \in \Omega} p(\omega) X(\omega),$$

and that the sum is finite.

1. For $a, b \in \mathbb{R}$,

$$\begin{aligned} E_{P_p}(aX + bY) &= \sum_{\omega \in \Omega} p(\omega)(aX(\omega) + bY(\omega)) \\ &= a \sum_{\omega \in \Omega} p(\omega) X(\omega) + b \sum_{\omega \in \Omega} p(\omega) Y(\omega) = aE_{P_p}(X) + bE_{P_p}(Y) \end{aligned}$$

where we used linearity of finite sums.

2. For $X \leq Y$,

$$E_{P_p}(X) = \sum_{\omega \in \Omega} p(\omega) X(\omega) \leq \sum_{\omega \in \Omega} p(\omega) Y(\omega) = E_{P_p}(Y),$$

as $a \leq a'$ and $b \leq b'$ implies $a + b \leq a' + b'$, and $p(\omega)X(\omega) \leq p(\omega)Y(\omega)$ as $p(\omega) \geq 0$ and $X(\omega) \leq Y(\omega)$ for all $\omega \in \Omega$.

Solution to Exercise 2.17. 1. This is the probability that $X \geq 10$, which is given by

$$P(X \geq 10) = \int_{-\infty}^{\infty} f_X(x) \mathbb{1}_{[10, +\infty)}(x) dx = \frac{1}{10} \int_{10}^{+\infty} e^{-x/10} dx = [-e^{-x/10}]_{10}^{+\infty} = e^{-1}.$$

2. This is given by the expected value of X :

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f_X(x) dx = \frac{1}{10} \int_0^{\infty} x e^{-x/10} dx = [x e^{-x/10}]_0^{\infty} + \int_0^{\infty} e^{-x/10} dx \\ &= [-10 e^{-x/10}]_0^{\infty} = 10, \end{aligned}$$

where we used integration by part.

Solution to Exercise 2.18. The casino has to ask for the expected gain of the player in order not to be at lose on average. The player can chose when to stop: so there is a number N (chosen by the player) such that he will stop as soon as he will have won N times (if he wins that many times at all). For a fixed N , the average gain of the player is

$$2^N G_0 2^{-N} + 0(1 - 2^{-N}) = 2$$

as there is a probability 2^{-N} that he will win N consecutive coin flips and $1 - 2^{-N}$ that he loses before that. So, independently of the value of N , the expected gain is 2, so the casino should ask to pay at least $x = 2$ to play the game.

Solution to Exercise 2.19. 1. One has that $f(x, y) \geq 0$ for all $x, y \in \mathbb{R}$. We then need to check that f integrates to 1.

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dy dx &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-(x-1.7)^2/0.08} e^{-(y-100(x-1))^2/50} dy dx \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(x-1.7)^2/0.08} \int_{-\infty}^{\infty} e^{-(y')^2/2} 5 \cdot dy' dx \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(x')^2/2} 0.2 \cdot dx' \int_{-\infty}^{\infty} e^{-(y')^2/2} 5 \cdot dy' \\ &= \frac{1}{2\pi} \left(\int_{-\infty}^{\infty} e^{-x^2/2} dx \right)^2 = 1. \end{aligned}$$

where we changed variables: $y' = (y - 100(x - 1))/5$, and $x' = (x - 1.7)/0.2$ and used the indication.

2. X_1 is the first marginal of X . So, its density is given by

$$\begin{aligned} f_{X_1}(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(x-1.7)^2/0.08} e^{-(y-100(x-1))^2/50} dy \\ &= \frac{1}{2\pi} e^{-(x-1.7)^2/0.08} \int_{-\infty}^{\infty} e^{-(y')^2/2} 5 \cdot dy' = \frac{5}{\sqrt{2\pi}} e^{-(x-1.7)^2/0.08} \end{aligned}$$

where we changed variable to $y' = (y - 100(x - 1))/5$.

3. The average height in that population is given by the expected value of X_1 which is (using the previous point):

$$\begin{aligned}\int_{-\infty}^{\infty} x f_{X_1}(x) dx &= \frac{5}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-(x-1.7)^2/0.08} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{x'}{5} + 1.7\right) e^{-(x')^2/2} dx' \\ &= 1.7 + \frac{1}{5\sqrt{2\pi}} \int_{-\infty}^{\infty} x' e^{-(x')^2/2} dx' = 1.7\end{aligned}$$

where we changed variable to $x' = (x - 1.7)/0.2$, used the indication, and used that the integral of an odd function on a symmetric interval is 0.

4. The average weight in that population is given by the expected value of X_2 which is

$$\begin{aligned}\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dy dx &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y e^{-(x-1.7)^2/0.08} e^{-(y-100(x-1))^2/50} dy dx \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(x')^2/2} \int_{-\infty}^{\infty} (5y' + 100(0.2 \cdot x' + 1.7 - 1)) e^{-(y')^2/2} dy' dx' \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-x^2/2} \int_{-\infty}^{\infty} (5y + 20x + 70) e^{-y^2/2} dy dx \\ &= 70 + \frac{20}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-x^2/2} dx + \frac{5}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y e^{-y^2/2} dy = 70,\end{aligned}$$

where we changed variable to $y' = (y - 100(x - 1))/5$ and $x' = (x - 1.7)/0.2$.

5. This is the probability that $X_1 \geq 1.8$ and that $X_2 \leq 70$. This gives the expression

$$P(X_1 \geq 1.8, X_2 \leq 70) = P(X \in [1.8, +\infty) \times (-\infty, 70]) = \int_{1.8}^{+\infty} \int_{-\infty}^{70} f(x, y) dy dx.$$

Solution to Exercise 2.20. 1. $P(X_1 \leq X_2) = P(X \in \{(1, 1), (0, 0)\}) = \frac{2}{3}$.

2. X_1, X_2 both take values in $\{0, 1\}$ with probability 1. Their laws are thus given by

$$\begin{aligned}P(X_1 = 1) &= 1 - P(X_1 = 0) = P(X = (1, 1)) = \frac{1}{3}, \\ P(X_2 = 1) &= 1 - P(X_2 = 0) = P(X \in \{(0, 1), (1, 1)\}) = \frac{2}{3}.\end{aligned}$$

3. $E(Y) = E(X_1 X_2) = P(X = (0, 0)) \cdot 0 + P(X = (0, 1)) \cdot 0 + P(X = (1, 1)) \cdot 1 = \frac{1}{3}$.

Solution to Exercise 2.21. 1. X_1 is the first marginal of X . Its density is thus given by

$$\begin{aligned}f_{X_1}(x) &= \int_{-\infty}^{+\infty} f_X(x, y) dy = \frac{15}{31} \mathbb{1}_{[0,1]}(x) \int_{-\infty}^{+\infty} \mathbb{1}_{[0,1]}(y) (x+y)^4 dy \\ &= \frac{15}{31} \mathbb{1}_{[0,1]}(x) \int_0^1 (x+y)^4 dy = \frac{15}{31} \mathbb{1}_{[0,1]}(x) \left[\frac{1}{5} (x+y)^5 \right]_0^1 = \frac{3}{31} \mathbb{1}_{[0,1]}(x) ((x+1)^5 - x^5)\end{aligned}$$

2. We have that

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{=:M} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}}_{=:M^{-1}} \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}.$$

In particular, Corollary 2.2.4 gives that the density for Y is

$$f_Y(y) = f_X(M^{-1}y) |\det(M^{-1})| = \frac{1}{2} f_X\left(\frac{y_1+y_2}{2}, \frac{y_1-y_2}{2}\right) = \frac{15}{62} \mathbb{1}_{[0,1]}\left(\frac{y_1+y_2}{2}\right) \mathbb{1}_{[0,1]}\left(\frac{y_1-y_2}{2}\right) y_1^4$$

3. We have that

$$Z = (Y_1^2, Y_2^3).$$

So, considering $\varphi : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}_+ \times \mathbb{R}$, $\varphi(x, y) = (x^2, y^3)$, we have $Z = \varphi(Y)$ and

$$\varphi^{-1}(x, y) = (\sqrt{x}, y^{1/3}), \quad D_\varphi(x, y) = \begin{pmatrix} 2x & 0 \\ 0 & 3y^2 \end{pmatrix}.$$

So, $\det D_\varphi(x, y) = 6xy^2$, and thus Corollary 2.2.4 gives that the density for Z is

$$\begin{aligned} f_Z(x, y) &= f_Y(\sqrt{x}, y^{1/3}) \frac{1}{|\det D_\varphi(\sqrt{x}, y^{1/3})|} \\ &= \frac{1}{6|\sqrt{x}y^{2/3}|} \frac{15}{62} \mathbb{1}_{[0,1]}\left(\frac{\sqrt{x}+y^{1/3}}{2}\right) \mathbb{1}_{[0,1]}\left(\frac{\sqrt{x}-y^{1/3}}{2}\right) x^2 \\ &= \frac{5x^{3/2}}{124|y|^{2/3}} \mathbb{1}_{[0,2]}(\sqrt{x} + y^{1/3}) \mathbb{1}_{[0,2]}(\sqrt{x} - y^{1/3}). \end{aligned}$$

Solution to Exercise 2.22. In each case, we apply Corollary 2.2.4 with $\varphi = g$.

1. We have $U = \mathbb{R}$, $V = \mathbb{R}$, $\varphi(x) = x^3$, $\varphi^{-1}(x) = x^{1/3}$, and thus

$$f_Y(x) = \frac{1}{3|x|^{2/3}} \frac{1}{\pi(1+x^{2/3})}.$$

2. We have $U = (0, +\infty)$, $V = (0, +\infty)$, $\varphi(x) = 25x^2$, $\varphi^{-1}(x) = \frac{1}{5}x^{1/2}$, and thus

$$f_Y(x) = \frac{1}{10|x|^{1/2}} \mathbb{1}_{[0,+\infty)}(x) e^{-\sqrt{x}/5}.$$

3. We have $U = (0, \pi/2)$, $V = (0, 1)$, $\varphi(x) = \sin(x)$, $\varphi^{-1}(x) = \arcsin(x)$, and thus

$$f_Y(x) = |\arcsin'(x)| \frac{2}{\pi} \mathbb{1}_{[0,\pi/2]}(\arcsin(x)) = \frac{1}{\sqrt{1-x^2}} \frac{2}{\pi} \mathbb{1}_{[0,1]}(x).$$

Solution to Exercise 2.23. The transfer theorem (Theorem 2.2.2) gives that the p th moment of a continuous random variable X is given by

$$E(X^p) = \int_{-\infty}^{\infty} f_X(x) x^p dx,$$

and similarly for discrete random variables.

1. We have that the first moment of X is

$$E(X) = \int_{-\infty}^{\infty} x \frac{9}{10} \mathbb{1}_{[1,10]}(x) \frac{1}{x^2} dx = \frac{9}{10} \int_1^{10} \frac{1}{x} dx = \frac{9}{10} \ln(10).$$

The second moment is

$$E(X^2) = \int_{-\infty}^{\infty} x^2 \frac{9}{10} \mathbb{1}_{[1,10]}(x) \frac{1}{x^2} dx = \frac{9}{10} \int_1^{10} dx = \frac{81}{10}.$$

And the third moment is

$$E(X^3) = \int_{-\infty}^{\infty} x^3 \frac{9}{10} \mathbb{1}_{[1,10]}(x) \frac{1}{x^2} dx = \frac{9}{10} \int_1^{10} x dx = \frac{9 \cdot 99}{20}.$$

a continuous random variable with density $f_X(x) = \frac{9}{10} \mathbb{1}_{[1,10]}(x) \frac{1}{x^2}$.

2. We have that $P(X \in \{-1, 0, 2\}) = 1$ so,

$$\begin{aligned} E(X) &= (-1) \cdot P(X = -1) + 0 \cdot P(X = 0) + 2P(X = 2) = \frac{1}{3}(-1 + 0 + 2) = \frac{1}{3}, \\ E(X^2) &= (-1)^2 \cdot P(X = -1) + 0^2 \cdot P(X = 0) + 2^2 P(X = 2) = \frac{1}{3}(1 + 0 + 4) = \frac{5}{3}, \\ E(X^3) &= (-1)^3 \cdot P(X = -1) + 0^3 \cdot P(X = 0) + 2^3 P(X = 2) = \frac{1}{3}(-1 + 0 + 8) = \frac{7}{3}. \end{aligned}$$

3. We have that $P(X \in \{-1, 1\}) = 1$ so,

$$\begin{aligned} E(X) &= 1 \cdot P(X = 1) - 1 \cdot P(X = -1) = \frac{1}{2} - \frac{1}{2} = 0, \\ E(X^2) &= 1^2 \cdot P(X = 1) + (-1)^2 \cdot P(X = -1) = \frac{1}{2} + \frac{1}{2} = 1, \\ E(X^3) &= 1^3 \cdot P(X = 1) + (-1)^3 \cdot P(X = -1) = \frac{1}{2} - \frac{1}{2} = 0. \end{aligned}$$

4. First note that as $x \frac{1}{2} e^{-|x|}$ and $x^3 \frac{1}{2} e^{-|x|}$ are odd functions, we have $E(X) = E(X^3) = 0$. Let us finally compute the second moment of X .

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{+\infty} x^2 f_X(x) dx = \frac{1}{2} \int_{-\infty}^{+\infty} x^2 e^{-|x|} dx = \int_0^{+\infty} x^2 e^{-x} dx \\ &= [-x^2 e^{-x}]_0^{\infty} + 2 \int_0^{+\infty} x e^{-x} dx = 2[-x e^{-x}]_0^{\infty} + 2 \int_0^{+\infty} e^{-x} dx = 2[-e^{-x}]_0^{\infty} = 2. \end{aligned}$$

Solution to Exercise 2.24. Let $C = E(e^{\delta|X|}) < \infty$. We have that for any $x \geq 0$, and any $p \in \mathbb{N}$,

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \geq \frac{x^p}{p!}.$$

So, for $p \in \mathbb{N}$, $e^{\delta|X|} \geq \frac{\delta^p |X|^p}{p!}$. In particular,

$$C = E(e^{\delta|X|}) \geq E\left(\frac{\delta^p |X|^p}{p!}\right) = \frac{\delta^p}{p!} E(|X|^p).$$

So,

$$E(|X|^p) \leq C p! \delta^{-p} < \infty.$$

Solution to Exercise 2.25. 1. Let $B \subset A \subset \Omega$. We have that the probability of B under $P(\cdot | A)$ is (using the definition):

$$P(B | A) = \frac{P(B \cap A)}{P(A)} = \frac{|B \cap A| \cdot |\Omega|}{|\Omega| \cdot |A|} = \frac{|B|}{|A|}$$

as we have $B \subset A$, so $B \cap A = B$.

2. Our realisation set is $\Omega = \{F_1, F_2, \dots, F_6\}$ (F_i stands for “face with i points”). We then compute the probability of each realisation in Ω under $P(\cdot | \text{even})$: recalling that $\text{even} = \{F_2, F_4, F_6\}$,

$$P(\{F_i\} | \text{even}) = \frac{P(\{F_i\} \cap \text{even})}{P(\text{even})} = \begin{cases} \frac{1/6}{1/2} & \text{if } i \text{ is even,} \\ 0 & \text{if } i \text{ is odd.} \end{cases}$$

So, the probability mass function

$$p(F_i) = \begin{cases} \frac{1}{3} & \text{if } i \text{ is even,} \\ 0 & \text{if } i \text{ is odd,} \end{cases}$$

is the probability mass function associated to the probability measure $P(\cdot | \text{even})$.

Solution to Exercise 2.26. Denote T the time that Nicole waits, H_A the event that she goes to hairdresser A, H_B the event that she goes to hairdresser B, and D the result of the dice. From the pieces of information we have, we get

$$P(H_A) = P(D \in \{5, 6\}) = \frac{2}{6}, \quad P(H_B) = P(D \in \{1, 2, 3, 4\}) = \frac{4}{6},$$

$$E(T | H_A) = \int_0^{30} \frac{1}{30} x dx = 15, \quad E(T | H_B) = \int_0^{20} \frac{1}{20} x dx = 10.$$

1. We want to compute $P(T \geq 25 | D = 5)$. Conditionally on H_A , T is a uniform random variable on $[0, 30]$, so using that $\{D = 5\} \subset H_A$,

$$P(T \geq 25 | D = 5) = P(T \geq 25 | D = 5, H_A) = \int_0^{30} \frac{1}{30} \mathbb{1}_{[25, +\infty)}(x) dx = \frac{1}{30} \int_{25}^{30} dx = \frac{1}{6}.$$

2. As $H_A \cap H_B = \emptyset$ (Nicole goes to at most one of the two hairdresser), we have $\mathbb{1}_{H_A} + \mathbb{1}_{H_B} \leq 1$. On the other hand, she goes to at least one the two so $H_A \cup H_B = \Omega$, and $\mathbb{1}_{H_A} + \mathbb{1}_{H_B} \geq 1$. So, $1 = \mathbb{1}_{H_A} + \mathbb{1}_{H_B}$, and thus

$$\begin{aligned} P(T \geq 15) &= E(\mathbb{1}_{\{T \geq 15\}}(\mathbb{1}_{H_A} + \mathbb{1}_{H_B})) \\ &= P(T \geq 15, H_A) + P(T \geq 15, H_B) \\ &= P(T \geq 15 | H_A)P(H_A) + P(T \geq 15 | H_B)P(H_B) \\ &= \frac{2}{6} \int_0^{30} \frac{1}{30} \mathbb{1}_{[15, +\infty)}(x) dx + \frac{4}{6} \int_0^{20} \frac{1}{20} \mathbb{1}_{[15, +\infty)}(x) dx \\ &= \frac{1}{3} \frac{15}{30} + \frac{2}{3} \frac{5}{20} = \frac{1}{3}. \end{aligned}$$

3. We use the definition of conditional probability twice:

$$\begin{aligned}
 P(D = 4 | T \geq 15) &= \frac{P(T \geq 15, D = 4)}{P(T \geq 15)} \\
 &= \frac{P(T \geq 15 | D = 4)P(D = 4)}{P(T \geq 15)} \\
 &= \frac{1/6}{1/3} P(T \geq 15 | D = 4, H_B) = \frac{1}{2} \int_0^{20} \frac{1}{20} \mathbb{1}_{[15, +\infty)}(x) dx = \frac{1}{8}.
 \end{aligned}$$

Solution to Exercise 2.27. The second reasoning is incorrect. The problem lies in the modelisation step: in the first case, we define everything from the start by fixing the total amount of money in the two envelopes. The second reasoning assumes that the amount is random (which can sound reasonable as we do not know how much money there is in total). Letting A be the amount in the envelope you first pick, the second reasoning gives the money distributions $(A, 2A)$ and $(A, A/2)$ to be equally likely. But we do not know which envelop we picked, so if we assume that these two situations are equally likely, we picked the envelop containing $2A$ with probability $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$ (proba $\frac{1}{2}$ of picking situation $(A, 2A)$ times proba $\frac{1}{2}$ of picking the envelop containing A)... which is a contradiction with the fact that A was the amount of money in the envelop you picked from the start, so that you should have a probability 1 of picking an envelop containing A .

The only value of A not leading to a contradiction is $A = 0$, which is the only value of the total money for which the two reasoning give the same solution!

Solution to Exercise 2.28. 1. Let $Y_1, \dots, Y_n$ be the random variables corresponding to the status of each components: $Y_i = 1$ if component i is working, and $Y_i = 0$ else. As the components are all independent from the others, for any $\epsilon_1, \dots, \epsilon_n \in \{0, 1\}$,

$$P(Y_1 = \epsilon_1, \dots, Y_n = \epsilon_n) = \prod_{i=1}^n P(Y_i = \epsilon_i) = \prod_{i=1}^n p^{\epsilon_i} (1-p)^{1-\epsilon_i}$$

In particular, for any sequence $\epsilon_1, \dots, \epsilon_n \in \{0, 1\}$ containing exactly k 1's,

$$P(Y_1 = \epsilon_1, \dots, Y_n = \epsilon_n) = p^k (1-p)^{n-k}.$$

Then, we have that the number of working component is given by $X = \sum_{i=1}^n Y_i \in \{0, 1, \dots, n\}$. The probability of $X = k$ is then given by the probability that exactly k of the Y_i 's are 1. There are $\binom{n}{k}$ possibilities for choosing which of the Y_i 's give 1, and by the previous computation, the probability of each choice is $p^k (1-p)^{n-k}$, so

$$P(X = k) = \mathbb{1}_{\{0, \dots, n\}}(k) \binom{n}{k} p^k (1-p)^{n-k}.$$

This is a *Binomial distribution*.

2. The probability of working for the 3 components system is

$$\binom{3}{2} p^2 (1-p) + \binom{3}{3} p^3 = 3p^2 (1-p) + p^3.$$

The probability of working for the 5 components system is

$$\binom{5}{3}p^3(1-p)^2 + \binom{5}{4}p^4(1-p) + \binom{5}{5}p^5 = 10p^3(1-p)^2 + 5p^4(1-p) + p^5.$$

We thus want to find for which values of p we have

$$\begin{aligned} 10p^3(1-p)^2 + 5p^4(1-p) + p^5 &> 3p^2(1-p) + p^3 \\ \iff (1-p)(10p(1-p) + 5p^2 - 3) + p^3 - p &> 0 \\ \iff (1-p)(10p(1-p) + 5p^2 - 3) + p(p^2 - 1) &> 0 \\ \iff (1-p)(10p(1-p) + 5p^2 - 3) - p(1-p)(p+1) &> 0 \\ \iff 10p(1-p) + 5p^2 - 3 - p(p+1) &> 0 \\ \iff 9p - 6p^2 - 3 &> 0 \\ \iff 3p - 2p^2 - 1 &> 0. \end{aligned}$$

As $3p - 2p^2 - 1 = (2p - 1)(1 - p)$, the last condition is equivalent to $p > 1/2$.

Solution to Exercise 2.29. Denote W the time Bob waits and T the time duration of the meeting.

1. f has to be a density function, so $f(x)$ should be non-negative, so $c \geq 0$. Then, f must integrate to 1, which implies

$$c^{-1} = \int_{-\infty}^{+\infty} \mathbb{1}_{[0,+\infty)}(x)e^{-2x}dx = \int_0^{+\infty} e^{-2x}dx = \left[-\frac{1}{2}e^{-2x}\right]_0^{\infty} = \frac{1}{2}$$

so $c = 2$.

2. The meeting duration is uniform over $[5, 15]$, so a density for the meeting duration T is

$$f_T(x) = \frac{1}{10}\mathbb{1}_{[5,15]}(x)$$

as the length of $[5, 15]$ is 10.

3. It is given by the expectation of $W + T$,

$$\begin{aligned} E(W + T) &= E(W) + E(T) = \int_{-\infty}^{+\infty} xf(x)dx + \int_{-\infty}^{+\infty} xf_T(x)dx \\ &= 2 \int_0^{+\infty} xe^{-2x}dx + \frac{1}{10} \int_5^{15} xdx = \frac{1}{2} + 10 = 10.5. \end{aligned}$$

4. To survive, Bob must leave the dentist office before 10:20, which is equivalent to say that the waiting time + meeting time should be less than 20. We thus want

to compute $P(W + T < 20)$.

$$\begin{aligned}
P(W + T < 20) &= \int_{-\infty}^{+\infty} dw \int_{-\infty}^{+\infty} dt f(w) f_T(t) \mathbb{1}_{[0,20)}(w+t) \\
&= \frac{2}{10} \int_0^{+\infty} dw \int_5^{15} dt e^{-2w} \mathbb{1}_{[0,20)}(w+t) \\
&= \frac{2}{10} \int_5^{15} dt \int_0^{+\infty} dw e^{-2w} \mathbb{1}_{[-t,20-t)}(w) \\
&= \frac{2}{10} \int_5^{15} dt \int_0^{20-t} dw e^{-2w} \\
&= \frac{1}{10} \int_5^{15} dt (1 - e^{-2(20-t)}) \\
&= 1 - \frac{e^{-40}}{10} \int_5^{15} dt e^{2t} \\
&= 1 - \frac{e^{-40}}{10} \frac{1}{2} (e^{30} - e^{10}) = 1 - \frac{1}{20} (e^{-10} - e^{-30}) \approx 0.99995.
\end{aligned}$$

Solution to Exercise 2.30. 1. We have $P(A) = P(B) = P(C) = \frac{2}{4} = \frac{1}{2}$. Then,

$$\begin{aligned}
P(A \cap B) &= P(\{1\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A)P(B), \\
P(A \cap C) &= P(\{2\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A)P(C), \\
P(B \cap C) &= P(\{3\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(B)P(C).
\end{aligned}$$

2. We have

$$P(A \cap B \cap C) = P(\emptyset) = 0 \neq \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = P(A)P(B)P(C),$$

so A, B, C are not forming an independent family.

3. We can take the random variables

$$X_1 = \mathbb{1}_A, \quad X_2 = \mathbb{1}_B, \quad X_3 = \mathbb{1}_C.$$

Solution to Exercise 2.31. 1. As X, Y are independent, $E(f(X)g(Y)) = E(f(X))E(g(Y))$ for any functions f, g . Applying this with $f(x) = g(x) = e^{tx}$, we get

$$M_{X+Y}(t) = E(e^{(X+Y)t}) = E(e^{Xt}e^{Yt}) = E(e^{Xt})E(e^{Yt}) = M_X(t)M_Y(t).$$

2. Same reasoning as in the previous point using the “expectation of product of functions” definition of being an independent family.

Solution to Exercise 2.32. 1. As X, Y are independent $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for any $A, B \subset \mathbb{R}$. In particular,

$$P(\max(X, Y) \leq t) = P(X \leq t, Y \leq t) = P(X \leq t)P(Y \leq t) = F_X(t)F_Y(t).$$

2. We use the same reasoning as in the previous point: as $X_1, \dots, X_n$ form an independent family, $P(X_1 \in A_1, \dots, X_n \in A_n) = P(X_1 \in A_1) \dots P(X_n \in A_n)$ for any $A_1, \dots, A_n \subset \mathbb{R}$. In particular,

$$\begin{aligned} P\left(\max_{i=1, \dots, n} X_i \leq t\right) &= P(X_1 \leq t, X_2 \leq t, \dots, X_n \leq t) \\ &= P(X_1 \leq t)P(X_2 \leq t) \dots P(X_n \leq t) = \prod_{i=1}^n F_{X_i}(t). \end{aligned}$$

Solution to Exercise 2.33. Let us first compute the expected treatment time of team $i \in \{1, 2, 3\}$. It is given by

$$E(T_i) = i \int_0^\infty x e^{-ix} dx = \frac{1}{i}.$$

Let U be a uniform random variable on $\{1, 2, 3\}$ independent of T_1, T_2, T_3 , which represent the team assigned to the request. We then have that $X = T_U$.

1. Using the formula of total probability (total expectation in this case), we get

$$E(X) = E(T_U) = \sum_{i=1}^3 E(T_i | U = i) P(U = i).$$

Then, we know that $P(U = i) = 1/3$ for $i = 1, 2, 3$, and that T_i is independent of U , so

$$E(T_i | U = i) = \frac{E(T_i \mathbf{1}_{\{i\}}(U))}{P(U = i)} = \frac{E(T_i) E(\mathbf{1}_{\{i\}}(U))}{P(U = i)} = \frac{E(T_i) P(U = i)}{P(U = i)} = E(T_i) = \frac{1}{i}.$$

Putting things together,

$$E(X) = \sum_{i=1}^3 \frac{1}{i} \cdot \frac{1}{3} = \frac{11}{18}.$$

2. Using the formula of total probability, we get

$$P(X \leq 1) = P(T_U \leq 1) = \sum_{i=1}^3 P(T_i \leq 1 | U = i) P(U = i).$$

As in the previous point, T_i being independent from U gives that

$$P(T_i \leq 1 | U = i) = P(T_i \leq 1) = i \int_0^1 e^{-ix} dx = [-e^{-ix}]_0^1 = 1 - e^{-i}.$$

So, putting things together,

$$P(X \leq 1) = \sum_{i=1}^3 (1 - e^{-i}) \cdot \frac{1}{3} = 1 - \frac{1}{3}(e^{-1} + e^{-2} + e^{-3}) \approx 0.8157.$$

3. We know that $X > 1$ and we want to know the probability that $U = i$ for $i \in \{1, 2, 3\}$. So, we want to compute $P(U = i | X > 1)$. Using Bayes formula,

$$\begin{aligned} P(U = i | X > 1) &= \frac{P(X > 1 | U = i)P(U = i)}{P(X > 1)} \\ &= \frac{P(T_i > 1 | U = i)^{\frac{1}{3}}}{1 - P(X \leq 1)} = \frac{P(T_i > 1)}{e^{-1} + e^{-2} + e^{-3}} \end{aligned}$$

where we used the previous point, that under the event $U = i$, $X = T_U = T_i$, and that T_i and U are independent. Now, using the computation of the previous point,

$$P(T_i > 1) = 1 - P(T_i \leq 1) = 1 - 1 + e^{-i} = e^{-i}.$$

So,

$$P(U = i | X > 1) = \frac{e^{-i}}{e^{-1} + e^{-2} + e^{-3}}.$$

Solution to Exercise 2.34. Note that W is neither discrete nor continuous. Nevertheless, using the formula of total probability (total expectation in this case),

$$\begin{aligned} E(W) &= E(ZX + (1 - Z)Y) = P(Z = 1)E(X | Z = 1) + P(Z = 0)E(Y | Z = 0) \\ &= \frac{1}{2}E(X) + \frac{1}{2}E(Y) = \frac{1}{2}\left(\frac{1}{4} \cdot (-1) + \frac{1}{4} \cdot 3 + \frac{1}{2} \cdot 0 + \int_0^{+\infty} \frac{3x}{(x+1)^4} dx\right) \\ &= \frac{1}{2}\left(\frac{1}{2} + \int_0^{+\infty} \frac{1}{(x+1)^3} dx\right) = \frac{1}{2}, \end{aligned}$$

as Z is independent from X, Y .

Solution to Exercise 2.35. Denote Bag the bag that you pick. Denote $M_1, M_2, \dots$ the colours of the marble that you pick first, second,...

1. We know that $M_1 = \text{blue}$, we thus want to compute $P(\text{Bag} = 1 | M_1 = \text{blue})$. We first compute

$$P(M_1 = \text{blue} | \text{Bag} = 1) = \frac{10}{30}, \quad P(M_1 = \text{blue} | \text{Bag} = 2) = \frac{19}{30}.$$

So, using Bayes' law, and the formula of total probability,

$$\begin{aligned} P(\text{Bag} = 1 | M_1 = \text{blue}) &= \frac{P(M_1 = \text{blue} | \text{Bag} = 1)P(\text{Bag} = 1)}{P(M_1 = \text{blue})} \\ &= \frac{P(M_1 = \text{blue} | \text{Bag} = 1)P(\text{Bag} = 1)}{P(M_1 = \text{blue} | \text{Bag} = 1)P(\text{Bag} = 1) + P(M_1 = \text{blue} | \text{Bag} = 2)P(\text{Bag} = 2)} \\ &= \frac{\frac{10}{30}}{\frac{10}{30} + \frac{19}{30}} = \frac{10}{29} \end{aligned}$$

as $P(\text{Bag} = 1) = P(\text{Bag} = 2) = 1/2$.

2. We use the formula of total probability:

$$P(M_2 = \text{red} \mid M_1 = \text{blue}) = P(M_2 = \text{red} \mid M_1 = \text{blue}, \text{Bag} = 1)P(\text{Bag} = 1 \mid M_1 = \text{blue}) \\ + P(M_2 = \text{red} \mid M_1 = \text{blue}, \text{Bag} = 2)P(\text{Bag} = 2 \mid M_1 = \text{blue}).$$

Now, using the first point,

$$P(\text{Bag} = 1 \mid M_1 = \text{blue}) = \frac{10}{29}, \\ P(\text{Bag} = 2 \mid M_1 = \text{blue}) = 1 - P(\text{Bag} = 1 \mid M_1 = \text{blue}) = \frac{19}{29}.$$

Moreover, counting the balls of each colour in each bags, we have

$$P(M_2 = \text{red} \mid M_1 = \text{blue}, \text{Bag} = 1) = \frac{10}{29}, \\ P(M_2 = \text{red} \mid M_1 = \text{blue}, \text{Bag} = 2) = \frac{3}{29}.$$

So,

$$P(M_2 = \text{red} \mid M_1 = \text{blue}) = \frac{10}{29} \cdot \frac{10}{29} + \frac{3}{29} \cdot \frac{19}{29} = \frac{157}{841}.$$

3. We now know $M_1 = M_2 = \text{blue}$. We proceed as in the first point: we first compute

$$P(M_1 = M_2 = \text{blue} \mid \text{Bag} = 1) = \frac{10 \cdot 9}{30 \cdot 29}, \quad P(M_1 = M_2 = \text{blue} \mid \text{Bag} = 2) = \frac{19 \cdot 18}{30 \cdot 29}.$$

Then, using Bayes' law, the formula of total probability, and $P(\text{Bag} = 1) = P(\text{Bag} = 2) = 1/2$,

$$P(\text{Bag} = 1 \mid M_1 = M_2 = \text{blue}) \\ = \frac{P(M_1 = M_2 = \text{blue} \mid \text{Bag} = 1)}{P(M_1 = M_2 = \text{blue} \mid \text{Bag} = 1) + P(M_1 = M_2 = \text{blue} \mid \text{Bag} = 2)} \\ = \frac{\frac{10 \cdot 9}{30 \cdot 29}}{\frac{10 \cdot 9}{30 \cdot 29} + \frac{19 \cdot 18}{30 \cdot 29}} = \frac{45}{216}$$

4. We proceed as in the second point: using the third point,

$$P(\text{Bag} = 1 \mid M_1 = M_2 = \text{blue}) = \frac{45}{216}, \\ P(\text{Bag} = 2 \mid M_1 = M_2 = \text{blue}) = 1 - \frac{45}{216} = \frac{171}{216}.$$

Moreover, counting the balls of each colour in each bags, we have

$$P(M_3 = \text{red} \mid M_1 = M_2 = \text{blue}, \text{Bag} = 1) = \frac{10}{28}, \\ P(M_3 = \text{red} \mid M_1 = M_2 = \text{blue}, \text{Bag} = 2) = \frac{3}{28}.$$

So, by the formula of total probability (as in the second point)

$$P(M_3 = \text{red} \mid M_1 = M_2 = \text{blue}) = \frac{10}{28} \cdot \frac{45}{216} + \frac{3}{28} \cdot \frac{171}{216} = \frac{963}{28 \cdot 216} = \frac{107}{672} \approx 0.16.$$

5. This probability is the probability that you picked bag 2 given what you already saw: that is $M_1 = M_2 = \text{blue}$. From the computations of the previous point, this probability is equal to

$$P(\text{Bag} = 2 \mid M_1 = M_2 = \text{blue}) = \frac{171}{216} \approx 0.79.$$

6. This probability is the probability that you picked bag 2 given that $M_1 = M_2 = \text{blue}$ and that $M_3 = \text{red}$. This is given by (again using Bayes' law)

$$\begin{aligned} & P(\text{Bag} = 2 \mid M_1 = M_2 = \text{blue}, M_3 = \text{red}) \\ &= \frac{P(M_3 = \text{red} \mid M_1 = M_2 = \text{blue}, \text{Bag} = 2)P(\text{Bag} = 2 \mid M_1 = M_2 = \text{blue})}{P(M_3 = \text{red} \mid M_1 = M_2 = \text{blue})} \\ &= \frac{\frac{3}{28} \frac{171}{216}}{\frac{107}{672}} = \frac{57}{107} \approx 0.53, \end{aligned}$$

where we used the quantities computed at the fourth point.

Solution to Exercise 2.36. 1., 3., 4.

The justification for 4. goes as follows. Every element of $\mathbb{Q}$ can be written as p/q with $p \in \mathbb{Z}$ and $q \in \mathbb{N}^*$. The event $\{X \in \mathbb{Q}\}$ can thus be written as

$$\{X \in \mathbb{Q}\} = \{\exists (p, q) \in \mathbb{Z} \times \mathbb{N}^* : X = \frac{p}{q}\} = \cup_{(p, q) \in \mathbb{Z} \times \mathbb{N}^*} \{X = \frac{p}{q}\}.$$

Now, by countable σ -sub-additivity (recall that $\mathbb{Z} \times \mathbb{N}^*$ is countable) of probability measures,

$$P(X \in \mathbb{Q}) = P(\cup_{(p, q) \in \mathbb{Z} \times \mathbb{N}^*} \{X = \frac{p}{q}\}) \leq \sum_{(p, q) \in \mathbb{Z} \times \mathbb{N}^*} P(X = \frac{p}{q}) = \sum_{(p, q) \in \mathbb{Z} \times \mathbb{N}^*} 0 = 0.$$

So, $P(X \notin \mathbb{Q}) = 1 - P(X \in \mathbb{Q}) = 1$.

Solution to Exercise 2.37. 1. We have that for any $x, y \in \{0, 1\}$,

$$2xy - x - y + 1 = \begin{cases} 1 & \text{if } x = y, \\ 0 & \text{else.} \end{cases}$$

So, for any $a, b \in \{0, 1\}$,

$$\begin{aligned} & P(X = a, Y = b) - P(X = a)P(Y = b) \\ &= E((2aX - X - a + 1)(2bY - Y - b + 1)) \\ &\quad - E(2aX - X - a + 1)E(2bY - Y - b + 1) \\ &= E(XY)(2a - 1)(2b - 1) + E(X)(2a - 1)(b - 1) + (1 - a)E(Y)(2b - 1) \\ &\quad + (1 - a)(1 - b) - E(X)E(Y)(2a - 1)(2b - 1) - E(X)(2a - 1)(b - 1) \\ &\quad - (1 - a)E(Y)(2b - 1) - (1 - a)(1 - b) \\ &= (2a - 1)(2b - 1)\text{Cov}(X, Y) = 0 \end{aligned}$$

where we used that $P(X \in \{0, 1\}) = P(Y \in \{0, 1\}) = 1$.

2. As $P(X \in \{0, 1\}) = 1 = P(Y \in \{0, 1\})$, we have that (X, Y) is a discrete random vector with $P((X, Y) \in \{0, 1\}^2) = 1$. So, for any $A, B \subset \mathbb{R}$

$$\begin{aligned} & P(X \in A, Y \in B) \\ &= P(X = 0, Y = 0)\mathbb{1}_{A \times B}(0, 0) + P(X = 1, Y = 0)\mathbb{1}_{A \times B}(1, 0) \\ &\quad + P(X = 0, Y = 1)\mathbb{1}_{A \times B}(0, 1) + P(X = 1, Y = 1)\mathbb{1}_{A \times B}(1, 1) \\ &= P(X = 0)P(Y = 0)\mathbb{1}_A(0)\mathbb{1}_B(0) + P(X = 1)P(Y = 0)\mathbb{1}_A(1)\mathbb{1}_B(0) \\ &\quad + P(X = 0)P(Y = 1)\mathbb{1}_A(0)\mathbb{1}_B(1) + P(X = 1)P(Y = 1)\mathbb{1}_A(1)\mathbb{1}_B(1) \\ &= P(X \in A)P(Y \in B). \end{aligned}$$

by the first point.

Solution to Exercise 2.38. We detail the calculations of the first point, the others follow the exact same scheme: expand, use linearity, re-combine. First, expanding the square and using linearity of the expectation,

$$\begin{aligned}\text{Var}(X) &= E((X - E(X))^2) = E(X^2 - 2XE(X) + E(X)^2) \\ &= E(X^2) - 2E(X)^2 + E(X)^2 = E(X^2) - E(X)^2.\end{aligned}$$

Then, using the definition,

$$\text{Cov}(X, X) = E((X_E(X))(X - E(X))) = E((X_E(X))^2) = \text{Var}(X).$$

Finally, as X, X' are independent,

$$\begin{aligned}E((X - X')^2) &= E(X^2 - 2XX' + (X')^2) = E(X^2) - 2E(XX') + E((X')^2) \\ &= 2E(X^2) - 2E(X)E(X') = 2(E(X^2) - E(X)^2) = 2\text{Var}(X),\end{aligned}$$

as $X \stackrel{\text{Law}}{=} X'$ implies that $E(X) = E(X')$, and $E(X^2) = E((X')^2)$.

Solution to Exercise 2.39. 1. We compute

$$\begin{aligned}E(X) &= 2 \cdot P((X, Y) = (2, 1)) + 1 \cdot P((X, Y) = (1, 0)) + 0 \cdot P((X, Y) = (0, 0)) = 1 \\ E(Y) &= 1 \cdot P((X, Y) = (2, 1)) + 0 \cdot P((X, Y) = (1, 0)) + 0 \cdot P((X, Y) = (0, 0)) = \frac{1}{4}, \\ E(XY) &= 2 \cdot 1 \cdot P((X, Y) = (2, 1)) + 1 \cdot 0 \cdot P((X, Y) = (1, 0)) \\ &\quad + 0 \cdot 0 \cdot P((X, Y) = (0, 0)) = \frac{1}{2}.\end{aligned}$$

Thus,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{2} - 1 \cdot \frac{1}{4} = \frac{1}{4}.$$

2. Proceed as in the previous point:

$$\begin{aligned}E(X) &= 0 \cdot \frac{1}{15} + 2 \cdot \frac{7}{15} + 5 \cdot \frac{4}{15} + 1 \cdot \frac{2}{15} + 0 \cdot \frac{1}{15} = \frac{12}{5}, \\ E(Y) &= 1 \cdot \frac{1}{15} + 3 \cdot \frac{7}{15} + 5 \cdot \frac{4}{15} + 0 \cdot \frac{2}{15} + 0 \cdot \frac{1}{15} = \frac{14}{5}, \\ E(XY) &= 0 \cdot \frac{1}{15} + 6 \cdot \frac{7}{15} + 25 \cdot \frac{4}{15} + 0 \cdot \frac{2}{15} + 0 \cdot \frac{1}{15} = \frac{142}{15}.\end{aligned}$$

Thus,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{142}{15} - \frac{12}{5} \cdot \frac{14}{5} = \frac{206}{75}.$$

3. We note that $f_{(X,Y)}$ is invariant under the exchange of x, y , for $(X, Y) \stackrel{\text{Law}}{=} (Y, X)$ and thus $X \stackrel{\text{Law}}{=} Y$, so $E(X) = E(Y)$. We then compute

$$\begin{aligned}E(X) = E(Y) &= 96 \int_1^\infty dx \int_1^\infty dy \frac{x}{(x+y)^5} \\ &= \frac{96}{4} \int_1^\infty dx \frac{x}{(x+1)^4} \\ &= \frac{96}{4} \left(\left[\frac{-x}{3(x+1)^3} \right]_1^\infty + \frac{1}{3} \int_1^\infty dx \frac{1}{(x+1)^3} \right) \\ &= \frac{96}{4} \left(\frac{1}{24} + \frac{1}{24} \right) = 2.\end{aligned}$$

Also,

$$E(XY) = 96 \int_1^\infty dx \int_1^\infty dy \frac{xy}{(x+y)^5} = 5$$

by successive integration by parts (or asking WolframAlpha). Thus,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 5 - 2 \cdot 2 = 1.$$

Solution to Exercise 2.40. Let us formalize the situation. Given the numbers, the picked processor model is A with probability $\frac{135}{410}$, B with probability $\frac{208}{410}$, and C with probability $\frac{67}{410}$. In particular, writing X for the price of the picked processor and Y for its expected lifetime, we have that (X, Y) is a discrete random vector with

$$\begin{aligned} P((X, Y) = (537, 5.6)) &= \frac{135}{410}, & P((X, Y) = (425, 6)) &= \frac{208}{410}, \\ P((X, Y) = (368, 3.7)) &= \frac{67}{410}. \end{aligned}$$

In particular, we have:

$$\begin{aligned} E(X) &= 537 \cdot \frac{135}{410} + 425 \cdot \frac{208}{410} + 368 \cdot \frac{67}{410} = \frac{185551}{410}, \\ E(Y) &= 5.6 \cdot \frac{135}{410} + 6 \cdot \frac{208}{410} + 3.7 \cdot \frac{67}{410} = \frac{22519}{4100}, \\ E(XY) &= 537 \cdot 5.6 \cdot \frac{135}{410} + 425 \cdot 6 \cdot \frac{208}{410} + 368 \cdot 3.7 \cdot \frac{67}{410} = \frac{10275992}{4100}. \end{aligned}$$

So,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{10275992}{4100} - \frac{185551}{410} \cdot \frac{22519}{4100} \approx 20.66.$$

Solution to Exercise 2.41. (A) with (c), (B) with (a), and (C) with (b).

Solution to Exercise 2.42. In each cases, we need to compute the covariance between X and Y , as well as the variance of X and of Y .

1. We have

$$\begin{aligned} \text{Cov}(X, Y) &= \text{Cov}(X, aX + b) = a\text{Var}(X), \\ \text{Var}(Y) &= \text{Var}(aX + b) = a^2\text{Var}(X). \end{aligned}$$

Thus,

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{a\text{Var}(X)}{\sqrt{a^2\text{Var}(X)^2}} = \frac{a}{|a|} \in \{-1, 1\}.$$

2. We have

$$\begin{aligned} E(X) &= \frac{1}{2}, & E(Y) &= P(X \leq 1/2) = \frac{1}{2}, & E(XY) &= \int_0^{1/2} x dx = \frac{1}{8}, \\ \text{Cov}(X, Y) &= \frac{1}{8} - \frac{1}{2} \cdot \frac{1}{2} = -\frac{1}{8}, \\ \text{Var}(X) &= E(X^2) - E(X)^2 = \int_0^1 x^2 dx - \frac{1}{4} = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}, \\ \text{Var}(Y) &= E(Y^2) - E(Y)^2 = E(Y) - E(Y)^2 = \frac{1}{4}. \end{aligned}$$

as $Y^2 = \mathbb{1}_{[0,1/2]}(X) \cdot \mathbb{1}_{[0,1/2]}(X) = Y$. Thus,

$$\rho_{XY} = \frac{-\frac{1}{8}}{\sqrt{\frac{1}{12} \cdot \frac{1}{4}}} = -\sqrt{\frac{48}{64}} = -\frac{\sqrt{3}}{2}.$$

3. We have

$$E(U) = E(V) = \frac{1}{2}, \quad E(U^2) = E(V^2) = \int_0^1 x^2 dx = \frac{1}{3},$$

$$\begin{aligned} \text{Cov}(X, Y) &= \text{Cov}(U + V, UV) = \text{Cov}(U, UV) + \text{Cov}(V, UV) \\ &= 2E(U^2V) - 2E(U)E(UV) = 2E(U^2)E(V) - 2E(U)^2E(V) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}, \\ \text{Var}(X) &= \text{Var}(U + V) = \text{Var}(U) + \text{Var}(V) + \text{Cov}(U, V) = 2(E(U^2) - E(U)^2) = \frac{1}{6}, \\ \text{Var}(Y) &= \text{Var}(UV) = E(U^2V^2) - E(UV)^2 = E(U^2)E(V^2) - E(U)^2E(V)^2 = \frac{1}{9} - \frac{1}{16} = \frac{7}{144}. \end{aligned}$$

as U, V are independent and identically distributed. Thus,

$$\rho_{XY} = \frac{\frac{1}{12}}{\sqrt{\frac{1}{6} \cdot \frac{7}{144}}} = \sqrt{\frac{6 \cdot 144}{7 \cdot 144}} = \sqrt{\frac{6}{7}}.$$

4. We have

$$\begin{aligned} E(X) &= \frac{3}{2}, \quad E(Y) = E(X^2) = 3, \quad E(XY) = E(X^3) = \frac{27}{4}, \quad E(Y^2) = E(X^4) = \frac{81}{5}, \\ \text{Cov}(X, Y) &= E(XY) - E(X)E(Y) = \frac{27}{4} - \frac{3}{2} \cdot 3 = \frac{9}{4}, \\ \text{Var}(X) &= E(X^2) - E(X)^2 = \frac{3}{4}, \\ \text{Var}(Y) &= E(Y^2) - E(Y)^2 = \frac{81}{5} - 9 = \frac{36}{5}. \end{aligned}$$

Thus,

$$\rho_{XY} = \frac{\frac{9}{4}}{\sqrt{\frac{3}{4} \cdot \frac{36}{5}}} = \frac{\sqrt{15}}{4}.$$

5. As $f_{(X,Y)}(x, y) = f_{(X,Y)}(y, x)$, we have $(X, Y) \stackrel{\text{Law}}{=} (Y, X)$ and in particular, $X \stackrel{\text{Law}}{=} Y$. Moreover, $f_{(X,Y)}(-x, y) = f_{(X,Y)}(x, y)$ for every y , thus $x \mapsto xf_{(X,Y)}(x, y)$ is an odd function so

$$E(X) = E(Y) = \frac{1}{2\pi} \int_{\mathbb{R}} dx \int_{\mathbb{R}} dy e^{-\sqrt{x^2+y^2}} x = 0,$$

and

$$E(XY) = \frac{1}{2\pi} \int_{\mathbb{R}} dx \int_{\mathbb{R}} dy e^{-\sqrt{x^2+y^2}} xy = 0.$$

Thus, $\text{Cov}(X, Y) = 0$, and $\rho_{XY} = 0$.

Solution to Exercise 2.43. 1. As the X_i 's only take values in $\{0, 1\}$, the variable $Y = \sum_{k=1}^n$ only takes values in $\{0, 1, \dots, n\}$. We just need to check that for any

$k \in \{0, 1, \dots, n\}$, $P(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}$. Now, for any set $I \subset \{1, \dots, n\}$ with $|I| = k$, we have

$$\begin{aligned} P(\cap_{i \in I} \{X_i = 1\}, \cap_{j \in I^c} \{X_j = 0\}) &= \prod_{i \in I} P(X_i = 1) \cdot \prod_{j \in I^c} P(X_j = 0) \\ &= p^{|I|} (1-p)^{n-|I|} = p^k (1-p)^{n-k}, \end{aligned}$$

where we used independence between the X_i 's. Now, the event $Y = k$ is the event that there is exactly one set $I \subset \{1, \dots, n\}$ with $|I| = k$ such that $(\cap_{i \in I} \{X_i = 1\}) \cap (\cap_{j \in I^c} \{X_j = 0\})$ occurs. The probability this event for a given I is $p^k (1-p)^{n-k}$ as computed previously, and there are $\binom{n}{k}$ subsets of $\{1, \dots, n\}$ with size k , so

$$P(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

as wanted.

2. We can use the previous point. Let $X_1, \dots, X_n$ be i.i.d. random variables with $X_i \sim \text{Bern}(p)$. By the previous point, $X \stackrel{\text{Law}}{=} \sum_{i=1}^n X_i$, so

$$E(X) = E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i) = \sum_{i=1}^n p = np,$$

where we used the properties of Bernoulli random variables. Also,

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i,j=1}^n \text{Cov}(X_i, X_j) = \sum_{i=1}^n \text{Var}(X_i) = np(1-p),$$

where we used that the X_i 's are independent in the third equality, and where we again used the properties of Bernoulli random variables in the last equality. Finally,

$$E(e^{tX}) = E\left(\exp\left(t \sum_{i=1}^n X_i\right)\right) = E\left(\prod_{i=1}^n e^{tX_i}\right) = \prod_{i=1}^n E(e^{tX_i}) = (1 + p(e^t - 1))^n$$

where we used that the X_i 's are independent in the third equality, and where we again used the properties of Bernoulli random variables in the last equality.

Solution to Exercise 2.44. 1. We first compute $E(X)$:

$$\begin{aligned} E(X) &= \sum_{n=1}^{\infty} p(1-p)^{n-1} n = p \sum_{n=1}^{\infty} x^{n-1} n \Big|_{x=1-p} = p \frac{d}{dx} \sum_{n=0}^{\infty} x^n \Big|_{x=1-p} \\ &= p \frac{d}{dx} \frac{1}{1-x} \Big|_{x=1-p} = p \frac{1}{(1-x)^2} \Big|_{x=1-p} = p \frac{1}{p^2} = \frac{1}{p}. \end{aligned}$$

Then, we compute the second moment $E(X^2)$: using the computation of the expectation, we have

$$\begin{aligned}
E(X^2) &= \sum_{n=1}^{\infty} p(1-p)^{n-1} n^2 = \sum_{n=1}^{\infty} p(1-p)^{n-1} n(n+1) - \sum_{n=1}^{\infty} p(1-p)^{n-1} n \\
&= \sum_{n=1}^{\infty} p(1-p)^{n-1} n(n+1) - \frac{1}{p} = p \sum_{n=1}^{\infty} x^{n-1} n(n+1) \Big|_{x=1-p} - \frac{1}{p} \\
&= p \frac{d^2}{dx^2} \sum_{n=0}^{\infty} x^{n+1} \Big|_{x=1-p} - \frac{1}{p} = p \frac{d^2}{dx^2} \frac{x}{1-x} \Big|_{x=1-p} - \frac{1}{p} \\
&= p \frac{2}{(1-x)^3} \Big|_{x=1-p} - \frac{1}{p} = p \frac{2}{p^3} - \frac{1}{p} = \frac{2-p}{p^2}.
\end{aligned}$$

as

$$\frac{d^2}{dx^2} \frac{x}{1-x} = \frac{d}{dx} \left(\frac{1}{1-x} + \frac{x}{(1-x)^2} \right) = \frac{1}{(1-x)^2} + \frac{1}{(1-x)^2} + \frac{2x}{(1-x)^3} = \frac{2}{(1-x)^3}.$$

The variance is then given by

$$\text{Var}(X) = E(X^2) - E(X)^2 = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}.$$

2. For $t \geq |\ln(1-p)| = -\ln(1-p)$, we have

$$\begin{aligned}
E(e^{tX}) &= \sum_{n=1}^{\infty} e^{tn} p(1-p)^{n-1} \geq \frac{p}{1-p} \sum_{n=1}^{\infty} e^{-\ln(1-p)n} (1-p)^n \\
&= \frac{p}{1-p} \sum_{n=1}^{\infty} (1-p)^{-n} (1-p)^n = \frac{p}{1-p} \sum_{n=1}^{\infty} 1 = +\infty.
\end{aligned}$$

Now, for $t < -\ln(1-p)$,

$$\begin{aligned}
E(e^{tX}) &= \sum_{n=1}^{\infty} e^{tn} p(1-p)^{n-1} = p e^t \sum_{n=1}^{\infty} (e^t(1-p))^{n-1} \\
&= p e^t \sum_{n=0}^{\infty} (e^t(1-p))^n = p e^t \frac{1}{1 - e^t(1-p)},
\end{aligned}$$

where we were able to use the geometric series as $e^t(1-p) < 1$ by the choice of t .

3. Let $X_1, X_2, \dots$ be an i.i.d. sequence of Bernoulli random variables with parameter p . Define

$$Y = 1 + \sum_{n \geq 1} \prod_{i=1}^n (1 - X_i).$$

Then, as $\prod_{i=1}^n (1 - X_i)$ takes values in $\{0, 1\}$ (it is 1 if $X_i, i = 1, \dots, n$ all take value 0, and is 0 otherwise), one has that $P(Y \in \mathbb{N}^*) = 1$, as does a geometric random

variable. It is thus sufficient to check that $P(Y = n) = p(1 - p)^n$ for all $n \in \mathbb{N}^*$. For $n \geq 1$, we have that $Y = n$ if and only if $\prod_{i=1}^k (1 - X_i) = 1$ for all $k \leq n - 1$ and $\prod_{i=1}^n (1 - X_i) = 0$. This happens if and only if $X_1 = X_2 = \dots = X_{n-1} = 0$, and $X_n = 1$. Thus, using independence,

$$\begin{aligned} P(Y = n) &= P(X_1 = \dots = X_{n-1} = 0, X_n = 1) \\ &= P(X_1 = 0) \dots P(X_{n-1} = 0) P(X_n = 1) = (1 - p)^{n-1} p, \end{aligned}$$

which is indeed the probability that a geometric of parameter p takes the value n .

Solution to Exercise 2.45. 1. We use a slightly different trick than in the previous exercises: we first compute the moment generating function.

$$M_X(t) = E(e^{tX}) = \sum_{n=0}^{\infty} e^{-\lambda} \frac{\lambda^n}{n!} e^{tn} = \sum_{n=0}^{\infty} e^{-\lambda} \frac{(\lambda e^t)^n}{n!} = e^{\lambda(e^t - 1)}$$

where we used the exponential series: $\sum_{n=0}^{\infty} \frac{z^n}{n!} = e^z$ for all $z \in \mathbb{R}$. We then use that

$$E(X) = M'_X(0), \quad E(X^2) = M''_X(0),$$

by Theorem 2.2.5. This gives

$$\begin{aligned} E(X) &= e^{\lambda(e^t - 1)} \lambda e^t \Big|_{t=0} = \lambda, \\ E(X^2) &= \left(e^{\lambda(e^t - 1)} \lambda e^t + e^{\lambda(e^t - 1)} \lambda^2 e^{2t} \right) \Big|_{t=0} = \lambda + \lambda^2. \end{aligned}$$

Thus, the variance is then given by

$$\text{Var}(X) = E(X^2) - E(X)^2 = \lambda + \lambda^2 - \lambda^2 = \lambda.$$

2. We first show that 1. implies 2.: suppose $X \sim \text{Poi}(\lambda)$, then, $P(X = 0) = e^{-\lambda}$, and for $k \in \mathbb{N}$,

$$\frac{P(X = k + 1)}{P(X = k)} = \frac{e^{-\lambda} \frac{\lambda^{k+1}}{(k+1)!}}{e^{-\lambda} \frac{\lambda^k}{k!}} = \frac{\lambda}{k + 1}.$$

We then show that 2. implies 1.: suppose that X is a random variable such that $P(X = 0) = e^{-\lambda}$ and for all $k \in \mathbb{N}$, $\frac{P(X = k+1)}{P(X = k)} = \frac{\lambda}{k+1}$. Then, for any $k \geq 1$,

$$\begin{aligned} P(X = k) &= \frac{P(X = k)}{P(X = k - 1)} P(X = k - 1) = \dots \\ &= \frac{P(X = k)}{P(X = k - 1)} \frac{P(X = k - 1)}{P(X = k - 2)} \dots \frac{P(X = 1)}{P(X = 0)} P(X = 0) \\ &= \frac{\lambda}{k} \frac{\lambda}{k - 1} \dots \frac{\lambda}{1} e^{-\lambda} = \frac{\lambda^k}{k!} e^{-\lambda}, \end{aligned}$$

which is indeed the probability that a Poisson random variable take value k .

Solution to Exercise 2.46. 1. Start with the expectation. We have

$$2 \sum_{k=1}^n k = \sum_{k=1}^n k + \sum_{k=1}^n (n+1-k) = \sum_{k=1}^n (k + (n+1-k)) = n(n+1),$$

so $\sum_{k=1}^n k = \frac{n(n+1)}{2}$. In particular,

$$E(X) = \sum_{k=0}^n \frac{1}{n+1} k = \frac{1}{n+1} \sum_{k=1}^n k = \frac{n}{2}.$$

Then, we show $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ by induction over n . The case $n = 1$ is

$$1 = \frac{(1+1)(2+1)}{6}.$$

Suppose the equality true for n and let us prove it for $n+1$.

$$\begin{aligned} \sum_{k=1}^{n+1} k^2 &= (n+1)^2 + \sum_{k=1}^n k^2 \\ &= (n+1)^2 + \frac{n(n+1)(2n+1)}{6} \\ &= \frac{(n+1)(n(2n+1) + 6(n+1))}{6} \\ &= \frac{(n+1)(2n^2 + 7n + 6)}{6} = \frac{(n+1)(n+2)(2n+3)}{6}, \end{aligned}$$

where we used the induction hypotheses in the second line. Using this, we obtain

$$E(X^2) = \sum_{k=0}^n \frac{1}{n+1} k^2 = \frac{1}{n+1} \sum_{k=1}^n k^2 = \frac{n(2n+1)}{6}.$$

So, the variance is given by

$$\text{Var}(X) = E(X^2) - E(X)^2 = \frac{n(2n+1)}{6} - \frac{n^2}{4} = \frac{4n(2n+1) - 6n^2}{24} = \frac{n^2 + 2n}{12}.$$

Finally, let us compute the moment generating function. For $t \in \mathbb{R}$,

$$E(e^{tX}) = \sum_{k=0}^n \frac{1}{n+1} e^{tk} = \frac{1}{n+1} \frac{1 - e^{(n+1)t}}{1 - e^t} = \frac{e^{(n+1)t} - 1}{(n+1)(e^t - 1)}$$

where we used the truncated geometric series identity.

2. *Solution 1:* Proceed exactly as in the first point:

$$\begin{aligned} E(Y) &= \sum_{k=1}^n \frac{1}{n} k = \frac{n+1}{2}, \quad E(Y^2) = \sum_{k=1}^n \frac{1}{n} k^2 = \frac{(n+1)(2n+1)}{6}, \\ \text{Var}(Y) &= E(Y^2) - E(Y)^2 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} \\ &= (n+1) \frac{2(2n+1) - 3(n+1)}{12} = (n+1) \frac{n-1}{12} = \frac{n^2 - 1}{12}, \\ E(e^{tY}) &= \sum_{k=1}^n \frac{1}{n} e^{tk} = \frac{1}{n} \frac{e^t - e^{t(n+1)}}{1 - e^t} = \frac{e^{nt} - 1}{n(1 - e^{-t})}. \end{aligned}$$

Solution 2: note that if $X \sim \text{Uni}(\{0, \dots, n-1\})$ then $X+1 \sim \text{Uni}(\{1, \dots, n\})$ and apply the first point.

Solution to Exercise 2.47. 1. We have

$$E(X) = \int_{-\infty}^{+\infty} x f_X(x) dx = \frac{1}{b-a} \int_a^b x dx = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{b+a}{2}.$$

In the same fashion,

$$E(X^2) = \int_{-\infty}^{+\infty} x^2 f_X(x) dx = \frac{1}{b-a} \int_a^b x^2 dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)}.$$

So,

$$\begin{aligned} \text{Var}(X) &= E(X^2) - E(X)^2 = \frac{b^3 - a^3}{3(b-a)} - \frac{(b+a)^2}{4} = \frac{4b^3 - 4a^3 - 3(b+a)^2(b-a)}{12(b-a)} \\ &= \frac{4b^3 - 4a^3 - 3(b+a)(b^2 - a^2)}{12(b-a)} = \frac{4b^3 - 4a^3 - 3b^3 + 3ba^2 - 3ab^2 + 3a^3}{12(b-a)} \\ &= \frac{b^3 - a^3 + 3ba^2 - 3ab^2}{12(b-a)} = \frac{(b-a)^3}{12(b-a)} = \frac{(b-a)^2}{12} \end{aligned}$$

Finally, for $t \neq 0$,

$$E(e^{tX}) = \int_{-\infty}^{+\infty} e^{tx} f_X(x) dx = \frac{1}{b-a} \int_a^b e^{tx} dx = \frac{1}{b-a} \left[\frac{e^{tx}}{t} \right]_a^b = \frac{e^{tb} - e^{ta}}{t(b-a)}.$$

2. As X is a uniform random variable on $[a, d]$, for any $a \leq b < c \leq d$,

$$P(X \in [b, c]) = \frac{c-b}{d-a}.$$

So, for any $a < b < t_1 < t_2 < c < d$,

$$\begin{aligned} P(X \in [t_1, t_2] \mid X \in [b, c]) &= \frac{P(X \in [t_1, t_2] \cap [b, c])}{P(X \in [b, c])} = \frac{P(X \in [t_1, t_2])}{P(X \in [b, c])} \\ &= \frac{t_2 - t_1}{d-a} \frac{d-a}{c-b} = \frac{t_2 - t_1}{c-b}. \end{aligned}$$

Solution to Exercise 2.48. 1. First using the change of variable $y = (x - \mu)/\sigma$

$$\int_{-\infty}^{+\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \sigma \int_{-\infty}^{+\infty} e^{-\frac{y^2}{2}} dy.$$

Now, using the hint, we compute

$$\begin{aligned} \left(\int_{-\infty}^{+\infty} e^{-\frac{y^2}{2}} dy \right)^2 &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-\frac{y^2+x^2}{2}} dy dx \\ &= \int_0^\infty \int_{-\pi}^\pi r e^{-\frac{r^2}{2}} d\theta dr = 2\pi \int_0^\infty r e^{-\frac{r^2}{2}} dr \end{aligned}$$

where we used polar coordinates $(x, y) = (r \cos(\theta), r \sin(\theta))$, see Remark 2.2.6. Finally, we compute using direct integration (note that $\frac{d}{dr}e^{-r^2/2} = -re^{-r^2/2}$):

$$\int_0^\infty re^{-\frac{r^2}{2}} dr = \left[-e^{-\frac{r^2}{2}} \right]_0^{+\infty} = 1.$$

Gathering everything, we obtain the wanted result.

2. *Solution 1 (direct computation)*: first, using $y = x - \mu$ and the previous point,

$$\begin{aligned} E(X) &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{+\infty} xe^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{+\infty} (y + \mu)e^{-\frac{y^2}{2\sigma^2}} dy \\ &= \mu + \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{+\infty} ye^{-\frac{y^2}{2\sigma^2}} dy = \mu, \end{aligned}$$

as the last integral is the integral of an odd function over a symmetric domain. Then, using again $y = \frac{x-\mu}{\sigma}$

$$\begin{aligned} \text{Var}(X) &= E((X - E(X))^2) = E((X - \mu)^2) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{+\infty} (x - \mu)^2 e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} y^2 e^{-\frac{y^2}{2}} dy \\ &= \frac{\sigma^2}{\sqrt{2\pi}} \underbrace{\left[-ye^{-\frac{y^2}{2}} \right]_{-\infty}^{+\infty}}_{=0} + \frac{\sigma^2}{\sqrt{2\pi}} \underbrace{\int_{-\infty}^{+\infty} e^{-\frac{y^2}{2}} dy}_{=\sqrt{2\pi}} \end{aligned}$$

where we used integration by part (primitive $ye^{-\frac{y^2}{2}}$ and differentiate y) in the fifth equality, and the first point. Let us now compute the moment generating function. Using again $y = \frac{x-\mu}{\sigma}$,

$$\begin{aligned} E(e^{tX}) &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{+\infty} e^{tx} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{e^{t\mu}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{\sigma ty} e^{-\frac{y^2}{2}} dy \\ &= \frac{e^{t\mu + \sigma^2 \frac{t^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{(y-\sigma t)^2}{2}} dy = \frac{e^{t\mu + \sigma^2 \frac{t^2}{2}}}{\sqrt{2\pi}} \underbrace{\int_{-\infty}^{+\infty} e^{-\frac{y^2}{2}} dy}_{=\sqrt{2\pi}} \end{aligned}$$

where we used another shift of the variable y , and a completion of the square:

$$\sigma ty - \frac{y^2}{2} = -\frac{1}{2}(\sigma^2 t^2 - 2\sigma ty + y^2) + \frac{\sigma^2 t^2}{2} = -\frac{1}{2}(y - \sigma t)^2 + \frac{\sigma^2 t^2}{2}$$

in the third equality.

Solution 2 (use Lemma 2.6.7): first, let $Z \sim \mathcal{N}(0, 1)$. Then,

$$E(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} dz ze^{-z^2/2} = 0,$$

as it is the (absolutely convergent) integral of an odd function over a symmetric domain. Also,

$$\begin{aligned}\text{Var}(Z) &= E(Z^2) - E(Z)^2 = E(Z^2) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} dz z^2 e^{-z^2/2} \\ &= \frac{1}{\sqrt{2\pi}} \left([-ze^{-z^2/2}]_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} dz e^{-z^2/2} \right) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} dz e^{-z^2/2} = 1,\end{aligned}$$

where we used integration by parts in the second line, and the first point in the last. Finally,

$$\begin{aligned}E(e^{tZ}) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} dz e^{tz - z^2/2} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} dz e^{\frac{t^2}{2} - \frac{(t-z)^2}{2}} \\ &= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} dy e^{-y^2/2} = e^{t^2/2},\end{aligned}$$

as $tz - \frac{z^2}{2} = \frac{t^2}{2} - \frac{(t-z)^2}{2}$ by completion of the square, and we used $y = z - t$ and the first point. Now, we can use Lemma 2.6.7 to get that $X \stackrel{\text{Law}}{=} \sigma Z + \mu$ and thus

$$\begin{aligned}E(X) &= \mu + \sigma E(Z) = \mu, \quad \text{Var}(X) = \text{Var}(\sigma Z + \mu) = \sigma^2 \text{Var}(Z) = \sigma^2, \\ E(e^{tX}) &= E(e^{\mu t + \sigma t Z}) = e^{\mu t} E(e^{\sigma t Z}) = e^{\mu t + \sigma^2 t^2/2}.\end{aligned}$$

3. First, let m be odd. Then, the function $x \mapsto x^m e^{-x^2/2}$ is odd. In particular,

$$E(X^m) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x^m e^{-\frac{x^2}{2}} dx = 0$$

as it is the integral of an odd function over a symmetric domain. Now, for $m = 2n$ even, define the sequence

$$a_n = E(X^{2n}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x^{2n} e^{-\frac{x^2}{2}} dx.$$

We know from the previous point that $a_1 = E(X^2) = 1$ (as we have $\mu = 0$ and $\sigma^2 = 1$). Now, using an integration by part, for $n \geq 2$,

$$\begin{aligned}a_n &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \underbrace{x^{2n-1}}_{=g(x)} \underbrace{x e^{-\frac{x^2}{2}}}_{=f'(x)} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[-x^{2n-1} e^{-\frac{x^2}{2}} \right]_{-\infty}^{+\infty} + \frac{(2n-1)}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x^{2n-2} e^{-\frac{x^2}{2}} dx \\ &= \frac{(2n-1)}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x^{2n-2} e^{-\frac{x^2}{2}} dx = (2n-1) a_{n-1}.\end{aligned}$$

We then prove then wanted result, which is $a_n = \frac{(2n)!}{2^n n!}$, by induction. Then case $n = 1$ follows from

$$\frac{2!}{2^1 1!} = \frac{2}{2} = 1 = a_1.$$

Then, suppose that the equality holds for $n \geq 1$, and let us show the equality for $n + 1$. By the relation we derived on the sequence (and the fact that $n + 1 \geq 2$),

$$a_{n+1} = (2(n+1) - 1)a_n = (2n+1) \frac{(2n)!}{2^n n!}$$

by the induction hypotheses. Now, $(2n+1) = \frac{(2n+2)(2n+1)}{2(n+1)}$, so

$$a_{n+1} = \frac{(2n+2)(2n+1)}{2(n+1)} \frac{(2n)!}{2^n n!} = \frac{(2(n+1))!}{2^{n+1}(n+1)!},$$

which is the wanted induction step.

Solution to Exercise 2.49. 1. We have that $\frac{d}{dx}e^{-\lambda x} = -\lambda e^{-\lambda x}$, so

$$\int_0^{+\infty} e^{-\lambda x} dx = \left[-\frac{1}{\lambda} e^{-\lambda x} \right]_0^{+\infty} = \frac{1}{\lambda}.$$

2. Using integration by parts, we have

$$E(X) = \lambda \int_0^{+\infty} x e^{-\lambda x} dx = \lambda \underbrace{\left[-\frac{x}{\lambda} e^{-\lambda x} \right]_0^{+\infty}}_{=0} + \lambda \underbrace{\int_0^{+\infty} \frac{1}{\lambda} e^{-\lambda x} dx}_{=\frac{1}{\lambda}},$$

by the first point. Then, again using integration by parts,

$$E(X^2) = \lambda \int_0^{+\infty} x^2 e^{-\lambda x} dx = \lambda \left[-\frac{x^2}{\lambda} e^{-\lambda x} \right]_0^{+\infty} + 2 \int_0^{+\infty} x e^{-\lambda x} dx = \frac{2}{\lambda} E(X) = \frac{2}{\lambda^2}.$$

Thus,

$$\text{Var}(X) = E(X^2) - E(X)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

3. We have that for $t \geq \lambda$,

$$E(e^{tX}) = \lambda \int_0^{+\infty} e^{tx} e^{-\lambda x} dx \geq \lambda \int_0^{+\infty} 1 dx = +\infty,$$

as $t - \lambda \geq 0$, so $e^{(t-\lambda)x} \geq 1$ for $x \geq 0$. Now, for $t < \lambda$,

$$E(e^{tX}) = \lambda \int_0^{+\infty} e^{tx} e^{-\lambda x} dx = \lambda \int_0^{+\infty} e^{-(\lambda-t)x} dx = \frac{\lambda}{\lambda - t}$$

as in the first point.

4. First, for $t \geq 0$, we compute,

$$P(X \geq t) = \lambda \int_0^{+\infty} \mathbf{1}_{[t, \infty)}(x) e^{-\lambda x} dx = \lambda \int_t^{+\infty} e^{-\lambda x} dx = \lambda \left[-\frac{1}{\lambda} e^{-\lambda x} \right]_t^{+\infty} = e^{-\lambda t}.$$

Then, for $0 < a < b$,

$$\begin{aligned} P(X \geq b | X \geq a) &= \frac{P(X \geq b, X \geq a)}{P(X \geq a)} = \frac{P(X \geq b)}{P(X \geq a)} \\ &= \frac{e^{-\lambda b}}{e^{-\lambda a}} = e^{-\lambda(b-a)} = P(X \geq b-a). \end{aligned}$$

Solution to Exercise 2.50. 1. To have $M = k$, we need that the sum of the first k variables is less than A , and that adding the $k + 1$ th makes the sum bigger than A . Formally,

$$\{M = k\} = \{S_k \leq A\} \cap \{S_{k+1} > A\} = \{S_k \leq A, X_{k+1} > A - S_k\}.$$

We can now compute the probability of this event.

$$\begin{aligned} P(M = k) &= P(S_k \leq A, X_{k+1} > A - S_k) \\ &= \int_0^\infty dx_1 e^{-x_1} \cdots \int_0^\infty dx_{k+1} e^{-x_{k+1}} \mathbb{1}_{x_1 + \cdots + x_k \leq A} \mathbb{1}_{x_{k+1} > A - (x_1 + \cdots + x_k)} \\ &= \int_0^\infty dx_1 e^{-x_1} \cdots \int_0^\infty dx_k e^{-x_k} \mathbb{1}_{x_1 + \cdots + x_k \leq A} P(X_{k+1} > A - (x_1 + \cdots + x_k)) \\ &= \int_0^\infty dx_1 e^{-x_1} \cdots \int_0^\infty dx_k e^{-x_k} \mathbb{1}_{x_1 + \cdots + x_k \leq A} e^{-A + x_1 + \cdots + x_k} \\ &= e^{-A} \int_0^\infty dx_1 \cdots \int_0^\infty dx_k \mathbb{1}_{x_1 + \cdots + x_k \leq A} \end{aligned}$$

where in the fourth line we used the fact that X_{k+1} is an exponential random variable with rate 1 so for $t > 0$,

$$P(X_{k+1} > t) = \int_t^\infty e^{-x} dx = e^{-t}.$$

To conclude, we need to compute the value of

$$a_k = \int_0^\infty dx_1 \cdots \int_0^\infty dx_k \mathbb{1}_{x_1 + \cdots + x_k \leq A}.$$

First, we have

$$a_0 = 1, \quad a_1 = \int_0^\infty dx_1 \mathbb{1}_{x_1 \leq A} = A.$$

Then, for $k \geq 2$, we make the change of variable

$$\begin{aligned} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_k \end{pmatrix} &= T \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} = \begin{pmatrix} x_1 \\ x_1 + x_2 \\ \vdots \\ x_1 + x_2 + \cdots + x_k \end{pmatrix} \\ T &= \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & 1 & 1 & \cdots & 1 \end{pmatrix}. \end{aligned}$$

We have $|\det T| = 1$ as it is a lower triangular matrix with 1's on the diagonal, so the change of variable formula gives

$$a_k = \int_0^\infty dx_1 \cdots \int_0^\infty dx_k \mathbb{1}_{x_1 + \cdots + x_k \leq A} = \int_0^A dy_k \int_0^{y_k} dy_{k-1} \cdots \int_0^{y_2} dy_1$$

as the image of $[0, +\infty)^k$ under T is $\{y \in [0, +\infty)^k : y_1 \leq y_2 \leq \dots \leq y_k\}$, and $y_k = x_1 + \dots + x_k \leq A$. To conclude, we have

$$\begin{aligned} a_k &= \int_0^A dy_k \int_0^{y_k} dy_{k-1} \dots \int_0^{y_2} dy_1 = \int_0^A dy_k \int_0^{y_k} dy_{k-1} \dots \int_0^{y_3} dy_2 y_2 \\ &= \int_0^A dy_k \int_0^{y_k} dy_{k-1} \dots \int_0^{y_4} dy_3 \frac{y_3^2}{2} \\ &= \dots = \int_0^A dy_k \frac{y_k^{k-1}}{(k-1)!} = \frac{A^k}{k!}. \end{aligned}$$

We obtained

$$P(M = k) = e^{-A} a_k = \begin{cases} e^{-A} & \text{if } k = 0, \\ e^{-A} \frac{A^k}{k!} & \text{if } k \in \mathbb{N}^*. \end{cases}$$

2. We recognize the Poisson law of parameter A , so $M \sim \text{Poi}(A)$. This observation justifies the use of Poisson random variables when we want to model “the number of occurrences of an event in a certain time duration” like the number of cars passing on a given road between 13 : 00 and 14 : 00. It also tells us that the parameter of the Poisson variable should be proportional (linear in) the length of the time interval.

Solution to Exercise 2.51. 1. As already observed, F_Z is a primitive of the density of Z , which is continuous. In particular, F_Z is differentiable and therefore continuous.

2. The function $F_Z : \mathbb{R} \rightarrow (0, 1)$ is continuous. Moreover, for $t < t'$,

$$F_Z(t') - F_Z(t) = \int_{-\infty}^{t'} \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx - \int_{-\infty}^t \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx = \int_t^{t'} \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx > 0,$$

so F_Z is strictly increasing. Moreover,

$$\lim_{t \rightarrow -\infty} F_Z(t) = 0, \quad \lim_{t \rightarrow +\infty} F_Z(t) = 1.$$

As seen in analysis I (consequence of the intermediate value theorem), F_Z is thus a bijection between $\mathbb{R}$ and $(0, 1)$. Denote its inverse $F_Z^{-1} : (0, 1) \rightarrow \mathbb{R}$.

3. It is sufficient to see that $F_Z^{-1}(U)$ has the same repartition function as Z . We thus write the definition: for $t \in \mathbb{R}$,

$$P(F_Z^{-1}(U) \leq t) = P(U \leq F_Z(t))$$

as F_Z is an increasing bijection. Finally, as $U \sim \text{Uni}([0, 1])$,

$$P(U \leq F_Z(t)) = \int_0^{F_Z(t)} du = F_Z(t).$$

So, we obtained that $F_{F_Z^{-1}(U)}(t) = F_Z(t)$ for all $t \in \mathbb{R}$, thus $F_Z^{-1}(U) \stackrel{\text{Law}}{=} Z$.

What did we use about Z ? Can we do the same procedure for other laws than the Gaussian? This kind of operation is extremely useful when sampling random numbers on a computer: it is sufficient to know how to sample a $\text{Uni}([0, 1])$ to be able to sample more complicated random variables by applying a suitable function!

Solution to Exercise 2.52. First, the CDF of $X \sim \text{Exp}(\lambda)$ is $F_X(t) = 0$ for $t \leq 0$, and

$$F_X(t) = P(X \leq t) = \int_0^t dx \lambda e^{-\lambda x} = 1 - e^{-\lambda t}$$

for $t > 0$.

1. We can compute the CDF of T_- : for $t \leq 0$, $P(T_- \leq t) = 0$, and for $t > 0$,

$$\begin{aligned} P(T_- \leq t) &= 1 - P(T_- > t) = 1 - P(X_1 > t, X_2 > t, \dots, X_n > t) \\ &= 1 - \prod_{i=1}^n P(X_i > t) = 1 - P(X_1 > t)^n = 1 - (e^{-\lambda t})^n = 1 - e^{-n\lambda t}, \end{aligned}$$

which is the CDF of an $\text{Exp}(n\lambda)$.

2. For $t \leq 0$, $F_{T_+}(t) = 0$ (as the X_i 's are almost surely positive), and for $t > 0$,

$$\begin{aligned} F_{T_+}(t) &= P(T_+ \leq t) = P(X_1 \leq t, X_2 \leq t, \dots, X_n \leq t) = \prod_{i=1}^n P(X_i \leq t) \\ &= P(X_1 \leq t)^n = (1 - e^{-\lambda t})^n. \end{aligned}$$

3. We have that F'_{T_+} is differentiable ($F'_{T_+}(x) = 0$ for $x \leq 0$, and $F'_{T_+}(x) = n(1 - e^{-\lambda x})^{n-1} \lambda e^{-\lambda x}$ for $x \geq 0$). It is therefore a density function for T_+ . We thus have

$$a_n := E(T_+) = \int_{-\infty}^{\infty} F'_{T_+}(x) x dx = \int_0^{\infty} n(1 - e^{-\lambda x})^{n-1} \lambda e^{-\lambda x} x dx \quad (\text{A.1})$$

as $F'_{T_+}(x) = 0$ for $x \leq 0$. We now compute this integral. The calculation is a bit convoluted, but it has the advantage of only needing exchange of integrals and finite sums, and is thus easy to justify.

We can now use a binomial expansion:

$$(1 - e^{-\lambda x})^{n-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} (-1)^k e^{-k\lambda x}$$

in (A.1) to obtain

$$\begin{aligned} a_n &= \sum_{k=0}^{n-1} \binom{n-1}{k} (-1)^k n \lambda \int_0^{\infty} e^{-(k+1)\lambda x} x dx \\ &= \sum_{k=0}^{n-1} \frac{(n-1)!}{k!(n-1-k)!} (-1)^k \frac{n}{(k+1)} \underbrace{\int_0^{\infty} (k+1) \lambda e^{-(k+1)\lambda x} x dx}_{\text{expectation of an } \text{Exp}((k+1)\lambda)}. \end{aligned}$$

Remembering that the expected value of an $\text{Exp}(\lambda)$ is $\frac{1}{\lambda}$, we obtain

$$a_n = \sum_{k=0}^{n-1} \frac{n!}{(k+1)!(n-(k+1))!} (-1)^k \frac{1}{\lambda(k+1)} = -\frac{1}{\lambda} \sum_{k=1}^n \binom{n}{k} (-1)^k \frac{1}{k},$$

We can then use the identity $\frac{1}{k} = \int_0^1 x^{k-1} dx$ to obtain

$$\begin{aligned}
a_n &= -\frac{1}{\lambda} \sum_{k=1}^n \binom{n}{k} (-1)^k \int_0^1 x^{k-1} dx \\
&= \frac{1}{\lambda} \int_0^1 \frac{1}{x} \left(1 - \sum_{k=0}^n \binom{n}{k} (-1)^k x^k \right) dx \\
&= \frac{1}{\lambda} \int_0^1 \frac{1}{x} (1 - (1-x)^n) dx \\
&= \frac{1}{\lambda} \int_0^1 \frac{1-u^n}{1-u} du = \frac{1}{\lambda} \int_0^1 \sum_{k=0}^{n-1} u^k du \\
&= \frac{1}{\lambda} \sum_{k=0}^{n-1} \int_0^1 u^k du = \frac{1}{\lambda} \sum_{k=0}^{n-1} \frac{1}{k+1} = \frac{1}{\lambda} \sum_{k=1}^n \frac{1}{k}
\end{aligned}$$

where we used the change of variable $u = 1 - x$, $dx = -du$, and the truncated geometric series identity

$$\sum_{k=0}^n r^k = \frac{1 - r^{n+1}}{1 - r}$$

for $r < 1$.

Solution to Exercise 2.53. Let $\sigma \geq 0$, $X \sim \mathcal{N}(0, \sigma^2)$. If $\sigma = 0$, the claim is empty: $X = 0$ almost surely and $E(Xf(X)) = 0$. Suppose $\sigma > 0$. Then, $X \stackrel{\text{Law}}{=} \sigma Z$ with $Z \sim \mathcal{N}(0, 1)$ (by Lemma 2.6.7), and for any $f : \mathbb{R} \rightarrow \mathbb{R}$ differentiable growing at most like $e^{|x|^{2-\epsilon}}$ at infinity,

$$\begin{aligned}
E(Xf(X)) &= \sigma E(Zf(\sigma Z)) = \sigma \int_{-\infty}^{\infty} dx \frac{e^{-x^2/2}}{\sqrt{2\pi}} x f(\sigma x) dx \\
&= \sigma \int_{-\infty}^{\infty} dx \frac{e^{-x^2/2}}{\sqrt{2\pi}} f'(\sigma x) \sigma dx \\
&= \sigma^2 E(f'(\sigma Z)) = \sigma^2 E(f'(X)).
\end{aligned}$$

where we used integration by parts: $\frac{d}{dx} e^{-x^2/2} = -x e^{-x^2/2}$, and the growth of f is killed by the decay of $e^{-x^2/2}$ at $\pm\infty$.

Solution to Exercise 2.54. 1. Let $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$. Then,

$$Y \equiv \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} := AX = \begin{pmatrix} a_{11}X_1 + a_{12}X_2 \\ a_{21}X_1 + a_{22}X_2 \end{pmatrix}.$$

By Lemma 2.7.1, Y is a Gaussian vector. Moreover, $E(Y_1) = a_{11}E(X_1) + a_{12}E(X_2) = 0$, and $E(Y_2) = a_{21}E(X_1) + a_{22}E(X_2) = 0$, so Y is centred. We need to find the values of the a_{ij} 's. We want

$$\text{Cov}(Y_i, Y_j) = \mathbb{1}_{i=j} \quad i, j = 1, 2. \quad (\text{A.2})$$

Now,

$$\begin{aligned}\text{Cov}(Y_i, Y_j) &= \text{Cov}(a_{i1}X_1 + a_{i2}X_2, a_{j1}X_1 + a_{j2}X_2) \\ &= a_{i1}a_{j1}\Sigma_{11} + a_{i1}a_{j2}\Sigma_{12} + a_{i2}a_{j1}\Sigma_{21} + a_{i2}a_{j2}\Sigma_{22} = (a_{i1} \ a_{i2})\Sigma \begin{pmatrix} a_{j1} \\ a_{j2} \end{pmatrix}.\end{aligned}$$

So, we can write (A.2) in matrix form:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = A\Sigma A^T.$$

There are several ways to proceed from here.

- We can solve the system of equation for the a_{ij} 's by hand (not advised...).
- Another way is to use Cholesky decomposition: decompose $\Sigma = LL^T$ with L a lower triangular matrix, and set $A = L^{-1}$. Then, $A\Sigma A^T = L^{-1}LL^T(L^{-1})^T = \text{Id}$ as wanted, as inverse of transpose is transpose of inverse. In this case the matrix L is given by $L = \begin{pmatrix} \sqrt{2} & 0 \\ \frac{1}{\sqrt{2}} & \sqrt{\frac{3}{2}} \end{pmatrix}$, which gives

$$A = L^{-1} = \frac{1}{\sqrt{3}} \begin{pmatrix} \sqrt{\frac{3}{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \sqrt{2} \end{pmatrix}.$$

- Another way (the most used one in probability) is to diagonalize $\Sigma = QDQ^T$ with Q an orthogonal matrix ($Q^{-1} = Q^T$), and D a diagonal matrix. This is possible as Σ is symmetric. Then, one can take $A = QD^{-1/2}Q^T$, which gives $A\Sigma A^T = QD^{-1/2}Q^T QDQ^T (QD^{-1/2}Q^T)^T = \text{Id}$ as wanted (you can ask your computer for the numeric values of Q, D , or compute them by hand).

2. As in the previous point, it is direct to see that $(Y_1, Y_2) \equiv (X_1 - X_2, X_1 + X_2)$ is centred. Let us compute the off-diagonal entries of its covariance matrix C :

$$\begin{aligned}C_{12} = C_{21} &= \text{Cov}(Y_1, Y_2) = \text{Cov}(X_1 - X_2, X_1 + X_2) \\ &= \Sigma_{11} + \Sigma_{12} - \Sigma_{21} - \Sigma_{22} = a + 1 - 1 + 1 - a^2 = -a^2 + a + 1,\end{aligned}$$

We want $-a^2 + a + 1 = 0$, i.e. $a^2 - a - 1 = 0$, which gives $a \in \{\frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}\}$ (solution of a degree 2 polynomial equation).

Solution to Exercise 2.55. We first have by bi-linearity of the covariance that

$$\text{Var}(R_n) = \text{Cov}(R_n, R_n) = \text{Cov}\left(\sum_{i=1}^n X_i^2, \sum_{i=1}^n X_i^2\right) = \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i^2, X_j^2).$$

On the one hand, for any $i = 1, \dots, n$, $E(X_i^2) = \Sigma_{ii}$ (as $E(X_i) = 0$ so $\text{Var}(X_i) = E(X_i^2)$). On the other hand, for $1 \leq i, j \leq n$,

$$E(X_i^2 X_j^2) = E(X_i X_i X_j^2) = \Sigma_{ii}E(X_j^2) + \Sigma_{ij}E(X_i X_j) = \Sigma_{ii}\Sigma_{jj} + \Sigma_{ij}^2,$$

by Lemma 2.7.3. Combining, we have

$$\text{Cov}(X_i^2, X_j^2) = E(X_i^2 X_j^2) - E(X_i^2)E(X_j^2) = \Sigma_{ii}\Sigma_{jj} + \Sigma_{ij}^2 - \Sigma_{ii}\Sigma_{jj} = \Sigma_{ij}^2.$$

Thus,

$$\text{Var}(R_n) = \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i^2, X_j^2) = \sum_{i=1}^n \sum_{j=1}^n \Sigma_{ij}^2.$$

Solution to Exercise 2.56. Let G denote the grade of the student.

1. From Markov's inequality, we get

$$P(G \geq 5.5) \leq \frac{E(G)}{5.5} = \frac{4.3}{5.5} \approx 0.78.$$

2. From Chebychev's inequality applied to the random variable $G - E(G) = G - 4.3$, we obtain

$$P(G \in [3.3, 5.3]) = P(|G - 4.3| \leq 1) = 1 - P(|G - E(G)| > 1) \geq 1 - \text{Var}(G) = 0.6.$$

3. Denote n the number of students in the class. Denote $G_1, \dots, G_n$ the grades of the students (we suppose that they are all independent and identically distributed). Set $\bar{G} = \frac{1}{n} \sum_{i=1}^n G_i$ the mean of the class. We have that the expected value of $\bar{G}$ is

$$E(\bar{G}) = \frac{1}{n} \sum_{i=1}^n E(G_i) = \frac{1}{n} n 4.3 = 4.3.$$

Its variance is then

$$\begin{aligned} \text{Var}(\bar{G}) &= \text{Cov}\left(\frac{1}{n} \sum_{i=1}^n G_i, \frac{1}{n} \sum_{i=1}^n G_i\right) = \frac{1}{n^2} \sum_{i,j=1}^n \text{Cov}(G_i, G_j) \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Cov}(G_i, G_i) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(G_i) = \frac{1}{n} 0.4, \end{aligned}$$

as $\text{Cov}(G_i, G_j) = 0$ for $i \neq j$ (by independence). We can then use Chebychev's inequality:

$$P(|\bar{G} - 4.3| \geq 0.3) \leq \frac{\frac{1}{n} 0.4}{0.09} = \frac{40}{9n}.$$

So, in order for the mean grade to have probability at least 0.9 to be in $[4, 4.6]$, it is sufficient to take n such that $\frac{40}{9n} < 0.1$, which amounts to take $n \geq 45$.

Solution to Exercise 2.57. 1. from the expression, F_X is continuous on $(-\infty, \alpha)$ and on $(\alpha, +\infty)$. Moreover,

$$\lim_{t \nearrow \alpha} F_X(t) = \lim_{t \nearrow \alpha} 0 = 0, \quad \lim_{t \searrow \alpha} F_X(t) = \lim_{t \searrow \alpha} \left(1 - \left(\frac{\alpha}{t}\right)^\theta\right) = 0,$$

so F_X is continuous (in particular, it is right-continuous). F_X is constant 0 on $(-\infty, \alpha]$, and $t \mapsto (\alpha/t)^\theta$ is decreasing to 0 on $(\alpha, +\infty)$, so we have the wanted monotonicity and limits.

2. We saw that the density function for X is given by the derivative of its CDF F_X . So, for $p > 0$,

$$E(X^p) = \int_{-\infty}^{+\infty} x^p F'_X(x) dx = \int_{\alpha}^{+\infty} x^p \theta \frac{\alpha^\theta}{x^{\theta+1}} dx$$

as F_X is constant on $(-\infty, \alpha]$. We then have

$$\int_{\alpha}^{+\infty} x^p \theta \frac{\alpha^\theta}{x^{\theta+1}} dx = \theta \alpha^\theta \int_{\alpha}^{+\infty} x^{p-\theta-1} dx,$$

which converges if and only if $\theta > p$. In particular, we have

$$E(X) = \frac{\theta}{\theta-1} \alpha \text{ if } \theta > 1, \quad E(X^2) = \frac{\theta}{\theta-2} \alpha^2 \text{ if } \theta > 2.$$

Thus,

$$\begin{aligned} \text{Var}(X) &= E(X^2) - E(X)^2 = \frac{\theta}{\theta-2} \alpha^2 - \frac{\theta^2}{(\theta-1)^2} \alpha^2 \\ &= \alpha^2 \theta \frac{(\theta-1)^2 - \theta(\theta-2)}{(\theta-2)(\theta-1)^2} \\ &= \frac{\alpha^2 \theta}{(\theta-2)(\theta-1)^2} \end{aligned}$$

and is finite only if $\theta > 2$.

3. For $c > \alpha$, and $t \in \mathbb{R}$, we have

$$P(X > c) = 1 - P(X \leq c) = 1 - F_X(c) = \frac{\alpha^\theta}{c^\theta},$$

and so, for $t > c$,

$$\begin{aligned} G_X(t) &:= P(X \leq t | X > c) = 1 - P(X > t | X > c) = 1 - \frac{P(X > t)}{P(X > c)} \\ &= 1 - \frac{\alpha^\theta}{t^\theta} \frac{c^\theta}{\alpha^\theta} = 1 - \frac{c^\theta}{t^\theta}. \end{aligned}$$

Then, $G_X(t) = 0$ for $t \leq c$. Now, realize that $G_X(t)$ is the CDF of X under $P(\cdot | X > c)$. And that it is the CDF of a Pareto random variable of parameter (c, θ) . Thus, if $\theta > 1$,

$$E(X | X > c) = \frac{\theta}{\theta-1} c.$$

Solution to Exercise 2.58. 1. We compute using linearity of the expectation:

$$E(S_n) = E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i) = np, \quad E(\bar{S}_n) = \frac{1}{n} E(S_n) = p.$$

2. We use bi-linearity of the covariance and the fact that $\text{Cov}(X_i, X_j) = 0$ for $i \neq j$ by independence:

$$\begin{aligned}\text{Cov}(S_5, S_{10}) &= \text{Cov}\left(\sum_{i=1}^5 X_i, \sum_{j=1}^{10} X_j\right) = \sum_{i=1}^5 \sum_{j=1}^{10} \text{Cov}(X_i, X_j) \\ &= \sum_{i=1}^5 \text{Var}(X_i) = 5p(1-p).\end{aligned}$$

3. By the weak Law of Large Numbers (Theorem 2.8.3), for any $\epsilon > 0$, and any $n \geq 1$,

$$P(|\bar{S}_n - E(X_1)| \geq \epsilon) \leq \frac{\text{Var}(X_1)}{\epsilon^2 n}.$$

Using $E(X_1) = p$, $\text{Var}(X_1) = p(1-p)$, and $\epsilon = 0.01$, we find that for any $n \geq 1$,

$$P(|\bar{S}_n - p| \geq 0.01) \leq \frac{p(1-p)}{10^{-4}n}.$$

We thus want n such that $10^4 \frac{p(1-p)}{n} \leq 0.01$, which amounts to take

$$n \geq 10^6 p(1-p).$$

Solution to Exercise 2.59. 1. Suppose $E(X^2) < \infty$. By Jensen's inequality (using that $x \mapsto \sqrt{x}$ is concave over $\mathbb{R}_+$), we have

$$E(|X|) = E(\sqrt{X^2}) \leq \sqrt{E(X^2)} < \infty.$$

2. The function $x \mapsto e^{tx}$ is convex for any $t \in \mathbb{R}$. Thus, by Jensen's inequality

$$M_X(t) = E(e^{tX}) \geq e^{tE(X)} = e^{t \cdot 0} = 1.$$

A.3 Solutions to Chapter 3

Solution to Exercise 3.1. First note that as $X \in \mathbb{Z}$ almost surely, F_X is constant on every interval of the form $(k, k+1)$, $k \in \mathbb{Z}$, and we have

$$P(X = k) = P(X \in [k, k+1)).$$

In particular, by the formula of total probability,

$$F_X(t) = \sum_{k=-\infty}^{\lfloor t \rfloor} P(X \in [k, k+1)) = \sum_{k=-\infty}^{\lfloor t \rfloor} P(X = k).$$

The discontinuity points of F_X are therefore $D := \{k \in \mathbb{Z} : P(X = k) > 0\}$.

We first show that 1. implies 2.

If we assume that $X_n \xrightarrow{\text{Law}} X$, we have that for every $t \in \mathbb{R} \setminus D$, $F_{X_n}(t) \rightarrow F_X(t)$. Moreover, as $X \in \mathbb{Z}$ almost surely, one has that for every $k \in \mathbb{Z}$, and every $\epsilon \in (0, 1)$,

$$P(X = k) = P(X \in (k - \epsilon, k + \epsilon]) = F_X(k + \epsilon) - F_X(k - \epsilon).$$

But, for $\epsilon > 0$, $k + \epsilon \notin D$ and $k - \epsilon \notin D$ so, on the one hand,

$$\begin{aligned} \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} P(X_n \in (k - \epsilon, k + \epsilon)) &\leq \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} P(X_n \in (k - \epsilon, k + \epsilon]) \\ &= \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} F_{X_n}(k + \epsilon) - F_{X_n}(k - \epsilon) \\ &= \lim_{\epsilon \searrow 0} F_X(k + \epsilon) - F_X(k - \epsilon) \\ &= \lim_{\epsilon \searrow 0} P(X = k) = P(X = k). \end{aligned}$$

On the other hand,

$$\begin{aligned} \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} P(X_n \in (k - \epsilon, k + \epsilon)) &\geq \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} P(X_n \in (k - \frac{\epsilon}{2}, k + \frac{\epsilon}{2}]) \\ &= \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} F_{X_n}(k + \frac{\epsilon}{2}) - F_{X_n}(k - \frac{\epsilon}{2}) \\ &= \lim_{\epsilon \searrow 0} F_X(k + \frac{\epsilon}{2}) - F_X(k - \frac{\epsilon}{2}) \\ &= \lim_{\epsilon \searrow 0} P(X = k) = P(X = k). \end{aligned}$$

So, $\lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} P(X_n \in (k - \epsilon, k + \epsilon)) = P(X = k)$.

We then show that 2. implies 1.

For $\epsilon > 0$, introduce

$$D_\epsilon = \bigcup_{k \in D} (k - \epsilon, k + \epsilon).$$

Here, our assumption is that for every $k \in \mathbb{Z}$,

$$\lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} P(X_n \in (k - \epsilon, k + \epsilon)) = P(X = k).$$

Let $t \in \mathbb{R} \setminus D$. We need to show that $\lim_{n \rightarrow \infty} F_{X_n}(t) = F_X(t)$.

We first show that for any $\epsilon, \delta > 0$, there exist $K = K(\delta)$, $N = N(\epsilon, \delta) > 0$, such that for any $n \geq N$,

$$P(|X_n| > K) \leq \delta, \quad P(X_n \notin D_\epsilon) \leq \delta.$$

Indeed, Let K be such that $P(|X| > K - 1) \leq \frac{\delta}{2}$. Let then $0 < \epsilon' < \epsilon$, N be such that for $n \geq N$,

$$\sum_{k=-K}^K |P(X_n \in (k - \epsilon', k + \epsilon')) - P(X = k)| \leq \frac{\delta}{2}.$$

Then, for $n \geq N$,

$$\begin{aligned} P(|X_n| > K) &= 1 - P(X_n \leq K) \leq 1 - \sum_{k=1-K}^{K-1} P(X_n \in (k - \epsilon, k + \epsilon)) \\ &\leq 1 + \frac{\delta}{2} - \sum_{k=1-K}^{K-1} P(X = k) = 1 + \frac{\delta}{2} - P(|X| \leq K - 1) = \frac{\delta}{2} + P(|X| > K - 1) \leq \delta. \end{aligned}$$

Also,

$$\begin{aligned} P(X_n \notin D_\epsilon) &\leq P(X_n \notin D_{\epsilon'}) = 1 - P(X_n \in D_{\epsilon'}) \leq 1 - P(X_n \in D_{\epsilon'}, |X_n| \leq K) \\ &\leq 1 - \sum_{k=-K}^K P(X_n \in (k - \epsilon', k + \epsilon')) \leq 1 + \frac{\delta}{2} - \sum_{k=-K}^K P(X = k) \\ &= 1 + \frac{\delta}{2} - P(|X| \leq K) = \frac{\delta}{2} + P(|X| > K) \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta. \end{aligned}$$

Now, on the one hand, for any $K > |t|$,

$$\begin{aligned} \lim_{n \rightarrow \infty} F_{X_n}(t) &\geq \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} P(X_n \leq t, X_n \in D_\epsilon, |X_n| \leq K + \epsilon) \\ &\geq \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} \sum_{k=-K}^{\lfloor t \rfloor} P(X_n \in (k - \epsilon, k + \epsilon)) = \sum_{k=-K}^{\lfloor t \rfloor} P(X = k). \end{aligned}$$

K being arbitrary, we can take $K \rightarrow \infty$ to obtain

$$\lim_{n \rightarrow \infty} F_{X_n}(t) \geq \sum_{k=-\infty}^{\lfloor t \rfloor} P(X = k) = F_X(t).$$

Let us now prove the reverse inequality. Let $\delta > 0$. By the observation made a few lines above, we have that there is $K = K(\delta) > 0$ such that

$$\begin{aligned} \lim_{n \rightarrow \infty} F_{X_n}(t) &\leq \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} (P(X_n \leq t, X_n \in D_\epsilon, |X_n| \leq K) + P(X_n \notin D_\epsilon) + P(|X_n| > K)) \\ &\leq 2\delta + \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} P(X_n \leq t, X_n \in D_\epsilon, |X_n| \leq K) \\ &\leq 2\delta + \lim_{\epsilon \searrow 0} \lim_{n \rightarrow \infty} \sum_{k=-K}^{\lfloor t \rfloor} P(X_n \in (k - \epsilon, k + \epsilon)) \\ &= 2\delta + \sum_{k=-K}^{\lfloor t \rfloor} P(X = k) \leq 2\delta + \sum_{k=-\infty}^{\lfloor t \rfloor} P(X = k) = 2\delta + F_X(t). \end{aligned}$$

$\delta > 0$ being arbitrary, we can take $\delta \rightarrow 0$ to obtain $\lim_{n \rightarrow \infty} F_{X_n}(t) \leq F_X(t)$.

Solution to Exercise 3.2. 1. The CDF of a $\text{Uni}([-1, 0])$ is

$$F(t) = \begin{cases} 0 & \text{if } t \leq -1, \\ t + 1 & \text{if } -1 < t < 0, \\ 1 & \text{if } t \geq 0. \end{cases}$$

In particular, it is continuous everywhere. Now, the CDF of X_n is

$$F_{X_n}(t) = \begin{cases} 0 & \text{if } t \leq -1, \\ \frac{n(t+1)}{n+1} & \text{if } -1 < t < \frac{1}{n}, \\ 1 & \text{if } t \geq \frac{1}{n}. \end{cases}$$

We need to check that for every $t \in \mathbb{R}$, $F_{X_n}(t) \rightarrow F(t)$ as $n \rightarrow \infty$. This is obvious for $t \leq -1$. Now, for $t > 0$, we have that for any $n \geq \frac{1}{t}$,

$$|F_{X_n}(t) - F(t)| = |1 - 1| = 0.$$

Remains to show the case $t \in (-1, 0)$. In this case,

$$|F_{X_n}(t) - F(t)| = \left| \frac{n(t+1)}{n+1} - (t+1) \right| = \frac{1}{n+1} |n(t+1) - (n+1)(t+1)| = \frac{t+1}{n+1} \xrightarrow{n \rightarrow \infty} 0.$$

2. We will show that X_n converges in L^2 towards X , which implies the convergence in probability.

$$E((X_n - X)^2) = E((X + Y_n - X)^2) = E(Y_n^2) = \frac{1}{n} \xrightarrow{n \rightarrow \infty} 0.$$

3. We have that

$$P(|X - X_n| \geq 1) = P(|X - 1 + X| \geq 1) = P(|2X - 1| \geq 1) = 1,$$

as $X \in \{0, 1\}$ almost surely, and so $2X - 1 \in \{-1, 1\}$ almost surely, which implies $|2X - 1| = 1$ almost surely. Let $X \sim \text{Bern}(1/2)$ and $X_n = 1 - X$ for $n \geq 1$. In particular, X_n does not converge in probability towards X .

Solution to Exercise 3.3. We will use Theorem 3.1.1. We first recall the moment generating function of $Z \sim \mathcal{N}(0, p(1-p))$ and of $X \sim \text{Bern}(p)$:

$$M_Z(t) = \exp\left(\frac{p(1-p)t^2}{2}\right), \quad M_X(t) = 1 + p(e^t - 1).$$

We want to prove that for any $t \in \mathbb{R}$,

$$M_{\tilde{S}_n}(t) \xrightarrow{n \rightarrow \infty} M_Z(t).$$

First, we have

$$\begin{aligned} M_{\tilde{S}_n}(t) &= E\left(\exp\left(t \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - p)\right)\right) = E\left(\prod_{i=1}^n e^{t(X_i - p)/\sqrt{n}}\right) \\ &= \prod_{i=1}^n E\left(e^{t(X_i - p)/\sqrt{n}}\right) = \prod_{i=1}^n e^{-tp/\sqrt{n}} E\left(e^{tX_i/\sqrt{n}}\right) = e^{-tp\sqrt{n}} (M_X(t/\sqrt{n}))^n, \end{aligned}$$

as the X_i 's form an independent family of $\text{Bern}(p)$. This gives

$$M_{\tilde{S}_n}(t) = \exp(-tp\sqrt{n} + n \ln(M_X(t/\sqrt{n}))) = \exp\left(-tp\sqrt{n} + n \ln\left(1 + p(e^{t/\sqrt{n}} - 1)\right)\right).$$

Now, by a second order Taylor expansion around $z = 0$, we have

$$\ln(1 + z) = z - \frac{z^2}{2} + o(z^2).$$

So,

$$\ln\left(1 + p(e^{t/\sqrt{n}} - 1)\right) = p(e^{t/\sqrt{n}} - 1) - \frac{p^2(e^{t/\sqrt{n}} - 1)^2}{2} + o(p^2(e^{t/\sqrt{n}} - 1)^2).$$

Now, for n large, another second order Taylor expansion gives

$$e^{t/\sqrt{n}} - 1 = \frac{t}{\sqrt{n}} + \frac{t^2}{2n} + o(n^{-1}).$$

Thus,

$$\begin{aligned} \ln\left(1 + p(e^{t/\sqrt{n}} - 1)\right) &= \frac{pt}{\sqrt{n}} + \frac{pt^2}{2n} + o(n^{-1}) - \frac{p^2t^2}{2n} + o(n^{-1}) + o(n^{-1}) \\ &= \frac{pt}{\sqrt{n}} + \frac{p(1-p)t^2}{2n} + o(n^{-1}) \end{aligned}$$

Plugging this in the expression for $M_{\tilde{S}_n}(t)$, we get

$$M_{\tilde{S}_n}(t) = \exp\left(-tp\sqrt{n} + pt\sqrt{n} + \frac{p(1-p)t^2}{2} + o_n(1)\right) = \exp\left(\frac{p(1-p)t^2}{2} + o_n(1)\right),$$

taking $n \rightarrow \infty$, $o_n(1)$ goes to 0, and we are left with

$$\lim_{n \rightarrow \infty} M_{\tilde{S}_n}(t) = \exp\left(\frac{p(1-p)t^2}{2}\right) = M_Z(t),$$

as wanted.

Solution to Exercise 3.4. *Solution 1, use Lemma 3.1.2:* by Lemma 3.1.2, it is sufficient to show that for any $k \in \mathbb{N}$,

$$\lim_{n \rightarrow \infty} P(S_n = k) = e^{-\lambda} \frac{\lambda^k}{k!} = P(N = k).$$

Now, S_n is the sum of n Bernoulli random variables with parameter p_n , thus $S_n \sim \text{Bin}(n, p_n)$, so

$$P(S_n = k) = \binom{n}{k} p_n^k (1 - p_n)^{n-k} = \frac{1}{k!} (1 - p_n)^n \prod_{i=1}^k \frac{p_n}{1 - p_n} (n - k + i).$$

Now, we first have

$$(1 - p_n)^n = \exp(n \ln(1 - p_n)) = \exp(n(-p_n + o(n^{-1})))$$

by a first order Taylor expansion, as $\lim_{n \rightarrow \infty} p_n n = \lambda$, so $p_n \leq C/n$ for some $C > 0$. We get,

$$\lim_{n \rightarrow \infty} (1 - p_n)^n = \lim_{n \rightarrow \infty} \exp(n(-p_n + o(n^{-1}))) = \exp\left(-\lim_{n \rightarrow \infty} np_n\right) = e^{-\lambda}.$$

On the other hand, (recall k does not depend on n),

$$\begin{aligned} \lim_{n \rightarrow \infty} \prod_{i=1}^k \frac{p_n}{1 - p_n} (n - k + i) &= \prod_{i=1}^k \left(\lim_{n \rightarrow \infty} \frac{p_n}{1 - p_n} (n - k + i) \right) \\ &= \prod_{i=1}^k \left(\lim_{n \rightarrow \infty} \frac{p_n}{1 - p_n} n \left(1 - \frac{k+i}{n}\right) \right) = \prod_{i=1}^k \lambda = \lambda^k, \end{aligned}$$

as $np_n \rightarrow \lambda$ by assumption, and thus $1 - p_n \rightarrow 1$. Combining, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} P(S_n = k) &= \lim_{n \rightarrow \infty} \frac{1}{k!} (1 - p_n)^n \prod_{i=1}^k \frac{p_n}{1 - p_n} (n - k + i) \\ &= \frac{1}{k!} \left(\lim_{n \rightarrow \infty} (1 - p_n)^n \right) \left(\lim_{n \rightarrow \infty} \prod_{i=1}^k \frac{p_n}{1 - p_n} (n - k + i) \right) = \frac{1}{k!} \lambda^k e^{-\lambda} \end{aligned}$$

which is exactly what we wanted.

Solution 2, use the MGF: we have that for $t \in \mathbb{R}$, the moment generating function of a $\text{Poi}(\lambda)$ is $\exp(\lambda(e^t - 1))$. Now, the MGF of S_n is the moment generating function of a $\text{Bin}(p_n, n)$, which is

$$\begin{aligned} E(e^{tS_n}) &= (1 + p_n(e^t - 1))^n = \exp(n \ln(1 + p_n(e^t - 1))) \\ &= \exp(n(p_n(e^t - 1) + O(p_n^2))) \xrightarrow{n \rightarrow \infty} \exp(\lambda(e^t - 1)) \end{aligned}$$

where we used a Taylor expansion $\ln(1 + z) = z + O(z^2)$, which is justified as $p_n n \rightarrow \lambda$, so $p_n(e^t - 1) \rightarrow 0$. We obtained that for any $t \in \mathbb{R}$, $E(e^{tS_n}) \rightarrow \exp(\lambda(e^t - 1))$, so, by Theorem 3.1.1, $S_n \xrightarrow{\text{Law}} \text{Poi}(\lambda)$.

Solution to Exercise 3.5. Let $X_1, X_2, \dots$ be a sequence of random variables. Let X be a random variable.

1. Suppose $X_n \xrightarrow{\text{Proba}} X$. Let $t \in \mathbb{R}$ be such that F_X is continuous at t . We need to show that

$$\lim_{n \rightarrow \infty} F_{X_n}(t) = F_X(t).$$

First, for any $\epsilon > 0$,

$$\begin{aligned} F_{X_n}(t) &= P(X_n \leq t) \\ &= P(X_n \leq t, |X_n - X| < \epsilon) + P(X_n \leq t, |X_n - X| \geq \epsilon) \\ &= P(X_n \leq t, |X_n - X| < \epsilon, X \leq t + \epsilon) + P(X_n \leq t, |X_n - X| \geq \epsilon) \\ &\leq P(X \leq t + \epsilon) + P(|X_n - X| \geq \epsilon) \end{aligned}$$

where we used the formula of total probability and the (deterministic) fact that if $x \leq t$ and $|x - y| \leq \epsilon$, then $y \leq t + \epsilon$. Thus,

$$\limsup_{n \rightarrow \infty} F_{X_n}(t) \leq P(X \leq t + \epsilon) + \limsup_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = F_X(t + \epsilon)$$

as $X_n \xrightarrow{\text{Proba}} X$. Now, $\epsilon > 0$ is arbitrary, so we can take $\epsilon \searrow 0$ on both sides, and use continuity of F_X at t to obtain

$$\limsup_{n \rightarrow \infty} F_{X_n}(t) \leq \lim_{\epsilon \searrow 0} F_X(t + \epsilon) = F_X(t).$$

Let us prove the reverse inequality. We have

$$\begin{aligned} F_{X_n}(t) &= P(X_n \leq t) \\ &\geq P(X_n \leq t, |X_n - X| < \epsilon, X \leq t - \epsilon) \\ &= P(|X_n - X| < \epsilon, X \leq t - \epsilon) \\ &= P(X \leq t - \epsilon) - P(|X_n - X| \geq \epsilon, X \leq t - \epsilon) \\ &\geq P(X \leq t - \epsilon) - P(|X_n - X| \geq \epsilon) \end{aligned}$$

where we used the same deterministic fact as before. As $X_n \xrightarrow{\text{Proba}} X$, we have that for any $\epsilon > 0$,

$$\liminf_{n \rightarrow \infty} F_{X_n}(t) \geq P(X \leq t - \epsilon) - \liminf_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = F_X(t - \epsilon).$$

As $\epsilon > 0$ is arbitrary, we can take $\epsilon \searrow 0$ on both sides and use continuity of F_X at t to obtain

$$\liminf_{n \rightarrow \infty} F_{X_n}(t) \geq \lim_{\epsilon \searrow 0} F_X(t - \epsilon) = F_X(t).$$

So, for every $t \in \mathbb{R}$ such that F_X is continuous at t , $F_X(t) \leq \liminf_{n \rightarrow \infty} F_{X_n}(t) \leq \limsup_{n \rightarrow \infty} F_{X_n}(t) \leq F_X(t)$, and thus $\lim_{n \rightarrow \infty} F_{X_n}(t) = F_X(t)$, which is convergence in law.

2. Suppose $X_n \xrightarrow{L^p} X$. Then, by Chebychev's inequality (the moment version), one has that for any $\epsilon > 0$

$$P(|X - X_n| \geq \epsilon) \leq \frac{E(|X_n - X|^p)}{\epsilon^p}.$$

Now, as $X_n \xrightarrow{L^p} X$, $E(|X_n - X|^p) \rightarrow 0$ as $n \rightarrow \infty$. So, for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X - X_n| \geq \epsilon) = 0,$$

which is the definition of convergence in probability.

Solution to Exercise 3.6. 1. The X_i are bounded random variables. In particular, $E(|X_1|) < \infty$, so the (strong) LLN implies that

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} E(X_1).$$

So, we know that S_n behaves like $nE(X_1)$ for n large. Moreover, still as the X_i 's are bounded, $E(X_1^2) < \infty$ and thus the CLT implies that

$$\frac{S_n - nE(X_1)}{\sqrt{n}} \xrightarrow{\text{a.s.}} \mathcal{N}(0, \text{Var}(X_1)).$$

In particular, we can approximate, for n large, the risk that S_n deviates from $nE(X_1)$ by more than $K\sqrt{n}$:

$$P(|S_n - nE(X_1)| \geq K\sqrt{n}) = P\left(\left|\frac{S_n - nE(X_1)}{\sqrt{n}}\right| \geq K\right) \approx \frac{2}{\sqrt{2\pi\text{Var}(X_1)}} \int_K^\infty e^{-\frac{x^2}{2\text{Var}(X_1)}}.$$

2. We have that $P(X_1 = 10^k) = 2^{-k-1}$. So,

$$E(|X_1|) = \sum_{k=0}^{\infty} 10^k 2^{-k-1} = \frac{1}{2} \sum_{k=0}^{\infty} 5^k = +\infty.$$

We thus have that the X_i 's do not have a first or second moment, the LLN and the CLT thus say nothing about S_n .

3. We have that

$$E(|X_1|) = 2 \int_1^\infty x \cdot x^{-3} dx = 2 \int_1^\infty x^{-2} dx = 2[-x^{-1}]_1^\infty = 2,$$

so the X_i 's have a first moment. The (strong) LLN thus tells us that

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} E(X_1).$$

So, as in the first case, we know that S_n behaves like $nE(X_1)$ for n large. But, unlike in the first case,

$$E(X_1^2) = 2 \int_1^\infty x^2 \cdot x^{-3} dx = 2 \int_1^\infty x^{-1} dx = 2[\ln(x)]_1^\infty = +\infty,$$

so the X_i 's do not have a second moment. The CLT does not give any information in this case, and we thus cannot estimate the risk that S_n deviates from $nE(X_1)$!

Solution to Exercise 3.7. 1. We have that our random variables X_i are Bernoulli random variables. We can suppose them independent as being repeated experiment without clear link between them (we can assume that we do not test twice the same card). As X_i gives 1 with probability ρ , we have $E(X_i) = \rho$ for all i 's. A good way to guess ρ is to find a way to approximation of $E(X_1)$. This is provided by the sLLN:

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} E(X_1).$$

We can thus take $\frac{1}{n} \sum_{i=1}^n X_i$ as an approximation of ρ as long as we take n large enough.

2. As we know nothing on ρ , we can only assume that the variance of X_i is given by some number σ^2 satisfying

$$\sigma^2 = \text{Var}(X_i) = \rho(1 - \rho) \leq 1.$$

We want to find n such that

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - E(X_1)\right| > 0.01\right) \leq 0.05.$$

We can use that $\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - E(X_i))$ is approximately a centred Gaussian random variable with variance σ^2 . Thus,

$$\begin{aligned} P\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - E(X_1)\right| > 0.01\right) &= P\left(\left|\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - E(X_i))\right| > 0.01\sqrt{n}\right) \\ &\approx P(|\mathcal{N}(0, \sigma^2)| > 0.01\sqrt{n}) \\ &= P(|\mathcal{N}(0, 1)| > \frac{\sqrt{n}}{100\sigma}) \\ &\leq P(|\mathcal{N}(0, 1)| > \frac{\sqrt{n}}{100}). \end{aligned}$$

As we know nothing about σ except $\sigma \leq 1$, we used the worst case scenario in the last line. Using the indication, we want n such that

$$\frac{\sqrt{n}}{100} \geq 1.96 \iff n \geq 38416.$$

3. Let $\tilde{S}_n = \frac{1}{\sigma\sqrt{n}} \sum_{i=1}^n (X_i - E(X_i))$. Let $Z \sim \mathcal{N}(0, 1)$. We start as in the previous point and replace the use of $\approx$ by a suitable use of Theorem 3.2.5. Note that $E(|X_i|^3) \leq 1$ is the only information we have on the third moment of the X_i 's. We have that (again using a worst case scenario on σ)

$$\begin{aligned} P(|\tilde{S}_n| > \frac{\sqrt{n}}{\sigma 100}) &\leq P(|\tilde{S}_n| > \frac{\sqrt{n}}{100}) \\ &\leq P(\tilde{S}_n \leq -\frac{\sqrt{n}}{100}) + 1 - P(\tilde{S}_n \leq \frac{\sqrt{n}}{100}) \\ &\leq 2 \frac{0.5 \cdot E(|X_1|^3)}{\sqrt{n}} + P(Z \leq -\frac{\sqrt{n}}{100}) + 1 - P(Z \leq \frac{\sqrt{n}}{100}) \\ &\leq \frac{1}{\sqrt{n}} + P(|Z| > \frac{\sqrt{n}}{100}) \end{aligned}$$

where we used Theorem 3.2.5 in the third inequality. Using the hint, we can, for example, search for n such that

$$\frac{1}{\sqrt{n}} \leq 0.01 \text{ and } P(|Z| > \frac{\sqrt{n}}{100}) \leq 0.04.$$

For the second inequality, the given Gaussian tails yield $n \geq 231^2 = 53361$, and we can thus take

$$n = \max(10000, 53361) = 53361.$$

Solution to Exercise 3.8. 1. Using the density (3.1), we have that the MGF of a χ_n^2 at $t < \frac{1}{2}$ is given by

$$g_n(t) := M_R(t) = E(e^{tR}) = \int_0^\infty \frac{x^{\frac{n}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} e^{tx} dx.$$

Proceed by induction over n . Our induction Hypotheses at level n is: “for $t < \frac{1}{2}$, $g_n(t) = (1 - 2t)^{-\frac{n}{2}}$ ”. Start by proving the case $n = 1$.

$$\begin{aligned} g_1(t) &= \int_0^\infty \frac{x^{-\frac{1}{2}} e^{-\frac{x}{2}}}{2^{\frac{1}{2}} \Gamma(\frac{1}{2})} e^{tx} dx = \int_0^\infty \frac{x^{-\frac{1}{2}} e^{-(\frac{1}{2}-t)x}}{2^{\frac{1}{2}} \Gamma(\frac{1}{2})} dx \\ &= \int_0^\infty \sqrt{\frac{1}{2}-t} \frac{y^{-\frac{1}{2}} e^{-y}}{2^{\frac{1}{2}} \Gamma(\frac{1}{2}) \frac{1}{2}-t} dy \\ &= (\frac{1}{2}-t)^{-\frac{1}{2}} \frac{1}{2^{\frac{1}{2}} \Gamma(\frac{1}{2})} \int_0^\infty y^{\frac{1}{2}-1} e^{-y} dy = (1-2t)^{-\frac{1}{2}}, \end{aligned}$$

where we used the change of variable $y = (\frac{1}{2}-t)x$, and the definition of the Gamma function (recall Definition 3.6.1). Then prove the case $n = 2$.

$$\begin{aligned} g_2(t) &= \frac{1}{2\Gamma(1)} \int_0^\infty e^{-(\frac{1}{2}-t)x} dx = \frac{1}{2\Gamma(1)} \left[-\frac{1}{(\frac{1}{2}-t)} e^{-(\frac{1}{2}-t)x} \right]_0^\infty \\ &= \frac{1}{(1-2t)} \frac{1}{\Gamma(1)} = \frac{1}{(1-2t)}, \end{aligned}$$

as $\Gamma(1) = 1$ (this is the same computation as the normalisation of the density of $\text{Exp}(1)$ random variables). We then assume the induction Hypotheses at levels $1, \dots, n \geq 2$, and show it at level $n+1$.

$$\begin{aligned} g_{n+1}(t) &= \frac{1}{2^{\frac{n+1}{2}} \Gamma(\frac{n+1}{2})} \int_0^\infty \underbrace{x^{\frac{n+1}{2}-1}}_{\downarrow} \underbrace{e^{-(\frac{1}{2}-t)x}}_{\uparrow} dx \\ &= \frac{1}{2 \cdot 2^{\frac{n-1}{2}} \Gamma(\frac{n-1}{2}) \frac{n-1}{2}} \int_0^\infty x^{\frac{n-1}{2}} x^{\frac{n-1}{2}-1} \frac{1}{\frac{1}{2}-t} e^{-(\frac{1}{2}-t)x} dx \\ &= \frac{1}{1-2t} \frac{1}{2^{\frac{n-1}{2}} \Gamma(\frac{n-1}{2})} \int_0^\infty x^{\frac{n-1}{2}-1} e^{-(\frac{1}{2}-t)x} dx \\ &= \frac{1}{1-2t} g_{n-1}(t) = \frac{1}{1-2t} (1-2t)^{-\frac{n-1}{2}} = (1-2t)^{-\frac{n+1}{2}} \end{aligned}$$

where we used integration by parts, and the induction Hypotheses at level $n-1$.

2. Let $Z \sim \mathcal{N}(0, 1)$. Then, we know from Lemma 2.6.7 that $\frac{X}{\sigma} \stackrel{\text{Law}}{=} Z$. So, $M_{\frac{X^2}{\sigma^2}} = M_{Z^2}$. We then compute: for $t < \frac{1}{2}$, using the density of the standard Gaussian,

$$\begin{aligned} M_{Z^2}(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-\frac{x^2}{2}} e^{tx^2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-(1-2t)\frac{x^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-\frac{y^2}{2}} \frac{dy}{\sqrt{1-2t}} = \frac{1}{\sqrt{1-2t}}, \end{aligned}$$

where we used the change of variable $y = \sqrt{1-2t}x$, and used the value of the Gaussian integral from Exercise 2.48.

3. As the X_i 's are independent, so are the $\frac{X_i^2}{\sigma^2}$. Thus, as $S = \sum_{i=1}^n \frac{X_i^2}{\sigma^2}$ is a sum of independent variables, $M_S(t) = M_{\frac{X_1^2}{\sigma^2}}(t) M_{\frac{X_2^2}{\sigma^2}}(t) \dots M_{\frac{X_n^2}{\sigma^2}}(t) = M_{\frac{X_1^2}{\sigma^2}}^n$ as the X_i 's all have the same law. Thus, by the previous point, $M_S(t) = (1 - 2t)^{-\frac{n}{2}}$.
4. S has the MGF of a χ_n^2 , MGF characterize the law, so $S \sim \chi_n^2$.

Solution to Exercise 3.9. 1. $(Y_1, \dots, Y_n, \bar{X}_n)$ is a Gaussian vector as it is the image by a linear function of a Gaussian vector (by Lemma 2.7.1). To compute the covariance matrix, we first compute:

$$\begin{aligned}\text{Cov}(X_i, \bar{X}_n) &= \frac{1}{n} \sum_{j=1}^n \text{Cov}(X_i, X_j) = \frac{\sigma^2}{n}, \\ \text{Var}(\bar{X}_n) &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) = \frac{\sigma^2}{n}.\end{aligned}$$

as the X_i 's are independent. Its covariance matrix is then given by

$$\begin{aligned}\text{Cov}(Y_i, Y_j) &= \text{Cov}(X_i - \bar{X}_n, X_j - \bar{X}_n) \\ &= \text{Cov}(X_i, X_j) + \text{Var}(\bar{X}_n) - \text{Cov}(X_i, \bar{X}_n) - \text{Cov}(X_j, \bar{X}_n) \\ &= \sigma^2 \mathbb{1}_{i=j} + \frac{\sigma^2}{n} - \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = \sigma^2 \mathbb{1}_{i=j} - \frac{\sigma^2}{n},\end{aligned}$$

and

$$\text{Cov}(Y_i, \bar{X}_n) = \text{Cov}(X_i - \bar{X}_n, \bar{X}_n) = \text{Cov}(X_i, \bar{X}_n) - \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0.$$

We deduce that $\bar{X}_n$ is independent from $(Y_1, \dots, Y_n)$.

2. Expanding the square,

$$\begin{aligned}\sum_{i=1}^n (X_i - \bar{X}_n)^2 &= \sum_{i=1}^n X_i^2 + n\bar{X}_n^2 - 2 \sum_{i=1}^n X_i \bar{X}_n \\ &= \sum_{i=1}^n X_i^2 + n\bar{X}_n^2 - 2n\bar{X}_n^2 = \sum_{i=1}^n X_i^2 - n\bar{X}_n^2.\end{aligned}$$

3. R and T are independent as R is a function of $(Y_1, \dots, Y_n)$ and T is a function of $\bar{X}_n$, which are independent by the first point. Now, $S = R + T$, so $M_S(t) = M_R(t)M_T(t)$.
4. Now, from (3.3), $\frac{1}{\sigma^2} \sum_{i=1}^n X_i^2 \sim \chi_n^2$, and $\frac{n}{\sigma^2} \bar{X}_n^2 \sim \chi_1^2$ (as $\sqrt{n}\bar{X}_n \sim \mathcal{N}(0, \sigma^2)$ from the first point). Thus,

$$M_S(t) = (1 - 2t)^{-\frac{n}{2}}, \quad M_T(t) = (1 - 2t)^{-\frac{1}{2}} \quad \forall t < \frac{1}{2},$$

and so, from the previous point, $M_R(t) = \frac{M_S(t)}{M_T(t)} = (1 - 2t)^{-\frac{n-1}{2}}$ which is the MGF of a χ_{n-1}^2 .

5. We reduce to the case of centred Gaussians. Introduce $Z_i = X_i - \mu$. Then, $Z_1, \dots, Z_n$ is an i.i.d. sequence of $\mathcal{N}(0, \sigma^2)$. Moreover, $\bar{Z}_n = \bar{X}_n - \mu$. So,

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu + \mu - \bar{X}_n)^2 = \frac{1}{\sigma^2} \sum_{i=1}^n (Z_i - \bar{Z}_n)^2.$$

Applying the result derived from the previous points, we get $\frac{1}{\sigma^2} \sum_{i=1}^n (Z_i - \bar{Z}_n)^2 \sim \chi_{n-1}^2$, which is the wanted claim.

A.4 Solutions to Chapter 4

Solution to Exercise 4.1. 1. The sample is $(X_1, X_2, \dots, X_{30})$ and the sample size is 30.

2. Yes.

3. The realization is the sequence of numbers

12, 15, 19, 21, 13, 6, 34, 9, 18, 4, 34, 5, 5, 27, 14, 34, 8,
26, 7, 25, 2, 36, 28, 21, 27, 2, 6, 14, 35, 12.

4. In a fair game, we would have that $X_1, \dots, X_{30}$ are independent random variables with uniform distribution on $\{0, 1, \dots, 36\}$. In particular, using the mean value of a uniform, we have

$$E(\bar{X}_{30}) = \frac{1}{30} \sum_{i=1}^{30} E(X_i) = \frac{1}{30} \cdot 30 \cdot \frac{36}{2} = 18,$$

$$E(Y) = \frac{1}{30} \cdot 30 \cdot P(\text{Uni}(\{0, 1, \dots, 36\}) \in [10, 26]) = \frac{|\{10, 11, \dots, 26\}|}{37} = \frac{17}{37} \approx 0.459.$$

The values we obtain from the sample realization are

$$\bar{X}_{30} = \frac{519}{30} = 17.3, \quad Y = \frac{12}{30} = 0.4.$$

We can say that they are fairly close to the ones of a fair game.

5. We will compare the expected value of $\bar{X}_{30}, Y$ in a the case of a fair game (uniform) to the values obtained in the sample. By the LLN, if these values are far apart, there is a high chance that the distribution of the X_i was *not* uniform. Let us compute the obtained values in each casinos:

	$\bar{X}_{30}$	Y
Fair game expect.	18	0.459
Casino 1	$\frac{693}{30} = 23.1$	$\frac{19}{30} \approx 0.63$
Casino 2	$\frac{382}{30} \approx 12.7$	$\frac{18}{30} = 0.6$
Casino 3	$\frac{537}{30} = 17.9$	$\frac{30}{30} = 1$

We can see that the value of $\bar{X}_{30}$ is way too high in the first case, way too low in the second but close to the fair game one in the third casino. On the other hand, we can see that the value of Y is completely off in the third casino. We can reasonably assume that non of these casinos are playing fair (indeed, the numbers were generated using a $\text{Uni}(\{10, \dots, 36\})$ in the first case, a $\text{Uni}(\{0, \dots, 26\})$ in the second, and a $\text{Uni}(\{10, \dots, 26\})$ in the third).

To idea of comparing what we obtained in the experiment to some a priori guess is the central idea of *Hypotheses testing*.

Solution to Exercise 4.2. For $a > 0$, let

$$I(a) = \int_a^\infty e^{-x^2/2} dx \geq 0.$$

One has

$$I(a) = \int_a^\infty \frac{1}{x} x e^{-x^2/2} dx = \left[-\frac{1}{x} e^{-x^2/2} \right]_a^\infty + \int_a^\infty \frac{1}{x^2} e^{-x^2/2} dx = \frac{e^{-a^2/2}}{a} - \int_a^\infty \frac{1}{x^2} e^{-x^2/2} dx$$

where we used integration by part. As $\int_a^\infty \frac{1}{x^2} e^{-x^2/2} dx \geq 0$,

$$I(a) \leq \frac{e^{-a^2/2}}{a},$$

which is the wanted upper bound. For the lower bound, notice that $\int_a^\infty \frac{1}{x^2} e^{-x^2/2} dx \leq \frac{1}{a^2} I(a)$, which gives

$$I(a) \geq \frac{e^{-a^2/2}}{a} - \frac{1}{a^2} I(a).$$

Re-arranging, we obtain

$$I(a) \geq (1 + a^{-2})^{-1} \frac{e^{-a^2/2}}{a},$$

which is the wanted lower bound.

Solution to Exercise 4.3. 1. We look at a two-outcomes experiments, that we can label 0,1. We thus need to understand the probability of getting 1 (tails will be 1 minus that). This gives a parameter $\theta = p = P(\text{coin gives } 1) \in [0, 1]$, so $\Theta = [0, 1]$, and $(\mathbb{P}_\theta)_{\theta \in \Theta}$ is the collection of Bernoulli laws: $(\text{Bern}(p))_{p \in [0, 1]}$.

2. Reasoning as in the first point, we can encode the result with 1,2,3,4,5 or 6. The parameters will be $p_i = P(\text{dice rolls } i)$, $i = 1, \dots, 6$, the parameter space will be

$$\Theta = \{(p_1, \dots, p_6) \in [0, 1]^6 : p_1 + \dots + p_6 = 1\},$$

and the family of laws $(\mathbb{P}_\theta)_{\theta \in \Theta}$ is the family of probability measures on $\{1, 2, \dots, 6\}$.

3. We saw that we can model inter-arrival times using exponential random variables. So, we can take the parameter space to be $\Theta = (0, +\infty)$, as exponential random variables are determined by a positive number λ . The family of laws is then $(\text{Exp}(\lambda))_{\lambda > 0}$.

Solution to Exercise 4.4. 1. X_1 is an estimator of p : it depends only on $X_1, X_2, \dots$ and not on p . Moreover, it is unbiased as $E(X_1) = P(X_1 = 1) = p$.

2. $X_1 + X_3$ is again an estimator of p : it depends only on $X_1, X_2, \dots$ and not on p , but it is biased $E(X_1 + X_3) = 2p$.

3. $\frac{1}{n} \sum_{i=1}^n (X_i - p)^2$ is **not** an estimator of p as it depends on p .

4. Same as the previous point.

5. $\left(\frac{1}{n} \sum_{i=1}^n X_i\right)^2$ is an estimator of p^2 as it does not depend on p^2 . It is a biased estimator as

$$\begin{aligned} E\left(\left(\frac{1}{n} \sum_{i=1}^n X_i\right)^2\right) &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n E(X_i X_j) \\ &= \frac{1}{n^2} \left(\sum_{i=1}^n E(X_i^2) + 2 \sum_{1 \leq i < j \leq n} E(X_i) E(X_j) \right) \\ &= \frac{1}{n^2} \left(\sum_{i=1}^n E(X_i) + 2 \sum_{1 \leq i < j \leq n} p^2 \right) \\ &= \frac{1}{n^2} (pn + (n^2 - n)p^2) = p^2 + \frac{p}{n} - \frac{p^2}{n}. \end{aligned}$$

Solution to Exercise 4.5. 1. It is μ as

$$\begin{aligned} P(X \geq \mu) &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mu}^{\infty} e^{-(x-\mu)^2/2\sigma^2} = -\frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mu}^{-\infty} e^{-(y+\mu)^2/2\sigma^2} \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\mu} e^{-(y+\mu)^2/2\sigma^2} = P(X \leq \mu) \end{aligned}$$

where we used the change of variable $y = 2\mu - x$.

2. It is 0.5 as

$$P(X \leq 0.5) = \frac{1}{2} = P(X \geq 0.5).$$

3. Any number in $(34, 45)$ gives a median for X : indeed, if $34 < a < 45$,

$$P(X \leq a) = P(X \in \{-1, 34\}) = \frac{1}{2}, \quad P(X \geq a) = P(X \in \{45, 55\}) = \frac{1}{2}.$$

We can take 40 and 41.3345 as medians.

4. We can take 0 as a median: as $X \stackrel{\text{Law}}{=} -X$, we have,

$$P(X \leq 0) = P(-X \leq 0) = P(X \geq 0).$$

5. By the previous point, 0 is a median for X_1 (f even implies $X_1 \stackrel{\text{Law}}{=} -X_1$). We need to show that it is the only one. Let $a > 0$, we have, as $X_1 \stackrel{\text{Law}}{=} -X_1$ and 0 is a median for X_1

$$\begin{aligned} P(X_1 \leq a) - P(X_1 \geq a) &= P(X_1 \leq 0) + P(X_1 \in (0, a]) - P(X_1 \geq 0) + P(X_1 \in [0, a)) \\ &= 2P(X_1 \in [0, a]) = 2 \int_0^a f(x) dx > 0, \end{aligned}$$

as $f > 0$, where we used that X_1 is continuous which implies $P(X_1 = 0) = P(X_1 = a) = 0$. In particular, a is not a median for X_1 . Proceed in the same way for $a < 0$. Now, as $X_1 \stackrel{\text{Law}}{=} -X_1$, $E(X_1) = 0$ which is also the median of X_1 . The empirical median is an estimator of $E(X_1)$ as it does not depend on

$E(X_1)$, remains to show that it is un-biased (has expectation 0 in this case). Let $\tilde{X}_1, \dots, \tilde{X}_n$ be the increasing re-arrangement of $X_1, X_2, \dots, X_n$ defined in the definition of the empirical median. Then, as $(X_1, \dots, X_n) \stackrel{\text{Law}}{=} (-X_1, \dots, -X_n)$, we have that

$$(\tilde{X}_1, \dots, \tilde{X}_n) \stackrel{\text{Law}}{=} (-\tilde{X}_n, \dots, -\tilde{X}_1)$$

as taking the negative of an ordered sequence reverses the order. In particular, we obtained that the empirical median has the same law as -1 times the empirical median, thus the empirical median has expected value 0.

Solution to Exercise 4.6. We compute the expected value of the estimator.

$$\begin{aligned} E(\hat{\sigma}_n^2) &= E(\overline{X^2}_n) - E(\overline{X}_n^2) \\ &= \frac{1}{n} \sum_{i=1}^n E(X_i^2) - \frac{1}{n^2} \sum_{i,j=1}^n E(X_i X_j) \\ &= E(X_1^2) - \frac{1}{n^2} \sum_{i=1}^n E(X_i^2) - \frac{1}{n^2} \sum_{i,j=1}^n \mathbb{1}_{i \neq j} E(X_i) E(X_j) \\ &= E(X_1^2) - \frac{1}{n} E(X_1^2) - \frac{1}{n^2} n(n-1) E(X_1)^2 = \frac{n-1}{n} \text{Var}(X_1), \end{aligned}$$

where we used that the X_i 's are i.i.d., and that there are $n^2 - n = n(n-1)$ elements of $\{1, \dots, n\}^2$ with non-matching entries. The quantity we wanted to estimate is $\text{Var}(X_1)$, so we have a bias

$$\text{Bias}_\theta(\hat{\sigma}_n^2) = E(\hat{\sigma}_n^2) - \text{Var}(X_1) = -\frac{1}{n} \text{Var}(X_1) \neq 0.$$

From the previous computation, we also get that

$$\text{Bias}_\theta\left(\frac{n}{n-1} \hat{\sigma}_n^2\right) = \frac{n}{n-1} E(\hat{\sigma}_n^2) - \text{Var}(X_1) = 0.$$

As $E(X_1^2) < \infty$, the LLN (Theorem 3.2.3), implies that $\overline{X}_n \xrightarrow{\text{a.s.}} E(X_1)$, and that $\overline{X^2}_n \xrightarrow{\text{a.s.}} E(X_1^2)$ as $n \rightarrow \infty$. So, $\hat{\sigma}_n^2$ converges towards $\text{Var}(X_1)$ (almost surely and therefore in probability). As $\frac{n}{n-1} \rightarrow 1$, $\frac{n}{n-1} \hat{\sigma}_n^2$ also converges towards $\text{Var}(X_1)$.

Solution to Exercise 4.7. $\hat{\tau}_n$ is an estimator of the covariance as it only depend on the sample $(X_1, Y_1), \dots, (X_n, Y_n)$ and not on any other parameters. We then compute

its expected value.

$$\begin{aligned}
E(\hat{\tau}_n) &= E\left(\frac{1}{n} \sum_{i=1}^n X_i Y_i - \frac{1}{n} \left(\sum_{i=1}^n X_i\right) \frac{1}{n} \left(\sum_{i=1}^n Y_i\right)\right) \\
&= \frac{1}{n} E\left(\sum_{i=1}^n X_i Y_i\right) - \frac{1}{n^2} E\left(\left(\sum_{i=1}^n X_i\right) \left(\sum_{i=1}^n Y_i\right)\right) \\
&= \frac{1}{n} \sum_{i=1}^n E(X_i Y_i) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n E(X_i Y_j) \\
&= \frac{1}{n} \sum_{i=1}^n E(X_i Y_i) - \frac{1}{n^2} \sum_{i=1}^n E(X_i Y_i) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \mathbb{1}_{i \neq j} E(X_i) E(Y_j) \\
&= \frac{1}{n} \cdot n \cdot E(XY) - \frac{1}{n^2} \cdot n \cdot E(XY) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \mathbb{1}_{i \neq j} E(X) E(Y) \\
&= E(X_1 Y_1) - \frac{1}{n} \cdot E(XY) - \frac{1}{n^2} \cdot (n^2 - n) \cdot E(X) E(Y) \\
&= \frac{n-1}{n} E(XY) - \frac{n-1}{n} \cdot E(X) E(Y) = \frac{n-1}{n} \text{Cov}(X, Y),
\end{aligned}$$

where we used linearity of E in the first three lines, the fact that (X_i, Y_i) and (X_j, Y_j) are independent when $i \neq j$ in the fourth line, and the fact that $(X_1, Y_1), \dots, (X_n, Y_n)$ all have the same law which is the law of (X, Y) in the fifth line. In particular,

$$E(\hat{\tau}_n) - \text{Cov}(X, Y) = -\frac{1}{n} \text{Cov}(X, Y) \neq 0,$$

so $\hat{\tau}_n$ is a biased estimator of $\text{Cov}(X, Y)$, and

$$E\left(\frac{n}{n-1} \hat{\tau}_n\right) - \text{Cov}(X, Y) = 0,$$

so $\frac{n}{n-1} \hat{\tau}_n$ is an unbiased estimator of $\text{Cov}(X, Y)$.

Solution to Exercise 4.8. 1. We can use the empirical mean as estimator: this gives an estimated average coffee consumption of $\frac{7.08}{10} = 0.708$ litre per week.

2. We can use the empirical mean as estimator: this gives an estimated average grade of $\frac{41.5}{10} = 4.15$.

3. We can use the empirical variance to obtain $\frac{6.5634}{10} - 0.708^2 = 0.155076$, or its unbiased version to obtain $\frac{10}{9} \cdot 0.155076 = 0.17230\bar{6}$.

4. We can use the empirical covariance to obtain $\frac{33.81}{10} - 0.708 \cdot 4.15 = 0.4428$, or its unbiased version to obtain $\frac{10}{9} \cdot 0.4428 = 0.429$.

Solution to Exercise 4.9. Let $X \sim \text{Uni}([0, \theta])$. Start by computing $E(g(X)) = E(X^a)$. We have

$$E(X^a) = \frac{1}{\theta} \int_0^\theta x^a dx = \frac{1}{\theta} \left[\frac{x^{a+1}}{a+1} \right]_0^\theta = \frac{\theta^a}{a+1}.$$

We therefore have

$$\theta = ((a+1)E(X^a))^{1/a},$$

so we want to use the method of moments with $g(x) = x^a$, and $h(x) = ((a+1)x)^{1/a}$. This gives the family of estimators of θ : for $n \geq 1$

$$\hat{\theta}_n = h\left(\frac{1}{n} \sum_{i=1}^n g(X_i)\right) = \left(\frac{(a+1)}{n} \sum_{i=1}^n X_i^a\right)^{1/a}.$$

Solution to Exercise 4.10. We know that if $X \sim \text{Exp}(\lambda)$, we have $E(X) = \frac{1}{\lambda}$. On our size 14 sample, we can take the moment method estimator of λ given by the choices $g(x) = x$, $h(x) = \frac{1}{x}$, which gives

$$\hat{\lambda}_{14} = \left(\frac{1}{14} \sum_{i=1}^{14} X_i\right)^{-1}.$$

On the given realisation of our sample, this estimator gives the value

$$\hat{\lambda}_{14}(1.12, 5.67, \dots, 6.37) = \left(\frac{1}{14} \cdot 212.79\right)^{-1} \approx 0.06579.$$

Solution to Exercise 4.11. 1. Recall that the moment generating function of a $\mathcal{N}(\mu, \sigma^2)$ is given by, for $t \in \mathbb{R}$,

$$M_X(t) = E(e^{tX}) = e^{\mu t + \sigma^2 t^2 / 2}.$$

In particular, as $Y = e^X$,

$$E(Y) = E(e^X) = M_X(1) = e^{\mu + \sigma^2 / 2}, \quad E(Y^2) = E(e^{2X}) = M_X(2) = e^{2\mu + 2\sigma^2},$$

so

$$\text{Var}(Y) = E(Y^2) - E(Y)^2 = e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1).$$

2. Start with estimating $\psi = \mu + \frac{\sigma^2}{2}$. By the first point, we have that

$$e^\psi = E(Y) \text{ so } \psi = \ln(E(Y)).$$

We can use the method of moments with $g(x) = x$, $h(x) = \ln(x)$ to obtain the estimator

$$\hat{\psi}_n(Y_1, \dots, Y_n) = \ln\left(\frac{1}{n} \sum_{i=1}^n Y_i\right).$$

We can then notice that

$$\frac{E(Y^2)}{E(Y)^2} = e^{\sigma^2}.$$

Following the idea of the method of moments, we can then use the empirical mean to estimate $E(Y^2)$ and $E(Y)^2$, and find an estimator of σ^2 by taking the log of the ratio of those estimations:

$$\hat{\sigma}_n^2(Y_1, \dots, Y_n) = \ln\left(\frac{\frac{1}{n} \sum_{i=1}^n Y_i^2}{\left(\frac{1}{n} \sum_{i=1}^n Y_i\right)^2}\right).$$

Solution to Exercise 4.12. 1. We are considering the case $(\mathbb{P}_\theta)_{\theta \in \Theta} = (\text{Exp}(\lambda))_{\lambda > 0}$. We are in the case of continuous random variables with density $f(x) = \lambda \mathbb{1}_{[0, +\infty)}(x)e^{-\lambda x}$, so, for $x_1, \dots, x_n \geq 0$, the likelihood function is given by

$$\mathcal{L}(x_1, \dots, x_n | \lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}.$$

To maximize $\mathcal{L}(x_1, \dots, x_n | \lambda)$, we can equivalently minimize the negative log-likelihood function:

$$-\ln(\mathcal{L}(x_1, \dots, x_n | \lambda)) = -n \ln(\lambda) + \lambda \sum_{i=1}^n x_i.$$

We can notice that this function is strictly convex as its second derivative in λ is equal to $\frac{n}{\lambda^2} > 0$. It therefore has a unique minimum, and its minimum is found at the critical point which is the solution to

$$-\frac{n}{\lambda} + \sum_{i=1}^n x_i = 0 \iff \lambda = \frac{n}{\sum_{i=1}^n x_i}.$$

So,

$$\text{MLE}(X_1, \dots, X_n) = \frac{n}{\sum_{i=1}^n X_i}.$$

2. We are considering the case $(\mathbb{P}_\theta)_{\theta \in \Theta} = (\text{Bern}(p))_{p \in [0, 1]}$. We are in the case of discrete random variables taking values in $\{0, 1\}$, so, for $x_1, \dots, x_n \in \{0, 1\}$, the likelihood function is given by

$$\mathcal{L}(x_1, \dots, x_n | p) = \prod_{i=1}^n P(X_i = x_i) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = (1-p)^n \left(\frac{p}{1-p}\right)^{\sum_{i=1}^n x_i}.$$

It will be simpler to write $p = 1 - e^{-\beta}$ with $\beta \geq 0$. As in the first point, we can equivalently minimize the negative log-likelihood function:

$$-\ln(\mathcal{L}(x_1, \dots, x_n | 1 - e^{-\beta})) = n\beta - \ln(e^\beta - 1) \sum_{i=1}^n x_i.$$

This function is then convex in β (second derivative is $\frac{e^{-\beta}}{(1-e^{-\beta})^2} \sum_{i=1}^n x_i \geq 0$). We can again find the location of the minima via the critical point equation:

$$n = \frac{e^\beta}{e^\beta - 1} \sum_{i=1}^n x_i = \frac{1}{p} \sum_{i=1}^n x_i.$$

So,

$$\text{MLE}(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n X_i.$$

3. We are considering the case $(\mathbb{P}_\theta)_{\theta \in \Theta} = (\text{Uni}([0, \theta]))_{\theta > 0}$. We are in the case of continuous random variables with density $f(x) = \frac{1}{\theta} \mathbb{1}_{[0, \theta]}(x)$, so, for $x_1, \dots, x_n \geq 0$, the likelihood function is given by

$$\mathcal{L}(x_1, \dots, x_n | \theta) = \prod_{i=1}^n \frac{1}{\theta} \mathbb{1}_{[0, \theta]}(x_i) = \theta^{-n} \prod_{i=1}^n \mathbb{1}_{[0, \theta]}(x_i).$$

Note then that this function is 0 if there is $i \in \{1, \dots, n\}$ such that $x_i > \theta$. Moreover, if $\theta \geq \max_{i=1, \dots, n} x_i$, $\mathcal{L}(x_1, \dots, x_n | \theta) = \theta^{-n}$ which is decreasing in θ . The value θ maximizing the likelihood is thus

$$\text{MLE}(X_1, \dots, X_n) = \max_{i=1, \dots, n} X_i.$$

The time between cars is a continuous random variable, so we could model the situation using either uniform or the exponential laws. As we saw several times, the exponential is in general better suited for modelling waiting times, that is the choice we make here (but the choice of uniform is also correct). We thus have the estimation of λ given by

$$\text{MLE}(8.15, \dots, 4.54) = \frac{10}{90.74} = \frac{1}{9.074} \approx 0.11$$

Solution to Exercise 4.13. 1. $\text{Est}_1(X_1, \dots, X_n) = 2\bar{X}_n$ is an estimator of θ as it does not depend on θ . It is an un-biased estimator as

$$E(\text{Est}_1(X_1, \dots, X_n)) = \frac{2}{n} \sum_{i=1}^n E(X_i) = \frac{2}{n} \sum_{i=1}^n \frac{\theta}{2} = \theta.$$

2. Recall first that by Exercise 4.12,

$$\text{MLE}(X_1, \dots, X_n) = \max(X_1, \dots, X_n).$$

We first compute the CDF of $\text{Est}_2 = \max(X_1, \dots, X_n)$. For $t \leq 0$, we obviously have $F_{\text{Est}_2}(t) = 0$, and for $t \geq \theta$, $F_{\text{Est}_2}(t) = 1$. Then, for $t \in [0, \theta]$,

$$F_{\text{Est}_2}(t) = \prod_{i=1}^n F_{X_i}(t) = \frac{t^n}{\theta^n}$$

by Exercise 2.32, and the fact that $P(X \leq t) = \frac{t}{\theta}$ for $t \in [0, \theta]$ and $X \sim \text{Uni}([0, \theta])$. Then, we can recall that the repartition function is a primitive of the density function, so that we can find the density $f_{\text{Est}_2}(x)$ as a constant c to be determined plus $F'_{\text{Est}_2}(x) = \mathbb{1}_{(0, \theta)}(x) \frac{nx^{n-1}}{\theta^n}$. As

$$\int_0^\theta \frac{nx^{n-1}}{\theta^n} dx = 1,$$

we have that $c = 0$, and $f_{\text{Est}_2}(x) = F'_{\text{Est}_2}(x) = \mathbb{1}_{(0, \theta)}(x) \frac{nx^{n-1}}{\theta^n}$ as wanted.

3. We compute the expected value of Est_2 :

$$\begin{aligned} E(\text{Est}_2) &= E(\max(X_1, \dots, X_n)) = \int_0^\theta \frac{nx^{n-1}}{\theta^n} x dx \\ &= \frac{n}{\theta^n} \int_0^\theta x^n dx = \frac{n}{\theta^n} \left[\frac{1}{n+1} x^{n+1} \right]_0^\theta = \frac{n}{(n+1)} \theta, \end{aligned}$$

so $E(\text{Est}_2) - \theta = -\frac{1}{n+1} \theta \xrightarrow{n \rightarrow \infty} 0$ so Est_2 is biased but asymptotically un-biased.

4. We first compute $\text{MSE}_\theta(\text{Est}_1)$:

$$\begin{aligned} \text{MSE}_\theta(\text{Est}_1) &= E((2\bar{X}_n - \theta)^2) = E((2\bar{X}_n - 2E(\bar{X}_n))^2) \\ &= 4\text{Var}(\bar{X}_n) = \frac{4}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{4}{n^2} n \frac{\theta^2}{12} = \frac{\theta^2}{3n}. \end{aligned}$$

Then compute $\text{MSE}_\theta(\text{Est}_2)$:

$$\begin{aligned} \text{MSE}_\theta(\text{Est}_2) &= E((\text{Est}_2 - \theta)^2) = \int_{-\infty}^{+\infty} (x - \theta)^2 f_{\text{Est}_2}(x) dx \\ &= \frac{1}{\theta^n} \int_0^\theta \underbrace{(x - \theta)^2}_{\downarrow} \underbrace{nx^{n-1}}_{\uparrow} dx = \frac{1}{\theta^n} [(x - \theta)^2 x^n]_0^\theta - \frac{2}{\theta^n} \int_0^\theta (x - \theta)x^n dx \\ &= -\frac{2}{\theta^n} \left(\int_0^\theta x^{n+1} dx - \theta \int_0^\theta x^n dx \right) = -\frac{2}{\theta^n} \left(\frac{\theta^{n+2}}{n+2} - \theta \frac{\theta^{n+1}}{n+1} \right) = \frac{2\theta^2}{(n+1)(n+2)} \end{aligned}$$

where we integrated by part in the fourth equality.

5. If we look at the difference between the two risks, we find

$$\begin{aligned} \text{MSE}_\theta(\text{Est}_1) - \text{MSE}_\theta(\text{Est}_2) &= \frac{\theta^2}{3n} - \frac{2\theta^2}{(n+1)(n+2)} \\ &= \theta^2 \frac{(n+1)(n+2) - 6n}{3n(n+1)(n+2)} = \theta^2 \frac{n^2 - 3n + 2}{3n(n+1)(n+2)}. \end{aligned}$$

In particular, for any $n \geq 3$, the second estimator has a strictly smaller mean square error than the first, whilst being biased!

Solution to Exercise 4.14. 1. We need to find the values $t_1, \dots, t_5$ such that

$$P(X \leq t_k) = \frac{k}{5}, \quad k = 1, \dots, 5.$$

We have

$$P(X \leq t_k) = \int_0^{t_k} dx = t_k.$$

So, the 5-quantiles of X are $\frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, 1$.

2. We need to find the values $t_1, \dots, t_5$ such that

$$P(Y \leq t_k) = \frac{k}{5}, \quad k = 1, \dots, 5.$$

We have

$$P(Y \leq t_k) = \int_0^{t_k} e^{-x} dx = 1 - e^{-t_k}.$$

So, t_k is the solution to $1 - e^{-t_k} = \frac{k}{5}$. The 5-quantiles of Y are thus $-\ln(1 - \frac{1}{5})$, $-\ln(1 - \frac{2}{5})$, $-\ln(1 - \frac{3}{5})$, $-\ln(1 - \frac{4}{5})$, $+\infty$.

Solution to Exercise 4.15. 1. $P(X \leq 0) = 0.5$ so $P(X \in [0, t]) = 0.4$, which gives $t = 1.28$.

2. $P(X > t) = 0.05$ so $P(X \in [0, t]) = 0.45$, which gives $t = 1.65$.

3. $P(|X| > t) = 0.05$ so $P(X > t) = 0.025$ so $P(X \in [0, t]) = 0.475$, which gives $t = 1.96$.

4. By the properties of Gaussian, see Lemma 2.6.7,

$$P(Y \leq t) = P(Y - 2 \leq t - 2) = P\left(\frac{Y-2}{2} \leq \frac{t-2}{2}\right) = P\left(X \leq \frac{t-2}{2}\right).$$

So, we look for t such that $P(X \leq \frac{t-2}{2}) = 0.8$. First, let us see that we cannot find such t with $t \leq 2$. Indeed, in that case we have $P(X \leq \frac{t-2}{2}) \leq P(X \leq 0) = 0.5$. Now, for $t > 2$, $P(X \leq \frac{t-2}{2}) = 0.8$ is equivalent to $P(X \in [0, \frac{t-2}{2}]) = 0.3$, so, using Table C.2, $\frac{t-2}{2} = 0.84$, so $t = 3.68$.

Solution to Exercise 4.16. 1. We will prove it by computing the CDF of Z^2 . For $t \leq 0$, we have $F_{X^2}(t) = P(X^2 \leq t) = 0$, as for χ_1^2 random variables. For $t > 0$, we have

$$\begin{aligned} F_{X^2}(t) &= P(X^2 \leq t) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \mathbf{1}_{(-\infty, t]}(x^2) e^{-x^2/2} dx \\ &= \frac{1}{\Gamma(2^{-1})\sqrt{2}} 2 \int_0^{+\infty} \mathbf{1}_{[0, t]}(x^2) e^{-x^2/2} dx \\ &= \frac{1}{\Gamma(2^{-1})\sqrt{2}} 2 \int_0^{+\infty} \mathbf{1}_{[0, t]}(y) \frac{1}{2\sqrt{y}} e^{-y/2} dy \\ &= \int_{-\infty}^{+\infty} \mathbf{1}_{(-\infty, t]}(y) f_{\chi_1^2}(y) dy \end{aligned}$$

which is the repartition function of the χ_1^2 law, where we used that $x \mapsto \mathbf{1}_{[0, t]}(x^2) e^{-x^2/2}$ is even, and did the change of variable $y = x^2$, $dx = \frac{1}{2\sqrt{y}} dy$.

2. By the first point, if $Z \sim \mathcal{N}(0, 1)$, $Z^2 \sim \chi_1^2$. We need to show that for any $t \in \mathbb{R}$, the CDF of $\frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2$ converges to the CDF of a χ_1^2 , which is equivalent to show that for any $t \in \mathbb{R}$,

$$P\left(\frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2 \leq t\right) \xrightarrow{n \rightarrow \infty} P(Z^2 \leq t).$$

For $t < 0$, both sides are 0 for any n , so there is nothing to show. Let $t \geq 0$. Then,

$$P\left(\frac{1}{n}\left(\sum_{i=1}^n X_i\right)^2 \leq t\right) = P\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i \in [-\sqrt{t}, \sqrt{t}]\right) \xrightarrow{n \rightarrow \infty} P(Z \in [-\sqrt{t}, \sqrt{t}])$$

by the CLT as $\text{Var}(X_1) = 1$, $E(X_1) = 0$. Finally, $P(Z \in [-\sqrt{t}, \sqrt{t}]) = P(Z^2 \leq t)$ which is what we wanted. Now, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} P\left(\left(\sum_{i=1}^n X_i\right)^2 \in [2.26n, 2.5n]\right) &= \lim_{n \rightarrow \infty} P\left(\frac{1}{n}\left(\sum_{i=1}^n X_i\right)^2 \in [2.26, 2.5]\right) \\ &= P(Z^2 \in [2.26, 2.5]) = P(Z^2 \leq 2.5) - P(Z^2 \leq 2.26) = 0.886 - 0.867 = 0.019 \end{aligned}$$

by the previous convergence, the fact that the χ_1^2 law is continuous, and Table C.5.

3. We proceed similarly to the first point and compute the CDF of $X^2 + Y^2$. As in the first point, it is sufficient to consider $t \geq 0$. Then,

$$\begin{aligned} F_{X^2+Y^2}(t) &= P(X^2 + Y^2 \leq t) \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dy \mathbb{1}_{[0,t]}(X^2 + Y^2) e^{-x^2/2} e^{-y^2/2} \\ &= \frac{1}{2\pi} \int_0^{2\pi} d\theta \int_0^\infty dr \mathbb{1}_{[0,t]}(r^2) r e^{-r^2/2} \end{aligned}$$

where we used polar coordinates, see Remark 2.2.6:

$$x = r \cos(\theta), \quad y = r \sin(\theta), \quad dx dy = r d\theta dr,$$

so

$$x^2 + y^2 = r^2(\cos^2(\theta) + \sin^2(\theta)) = r^2.$$

Now, as $\Gamma(1) = 1$, $\int_0^{2\pi} d\theta = 2\pi$, and using the change of variable $s = r^2$, $dr = \frac{1}{2\sqrt{s}} ds$, the last expression is equal to

$$\frac{1}{\Gamma(1)} \int_0^\infty ds \mathbb{1}_{[0,t]}(s) \frac{\sqrt{s}}{2\sqrt{s}} e^{-s/2} = \int_{-\infty}^\infty ds \mathbb{1}_{(-\infty,t]}(s) f_{\chi_2^2}(s)$$

which is indeed the CDF of a χ_2^2 evaluated at t .

4. By the previous point, $X^2 + Y^2 \sim \chi_2^2$, so by Table C.6, $P(X^2 + Y^2 \leq 4.47) = 0.893$, so $C = 4.47$. Then, still by Table C.6, and the fact that $X^2 + Y^2$ is a continuous random variable,

$$P(X^2 + Y^2 \in [3, 4]) = P(X^2 + Y^2 \leq 4) - P(X^2 + Y^2 \leq 3) = 0.865 - 0.777 = 0.088.$$

Solution to Exercise 4.17. 1. We can estimate the percentage using the empirical mean: $\frac{19}{190} = \frac{1}{10} = 10\%$. More formally, we want to let X be a Bernoulli random variable that gives 1 if a uniformly sampled individual in the population

answers the question wrong and 0 else. We want to estimate the parameter of this law, $p \in [0, 1]$. We then write $X_i = 1$ if the i th person answers wrongly to the question, and $X_i = 0$ else. $X_1, \dots, X_{190}$ is then a 190-sample of law $\text{Bern}(p)$. As $p = E(X)$, we can use the empirical mean to estimate it. With the result of the experiment, this gives the estimation $\hat{p}(x_1, \dots, x_{190}) = \frac{19}{190} = \frac{1}{10}$.

2. We seek an interval $I(X_1, \dots, X_{190})$ such that the event $p \in I$ has probability $\frac{95}{100}$. We can use the CLT and Gaussian approximation (supposing that 190 is large enough) to say that

$$\frac{1}{\sqrt{190p(1-p)}} \sum_{i=1}^{190} (X_i - p) \approx \mathcal{N}(0, 1).$$

We can take an interval of the form

$$I(X_1, \dots, X_{190}) = [\hat{p} - \delta, \hat{p} + \delta].$$

To find a suitable δ , we compute

$$\begin{aligned} \sup_{p \in [0,1]} P(|\hat{p} - p| \geq \delta) &= \sup_{p \in [0,1]} P\left(\left|\frac{1}{190} \sum_{i=1}^{190} (X_i - p)\right| \geq \delta\right) \\ &= \sup_{p \in [0,1]} P\left(\left|\frac{1}{\sqrt{190p(1-p)}} \sum_{i=1}^{190} (X_i - p)\right| \geq \frac{\sqrt{190}\delta}{\sqrt{p(1-p)}}\right) \\ &\approx \sup_{p \in [0,1]} P\left(|Z| \geq \frac{\sqrt{190}\delta}{\sqrt{p(1-p)}}\right) \\ &= P(|Z| \geq 2\sqrt{190}\delta) \approx 2P(Z \geq 27.57\delta) \end{aligned}$$

where $Z \sim \mathcal{N}(0, 1)$. We then seek δ such that

$$2P(Z \geq 27.57\delta) = \frac{5}{100} \iff P(0 \leq Z \leq 27.57\delta) = 0.475.$$

Looking at the Gaussian table C.2, we find that $27.57\delta = 1.96$ solves this, so $\delta = 0.071$ does the job.

Solution to Exercise 4.18. We assume that an apricot weight follows a Gaussian law $\mathcal{N}(\mu, \sigma^2)$. The variance is a priori unknown, but we assume it is given by the empirical variance: the empirical mean of our sample realisation is $374, 8/10 = 37.48$, and the empirical variance is $215.036/10 = 21.5036 \approx 21.5$. We can thus make the assumption that the weight of an apricot is following a $\mathcal{N}(\mu, 21.5)$. As we know the weight of an apricot is non-negative, we can take as parameter space $\Theta = (0, +\infty)$, $\theta = \mu$. Using these assumptions, we can find a confidence interval for μ centred around the empirical mean: we look for $\delta > 0$ such that for all $\mu > 0$,

$$P(|\bar{X}_{10} - \mu| \geq \delta) = 10\%$$

where $X_1, \dots, X_{10}$ are i.i.d. $\mathcal{N}(\mu, 21.5)$. By the properties of Gaussian, this is the same as finding $\delta > 0$ such that

$$P(|Z| \geq \frac{\delta}{\sqrt{2.15}}) = 10\% \iff P(Z \geq \frac{\delta}{\sqrt{2.15}}) = 5\%$$

where $Z \sim \mathcal{N}(0, 1)$, as $\bar{X}_{10} - \mu \sim \mathcal{N}(0, 2.15)$. For such a δ (that we can find using tables or R), $[\bar{X}_{10} - \delta, \bar{X}_{10} + \delta]$ is a 90% confidence interval.

A possible meaning of “ $[\bar{X}_{10} - \delta, \bar{X}_{10} + \delta]$ is a 90% confidence interval for μ ” is that if we were to perform our sampling (picking 10 apricots at random) several times, we will see that the empirical mean of the sample is δ -close to the real mean 90% of the time. So our assumption $\mu \in [\bar{X}_{10} - \delta, \bar{X}_{10} + \delta]$ is correct in 90% of our sampling repetitions.

Solution to Exercise 4.19. The setup is that we have $X_1, \dots, X_n$ a sample of $\mathcal{N}(\mu, \sigma^2)$. We use the confidence interval

$$I(X_1, \dots, X_n) = [\bar{X}_n - \delta, \bar{X}_n + \delta]$$

for some $\delta > 0$ as a confidence interval for μ . The confidence level of our test is

$$P(\mu \in I(X_1, \dots, X_n)) = P(|\bar{X}_n - \mu| \leq \delta) = P(|\mathcal{N}(0, 1)| \leq \frac{\delta\sqrt{n}}{\sigma}).$$

In a realisation $x_1, \dots, x_{30}$ of our sample with $n = 30$, we find the interval $[6.04, 6.18]$. In particular, we have that $\bar{x}_{30} = 6.11$ (the midpoint) and $\delta = 0.07$. In particular, we know from the given information that

$$P(|\mathcal{N}(0, 1)| \leq \frac{0.07\sqrt{30}}{\sigma}) = 0.9.$$

so

$$P(\mathcal{N}(0, 1) \in [0, \frac{0.07\sqrt{30}}{\sigma}]) = 0.45. \quad (\text{A.3})$$

Looking at Table C.2, this gives $\frac{0.07\sqrt{30}}{\sigma} = 1.64$, which is $\sigma = \frac{0.07\sqrt{30}}{1.64}$.

1. If we want to divide the length of the confidence interval, i.e.: to have $\delta = 0.035$ instead of $\delta = 0.07$, we need n such that

$$P(|\mathcal{N}(0, 1)| \leq \frac{0.035\sqrt{n}}{\sigma}) = P(|\mathcal{N}(0, 1)| \leq \frac{0.07\sqrt{30}}{\sigma}) = 0.9,$$

we can thus take a sample size n such that $\frac{0.035\sqrt{n}}{\sigma} = \frac{0.07\sqrt{30}}{\sigma}$ so, $n = 120$.

2. If we want to obtain a 95% confidence interval with the same value $\delta = 0.07$, we need to take n such that

$$P(|\mathcal{N}(0, 1)| \leq \frac{0.07\sqrt{n}}{\sigma}) = 0.95.$$

using the value of σ we had, we need to solve

$$P(|\mathcal{N}(0, 1)| \leq \frac{1.64\sqrt{n}}{\sqrt{30}}) = 0.95,$$

which is equivalent to

$$P(\mathcal{N}(0, 1) \in [0, \frac{1.64\sqrt{n}}{\sqrt{30}}]) = 0.475.$$

Looking at Table C.2, we find that this is the same as asking $\frac{1.64\sqrt{n}}{\sqrt{30}} = 1.96$, so $n = 43$.

Solution to Exercise 4.20. 1. As $E(X) = \frac{\theta}{2}$, the condition “ $E(X) \geq 10$ ” is equivalent to $\theta \geq 20$, so we can take $\Theta_0 = [20, +\infty)$, $\Theta_1 = \Theta \setminus \Theta_0 = (0, 20)$.

2. For any $C > 23.45$.

3. We first compute the risk level α .

$$\alpha = \sup_{\theta \in \Theta_0} P(X \in D_C) = \sup_{\theta \geq 20} P(X < C) = \sup_{\theta \geq 20} \min\left(\frac{C}{\theta}, 1\right) = \min\left(\frac{C}{20}, 1\right).$$

In particular, if $C \geq 20$, the confidence level of the test is 0. We then compute the power of the test $1 - \beta$:

$$\begin{aligned} \beta &= \sup_{\theta \in \Theta_1} P(X \notin D_C) = \sup_{0 < \theta < 20} P(X \geq C) = \sup_{0 < \theta < 20} \max\left(\frac{\theta - C}{\theta}, 0\right) \\ &= \sup_{0 < \theta < 20} \max\left(1 - \frac{C}{\theta}, 0\right) = \max\left(1 - \frac{C}{20}, 0\right). \end{aligned}$$

In particular, if $C \geq 20$, the power of the test is 1.

Solution to Exercise 4.21. We first notice that

$$\mathbb{1}_{A_n}(i_1, \dots, i_n) = \prod_{j, l \in \{1, \dots, N\}: j \neq l} \mathbb{1}_{i_j \neq i_l} = \prod_{j=1}^n \prod_{l \in \{1, \dots, N\} \setminus j} \mathbb{1}_{i_j \neq i_l}.$$

Then, note that the number of sequences in $\{1, \dots, N\}^n$ with all their entries being different is $\frac{N!}{(N-n)!}$. Thus,

$$Q_{N,n}(A_n) = \frac{\frac{N!}{(N-n)!}}{N^n} = \frac{N!}{(N-n)! \cdot N^n}.$$

1. We have that for any event $B \subset \{1, \dots, N\}^n$,

$$\begin{aligned} Q_{N,n}(B | A_n) &= \frac{Q_{N,n}(B \cap A_n)}{Q_{N,n}(A_n)} \\ &= \frac{(N-n)! \cdot N^n}{N!} \sum_{(i_1, \dots, i_n) \in B} \mathbb{1}_{A_n}(i_1, \dots, i_n) q_{N,n}(i_1, \dots, i_n) \\ &= \frac{(N-n)! \cdot N^n}{N!} \sum_{(i_1, \dots, i_n) \in B} \left(\prod_{j=1}^n \prod_{l \in \{1, \dots, N\} \setminus j} \mathbb{1}_{i_j \neq i_l} \right) \frac{1}{N^n} \\ &= \sum_{(i_1, \dots, i_n) \in B} \frac{(N-n)!}{N!} \prod_{j=1}^n \prod_{l \in \{1, \dots, N\} \setminus j} \mathbb{1}_{i_j \neq i_l} \\ &= \sum_{(i_1, \dots, i_n) \in B} p_{N,n}(i_1, \dots, i_n) = P_{N,n}(B). \end{aligned}$$

2. First, for $x \in [0, 0.5]$,

$$1 - x = \exp(\ln(1 - x)) = \exp\left(\int_0^x \frac{-1}{1-s} ds\right) \geq \exp\left(-2 \int_0^x ds\right) = e^{-2x},$$

as $s \in [0, x] \subset [0, 0.5]$ implies $\frac{1}{1-s} \leq 2$. Then, for $x \geq 0$,

$$e^{-x} - 1 = - \int_0^x e^{-s} ds \geq - \int_0^x 2 ds = -2x$$

as $e^{-s} \leq 1$ for $s \geq 0$. So, for $x \geq 0$, $e^{-x} \geq 1 - x$. Now,

$$\begin{aligned} Q_{N,n}(A_n) &= \frac{N!}{(N-n)! \cdot N^n} = \prod_{k=0}^{n-1} \frac{N-k}{N} = \prod_{k=0}^{n-1} \left(1 - \frac{k}{N}\right) \geq \prod_{k=0}^{n-1} e^{-2k/N} \\ &= \exp\left(-\frac{2}{N} \sum_{k=0}^{n-1} k\right) = \exp\left(-\frac{1}{N}(n-1)n\right) \geq e^{-\frac{n^2}{N}} \geq 1 - \frac{n^2}{N} \end{aligned}$$

as $\frac{k}{N} < \frac{n}{N} \leq \frac{1}{2}$ by assumption.

Solution to Exercise 4.22. We divide the student into two groups of equal sizes: 30 are given the drug, and 30 are given a placebo or nothing at all. Denote $X_1, \dots, X_{30}$ the sample of the test results of students with the drug, and $Y_1, \dots, Y_{30}$ the sample of the test results of student without the drug. We assume that all test results are independent and Gaussian, all with the same variance. In mathematical terms, we assume that there are $\mu_X, \mu_Y \in \mathbb{R}$, and $\sigma^2 > 0$ such that $X_1, \dots, X_{30}, Y_1, \dots, Y_{30}$ form an independent sequence and

$$X_i \sim \mathcal{N}(\mu_X, \sigma^2), \quad i = 1, \dots, 30, \quad Y_i \sim \mathcal{N}(\mu_Y, \sigma^2), \quad i = 1, \dots, 30.$$

We want to perform a two-sample t-test for testing the Hypotheses H_0 “ $\mu_X = \mu_Y$ ” (so $\Theta = \mathbb{R}^2 \times (0, +\infty)$, $\theta = (\mu_X, \mu_Y, \sigma^2)$, and $\Theta_0 = \{(\mu, \mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}$). The test statistic is therefore

$$T(X_1, \dots, X_{30}, Y_1, \dots, Y_{30}) = \frac{\sqrt{30}(\bar{X}_{30} - \bar{Y}_{30})}{\sqrt{s_X^2 + s_Y^2}}$$

with

$$s_X^2 = \frac{1}{29} \sum_{i=1}^{30} (X_i - \bar{X}_{30})^2, \quad s_Y^2 = \frac{1}{29} \sum_{i=1}^{30} (Y_i - \bar{Y}_{30})^2.$$

Under the null Hypotheses H_0 , $\mu_X = \mu_Y$ and therefore, by our assumptions on the Gaussian nature of the variables, $T \sim \text{Student}_t(2 \cdot 30 - 2) = \text{Student}_t(58)$ under H_0 . We take a rejection region of the form

$$\{|T(X_1, \dots, X_{30}, Y_1, \dots, Y_{30})| \geq \delta\},$$

with δ such that $P(\text{Student}_t(58) \in [-\delta, \delta]) = 0.9$, as we want a confidence level 0.9. Using that the Student law is symmetric,

$$P(\text{Student}_t(58) \in [-\delta, \delta]) = 2P(\text{Student}_t(58) \in [0, \delta]).$$

Looking at the given numbers, we have that $\delta = 1.67$ does the job. We then look at the value of our test statistic on the given realisation of the samples.

$$T(x_1, \dots, x_{30}, y_1, \dots, y_{30}) \approx \frac{\sqrt{30}(\frac{1828}{30} - \frac{1200}{30})}{\sqrt{\frac{6880}{29} + \frac{5426}{29}}} = \frac{628 \cdot \sqrt{29}}{\sqrt{30} \cdot \sqrt{12306}} \approx 5.57$$

which is well above $\delta = 1.67$, so we reject the null Hypotheses “the drug has no influence on the test results”.

Solution to Exercise 4.23. 1. We compute using the density of the Fisher law:
for $m > 2$,

$$\begin{aligned}
E(X) &= \int_{-\infty}^{+\infty} x f_X(x) dx \\
&= \frac{1}{B(n/2, m/2)} \int_0^{+\infty} \left(\frac{nx}{nx+m} \right)^{n/2} \left(1 - \frac{nx}{nx+m} \right)^{m/2} dx \\
&= \frac{m}{n} \frac{1}{B(n/2, m/2)} \int_0^1 \underbrace{y^{\frac{n}{2}}}_{\downarrow} \underbrace{(1-y)^{\frac{m}{2}-2}}_{\uparrow} dy \\
&= \frac{mn}{n2(\frac{m}{2}-1) B(n/2, m/2)} \int_0^1 y^{\frac{n}{2}-1} (1-y)^{\frac{m}{2}-1} dy \\
&= \frac{m}{m-2} \frac{1}{B(n/2, m/2)} B(n/2, m/2) = \frac{m}{m-2},
\end{aligned}$$

where we used the change of variable $y = \frac{nx}{nx+m}$, $x = \frac{my}{n(1-y)}$, $dx = \frac{m}{n(1-y)^2} dy$, in the third line, and integrated by part in the fourth.

2. We start by computing the second moment of X using the same change of variable as in the previous point. For $m > 4$,

$$\begin{aligned}
E(X^2) &= \int_{-\infty}^{+\infty} x^2 f_X(x) dx \\
&= \frac{1}{B(n/2, m/2)} \int_0^{+\infty} x \left(\frac{nx}{nx+m} \right)^{n/2} \left(1 - \frac{nx}{nx+m} \right)^{m/2} dx \\
&= \frac{1}{B(n/2, m/2)} \int_0^1 \frac{my}{n(1-y)} y^{\frac{n}{2}} (1-y)^{\frac{m}{2}} \frac{m}{n(1-y)^2} dy \\
&= \frac{m^2}{n^2} \frac{1}{B(n/2, m/2)} \int_0^1 \underbrace{y^{\frac{n}{2}+1}}_{\downarrow} \underbrace{(1-y)^{\frac{m}{2}-3}}_{\uparrow} dy \\
&= \frac{m^2}{n^2} \frac{1}{B(n/2, m/2)} \left(\frac{n}{2} + 1 \right) \frac{1}{\frac{m}{2}-2} \int_0^1 \underbrace{y^{\frac{n}{2}}}_{\downarrow} \underbrace{(1-y)^{\frac{m}{2}-2}}_{\uparrow} dy \\
&= \frac{m^2(n+2)}{n^2(m-4) B(n/2, m/2)} \frac{n}{2} \frac{1}{\frac{m}{2}-1} \int_0^1 y^{\frac{n}{2}-1} (1-y)^{\frac{m}{2}-1} dy \\
&= \frac{m^2(n+2)}{n(m-4)(m-2)}
\end{aligned}$$

where we used integration by part in the fifth and sixth lines. We then compute the variance:

$$\text{Var}(X) = E(X^2) - E(X)^2 = \frac{m^2(n+2)}{n(m-4)(m-2)} - \frac{m^2}{(m-2)^2} = \frac{2m^2(m+n-2)}{n(m-2)^2(m-4)}.$$

3. We can use Chebychev's inequality and the previous point,

$$P(|X - E(X)| \geq \epsilon) \leq \frac{\text{Var}(X)}{\epsilon^2} = \frac{2n^2(2n-2)}{n(n-2)^2(n-4)\epsilon^2} = \frac{4n(n-1)}{(n-2)^2(n-4)\epsilon^2} \leq \frac{12}{(n-4)\epsilon^2}$$

as $n(n-1) \leq 3(n-2)^2$ for $n > 4$.

Solution to Exercise 4.24. We have 74 politicians divided into three groups of sizes 12, 26, 36. Denote $X_1, \dots, X_{12}$ a sample of the first group, $Y_1, \dots, Y_{26}$ a sample of the second group, and $Z_1, \dots, Z_{36}$ a sample of the third group. We make the assumption that weekly wine consumption is normally distributed and that the variance of an individual consumption is not depending on the individual. Mathematically, we assume that there are $\mu_{X,1}, \dots, \mu_{X,12} \in \mathbb{R}$, $\mu_{Y,1}, \dots, \mu_{Y,26} \in \mathbb{R}$, $\mu_{Z,1}, \dots, \mu_{Z,36} \in \mathbb{R}$, and $\sigma^2 > 0$ such that

$$\begin{aligned} X_i &\sim \mathcal{N}(\mu_{X,i}, \sigma^2), \quad i = 1, \dots, 12, \\ Y_i &\sim \mathcal{N}(\mu_{Y,i}, \sigma^2), \quad i = 1, \dots, 26, \\ Z_i &\sim \mathcal{N}(\mu_{Z,i}, \sigma^2), \quad i = 1, \dots, 36. \end{aligned}$$

We then use the test statistic

$$\begin{aligned} T(X_1, \dots, X_{12}, Y_1, \dots, Y_{26}, Z_1, \dots, Z_{36}) \\ = \frac{71(12(\bar{X}_{12} - \bar{\mu}_{74})^2 + 26(\bar{Y}_{26} - \bar{\mu}_{74})^2 + 36(\bar{Z}_{36} - \bar{\mu}_{74})^2)}{2(s_{X,12}^2 + s_{Y,26}^2 + s_{Z,36}^2)} \end{aligned}$$

where

$$\begin{aligned} \bar{X}_{12} &= \frac{1}{12} \sum_{i=1}^{12} X_i, \quad \bar{Y}_{26} = \frac{1}{26} \sum_{i=1}^{26} Y_i, \quad \bar{Z}_{36} = \frac{1}{36} \sum_{i=1}^{36} Z_i, \\ \bar{\mu}_{74} &= \frac{1}{74} \left(\sum_{i=1}^{12} X_i + \sum_{j=1}^{26} Y_j + \sum_{k=1}^{36} Z_k \right) = \frac{12\bar{X}_{12} + 26\bar{Y}_{26} + 36\bar{Z}_{36}}{74} \\ s_{X,12}^2 &= \sum_{i=1}^{12} (X_i - \bar{X}_{12})^2, \quad s_{Y,26}^2 = \sum_{i=1}^{26} (Y_i - \bar{Y}_{26})^2, \quad s_{Z,36}^2 = \sum_{i=1}^{36} (Z_i - \bar{Z}_{36})^2. \end{aligned}$$

By assumptions on the sample, $T \sim F(2, 71)$. As suggested in the description of the test, we take a rejection region of the form

$$\{T > \delta\}.$$

We now want a confidence level 0.9, which is achieved by picking δ such that $P(F(2, 71) \leq \delta) = 0.9$, which is given to be $\delta = 2.379 \approx 2.4$. We now perform the test on the given realisation. We have

$$\bar{x}_{12} = \frac{60.9}{12} \approx 5.1, \quad \bar{y}_{26} = \frac{88.4}{26} = 3.4, \quad \bar{z}_{36} = \frac{86.3}{36} \approx 2.4, \quad \bar{\mu}_{74} \approx \frac{12 \cdot 5.1 + 26 \cdot 3.4 + 36 \cdot 2.4}{74} \approx 3.2,$$

and, from the given data,

$$s_{x,12}^2 \approx 5.7, \quad s_{y,26}^2 = 28.6, \quad s_{z,36}^2 = 41.1.$$

So, on our realisation,

$$\begin{aligned} T(x_1, \dots, x_{12}, y_1, \dots, y_{26}, z_1, \dots, z_{36}) \\ \approx \frac{71(12(5.1 - 3.2)^2 + 26(3.4 - 3.2)^2 + 36(2.4 - 3.2)^2)}{2(5.7 + 28.6 + 41.1)} \approx 31.7 \end{aligned}$$

This is well above $\delta = 2.4$, so we reject the null Hypotheses.

Solution to Exercise 4.25. 1. Under H_0 , X_1 is a centred Gaussian, it is therefore a continuous symmetric random variable, so

$$P(X_1 < 0) = P(X_1 \leq 0) = P(X_1 \geq 0) = P(X_1 > 0) = 0.5.$$

2. The test statistic is

$$\begin{aligned} T(X_1, \dots, X_{50}) &= 50 \left(\frac{\left(\frac{\sum_{i=1}^{50} \mathbf{1}_{(-\infty, 0)}(X_i)}{50} - 0.5 \right)^2}{0.5} + \frac{\left(\frac{\sum_{i=1}^{50} \mathbf{1}_{[0, +\infty)}(X_i)}{50} - 0.5 \right)^2}{0.5} \right) \\ &= 100 \left(\left(\frac{\sum_{i=1}^{50} \mathbf{1}_{(-\infty, 0)}(X_i)}{50} - 0.5 \right)^2 + \left(\frac{\sum_{i=1}^{50} \mathbf{1}_{[0, +\infty)}(X_i)}{50} - 0.5 \right)^2 \right). \end{aligned}$$

There are $2 - 1 = 1$ degrees of freedom as we partitioned the space into two classes, and there are therefore $2 - 1$ parameters being estimated.

3. By assumption that 50 is large enough, we can assume that T follows a χ_1^2 law to determine a rejection region. As we want a confidence level 0.9, we chose a rejection region of the form

$$\{T(X_1, \dots, X_{50}) > \delta\},$$

with δ such that $P(Z \leq \delta) = 0.9$ with $Z \sim \chi_1^2$. Looking at Table C.5, we find that $\delta = 2.7$ does the job.

4. In the given realisation of the sample, we have that there are 35 non-negative numbers and 15 negative numbers. So, $\sum_{i=1}^{50} \mathbf{1}_{(-\infty, 0)}(x_i) = 15$, and $\sum_{i=1}^{50} \mathbf{1}_{[0, +\infty)}(x_i) = 35$, and thus

$$T(x_1, \dots, x_{50}) = 100 \left(\left(\frac{15}{50} - 0.5 \right)^2 + \left(\frac{35}{50} - 0.5 \right)^2 \right) = 100 \left(\frac{1}{25} + \frac{1}{25} \right) = 8$$

which is well above $\delta = 2.7$, so we reject H_0 in this case.

Appendix B

Revision exercises

This chapter contains exercises that are “typical exam questions”.

B.1 Revision Chapter 1

Exercise B.1. We have an alphabet of 12 different symbols.

1. How many passwords of 10 characters can you create using these symbols only?
2. How many passwords of 10 characters using at most once each symbol can you create using these symbols only?
3. How many passwords of 10 characters using these symbols only can you create that are such that every symbol present in the password is used exactly twice?

Solution in [B.5](#)

Exercise B.2. We need to form a committee of 3 men and 3 women (6 people in total), out of a group of 17 people containing 8 men and 9 women. How many different committee can you form if (the constraints of each point apply only to that point)

1. there are no constraints?
2. 2 of the men refuse to be part of the same committee?
3. 2 of the women refuse to be part of the same committee?
4. 1 man and 1 woman refuse to be part of the same committee?

Solution in [B.5](#)

Exercise B.3. You jump above three rivers to get to a lost temple. In each river, there is a crocodile who attempt to eat you when you jump above the river. The crocodile in the first river has a success rate of 15%. The crocodile in the second river has a success rate of 5%. The crocodile in the third river has a success rate of 30%. All probability are rounded to three digits.

1. What is the probability that you arrive to the lost temple alive?

A) 0.7 B) 0.575 C) 0.565 D) 0.585

2. What is the probability that you get eaten by the third crocodile?

A) 0.242 B) 0.3 C) 0.437 D) 0.255

Solution in [B.5](#)

B.2 Revision Chapter 2

Exercise B.4. Consider the set $\Omega = \{1, 2, 3, 4\}$. Which of the following functions $p : \Omega \rightarrow \mathbb{R}$ are probability mass functions on Ω ?

A) $p(i) = \frac{i}{10}$ B) $p(i) = \frac{(-1)^i i}{2}$ C) $p(i) = \frac{i^2}{31}$ D) $p(i) = \frac{1+(-1)^i}{4}$.

Solution in [B.5](#)

Exercise B.5. Let X, Y be two independent identically distributed continuous random variables with density function

$$f(x) = \mathbf{1}_{[0,+\infty)}(x)e^{-x}.$$

1. What is the cumulative distribution function of $\max(X, Y)$ ($F_{\max(X, Y)}$)?

2. Give the density function of the random variable $\max(X, Y)$.

Solution in [B.5](#)

Exercise B.6. Model the following random experiments using a discrete probability space.

1. Toss an unfair coin that falls on “head” five times more often as on “tail”.

2. Roll an unfair dice that rolls 6 twice as often as any other fixed number between 1 and 5.

Solution in [B.5](#)

Exercise B.7. Model the following random experiments using a continuous random variable in one of the following families:

- Exponential random variables: one parameter $\lambda > 0$, density $f_\lambda(x) = \mathbb{1}_{[0,+\infty)}(x)\lambda e^{-\lambda x}$.
 - Gaussian random variable: two parameters $\mu \in \mathbb{R}$, $\sigma^2 > 0$, density $f_{\mu,\sigma^2}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.
1. Pick an individual in a population uniformly at random and measure the difference between its height and 1.50 meters. You know for a fact that the average height in that population is 1.7 meters and that the average square of height is 3.61.
 2. You try to model the time in minutes it takes before being admitted to the hospital for a broken arm, and you know that the average time people have to wait is 52.456 minutes.

Solution in [B.5](#)

Exercise B.8. You have two identical boxes containing marbles with the same shape, texture, mass, volume,... In box 1, there are 23 blue, 12 green and 5 red. In box 2, there are 10 blue, 17 green and 13 red. You pick a box uniformly at random and then pick a marble in the box uniformly at random.

1. If the marble you picked is red, what is the probability that you picked box number 1?
2. If the marble you picked is blue, what is the probability that if you pick a second marble from the same box, this marble will also be blue? (the first marble is **not** put back in the box)

Solution in [B.5](#)

Exercise B.9. You have 3 fair dices with six faces each, a blue, a red, and a green one. The blue one has numbers 0, 0, 1, 3, 3, 4 on its faces, the red one has numbers 1, 1, 1, 1, 10, 10 on its faces, and the green one has numbers 0, 0, 0, 12, 20, 20 on its faces. You pick one of the dices uniformly at random and roll it three times. Let X_1, X_2, X_3 be the random variables giving the numbers you obtained in each roll.

1. What is the expected value of X_1 ?
2. What is the variance of X_1 ?

Suppose now that the sum of the three numbers you obtained is 12.

3. What is the probability that you picked the blue dice?

Solution in [B.5](#)

Exercise B.10. Let $X \sim \text{Exp}(2)$ (X continuous random variable with density $f_X(x) = 2e^{-2x}$). Let Y be a random variable with $E(Y) = 1$, $E(Y^2) = 3$. Let $Z \sim \text{Uni}(\{-1, 1\})$ (Z discrete random variable with $P(Z = 1) = P(Z = -1) = 0.5$). Suppose X, Y, Z form an independent family. Introduce the random variables

$$W_1 = XZ, \quad W_2 = X + Y, \quad W_3 = XYZ.$$

1. What is the value of $P(W_1 < 0)$?
2. Show that $E(X) = 0.5$, $E(X^2) = 0.5$?
3. What is the value of $\text{Var}(W_2)$?
4. Show that $E(W_3^2) = 1.5$. What is the value of $\text{Var}(W_3)$?
5. What is the value of $\text{Cov}(W_2, W_3)$?
6. Show that $P(|W_3| \geq 3) \leq \frac{1}{6}$.

Solution in [B.5](#)

Exercise B.11. Recall that there will be 28 days in February 2026. Let $X_1, \dots, X_{28}$ be the number of people breaking a leg in Glacier 3000 during each day of February. Let N be the number of sunny days during February 2026. Suppose that we take the following for granted:

- Nobody breaks a leg during non-sunny days.
- The number of broken legs during a day has the same law every day, the probability that the sky is sunny or not is the same every day.
- Each day is sunny or not independently of the other days.
- The number of broken legs during a day is independent of the number of broken legs during the other days, and of the sun status of the other days.
- $E(X_i | \text{day } i \text{ is sunny}) \leq 5$, and $E(N) \leq 14$.

Answer the following points.

1. Show that for every $i = 1, \dots, 28$, $P(\text{day } i \text{ is sunny}) \leq 0.5$.
2. Letting Z be the number of broken legs during February 2026 in Glacier 3000, show that $P(Z \geq 140) \leq 0.5$.

Solution in [B.5](#)

Exercise B.12. Let $X_1, \dots, X_{10}$ be identically distributed random variables, and let $Y_1, \dots, Y_{10}$ be identically distributed random variables (with possibly a different law than the one of the X_i 's). Suppose that

$$E(X_1) = 2, \quad E(X_1^2) \leq 9, \quad E(Y_1) = 3, \quad E(Y_1^2) \leq 16.$$

Answer the following points.

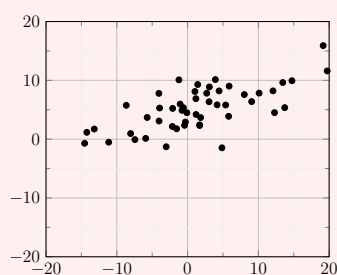
1. Show that for any $i, j \in \{1, \dots, 10\}$, $\text{Cov}(X_i, Y_j) \leq 6$.
2. Let $S = \sum_{i=1}^{10} X_i$, and $Z = \sum_{i=1}^{10} Y_i$. Show that $\text{Cov}(S, Z) \leq 600$.

Solution in [B.5](#)

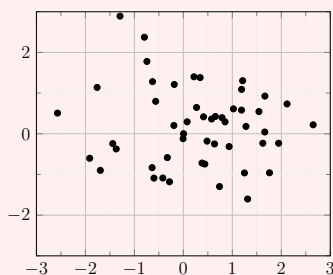
Exercise B.13. 1. Recall the definition of the coefficient of correlation, ρ_{XY} , between two random variables X, Y .

2. Associate each of the values of ρ_{XY} to a graph of samples from the law of (X, Y) .

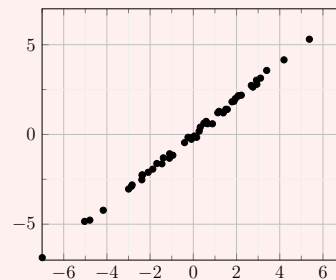
(A) $\rho_{XY} = 0.99$, (B) $\rho_{XY} = 0.4$, (C) $\rho_{XY} = 0$.



(a)



(b)



(c)

3. Suppose $|\rho_{XY}| = 1$. What can you say about the relation between X and Y ?
4. Suppose $\rho_{XY} = 1$, $E(X) = 0$, $E(Y) = 1$, $E(X^2) = 1$, $E(Y^2) = 2$. What can you say about the relation between X and Y ?

Solution in [B.5](#)

B.3 Revision Chapter 3

Exercise B.14. Is the following statement true: “ Suppose $X_1, X_2, \dots$ is a sequence of random variables with $E(|X_n|) \leq n^{-1}$. Then, X_n converges in probability towards 0.”

True False

Solution in [B.5](#)

Exercise B.15. Let $X_1, X_2, \dots$ be an independent sequence of identically distributed random variables. Suppose that X_1 is a continuous random variable with density f . In which of the following cases do we have that the strong Law of Large Numbers implies that $\frac{1}{n} \sum_{i=1}^n X_i$ converge almost-surely to $E(X_1)$ as $n \rightarrow \infty$?

- A) $f(x) = \frac{1}{2}e^{-|x|}$ B) $f(x) = \mathbf{1}_{[1,+\infty)}(x)x^{-2}$
 C) $f(x) = \frac{1}{100}\mathbf{1}_{[1,e^{100}]}(x)x^{-1}$ D) $f(x) = 2\mathbf{1}_{[1,+\infty)}(x)x^{-3}$

Solution in [B.5](#)

Exercise B.16. Let $X_1, X_2, \dots$ be an independent sequence of identically distributed random variables. Suppose that X_1 is a continuous random variable with density $f(x) = \frac{c}{1+|x|^3}$, where $c = \left(\int_{-\infty}^{\infty} \frac{1}{1+|x|^3} dx\right)^{-1}$. Is the following statement true or false: “the Central Limit Theorem implies that $\frac{1}{\sqrt{n\text{Var}(X_1)}} \sum_{i=1}^n X_i$ converges in law (as $n \rightarrow \infty$) towards a Gaussian $\mathcal{N}(0, 1)$ random variable”

True False

Solution in ??

Exercise B.17. Let $X_1, X_2, \dots$ be an independent sequence of identically distributed random variables. Suppose that

$$E(X_1) = 0, \quad E(X_1^2) = 1.$$

Let $S_n = \sum_{i=1}^n X_i$ for $n \geq 1$.

1. Compute $E(S_n^2)$.
2. Show that for any $\epsilon > 0$, and any $\alpha > 1/2$,

$$\lim_{n \rightarrow \infty} P(|n^{-\alpha} S_n| \geq \epsilon) = 0.$$

Solution in [B.5](#)

B.4 Revision Chapter 4

Exercise B.18. Let $\Theta = \mathbb{R} \times (0, +\infty)$ be the parameter space. Let $\theta = (\mu, \sigma^2) \in \Theta$. For $n \geq 1$, let $X_1, \dots, X_n$ be an n -sample of law $\mathcal{N}(\mu, \sigma^2)$ (continuous probability measure on $\mathbb{R}$ with density $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$).

Answer the following True/False question (True if the statement is always true in this context, False if it admits at least one counter-example.).

1. For $n \geq 1$, $Z_n = \frac{1}{n} \sum_{i=1}^n X_i^2$ is a convergent sequence of estimators of $f(\theta) = \sigma^2 + \mu^2$.

True

False

2. $\frac{1}{100} \sum_{i=1}^{100} (X_i - \mu)^2$ is an estimator of σ^2 .

True

False

3. $\frac{1}{98} \sum_{i=1}^{100} X_i^3$ is an unbiased estimator of $E(X_1^3)$.

True

False

Solution in [B.5](#)

Exercise B.19. Let $\Theta = (1, +\infty)$, and $X_1, \dots, X_n$ an n -sample of law $\mathbb{P}_\theta$, where $\mathbb{P}_\theta$, $\theta \in \Theta$, is the continuous law on $\mathbb{R}$ with density

$$f_\theta(x) = \mathbf{1}_{[1, \infty)}(x)(\theta - 1)x^{-\theta}.$$

Compute the maximum likelihood estimator of θ .

Solution in [B.5](#)

Exercise B.20. Let $X_1, \dots, X_{64}$ be a sample of law $\mathcal{N}(\mu, 4)$, $\mu \in \mathbb{R}$ (continuous law with density $\frac{1}{\sqrt{8\pi}}e^{-(x-\mu)^2/8}$). We want to find a confidence interval of level $1 - \alpha$ for $\mu = E(X_1)$ which is centred around the empirical mean.

1. What is the law of $\frac{X_1 - \mu}{2}$?
2. What is the law of $\sum_{i=1}^{64} X_i$?
3. Find the wanted confidence interval for $\alpha = 0.07$ using the table at the end of the exercise.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.4	0.419	0.421	0.422	0.424	0.425	0.426	0.428	0.429	0.431	0.432
1.5	0.433	0.434	0.436	0.437	0.438	0.439	0.441	0.442	0.443	0.444
1.6	0.445	0.446	0.447	0.448	0.449	0.451	0.452	0.453	0.454	0.454
1.7	0.455	0.456	0.457	0.458	0.459	0.460	0.461	0.462	0.462	0.463
1.8	0.464	0.465	0.466	0.466	0.467	0.468	0.469	0.469	0.470	0.471
1.9	0.471	0.472	0.473	0.473	0.474	0.474	0.475	0.476	0.476	0.477
2.0	0.477	0.478	0.478	0.479	0.479	0.480	0.480	0.481	0.481	0.482
2.1	0.482	0.483	0.483	0.483	0.484	0.484	0.485	0.485	0.485	0.486

Figure B.2: Table of $z \mapsto P(0 \leq Z \leq z)$ with $Z \sim \mathcal{N}(0, 1)$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.

Solution in [B.5](#)

Exercise B.21. We want to test whether the average duration of the Lausanne-Genève train travel using the IC is indeed what is announced on the SBB schedule, 49mn, or not. We assume that the duration of a fixed travel is Gaussian with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$. Let $X_1, \dots, X_n$ be an n -sample of travel durations (in minutes). We perform a one-sample t-test to test the null Hypotheses $H_0: \mu = 49$.

1. Recall the test statistic $T_n(X_1, \dots, X_n)$ of the one-sample t-test.

Under the null hypotheses, this statistic follows a $\text{Student}_t(n-1)$ law.

2. Using the table given at the end of the exercise, given a rejection region for this test with risk level 0.1 when the sample size $n = 100$.

You measure the duration times of 100 travels (i.e.: you find a realisation of a 100-sample) $x_1, \dots, x_{100}$ such that

$$\sum_{i=1}^{100} x_i = 4960, \quad \sum_{i=1}^{100} \left(x_i - \frac{1}{100} \sum_{j=1}^{100} x_j \right)^2 = 3939.62.$$

3. Based on the rejection region constructed at the previous point, do you reject the null Hypotheses?

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1	0.34	0.343	0.345	0.347	0.35	0.352	0.354	0.356	0.359	0.361
1.1	0.363	0.365	0.367	0.369	0.371	0.374	0.376	0.378	0.38	0.382
1.2	0.383	0.385	0.387	0.389	0.391	0.393	0.395	0.396	0.398	0.4
1.3	0.402	0.403	0.405	0.407	0.408	0.41	0.412	0.413	0.415	0.416
1.4	0.418	0.419	0.421	0.422	0.423	0.425	0.426	0.428	0.429	0.43
1.5	0.432	0.433	0.434	0.435	0.437	0.438	0.439	0.44	0.441	0.442
1.6	0.444	0.445	0.446	0.447	0.448	0.449	0.45	0.451	0.452	0.453
1.7	0.454	0.455	0.456	0.457	0.458	0.458	0.459	0.46	0.461	0.462
1.8	0.463	0.463	0.464	0.465	0.466	0.466	0.467	0.468	0.468	0.469
1.9	0.47	0.47	0.471	0.472	0.472	0.473	0.474	0.474	0.475	0.475
2	0.476	0.476	0.477	0.477	0.478	0.478	0.479	0.479	0.48	0.48
2.1	0.481	0.481	0.482	0.482	0.483	0.483	0.483	0.484	0.484	0.485

Figure B.3: Tables of $z \mapsto P(0 \leq X \leq z)$ with $X \sim \text{Student}_t(99)$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.

Solution in [B.5](#)

Exercise B.22. We have a 50-sample $X_1, \dots, X_{50}$ of law $\mathbb{P}$. We want to test whether $\mathbb{P}$ is a uniform law on $[0, 10]$ or not using a χ^2 adequacy test. We partition $\mathbb{R}$ into $\Omega_1 = (-\infty, 2.5)$, $\Omega_2 = [2.5, 5)$, $\Omega_3 = [5, 7.5)$, $\Omega_4 = [7.5, +\infty)$.

Let $T(X_1, \dots, X_{50})$ be the test statistic of the χ^2 -adequacy test for the null Hypotheses H_0 : “ $\mathbb{P}(\Omega_i) = P(U \in \Omega_i)$ for all $i = 1, \dots, 4$ ” where $U \sim \text{Uni}([0, 10])$ is a uniform random variable on $[0, 10]$. We use a rejection region of the form $\{T(X_1, \dots, X_{50}) > \delta\}$ with $\delta > 0$ so that $P(Z \leq \delta) = 0.95$ with $Z \sim \chi_3^2$.

You find the realisation

$(x_1, \dots, x_{50}) = (3.7, 5.4, 5.5, 4.3, 1.6, 3.9, 0.1, 6.9, 2.8, 4.6, 4.7, 5.0, 7.7, 5.2, 6.7, 4.4, 8.0, 2.9, 5.5, 5.1, 4.2, 2.5, 4.6, 2.6, 7.5, 5.4, 6.2, 6.5, 6.3, 5.2, 5.4, 7.9, 2.3, 1.6, 4.6, 7.9, 5.5, 6.4, 3.0, 5.9, 4.1, 5.9, 5.0, 5.7, 6.5, 4.1, 2.1, 4.7, 7.6, 9.5)$

Do you reject H_0 ?

Yes

No

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
7	0.928	0.928	0.929	0.929	0.929	0.93	0.93	0.93	0.931	0.931
7.1	0.931	0.932	0.932	0.932	0.932	0.933	0.933	0.933	0.934	0.934
7.2	0.934	0.935	0.935	0.935	0.935	0.936	0.936	0.936	0.937	0.937
7.3	0.937	0.937	0.938	0.938	0.938	0.938	0.939	0.939	0.939	0.94
7.4	0.94	0.94	0.94	0.941	0.941	0.941	0.941	0.942	0.942	0.942
7.5	0.942	0.943	0.943	0.943	0.943	0.944	0.944	0.944	0.944	0.945
7.6	0.945	0.945	0.945	0.946	0.946	0.946	0.946	0.947	0.947	0.947
7.7	0.947	0.948	0.948	0.948	0.948	0.949	0.949	0.949	0.949	0.949
7.8	0.95	0.95	0.95	0.95	0.951	0.951	0.951	0.951	0.951	0.952
7.9	0.952	0.952	0.952	0.953	0.953	0.953	0.953	0.953	0.954	0.954
8	0.954	0.954	0.954	0.955	0.955	0.955	0.955	0.955	0.956	0.956

Figure B.4: Table of $z \mapsto P(Z \leq z)$ with $Z \sim \chi_3^2$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.

Solution in [B.5](#)

Exercise B.23. We have two independent Gaussian samples, $X_1, \dots, X_n$, and $Y_1, \dots, Y_n$. We want to test the null Hypotheses H_0 : “ $\text{Var}(X_1) = \text{Var}(Y_1)$ ” using a F-test.

1. Recall the F-test statistic for equality of variances in our case.
2. Using the table given below, construct a rejection region using that statistic to obtain a test with confidence level 0.94 when $n = 100$.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.5	0.000	0.000	0.001	0.001	0.001	0.002	0.002	0.003	0.004	0.005
0.6	0.006	0.007	0.009	0.011	0.014	0.017	0.020	0.024	0.028	0.033
0.7	0.039	0.045	0.052	0.060	0.068	0.077	0.087	0.098	0.109	0.121
0.8	0.134	0.148	0.163	0.178	0.194	0.210	0.227	0.245	0.263	0.282
0.9	0.301	0.320	0.340	0.359	0.379	0.400	0.420	0.440	0.460	0.480
1.0	0.500	0.520	0.539	0.558	0.577	0.596	0.614	0.631	0.649	0.666
1.1	0.682	0.698	0.713	0.728	0.742	0.756	0.769	0.782	0.794	0.806
1.2	0.817	0.828	0.838	0.848	0.857	0.866	0.874	0.882	0.889	0.897
1.3	0.903	0.910	0.916	0.921	0.926	0.931	0.936	0.941	0.945	0.948
1.4	0.952	0.956	0.959	0.962	0.964	0.967	0.969	0.972	0.974	0.976
1.5	0.978	0.979	0.981	0.982	0.984	0.985	0.986	0.987	0.988	0.989

Figure B.5: Table of $z \mapsto P(Z \leq z)$ with $Z \sim F(99, 99)$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.

Solution in B.5

Exercise B.24. Let $X_1, \dots, X_n$ be an n -sample of some law $\mathbb{P}_\theta$, $\theta \in [1, 10]$. We are performing a Hypotheses test for the null hypotheses H_0 : “the parameter θ is less than 2” using the rejection region $\{T_n(X_1, \dots, X_n) > \delta\}$ where T_n is some statistic and $\delta > 0$ is some number.

1. Assuming that $T_n(X_1, \dots, X_n)$ is uniformly distributed on $[0, \theta]$, what is the value of δ needed to have a test with confidence level 90%?
2. Still assuming that $T_n(X_1, \dots, X_n)$ is uniformly distributed on $[0, \theta]$, what is the power of the test as a function of δ ?

Solution in B.5

Exercise B.25. For security protocol reasons, you need to understand what is the maximal temperature a lobbyist can survive during 24 hours. You test the maximal temperature of the lobbyists from 4 different countries: France, US, Switzerland, and Italy. We know for a fact (from physical considerations) that nobody can survive 100 degrees during 24h, and that everyone survives to 25 degrees during 24h. We assume that the maximal temperature someone can survive is randomly distributed in the population of each countries, and its law depends only on the average maximal temperature someone of this country can survive.

1. Based on those informations/assumptions, what is the parameter space Θ to be considered in the estimation of the maximal temperatures lobbyists from France, US, Switzerland, and Italy can survive?
2. What is the subset of the parameter space $\Theta_0 \subset \Theta$ corresponding to the null Hypotheses “Italy lobbyists are more resistant than US lobbyists that are more resistant than France lobbyists that are more resistant than Switzerland lobbyists”?

Solution in [B.5](#)

B.5 Solutions

Solution to Exercise B.1. 1. 12^{10}

2. $\frac{12!}{2}$

3. $\binom{12}{5} \cdot \binom{10}{2} \cdot \binom{8}{2} \cdot \binom{6}{2} \cdot \binom{4}{2}$

Solution to Exercise B.2. 1. $\binom{8}{3} \cdot \binom{9}{3}$

2. $\binom{9}{3} \cdot ((\binom{6}{3}) + 2 \cdot \binom{6}{2})$

3. $\binom{8}{3} \cdot ((\binom{7}{3}) + 2 \cdot \binom{7}{2})$

4. $\binom{7}{3} \cdot \binom{8}{3} + \binom{7}{2} \cdot \binom{8}{3} + \binom{7}{3} \cdot \binom{8}{2}$

Solution to Exercise B.3. 1. C)

2. A)

Solution to Exercise B.4. A) and D)

Short explanation: B) takes negative values, C) does not sum up to 1.

Solution to Exercise B.5. 1. For $t \in \mathbb{R}$, as X, Y are independent and identically distributed,

$$\begin{aligned} F_{\max(X,Y)}(t) &= P(\max(X, Y) \leq t) = P(X \leq t, Y \leq t) \\ &= P(X \leq t)P(Y \leq t) = P(X \leq t)^2. \end{aligned}$$

Now, $P(X \leq t) = 0$ if $t \leq 0$ as $f_X(x) = 0$ for $x \leq 0$. For $t > 0$, $P(X \leq t) = \int_0^t e^{-x} dx = 1 - e^{-t}$. So,

$$F_{\max(X,Y)}(t) = \mathbf{1}_{(0,+\infty)}(t)(1 - e^{-t})^2.$$

2. As the max of the positive random variables is a.s. positive, $f_{\max(X,Y)}(x) = 0$ for all $x \leq 0$. Then, the CDF is a primitive of the density function, so we can try $f_{\max(X,Y)}(x) = C + F'_{\max(X,Y)}(x)$ for some constant C . This gives for $x > 0$,

$$f_{\max(X,Y)}(x) = C + 2(1 - e^{-x})e^{-x}.$$

We compute the value of C . First, $C \geq 0$ as $f_{\max(X,Y)}(x) \geq 0$ for all x . Then, C must satisfy $\int_0^\infty (C + 2(1 - e^{-x})e^{-x})dx = 1$, so $C = 0$. We obtain

$$f_{\max(X,Y)}(x) = \mathbb{1}_{(0,+\infty)}(x)2(1 - e^{-x})e^{-x}.$$

Solution to Exercise B.6. 1. One can take as realisation space $\Omega = \{\text{Head}, \text{Tail}\}$. Then, the probability mass function $p : \Omega \rightarrow [0, 1]$ must satisfy

$$p(\text{Head}) + p(\text{Tail}) = 1, \quad p(\text{Head}) = 5p(\text{Tail}),$$

$$\text{so } p(\text{Head}) = \frac{5}{6}, \quad p(\text{Tail}) = \frac{1}{6}.$$

2. One can take as realisation space $\Omega = \{1, 2, 3, 4, 5, 6\}$. Then, the probability mass function $p : \Omega \rightarrow [0, 1]$ must satisfy

$$p(1) + p(2) + p(3) + p(4) + p(5) + p(6) = 1, \quad p(6) = 2p(i) \quad i = 1, \dots, 5.$$

so $p(1) = p(2) = p(3) = p(4) = p(5) = \frac{p(6)}{2}$. This gives

$$p(6) = \frac{2}{7}, \quad p(1) = p(2) = p(3) = p(4) = p(5) = \frac{1}{7}.$$

Solution to Exercise B.7. 1. We need to model a random number that can be both positive and negative, we therefore use a Gaussian random variable X . We need to find the values of

$$\mu = E(X), \quad \sigma^2 = \text{Var}(X).$$

X is equal to a random height minus 1.5. As the average height in the population is 1.7, we want $E(X + 1.5) = 1.7$, so $\mu = 0.2$. As the average square of height is 3.61, we want to have $E((X + 1.5)^2) = 3.61$, so

$$\begin{aligned} \sigma^2 = \text{Var}(X) &= \text{Var}(X + 1.5) = E((X + 1.5)^2) - E(X + 1.5)^2 \\ &= 3.61 - 1.7^2 = 3.61 - 2.89 = 0.72. \end{aligned}$$

2. We want to model a random positive number, and exponential random variables are well suited for waiting time. We use an exponential Y of parameter λ to determine. We have

$$E(Y) = \int_0^\infty y\lambda e^{-\lambda y} dy = [-ye^{-\lambda y}]_0^\infty + \int_0^\infty e^{-\lambda y} dy = [-\lambda^{-1}e^{-\lambda y}]_0^\infty = \frac{1}{\lambda}.$$

As the average waiting time is 52.456 minutes, we want to take $\lambda = \frac{1}{52.456}$.

Solution to Exercise B.8. Let $\text{Box}_1, \text{Box}_2$ be the event that box 1, resp. 2, is picked, and $1_R, 1_B, 1_G$ be the events that the first drawn marble is, respectively, red, blue, green. Let $2_R, 2_B, 2_G$ be the events that the second drawn marble is, respectively, red, blue, green.

1. We want to find $P(\text{Box}_1 | 1_R)$. By Bayes law and the formula of total probability,

$$\begin{aligned} P(\text{Box}_1 | 1_R) &= \frac{P(1_R | \text{Box}_1)P(\text{Box}_1)}{P(1_R)} \\ &= \frac{P(1_R | \text{Box}_1)P(\text{Box}_1)}{P(1_R | \text{Box}_1)P(\text{Box}_1) + P(1_R | \text{Box}_2)P(\text{Box}_2)} \\ &= \frac{\frac{5}{40} \cdot 0.5}{\frac{5}{40} \cdot 0.5 + \frac{13}{40} \cdot 0.5} = \frac{5}{5 + 13} = \frac{5}{18}. \end{aligned}$$

2. We want to find $P(2_B | 1_B)$. By the definition of conditional probability and the formula of total probability

$$\begin{aligned} P(2_B | 1_B) &= \frac{P(2_B, 1_B)}{P(1_B)} \\ &= \frac{P(2_B, 1_B | \text{Box}_1)P(\text{Box}_1) + P(2_B, 1_B | \text{Box}_2)P(\text{Box}_2)}{P(1_B | \text{Box}_1)P(\text{Box}_1) + P(1_B | \text{Box}_2)P(\text{Box}_2)} \\ &= \frac{\frac{23}{40} \cdot \frac{22}{39} \cdot 0.5 + \frac{10}{40} \cdot \frac{9}{39} \cdot 0.5}{\frac{23}{40} \cdot 0.5 + \frac{10}{40} \cdot 0.5} \\ &= \frac{23 \cdot \frac{22}{39} + 10 \cdot \frac{9}{39}}{23 + 10} = \frac{23 \cdot 22 + 90}{39 \cdot 33} = \frac{596}{1287} \approx 0.46 \end{aligned}$$

Solution to Exercise B.9. Let $\text{Dice}_B, \text{Dice}_R, \text{Dice}_G$ be the events that the picked dice is, respectively, blue, red, green.

1. By the formula of total expectation,

$$\begin{aligned} E(X_1) &= E(X_1 | \text{Dice}_B)P(\text{Dice}_B) + E(X_1 | \text{Dice}_R)P(\text{Dice}_R) \\ &\quad + E(X_1 | \text{Dice}_G)P(\text{Dice}_G) \\ &= \frac{0 + 0 + 1 + 3 + 3 + 4}{6} \cdot \frac{1}{3} + \frac{1 + 1 + 1 + 1 + 10 + 10}{6} \cdot \frac{1}{3} \\ &\quad + \frac{0 + 0 + 0 + 12 + 20 + 20}{6} \cdot \frac{1}{3} \\ &= \frac{11}{18} + \frac{24}{18} + \frac{52}{18} = \frac{87}{18} \approx 4.83. \end{aligned}$$

2. Let us first compute $E(X_1^2)$. Proceeding as in the first point,

$$\begin{aligned} E(X_1^2) &= E(X_1^2 | \text{Dice}_B)P(\text{Dice}_B) + E(X_1^2 | \text{Dice}_R)P(\text{Dice}_R) \\ &\quad + E(X_1^2 | \text{Dice}_G)P(\text{Dice}_G) \\ &= \frac{0 + 0 + 1 + 9 + 9 + 16}{6} \cdot \frac{1}{3} + \frac{1 + 1 + 1 + 1 + 100 + 100}{6} \cdot \frac{1}{3} \\ &\quad + \frac{0 + 0 + 0 + 144 + 400 + 400}{6} \cdot \frac{1}{3} \\ &= \frac{35}{18} + \frac{204}{18} + \frac{944}{18} = \frac{1183}{18}. \end{aligned}$$

So,

$$\text{Var}(X_1) = E(X_1^2) - E(X_1)^2 = \frac{1183}{18} - \frac{87^2}{18^2} = \frac{1183 \cdot 18 - 87^2}{324} \approx 42,36.$$

3. We want to compute $P(\text{Dice}_B | X_1 + X_2 + X_3 = 12)$. Introduce the event $A = \{X_1 + X_2 + X_3 = 12\}$. By Bayes formula, and the formula of total probability,

$$\begin{aligned} P(\text{Dice}_B | A) &= \frac{P(A | \text{Dice}_B)P(\text{Dice}_B)}{P(A)} \\ &= \frac{P(A | \text{Dice}_B)P(\text{Dice}_B)}{P(A | \text{Dice}_B)P(\text{Dice}_B) + P(A | \text{Dice}_R)P(\text{Dice}_R) + P(A | \text{Dice}_G)P(\text{Dice}_G)} \\ &= \frac{P(A | \text{Dice}_B)}{P(A | \text{Dice}_B) + P(A | \text{Dice}_R) + P(A | \text{Dice}_G)} \end{aligned}$$

as $P(\text{Dice}_B) = P(\text{Dice}_R) = P(\text{Dice}_G) = 1/3$. Now, the only way to get 12 with three rolls of the blue dice is to make a 4 each time, so

$$P(A | \text{Dice}_B) = P(X_1 = X_2 = X_3 = 4 | \text{Dice}_B) = \left(\frac{1}{6}\right)^3 = \frac{1}{6^3}.$$

Then, to get 12 with three rolls of the red dice, one has to make 10 once and twice 1, so

$$\begin{aligned} P(A | \text{Dice}_R) &= P(X_1 = X_2 = 1, X_3 = 10 | \text{Dice}_R) \\ &\quad + P(X_1 = X_3 = 1, X_2 = 10 | \text{Dice}_R) + P(X_3 = X_2 = 1, X_1 = 10 | \text{Dice}_R) \\ &= 3 \cdot \left(\frac{4}{6}\right)^2 \cdot \frac{2}{6} = \frac{96}{6^3}. \end{aligned}$$

Finally, to get 12 with three rolls of the green dice, one has to make 12 once and twice 0, so

$$\begin{aligned} P(A | \text{Dice}_G) &= P(X_1 = X_2 = 0, X_3 = 12 | \text{Dice}_G) \\ &\quad + P(X_1 = X_3 = 0, X_2 = 12 | \text{Dice}_G) + P(X_3 = X_2 = 0, X_1 = 12 | \text{Dice}_G) \\ &= 3 \cdot \left(\frac{3}{6}\right)^2 \cdot \frac{1}{6} = \frac{27}{6^3}. \end{aligned}$$

We therefore get

$$\begin{aligned} P(\text{Dice}_B | A) &= \frac{P(A | \text{Dice}_B)}{P(A | \text{Dice}_B) + P(A | \text{Dice}_R) + P(A | \text{Dice}_G)} \\ &= \frac{\frac{1}{6^3}}{\frac{1}{6^3} + \frac{96}{6^3} + \frac{27}{6^3}} = \frac{1}{1 + 96 + 27} = \frac{1}{124}. \end{aligned}$$

Solution to Exercise B.10. 1. $f_X(x) = 0$ for $x \leq 0$, so $P(X \leq 0) = 0$. So,

$$\begin{aligned} P(W_1 < 0) &= P(XZ < 0, X > 0) = P(Z < 0, X > 0) \\ &= P(Z = -1)P(X > 0) = 0.5 \end{aligned}$$

where we used that X, Z are independent.

2. As X is continuous with density f_X ,

$$\begin{aligned} E(X) &= \int_{-\infty}^{+\infty} f_X(x)xdx = 2 \int_0^{\infty} xe^{-2x}dx \\ &= [-xe^{-2x}]_0^{\infty} + \int_0^{\infty} e^{-2x}dx = 0 + [-e^{-2x}/2]_0^{\infty} = 0.5. \end{aligned}$$

In the same fashion,

$$\begin{aligned} E(X^2) &= 2 \int_0^{\infty} x^2 e^{-2x}dx = [-x^2 e^{-2x}]_0^{\infty} + 2 \int_0^{\infty} xe^{-2x}dx \\ &= 0 + [-xe^{-2x}]_0^{\infty} + \int_0^{\infty} e^{-2x}dx = 0 + [-e^{-2x}/2]_0^{\infty} = 0.5. \end{aligned}$$

3. As X, Y are independent, $\text{Var}(W_2) = \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$. Then, by the previous point,

$$\begin{aligned} E(X) &= 0.5, \quad E(X^2) = 0.5, \quad \text{Var}(X) = E(X^2) - E(X)^2 = 0.25, \\ \text{Var}(Y) &= E(Y^2) - E(Y)^2 = 3 - 1 = 2. \end{aligned}$$

So, $\text{Var}(W_2) = 0.25 + 2 = 2.25$.

4. We have that as X, Y, Z are independent, $E(W_3) = E(XYZ) = E(X)E(Y)E(Z) = 0$, as $E(Z) = 0.5 \cdot \dots \cdot 1 + 0.5 \cdot (-1) = 0$. Then, still as X, Y, Z are independent,

$$E(W_3^2) = E(X^2Y^2Z^2) = E(X^2)E(Y^2)E(Z^2) = 0.5 \cdot 3 \cdot 1 = 1.5$$

as $P(Z^2 = 1) = 1$. So, $\text{Var}(W_3) = E(W_3^2) - E(W_3)^2 = 1.5 - 0^2 = 1.5$.

5. As $E(W_3) = 0 = E(Z)$ as in the previous points

$$\begin{aligned} \text{Cov}(W_2, W_3) &= E(W_2W_3) - E(W_2)E(W_3) = E((X + Y)XYZ) \\ &= E((X + Y)XY)E(Z) = 0, \end{aligned}$$

as Z and $(X + Y)XY$ are independent.

6. By Chebychev's inequality,

$$P(|W_3| \geq 3) \leq \frac{E(|W_3|^2)}{9} = \frac{E(W_3^2)}{9} = \frac{1.5}{9} = \frac{1}{6}.$$

Solution to Exercise B.11. Let $Y_1, \dots, Y_{28}$ be random variables such that Y_i is 1 if February the i th is sunny, and 0 else. Then, we know from the assumptions that $((X_i, Y_i))_{i=1, \dots, 28}$ is an independent family of random vectors. Moreover, we have $Z = \sum_{i=1}^{28} X_i$, $E(X_i | Y_i = 1) \leq 5$, and $P(X_i = 0 | Y_i = 0) = 0$.

1. We have that the Y_i 's are identically distributed Bernoulli random variables of unknown parameter $p = P(Y_i = 1)$. We know that $14 \geq E(N) = \sum_{i=1}^{28} E(X_i) = 28p$. So $p \leq \frac{14}{28} = 0.5$.

2. We have that

$$\begin{aligned} E(Z) &= \sum_{i=1}^{28} E(X_i) = \sum_{i=1}^{28} (E(X_i | Y_i = 1)P(Y_i = 1) + E(X_i | Y_i = 0)P(Y_i = 0)) \\ &= \sum_{i=1}^{28} E(X_i | Y_i = 1)P(Y_i = 1) \leq \sum_{i=1}^{28} 5 \cdot 0.5 = 70. \end{aligned}$$

So, by Markov's inequality, $P(Z \geq 140) \leq \frac{70}{140} = 0.5$.

Solution to Exercise B.12. 1. We have that $\text{Cov}(X_i, Y_j) = E(X_i Y_j) - E(X_i)E(Y_j) = E(X_i Y_j) - 6$. Now, by the Cauchy-Schwarz inequality, $E(X_i Y_j) \leq \sqrt{E(X_i^2)E(Y_j^2)} \leq \sqrt{9 \cdot 16} = 12$. So, $\text{Cov}(X_i, Y_j) \leq 12 - 6 = 6$.

2. We have

$$\text{Cov}(S, Z) = \text{Cov}\left(\sum_{i=1}^{10} X_i, \sum_{j=1}^{10} Y_j\right) = \sum_{i=1}^{10} \sum_{j=1}^{10} \text{Cov}(X_i, Y_j) \leq 100 \cdot 6,$$

by the first point.

Solution to Exercise B.13. 1. $\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$.

2. (A) (c), (B) (a), (C) (b).

3. We can say that there are $a, b \in \mathbb{R}$ such that $Y = aX + b$.

4. We can say that $Y = X + 1$. Indeed, as in the previous point, there are $a, b \in \mathbb{R}$ such that $Y = aX + b$, and we thus have $1 = E(Y) = aE(X) + b = b$, so $b = 1$, and $2 = E(Y^2) = a^2 E(X^2) + 2aE(X) + 1 = a^2 + 1$, so $a \in \{-1, 1\}$. Finally, $\rho_{XY} > 0$, so $0 < \text{Cov}(X, Y) = \text{Cov}(X, aX + 1) = a\text{Var}(X)$, so $a > 0$, so $a = 1$.

Solution to Exercise B.14. True

Short explanation: by Markov's inequality, $P(|X_n| \geq \epsilon) \leq \frac{E(|X_n|)}{\epsilon} \xrightarrow{n \rightarrow \infty} 0$.

Solution to Exercise B.15. A), C), and D).

Short explanation: to apply the sLLN, one needs $E(|X_1|) < \infty$.

Solution to Exercise B.17. 1. We have that, as X_i, X_j are independent if $i \neq j$,

$$\begin{aligned} E(S_n^2) &= E\left(\left(\sum_{i=1}^n X_i\right)^2\right) = E\left(\sum_{i=1}^n \sum_{j=1}^n X_i X_j\right) = \sum_{i=1}^n \sum_{j=1}^n E(X_i X_j) \\ &= \sum_{i=1}^n E(X_i^2) + 2 \sum_{1 \leq i < j \leq n} E(X_i)E(X_j) = nE(X_1^2) = n. \end{aligned}$$

2. Let $\epsilon > 0$. By Chebychev's inequality applied to $|n^{-\alpha} S_n|$,

$$P(|n^{-\alpha} S_n| \geq \epsilon) \leq \frac{E(|n^{-\alpha} S_n|^2)}{\epsilon^2} = \frac{n^{-2\alpha} E(S_n^2)}{\epsilon^2} = \frac{n^{1-2\alpha}}{\epsilon^2} \xrightarrow{n \rightarrow \infty} 0,$$

as $\alpha > 1/2$, so $1 - 2\alpha < 0$.

Solution to Exercise B.18. 1. True

Short explanation: by the LLN, Z_n is a convergent sequence of estimators of $E(X_1^2) = \text{Var}(X_1) + E(X_1)^2 = \sigma^2 + \mu^2$.

2. False

Short explanation: the statistic depends on the parameter, so not an estimator.

3. False

Short explanation: $E(\frac{1}{98} \sum_{i=1}^{100} X_i^3) = \frac{100}{98} E(X_1^3) \neq E(X_1^3)$ whenever $E(X_1^3) \neq 0$ (which is the case when $\mu \neq 0$).

Solution to Exercise B.19. For an realization $x_1, \dots, x_n \in [1, +\infty]$ of $X_1, \dots, X_n$, we have that the likelihood function gives

$$\mathcal{L}(x_1, \dots, x_n | \theta) = \prod_{i=1}^n f_{\theta}(x_i) = (\theta - 1)^n \prod_{i=1}^n x_i^{-\theta}.$$

We look at the value of θ which maximizes this function. We can equivalently search for the value which minimizes $-\ln(\mathcal{L})$:

$$-\ln(\mathcal{L}(x_1, \dots, x_n | \theta)) = -n \ln(\theta - 1) + \theta \sum_{i=1}^n \ln(x_i).$$

The first derivative in θ of this is

$$\frac{d}{d\theta} -\ln(\mathcal{L}(x_1, \dots, x_n | \theta)) = \frac{-n}{\theta - 1} + \sum_{i=1}^n \ln(x_i).$$

The second derivative in θ of this is

$$\frac{d^2}{d\theta^2} -\ln(\mathcal{L}(x_1, \dots, x_n | \theta)) = \frac{n}{(\theta - 1)^2} > 0,$$

so the function is strictly convex, its unique minima is attained at the critical point θ_* which solves the equation

$$0 = \frac{-n}{\theta_* - 1} + \sum_{i=1}^n \ln(x_i).$$

This gives $\theta_* = 1 + \frac{n}{\sum_{i=1}^n \ln(x_i)}$. The maximum likelihood estimator is therefore given by

$$\text{MLE}(X_1, \dots, X_n) = 1 + \frac{n}{\sum_{i=1}^n \ln(X_i)}.$$

Solution to Exercise B.20. 1. As X_1 a Gaussian random variable with mean μ and variance 4, $\frac{X_1 - \mu}{2}$ is a Gaussian random variable with mean 0 and variance 1: $\frac{X_1 - \mu}{2} \sim \mathcal{N}(0, 1)$.

2. The sum of independent Gaussian random variables is again a Gaussian random variable. The mean is given by $E(\sum_{i=1}^{64} X_i) = 64E(X_1) = 64\mu$, and the variance is given by $\text{Var}(\sum_{i=1}^{64} X_i) = 64\text{Var}(X_1) = 64 \cdot 4 = 256$, so $\sum_{i=1}^{64} X_i \sim \mathcal{N}(64\mu, 256)$.

3. Letting $\bar{X}_{64} = \frac{1}{64} \sum_{i=1}^{64}$ denote the empirical mean, we are looking for a confidence interval of the form $[\bar{X}_{64} - \delta, \bar{X}_{64} + \delta]$ with $\delta > 0$ such that $P(\mu \in [\bar{X}_{64} - \delta, \bar{X}_{64} + \delta]) = 0.93$. Then,

$$\begin{aligned} P(\mu \in [\bar{X}_{64} - \delta, \bar{X}_{64} + \delta]) &= P(|\bar{X}_{64} - \mu| \leq \delta) \\ &= P\left(\left|\frac{\sqrt{64}(\bar{X}_{64} - \mu)}{\sqrt{\text{Var}(X_1)}}\right| \leq \frac{\sqrt{64} \cdot \delta}{\sqrt{\text{Var}(X_1)}}\right) \\ &= P\left(\left|\frac{8(\bar{X}_{64} - \mu)}{2}\right| \leq \frac{8 \cdot \delta}{2}\right) \end{aligned}$$

as $\text{Var}(X_1) = 4$. Now, by the properties of Gaussian random variables, $\frac{8(\bar{X}_{64} - \mu)}{2} \sim \mathcal{N}(0, 1)$, so, letting $Z \sim \mathcal{N}(0, 1)$, we want $\delta > 0$ such that

$$0.93 = P(|Z| \leq 4\delta) = 2P(0 \leq Z \leq 4\delta),$$

looking at the table, we find $4\delta = 1.81$, so $\delta = 0.4525$.

Solution to Exercise B.21. 1. The test statistic of the one-sample t-test is

$$T_n(X_1, \dots, X_n) = \frac{\sqrt{n}(\bar{X}_n - 49)}{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2}}$$

where $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ is the empirical mean.

2. The rejection region for this test is of the form $\{|T_n(X_1, \dots, X_n)| \geq \delta\}$ for some $\delta > 0$. We want to find such a δ satisfying $P(|T_{100}(X_1, \dots, X_{100})| \leq \delta) = 0.9$. Now, under our assumptions, $T_{100}(X_1, \dots, X_{100}) \sim \text{Student}_t(99)$. As Student random variables are symmetric, we have

$$P(|T_{100}(X_1, \dots, X_{100})| \leq \delta) = 2P(0 \leq T_{100}(X_1, \dots, X_{100}) \leq \delta).$$

Using the table given at the end of the exercise, we find that $\delta = 1.66$ does the job.

3. With the given realisation, we have

$$|T_{100}(x_1, \dots, x_{100})| = \frac{\sqrt{100}(\frac{4960}{100} - 49)}{\sqrt{\frac{3939.62}{99}}} = \frac{6}{\sqrt{\frac{3939.62}{99}}} \leq \frac{6}{\sqrt{36}} = 1 < 1.66$$

so we do not reject H_0 .

Solution to Exercise B.22. Yes

Short explanation: as $P(U \in \Omega_i) = 0.25$ for $i = 1, \dots, 4$, the test statistic is $T(X_1, \dots, X_{50}) = 50 \sum_{i=1}^4 \frac{(\frac{N_i(X_1, \dots, X_{50})}{50} - 0.25)^2}{0.25}$ with $N_i(X_1, \dots, X_{50})$ the number of X_j 's that are in Ω_i . In our case,

$$\begin{aligned} T(x_1, \dots, x_{50}) &= \frac{50}{0.25} \left(\left(\frac{5}{50} - 0.25 \right)^2 + \left(\frac{17}{50} - 0.25 \right)^2 + \left(\frac{21}{50} - 0.25 \right)^2 + \left(\frac{7}{50} - 0.25 \right)^2 \right) \\ &= \frac{50}{0.25} \left(\left(\frac{15}{100} \right)^2 + \left(\frac{9}{100} \right)^2 + \left(\frac{17}{100} \right)^2 + \left(\frac{11}{100} \right)^2 \right) = \frac{716}{50} = 14.32. \end{aligned}$$

From the table, we get that the rejection region is $\{T(X_1, \dots, X_{50}) > 5.98\}$.

Solution to Exercise B.23. 1. As the two sample sizes are equal, the test statistic is $T_n = \frac{\sum_{i=1}^n (X_i - \bar{X}_n)^2}{\sum_{i=1}^n (Y_i - \bar{Y}_n)^2}$ where $\bar{X}_n, \bar{Y}_n$ are the empirical means.

2. We can use for example a rejection region of the form $\{T_{100} \notin [a, b]\}$ for $0 < a < b$ with $P(T_{100} < a) = 0.03 = P(T_{100} > b)$. Under our assumptions (Gaussian), $T_{100} \sim F(99, 99)$, so the table gives $a = 0.69$ and $b = 1.47$, so we can take the rejection region $\{T_{100} \notin [0.69, 1.47]\}$.

Solution to Exercise B.24. 1. $\delta = 1.8$: we want $\sup_{\theta \in [1, 2]} P(T_n > \delta) = 0.1$, and we have $P(T_n > \delta) = \max(0, \frac{\theta - \delta}{\theta})$ by assumption.

2. $\max(0, 1 - \frac{\delta}{2})$: the max probability of type II error is $\beta = \sup_{\theta \in (2, 10]} P(T_n \leq \delta)$, and $P(T_n \leq \delta) = \min(1, \frac{\delta}{\theta})$ by assumption.

Solution to Exercise B.25. 1. We need to estimate one temperature per country, and we know that each of these temperature is between 25 and 100, so we can take $\Theta = [25, 100]^{\{I, F, US, S\}}$.

2. $\Theta_0 = \{(t_I, t_F, t_{US}, t_S) \in \Theta : t_I \geq t_{US} \geq t_F \geq t_S\}$.

Appendix C

Repartition Tables

C.1 Gaussian

Recall that the $\mathcal{N}(0, 1)$ distribution is a continuous distribution with density

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

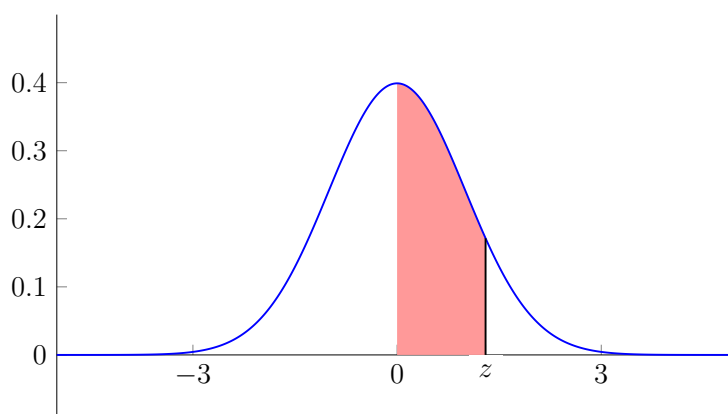


Figure C.1: The area in red is $P(0 \leq X \leq z)$ with $X \sim \mathcal{N}(0, 1)$.

When using the next table, it is useful to remember that if $X \sim \mathcal{N}(0, 1)$ and $t \geq 0$,

$$\begin{aligned} P(X \geq t) &= P(X > t) = 1 - P(X \leq t), \\ P(X \leq t) &= P(0 \leq X \leq t) + P(X \leq 0) = \frac{1}{2} + P(0 \leq X \leq t), \\ P(|X| \leq t) &= P(0 \leq X \leq t) + P(-t \leq X \leq 0) = 2P(0 \leq X \leq t). \end{aligned}$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.000	0.004	0.008	0.012	0.016	0.020	0.024	0.028	0.032	0.036
0.1	0.040	0.044	0.048	0.052	0.056	0.060	0.064	0.067	0.071	0.075
0.2	0.079	0.083	0.087	0.091	0.095	0.099	0.103	0.106	0.110	0.114
0.3	0.118	0.122	0.126	0.129	0.133	0.137	0.141	0.144	0.148	0.152
0.4	0.155	0.159	0.163	0.166	0.170	0.174	0.177	0.181	0.184	0.188
0.5	0.191	0.195	0.198	0.202	0.205	0.209	0.212	0.216	0.219	0.222
0.6	0.226	0.229	0.232	0.236	0.239	0.242	0.245	0.249	0.252	0.255
0.7	0.258	0.261	0.264	0.267	0.270	0.273	0.276	0.279	0.282	0.285
0.8	0.288	0.291	0.294	0.297	0.300	0.302	0.305	0.308	0.311	0.313
0.9	0.316	0.319	0.321	0.324	0.326	0.329	0.331	0.334	0.336	0.339
1.0	0.341	0.344	0.346	0.348	0.351	0.353	0.355	0.358	0.360	0.362
1.1	0.364	0.367	0.369	0.371	0.373	0.375	0.377	0.379	0.381	0.383
1.2	0.385	0.387	0.389	0.391	0.393	0.394	0.396	0.398	0.400	0.401
1.3	0.403	0.405	0.407	0.408	0.410	0.411	0.413	0.415	0.416	0.418
1.4	0.419	0.421	0.422	0.424	0.425	0.426	0.428	0.429	0.431	0.432
1.5	0.433	0.434	0.436	0.437	0.438	0.439	0.441	0.442	0.443	0.444
1.6	0.445	0.446	0.447	0.448	0.449	0.451	0.452	0.453	0.454	0.454
1.7	0.455	0.456	0.457	0.458	0.459	0.460	0.461	0.462	0.462	0.463
1.8	0.464	0.465	0.466	0.466	0.467	0.468	0.469	0.469	0.470	0.471
1.9	0.471	0.472	0.473	0.473	0.474	0.474	0.475	0.476	0.476	0.477
2.0	0.477	0.478	0.478	0.479	0.479	0.480	0.480	0.481	0.481	0.482
2.1	0.482	0.483	0.483	0.483	0.484	0.484	0.485	0.485	0.485	0.486
2.2	0.486	0.486	0.487	0.487	0.487	0.488	0.488	0.488	0.489	0.489
2.3	0.489	0.490	0.490	0.490	0.490	0.491	0.491	0.491	0.491	0.492
2.4	0.492	0.492	0.492	0.492	0.493	0.493	0.493	0.493	0.493	0.494

Figure C.2: Tables of $z \mapsto P(0 \leq X \leq z)$ with $X \sim \mathcal{N}(0, 1)$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.

C.2 Chi-squared

Recall that the χ_k^2 distribution is a continuous distribution with density

$$f(x) = \mathbb{1}_{[0,+\infty)}(x) \frac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{k}{2}} \Gamma(\frac{k}{2})}.$$

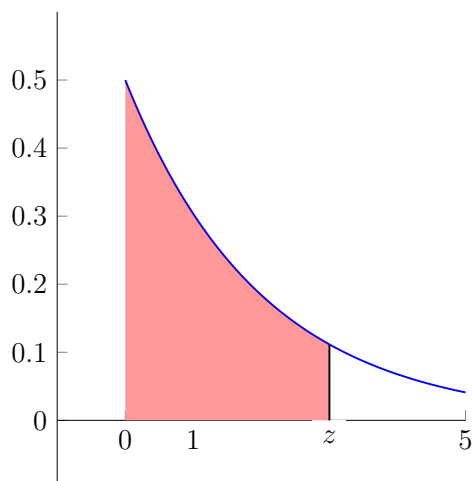


Figure C.3: The area in red is $P(0 \leq X \leq z)$ with $X \sim \chi_2^2$.

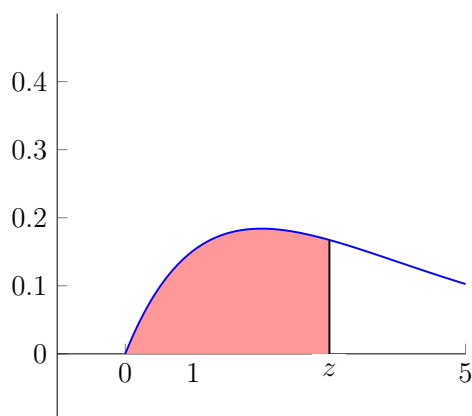


Figure C.4: The area in red is $P(0 \leq X \leq z)$ with $X \sim \chi_4^2$.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0	0.08	0.112	0.138	0.159	0.177	0.194	0.209	0.223	0.236
0.1	0.248	0.26	0.271	0.282	0.292	0.301	0.311	0.32	0.329	0.337
0.2	0.345	0.353	0.361	0.368	0.376	0.383	0.39	0.397	0.403	0.41
0.3	0.416	0.422	0.428	0.434	0.44	0.446	0.451	0.457	0.462	0.468
0.4	0.473	0.478	0.483	0.488	0.493	0.498	0.502	0.507	0.512	0.516
0.5	0.52	0.525	0.529	0.533	0.538	0.542	0.546	0.55	0.554	0.558
0.6	0.561	0.565	0.569	0.573	0.576	0.58	0.583	0.587	0.59	0.594
0.7	0.597	0.601	0.604	0.607	0.61	0.614	0.617	0.62	0.623	0.626
0.8	0.629	0.632	0.635	0.638	0.641	0.643	0.646	0.649	0.652	0.655
0.9	0.657	0.66	0.663	0.665	0.668	0.67	0.673	0.675	0.678	0.68
1.0	0.683	0.685	0.687	0.69	0.692	0.694	0.697	0.699	0.701	0.704
1.1	0.706	0.708	0.71	0.712	0.714	0.716	0.719	0.721	0.723	0.725
1.2	0.727	0.729	0.731	0.733	0.735	0.736	0.738	0.74	0.742	0.744
1.3	0.746	0.748	0.749	0.751	0.753	0.755	0.756	0.758	0.76	0.762
1.4	0.763	0.765	0.767	0.768	0.77	0.771	0.773	0.775	0.776	0.778
1.5	0.779	0.781	0.782	0.784	0.785	0.787	0.788	0.79	0.791	0.793
1.6	0.794	0.796	0.797	0.798	0.8	0.801	0.802	0.804	0.805	0.806
1.7	0.808	0.809	0.81	0.812	0.813	0.814	0.815	0.817	0.818	0.819
1.8	0.82	0.821	0.823	0.824	0.825	0.826	0.827	0.829	0.83	0.831
1.9	0.832	0.833	0.834	0.835	0.836	0.837	0.838	0.84	0.841	0.842
2.0	0.843	0.844	0.845	0.846	0.847	0.848	0.849	0.85	0.851	0.852
2.1	0.853	0.854	0.855	0.856	0.856	0.857	0.858	0.859	0.86	0.861
2.2	0.862	0.863	0.864	0.865	0.866	0.866	0.867	0.868	0.869	0.87
2.3	0.871	0.871	0.872	0.873	0.874	0.875	0.876	0.876	0.877	0.878
2.4	0.879	0.879	0.88	0.881	0.882	0.882	0.883	0.884	0.885	0.885
2.5	0.886	0.887	0.888	0.888	0.889	0.89	0.89	0.891	0.892	0.892
2.6	0.893	0.894	0.894	0.895	0.896	0.896	0.897	0.898	0.898	0.899
2.7	0.9	0.9	0.901	0.902	0.902	0.903	0.903	0.904	0.905	0.905
2.8	0.906	0.906	0.907	0.907	0.908	0.909	0.909	0.91	0.91	0.911
2.9	0.911	0.912	0.913	0.913	0.914	0.914	0.915	0.915	0.916	0.916
3.0	0.917	0.917	0.918	0.918	0.919	0.919	0.92	0.92	0.921	0.921
3.1	0.922	0.922	0.923	0.923	0.924	0.924	0.925	0.925	0.925	0.926
3.2	0.926	0.927	0.927	0.928	0.928	0.929	0.929	0.929	0.93	0.93
3.3	0.931	0.931	0.932	0.932	0.932	0.933	0.933	0.934	0.934	0.934
3.4	0.935	0.935	0.936	0.936	0.936	0.937	0.937	0.938	0.938	0.938
3.5	0.939	0.939	0.939	0.94	0.94	0.94	0.941	0.941	0.942	0.942
3.6	0.942	0.943	0.943	0.943	0.944	0.944	0.944	0.945	0.945	0.945
3.7	0.946	0.946	0.946	0.947	0.947	0.947	0.948	0.948	0.948	0.948
3.8	0.949	0.949	0.949	0.95	0.95	0.95	0.951	0.951	0.951	0.951
3.9	0.952	0.952	0.952	0.953	0.953	0.953	0.953	0.954	0.954	0.954
4.0	0.954	0.955	0.955	0.955	0.956	0.956	0.956	0.956	0.957	0.957
4.1	0.957	0.957	0.958	0.958	0.958	0.958	0.959	0.959	0.959	0.959
4.2	0.96	0.96	0.96	0.96	0.961	0.961	0.961	0.961	0.961	0.962
4.3	0.962	0.962	0.962	0.963	0.963	0.963	0.963	0.963	0.964	0.964
4.4	0.964	0.964	0.964	0.965	0.965	0.965	0.965	0.966	0.966	0.966

Figure C.5: Tables of $z \mapsto P(X \leq z) = P(0 \leq X \leq z)$ with $X \sim \chi_1^2$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
2.0	0.632	0.634	0.636	0.638	0.639	0.641	0.643	0.645	0.647	0.648
2.1	0.65	0.652	0.654	0.655	0.657	0.659	0.66	0.662	0.664	0.665
2.2	0.667	0.669	0.67	0.672	0.674	0.675	0.677	0.679	0.68	0.682
2.3	0.683	0.685	0.687	0.688	0.69	0.691	0.693	0.694	0.696	0.697
2.4	0.699	0.7	0.702	0.703	0.705	0.706	0.708	0.709	0.711	0.712
2.5	0.713	0.715	0.716	0.718	0.719	0.721	0.722	0.723	0.725	0.726
2.6	0.727	0.729	0.73	0.732	0.733	0.734	0.736	0.737	0.738	0.739
2.7	0.741	0.742	0.743	0.745	0.746	0.747	0.748	0.75	0.751	0.752
2.8	0.753	0.755	0.756	0.757	0.758	0.759	0.761	0.762	0.763	0.764
2.9	0.765	0.767	0.768	0.769	0.77	0.771	0.772	0.773	0.775	0.776
3.0	0.777	0.778	0.779	0.78	0.781	0.782	0.783	0.785	0.786	0.787
3.1	0.788	0.789	0.79	0.791	0.792	0.793	0.794	0.795	0.796	0.797
3.2	0.798	0.799	0.8	0.801	0.802	0.803	0.804	0.805	0.806	0.807
3.3	0.808	0.809	0.81	0.811	0.812	0.813	0.814	0.815	0.815	0.816
3.4	0.817	0.818	0.819	0.82	0.821	0.822	0.823	0.824	0.824	0.825
3.5	0.826	0.827	0.828	0.829	0.83	0.831	0.831	0.832	0.833	0.834
3.6	0.835	0.836	0.836	0.837	0.838	0.839	0.84	0.84	0.841	0.842
3.7	0.843	0.844	0.844	0.845	0.846	0.847	0.847	0.848	0.849	0.85
3.8	0.85	0.851	0.852	0.853	0.853	0.854	0.855	0.856	0.856	0.857
3.9	0.858	0.858	0.859	0.86	0.861	0.861	0.862	0.863	0.863	0.864
4.0	0.865	0.865	0.866	0.867	0.867	0.868	0.869	0.869	0.87	0.871
4.1	0.871	0.872	0.873	0.873	0.874	0.874	0.875	0.876	0.876	0.877
4.2	0.878	0.878	0.879	0.879	0.88	0.881	0.881	0.882	0.882	0.883
4.3	0.884	0.884	0.885	0.885	0.886	0.886	0.887	0.888	0.888	0.889
4.4	0.889	0.89	0.89	0.891	0.891	0.892	0.892	0.893	0.894	0.894
4.5	0.895	0.895	0.896	0.896	0.897	0.897	0.898	0.898	0.899	0.899
4.6	0.9	0.9	0.901	0.901	0.902	0.902	0.903	0.903	0.904	0.904
4.7	0.905	0.905	0.906	0.906	0.907	0.907	0.907	0.908	0.908	0.909
4.8	0.909	0.91	0.91	0.911	0.911	0.912	0.912	0.912	0.913	0.913
4.9	0.914	0.914	0.915	0.915	0.915	0.916	0.916	0.917	0.917	0.918
5.0	0.918	0.918	0.919	0.919	0.92	0.92	0.92	0.921	0.921	0.922
5.1	0.922	0.922	0.923	0.923	0.923	0.924	0.924	0.925	0.925	0.925
5.2	0.926	0.926	0.926	0.927	0.927	0.928	0.928	0.928	0.929	0.929
5.3	0.929	0.93	0.93	0.93	0.931	0.931	0.931	0.932	0.932	0.932
5.4	0.933	0.933	0.933	0.934	0.934	0.934	0.935	0.935	0.935	0.936
5.5	0.936	0.936	0.937	0.937	0.937	0.938	0.938	0.938	0.939	0.939
5.6	0.939	0.939	0.94	0.94	0.94	0.941	0.941	0.941	0.942	0.942
5.7	0.942	0.942	0.943	0.943	0.943	0.944	0.944	0.944	0.944	0.945
5.8	0.945	0.945	0.946	0.946	0.946	0.946	0.947	0.947	0.947	0.947
5.9	0.948	0.948	0.948	0.948	0.949	0.949	0.949	0.949	0.95	0.95
6.0	0.95	0.95	0.951	0.951	0.951	0.951	0.952	0.952	0.952	0.952

Figure C.6: Tables of $z \mapsto P(X \leq z) = P(0 \leq X \leq z)$ with $X \sim \chi^2_2$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
5.0	0.828	0.829	0.83	0.83	0.831	0.832	0.833	0.833	0.834	0.835
5.1	0.835	0.836	0.837	0.837	0.838	0.839	0.84	0.84	0.841	0.842
5.2	0.842	0.843	0.844	0.844	0.845	0.846	0.846	0.847	0.848	0.848
5.3	0.849	0.85	0.85	0.851	0.851	0.852	0.853	0.853	0.854	0.855
5.4	0.855	0.856	0.856	0.857	0.858	0.858	0.859	0.86	0.86	0.861
5.5	0.861	0.862	0.863	0.863	0.864	0.864	0.865	0.865	0.866	0.867
5.6	0.867	0.868	0.868	0.869	0.869	0.87	0.871	0.871	0.872	0.872
5.7	0.873	0.873	0.874	0.874	0.875	0.876	0.876	0.877	0.877	0.878
5.8	0.878	0.879	0.879	0.88	0.88	0.881	0.881	0.882	0.882	0.883
5.9	0.883	0.884	0.884	0.885	0.885	0.886	0.886	0.887	0.887	0.888
6.0	0.888	0.889	0.889	0.89	0.89	0.891	0.891	0.892	0.892	0.893
6.1	0.893	0.894	0.894	0.895	0.895	0.895	0.896	0.896	0.897	0.897
6.2	0.898	0.898	0.899	0.899	0.9	0.9	0.9	0.901	0.901	0.902
6.3	0.902	0.903	0.903	0.903	0.904	0.904	0.905	0.905	0.905	0.906
6.4	0.906	0.907	0.907	0.908	0.908	0.908	0.909	0.909	0.91	0.91
6.5	0.91	0.911	0.911	0.912	0.912	0.912	0.913	0.913	0.913	0.914
6.6	0.914	0.915	0.915	0.915	0.916	0.916	0.916	0.917	0.917	0.918
6.7	0.918	0.918	0.919	0.919	0.919	0.92	0.92	0.92	0.921	0.921
6.8	0.921	0.922	0.922	0.922	0.923	0.923	0.924	0.924	0.924	0.925
6.9	0.925	0.925	0.926	0.926	0.926	0.926	0.927	0.927	0.927	0.928
7.0	0.928	0.928	0.929	0.929	0.929	0.93	0.93	0.93	0.931	0.931
7.1	0.931	0.932	0.932	0.932	0.932	0.933	0.933	0.933	0.934	0.934
7.2	0.934	0.935	0.935	0.935	0.935	0.936	0.936	0.936	0.937	0.937
7.3	0.937	0.937	0.938	0.938	0.938	0.938	0.939	0.939	0.939	0.94
7.4	0.94	0.94	0.94	0.941	0.941	0.941	0.941	0.942	0.942	0.942
7.5	0.942	0.943	0.943	0.943	0.943	0.944	0.944	0.944	0.944	0.945
7.6	0.945	0.945	0.945	0.946	0.946	0.946	0.946	0.947	0.947	0.947
7.7	0.947	0.948	0.948	0.948	0.948	0.949	0.949	0.949	0.949	0.949
7.8	0.95	0.95	0.95	0.95	0.951	0.951	0.951	0.951	0.951	0.952
7.9	0.952	0.952	0.952	0.953	0.953	0.953	0.953	0.953	0.954	0.954
8.0	0.954	0.954	0.954	0.955	0.955	0.955	0.955	0.955	0.956	0.956

Figure C.7: Tables of $z \mapsto P(X \leq z) = P(0 \leq X \leq z)$ with $X \sim \chi^2_3$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.

C.3 Student-t

Recall that the Student-t distribution with parameter $\nu > 0$ is a continuous distribution with density

$$f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}.$$

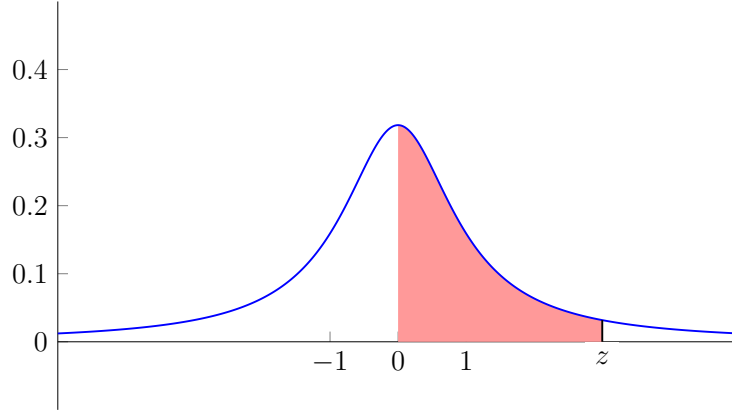


Figure C.8: The area in red is $P(0 \leq X \leq z)$ with $X \sim \text{Student}_t(1)$.

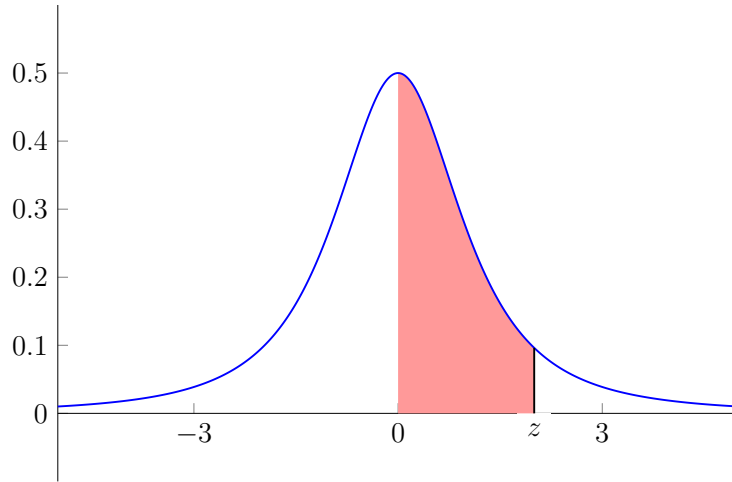


Figure C.9: The area in red is $P(0 \leq X \leq z)$ with $X \sim \text{Student}_t(2)$.

When using the next table, it is useful to remember that if $X \sim \text{Student}_t(\nu)$ and $t \geq 0$,

$$\begin{aligned} P(X \geq t) &= P(X > t) = 1 - P(X \leq t), \\ P(X \leq t) &= P(0 \leq X \leq t) + P(X \leq 0) = \frac{1}{2} + P(0 \leq X \leq t), \\ P(|X| \leq t) &= P(0 \leq X \leq t) + P(-t \leq X \leq 0) = 2P(0 \leq X \leq t). \end{aligned}$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
2.0	0.352	0.353	0.354	0.354	0.355	0.356	0.356	0.357	0.357	0.358
2.1	0.359	0.359	0.360	0.360	0.361	0.361	0.362	0.363	0.363	0.364
2.2	0.364	0.365	0.365	0.366	0.366	0.367	0.367	0.368	0.368	0.369
2.3	0.369	0.370	0.370	0.371	0.371	0.372	0.372	0.373	0.373	0.374
2.4	0.374	0.375	0.375	0.376	0.376	0.377	0.377	0.378	0.378	0.378
2.5	0.379	0.379	0.380	0.380	0.381	0.381	0.381	0.382	0.382	0.383
2.6	0.383	0.384	0.384	0.384	0.385	0.385	0.386	0.386	0.386	0.387
2.7	0.387	0.387	0.388	0.388	0.389	0.389	0.389	0.390	0.390	0.390
2.8	0.391	0.391	0.392	0.392	0.392	0.393	0.393	0.393	0.394	0.394
2.9	0.394	0.395	0.395	0.395	0.396	0.396	0.396	0.397	0.397	0.397
3.0	0.398	0.398	0.398	0.399	0.399	0.399	0.399	0.400	0.400	0.400
3.1	0.401	0.401	0.401	0.402	0.402	0.402	0.402	0.403	0.403	0.403
3.2	0.404	0.404	0.404	0.404	0.405	0.405	0.405	0.406	0.406	0.406
3.3	0.406	0.407	0.407	0.407	0.407	0.408	0.408	0.408	0.408	0.409
3.4	0.409	0.409	0.409	0.410	0.410	0.410	0.410	0.411	0.411	0.411
3.5	0.411	0.412	0.412	0.412	0.412	0.413	0.413	0.413	0.413	0.414
3.6	0.414	0.414	0.414	0.414	0.415	0.415	0.415	0.415	0.416	0.416
3.7	0.416	0.416	0.416	0.417	0.417	0.417	0.417	0.417	0.418	0.418
3.8	0.418	0.418	0.419	0.419	0.419	0.419	0.419	0.420	0.420	0.420
3.9	0.420	0.420	0.420	0.421	0.421	0.421	0.421	0.421	0.422	0.422
4.0	0.422	0.422	0.422	0.423	0.423	0.423	0.423	0.423	0.423	0.424
4.1	0.424	0.424	0.424	0.424	0.425	0.425	0.425	0.425	0.425	0.425
4.2	0.426	0.426	0.426	0.426	0.426	0.426	0.427	0.427	0.427	0.427
4.3	0.427	0.427	0.428	0.428	0.428	0.428	0.428	0.428	0.429	0.429
4.4	0.429	0.429	0.429	0.429	0.429	0.430	0.430	0.430	0.430	0.430
4.5	0.430	0.431	0.431	0.431	0.431	0.431	0.431	0.431	0.432	0.432
4.6	0.432	0.432	0.432	0.432	0.432	0.433	0.433	0.433	0.433	0.433
4.7	0.433	0.433	0.434	0.434	0.434	0.434	0.434	0.434	0.434	0.434
4.8	0.435	0.435	0.435	0.435	0.435	0.435	0.435	0.436	0.436	0.436
4.9	0.436	0.436	0.436	0.436	0.436	0.437	0.437	0.437	0.437	0.437
5.0	0.437	0.437	0.437	0.438	0.438	0.438	0.438	0.438	0.438	0.438
5.1	0.438	0.438	0.439	0.439	0.439	0.439	0.439	0.439	0.439	0.439
5.2	0.440	0.440	0.440	0.440	0.440	0.440	0.440	0.440	0.440	0.441
5.3	0.441	0.441	0.441	0.441	0.441	0.441	0.441	0.441	0.442	0.442
5.4	0.442	0.442	0.442	0.442	0.442	0.442	0.442	0.442	0.443	0.443
5.5	0.443	0.443	0.443	0.443	0.443	0.443	0.443	0.443	0.444	0.444
5.6	0.444	0.444	0.444	0.444	0.444	0.444	0.444	0.444	0.445	0.445
5.7	0.445	0.445	0.445	0.445	0.445	0.445	0.445	0.445	0.445	0.446
5.8	0.446	0.446	0.446	0.446	0.446	0.446	0.446	0.446	0.446	0.446
5.9	0.447	0.447	0.447	0.447	0.447	0.447	0.447	0.447	0.447	0.447
6.0	0.447	0.448	0.448	0.448	0.448	0.448	0.448	0.448	0.448	0.448
6.1	0.448	0.448	0.448	0.449	0.449	0.449	0.449	0.449	0.449	0.449
6.2	0.449	0.449	0.449	0.449	0.449	0.449	0.450	0.450	0.450	0.450

Figure C.10: Tables of $z \mapsto P(0 \leq X \leq z)$ with $X \sim \text{Student}_t(1)$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.0	0.289	0.291	0.292	0.294	0.296	0.298	0.300	0.302	0.303	0.305
1.1	0.307	0.309	0.310	0.312	0.314	0.315	0.317	0.319	0.320	0.322
1.2	0.323	0.325	0.327	0.328	0.330	0.331	0.333	0.334	0.336	0.337
1.3	0.338	0.340	0.341	0.343	0.344	0.345	0.347	0.348	0.349	0.350
1.4	0.352	0.353	0.354	0.356	0.357	0.358	0.359	0.360	0.361	0.363
1.5	0.364	0.365	0.366	0.367	0.368	0.369	0.370	0.372	0.373	0.374
1.6	0.375	0.376	0.377	0.378	0.379	0.380	0.381	0.382	0.383	0.383
1.7	0.384	0.385	0.386	0.387	0.388	0.389	0.390	0.391	0.391	0.392
1.8	0.393	0.394	0.395	0.396	0.396	0.397	0.398	0.399	0.400	0.400
1.9	0.401	0.402	0.403	0.403	0.404	0.405	0.405	0.406	0.407	0.408
2.0	0.408	0.409	0.410	0.410	0.411	0.412	0.412	0.413	0.413	0.414
2.1	0.415	0.415	0.416	0.417	0.417	0.418	0.418	0.419	0.419	0.420
2.2	0.421	0.421	0.422	0.422	0.423	0.423	0.424	0.424	0.425	0.425
2.3	0.426	0.426	0.427	0.427	0.428	0.428	0.429	0.429	0.430	0.430
2.4	0.431	0.431	0.432	0.432	0.433	0.433	0.433	0.434	0.434	0.435
2.5	0.435	0.436	0.436	0.436	0.437	0.437	0.438	0.438	0.438	0.439
2.6	0.439	0.440	0.440	0.440	0.441	0.441	0.441	0.442	0.442	0.443
2.7	0.443	0.443	0.444	0.444	0.444	0.445	0.445	0.445	0.446	0.446
2.8	0.446	0.447	0.447	0.447	0.448	0.448	0.448	0.449	0.449	0.449
2.9	0.449	0.450	0.450	0.450	0.451	0.451	0.451	0.451	0.452	0.452
3.0	0.452	0.453	0.453	0.453	0.453	0.454	0.454	0.454	0.454	0.455
3.1	0.455	0.455	0.455	0.456	0.456	0.456	0.456	0.457	0.457	0.457
3.2	0.457	0.458	0.458	0.458	0.458	0.458	0.459	0.459	0.459	0.459
3.3	0.460	0.460	0.460	0.460	0.460	0.461	0.461	0.461	0.461	0.461
3.4	0.462	0.462	0.462	0.462	0.462	0.463	0.463	0.463	0.463	0.463
3.5	0.464	0.464	0.464	0.464	0.464	0.464	0.465	0.465	0.465	0.465
3.6	0.465	0.466	0.466	0.466	0.466	0.466	0.466	0.467	0.467	0.467
3.7	0.467	0.467	0.467	0.468	0.468	0.468	0.468	0.468	0.468	0.468
3.8	0.469	0.469	0.469	0.469	0.469	0.469	0.469	0.470	0.470	0.470
3.9	0.470	0.470	0.470	0.470	0.471	0.471	0.471	0.471	0.471	0.471
4.0	0.471	0.472	0.472	0.472	0.472	0.472	0.472	0.472	0.472	0.473
4.1	0.473	0.473	0.473	0.473	0.473	0.473	0.473	0.474	0.474	0.474
4.2	0.474	0.474	0.474	0.474	0.474	0.474	0.475	0.475	0.475	0.475
4.3	0.475	0.475	0.475	0.475	0.475	0.476	0.476	0.476	0.476	0.476
4.4	0.476	0.476	0.476	0.476	0.476	0.477	0.477	0.477	0.477	0.477
4.5	0.477	0.477	0.477	0.477	0.477	0.477	0.478	0.478	0.478	0.478
4.6	0.478	0.478	0.478	0.478	0.478	0.478	0.478	0.479	0.479	0.479
4.7	0.479	0.479	0.479	0.479	0.479	0.479	0.479	0.479	0.479	0.480
4.8	0.480	0.480	0.480	0.480	0.480	0.480	0.480	0.480	0.480	0.480
4.9	0.480	0.480	0.481	0.481	0.481	0.481	0.481	0.481	0.481	0.481

Figure C.11: Tables of $z \mapsto P(0 \leq X \leq z)$ with $X \sim \text{Student}_t(2)$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.0	0.304	0.307	0.309	0.311	0.313	0.315	0.317	0.318	0.320	0.322
1.1	0.324	0.326	0.328	0.330	0.331	0.333	0.335	0.337	0.338	0.340
1.2	0.342	0.344	0.345	0.347	0.348	0.350	0.352	0.353	0.355	0.356
1.3	0.358	0.359	0.361	0.362	0.364	0.365	0.366	0.368	0.369	0.371
1.4	0.372	0.373	0.375	0.376	0.377	0.379	0.380	0.381	0.382	0.384
1.5	0.385	0.386	0.387	0.388	0.389	0.391	0.392	0.393	0.394	0.395
1.6	0.396	0.397	0.398	0.399	0.400	0.401	0.402	0.403	0.404	0.405
1.7	0.406	0.407	0.408	0.409	0.410	0.411	0.412	0.413	0.413	0.414
1.8	0.415	0.416	0.417	0.418	0.418	0.419	0.420	0.421	0.422	0.422
1.9	0.423	0.424	0.425	0.425	0.426	0.427	0.428	0.428	0.429	0.430
2.0	0.430	0.431	0.432	0.432	0.433	0.434	0.434	0.435	0.435	0.436
2.1	0.437	0.437	0.438	0.438	0.439	0.440	0.440	0.441	0.441	0.442
2.2	0.442	0.443	0.443	0.444	0.445	0.445	0.446	0.446	0.447	0.447
2.3	0.448	0.448	0.448	0.449	0.449	0.450	0.450	0.451	0.451	0.452
2.4	0.452	0.452	0.453	0.453	0.454	0.454	0.455	0.455	0.455	0.456
2.5	0.456	0.457	0.457	0.457	0.458	0.458	0.458	0.459	0.459	0.459
2.6	0.460	0.460	0.460	0.461	0.461	0.462	0.462	0.462	0.462	0.463
2.7	0.463	0.463	0.464	0.464	0.464	0.465	0.465	0.465	0.466	0.466
2.8	0.466	0.466	0.467	0.467	0.467	0.467	0.468	0.468	0.468	0.468
2.9	0.469	0.469	0.469	0.469	0.470	0.470	0.470	0.470	0.471	0.471
3.0	0.471	0.471	0.472	0.472	0.472	0.472	0.473	0.473	0.473	0.473
3.1	0.473	0.474	0.474	0.474	0.474	0.474	0.475	0.475	0.475	0.475
3.2	0.475	0.476	0.476	0.476	0.476	0.476	0.476	0.477	0.477	0.477
3.3	0.477	0.477	0.477	0.478	0.478	0.478	0.478	0.478	0.478	0.479
3.4	0.479	0.479	0.479	0.479	0.479	0.480	0.480	0.480	0.480	0.480
3.5	0.480	0.480	0.481	0.481	0.481	0.481	0.481	0.481	0.481	0.481
3.6	0.482	0.482	0.482	0.482	0.482	0.482	0.482	0.482	0.483	0.483
3.7	0.483	0.483	0.483	0.483	0.483	0.483	0.484	0.484	0.484	0.484
3.8	0.484	0.484	0.484	0.484	0.484	0.485	0.485	0.485	0.485	0.485
3.9	0.485	0.485	0.485	0.485	0.485	0.486	0.486	0.486	0.486	0.486
4.0	0.486	0.486	0.486	0.486	0.486	0.486	0.487	0.487	0.487	0.487
4.1	0.487	0.487	0.487	0.487	0.487	0.487	0.487	0.487	0.488	0.488
4.2	0.488	0.488	0.488	0.488	0.488	0.488	0.488	0.488	0.488	0.488
4.3	0.488	0.489	0.489	0.489	0.489	0.489	0.489	0.489	0.489	0.489
4.4	0.489	0.489	0.489	0.489	0.489	0.489	0.490	0.490	0.490	0.490
4.5	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490
4.6	0.490	0.490	0.490	0.491	0.491	0.491	0.491	0.491	0.491	0.491
4.7	0.491	0.491	0.491	0.491	0.491	0.491	0.491	0.491	0.491	0.491
4.8	0.491	0.491	0.491	0.492	0.492	0.492	0.492	0.492	0.492	0.492
4.9	0.492	0.492	0.492	0.492	0.492	0.492	0.492	0.492	0.492	0.492

Figure C.12: Tables of $z \mapsto P(0 \leq X \leq z)$ with $X \sim \text{Student}_t(3)$. The value z associated with the (i, j) entry in the table is the sum of the i th entry of the first line with the j th entry of the first column.